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Measurements of inclusive $B \rightarrow X_u \ell \nu$ decays and $|V_{ub}|$ with hadronic tagging

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Abstract

This analysis note presents the analysis details of the inclusive charmless semileptonic $B \rightarrow X_u \ell \nu$ decays. The study makes use of hadronic tagging and is performed on the LS1 data set. The $B \rightarrow X_u \ell \nu$ partial branching fraction and $|V_{ub}|$ in various phase-space regions are measured.

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1. Change Log

v2.0:

- Rephrased parts of the text for the sake of clarity
- Now using the full available MC15_ri data set
 - Updated continuum suppression MVA classifier
 - Updated $B \rightarrow X_c \ell \nu$ MVA classifier
 - Updated fit
 - Updated plots
- A linearity check of the fit was added
- A bug with the γ_S systematic calculation was fixed
- The description of the correction factor extraction has not been updated as a new strategy is currently being tested
- Added Table XVI: it lists the various fiducial regions used for the fit together with their corresponding acceptances
- Added Tables XVIII to XXII: the set of fits intended to be used for signal extraction together with their corresponding Asimov results is given
- A plan for the extraction of $|V_{ub}|$ was added

v3.0:

- Added information about train and test samples for MVAs
- Removed the kaon multiplicities from Tab. IX
- Added description of the exact procedure used to correct the $B \rightarrow X_c \ell \nu$ shape in Section 7.3.3
- A precise description of the new signal extraction setup is given in Section 8
- Updated all plots and branching fractions obtained from the fit in Section 8
- Added extracted values of $|V_{ub}|$ in Section 8.9
- Added distributions of $X_c \ell \nu$ suppression MVA input features in Appendix C
- Added information about `pyhf` as a fitting tool and how we use it in this analysis to extract the signal in Appendix E
- Added plots from all fits in Appendix F
- Added information about the impact of K_L on the analysis in Appendix G

v4.0:

- The recommended value of $f^{\pm/00}$ is now used; value obtained from Belle measurement [1] rather than PDG value
- The $B \rightarrow \eta \ell \nu$ was updated based on the latest PDG average
- The exclusive $B \rightarrow X_c \ell \nu$ branching fractions were updated based on the latest HFLAV public averages (2021)
- A section describing the systematic uncertainties related to the continuum calibration was added
- A table summarising all fit parameters was added to Appendix E
- The `nCDCHits` > 0 requirement for leptons was removed and hadron ID correction factors were computed with a `nCDCHits` > 20 requirement
- A description of the high-momentum $\pi^\pm$ selection was added in Section 3.2.3
- Descriptions of the X_u hadronisation modelling and fit-related uncertainties were added in Section 6
- Added in Appendix H plots to illustrate the efficiency and fake rate of kaon and pion ID for the likelihood-based and the neural-network-based scores
- Fits 2 and 6 were dropped as their output was estimated to not be as meaningful as for other fits. The numbering of other fits is kept the same to allow for easy comparison with previous versions of the analysis note.

v4.1:

- Added a fit systematics breakdown in Table XXIV
- Added data-MC comparison of p_ℓ^B , M_X and q^2 using off-resonance samples in Figure 11
- Added data-MC comparison of p_ℓ^B , M_X and q^2 in the wrong-sign-lepton region in Figure 32
- Updated plots and description in Appendix G
- Documented in Section 8.10 and Appendix I the following test proposed during working group review: repeat the fit procedure with data using only the control region $\text{CR}_{0,low}$ split in two in order to evaluate the $B \rightarrow X_c \ell \nu$ correction procedure

v4.2:

- Corrected Sections 3.2.1 and 3.2.5 as the events with multiple signal and ROE lepton candidates are now rejected
- Added low-lepton-momentum region plots in Appendix J
- A small bug was fixed when computing the input signal BR for the fit
- Documented the split lepton flavour e/μ in Section 8.11.2 and Appendix K

- Performed two additional split sample fits as discussed in Section 8.1.1.3
- Added Section 9 dedicated to the unblinded results; for now, a proposed unblinding plan is detailed in that Section
- Updated continuum reweighting plots in Section 4 and added plots of all variables used for training in Appendix L
- Added plots of the track and cluster multiplicities in Appendix M in order to study the topology differences between $B \rightarrow X_u \ell \nu$ and $B \rightarrow X_c \ell \nu$ events
- Added in Appendix N a study of a sample with an FEI cut set at $\mathcal{P}_{\text{FEI}} > 0.001$ instead of 0.01 as in our nominal sample
- Added in Section 7 details about the data and MC efficiencies for various cuts

v4.3

- A small bug in the template definition was fixed and the stability of fits was reconsidered: fit 7 and fit 1 with lepton flavour split are dropped from the list of results; in fit 3 and 5, the "other backgrounds" template normalisation is fixed because of the relatively low yields in the considered regions
- Added a test of the fit setup using $\text{CR}_{K,\text{low}}$ and $\text{CR}_{K,\text{high}}$ in Section 8.10.2
- Added a summary of the discussion with Peter Skands regarding the α_s parameter in Appendix O
- Added plots of the efficiency as a function of p_ℓ^B , M_X and q^2 in Appendix P

v4.4

- Use the 2D variable $p_\ell^B:q^2$ for the nominal fit; replaced figures and rephrased text wherever relevant
- Use the $\text{CR}_{K,\text{high}}$ region to produce pseudo-data for the signal region
- Rephrased the MC and data efficiency discussion in Section 7
- Show full MC efficiency in Appendix P using generator level samples

v4.5

- Updated $\mathcal{B}(B \rightarrow X_c \ell \nu)$ based on HFLAV 2023 and updated all relevant fit results
- Added section 8.10.3 to document the $\text{CR}_{0,\text{low}}\text{-}\text{CR}_{K,\text{high}}$ fit

v4.6

- Documented the fit projection in Appendix E.4
- Documented the X_u fragmentation in Section 6.1.4
- Updated unblinding plan based on discussion with RC (Section 9)

111 v5.0

- 112 • The first unboxing step was performed. The results of the $\text{CR}_{0,low}$ - VR_2 fits are
113 documented in Section 9.2 and Appendix Q.

114 v5.1

- 115 • The second unboxing step was performed. The results and the added corrections and
116 uncertainties are documented in Section 9.3 and Appendix S.

117 v5.2

- 118 • The final unboxing step was performed. The results are documented in Section 9.4.

119 v5.3

- 120 • During CWR2, a question was raised about the shift in the fraction of kaon events
121 in inclusive $B \rightarrow X_u \ell \nu$ events caused by the hybrid procedure. The results of the
122 conducted tests are documented in Appendix O.

123 CONTENTS

124	1. Change Log	2
125	1. Introduction	9
126	2. Modelling	10
127	1. Data	10
128	2. Monte Carlo simulation	10
129	1. $B \rightarrow X_u \ell \nu$	10
130	2. Hybrid model	11
131	3. BLNP model issues	12
132	4. $B \rightarrow X_c \ell \nu$	13
133	5. Continuum	15
134	6. Other	16
135	3. Charm Decay Modelling	16
136	3. Event selection and Reconstruction	19
137	1. Tag Side	19
138	1. Continuum Suppression Masks	19
139	2. Signal Side	20
140	1. Rest Of Event	20
141	2. Slow π	21
142	3. High-momentum $\pi^\pm$	21
143	4. Kaons	22
144	5. Signal Lepton	22
145	6. J/ψ and Photon Conversion Veto	23
146	3. Other selections	24

147	4. Continuum Modelling	25
148	1. Input Variables	25
149	1. Classifier Training	26
150	2. Variable Importance	28
151	3. Uncertainty	29
152	4. Results	29
153	5. Multivariate selection	32
154	1. Classifiers and training	32
155	1. Continuum suppression	32
156	2. $B \rightarrow X_c \ell \nu$ background	34
157	2. MVA-induced model dependence	41
158	6. Corrections and Systematics	43
159	1. Slow π efficiency	43
160	2. Tracking efficiency	44
161	3. Form factors	44
162	1. $B \rightarrow X_u \ell \nu$ form factors	45
163	2. $B \rightarrow X_c \ell \nu$ form factors	46
164	4. Charged particle identification	47
165	5. FEI tagging	48
166	6. Branching fractions	48
167	7. Hybrid parameter variations	49
168	8. Inclusive model differences	49
169	9. K_S^0 efficiency	50
170	10. $s\bar{s}$ fragmentation	50
171	11. MC statistics	50
172	12. $f^{\pm/0}$	50
173	13. Continuum calibration	50
174	14. X_u hadronisation modelling	50
175	15. Uncertainties related to the fitting procedures	51
176	7. Data-MC agreement	52
177	1. Kinematic distributions	53
178	2. Sample compositions	53
179	3. Data-MC disagreement	53
180	1. Origin of the data-MC disagreement	55
181	2. Normalisation mismodelling	59
182	3. Shape mismodelling	64
183	8. Signal extraction	66
184	1. pyhf	66
185	2. Fit templates, free-floating parameters and phase-space regions	66
186	3. Fit parameters	68
187	1. Implementation of systematic uncertainties within the fit	69
188	4. Symmetrising	71
189	5. List of fits	71
190	6. $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu)$ measurement	73

191	1. Asimov test	73
192	2. Systematics breakdown	77
193	7. Fitter validation	79
194	1. Toy fit	79
195	2. Linearity check	79
196	8. $B \rightarrow X_c \ell \nu$ factors	82
197	9. $ V_{ub} $ measurement	83
198	10. Fits with control regions	86
199	1. Fit with $\text{CR}_{0,low}$ split	86
200	2. Fit with $\text{CR}_{K,low}$ and $\text{CR}_{K,high}$	86
201	3. Fit with $\text{CR}_{0,low}$ and $\text{CR}_{K,high}$	87
202	11. Split sample fits	88
203	1. Fit with split B charge: $\Delta\mathcal{B}(B^0 \rightarrow X_u \ell \nu)$, $\Delta\mathcal{B}(B^\pm \rightarrow X_u \ell \nu)$	88
204	2. Fit with split lepton flavour: $\Delta\mathcal{B}(B \rightarrow X_u e \nu)$, $\Delta\mathcal{B}(B \rightarrow X_u \mu \nu)$	88
205	3. Additional split sample fits: split missing momentum θ angle and split lepton	
206	charge	90
207	9. Box opening	97
208	1. Proposed unblinding strategy	97
209	2. Step 1: $\text{CR}_{0,low}$ - VR_2 fit	97
210	3. Step 2: signal region plots and first checks	102
211	4. Final unboxing step: perform the three nominal fits	107
212	A. BLNP modelling	116
213	B. Continuum suppression MVA input features	119
214	C. $X_c \ell \nu$ suppression MVA input features	128
215	D. Data - MC comparison in SB1 in different channels	131
216	E. pyhf	135
217	1. Notation	135
218	2. Statistical model	136
219	3. Modifiers	136
220	4. Postfit projections	137
221	F. Plots from fits to all variables in all regions	140
222	1. q^2 fit	140
223	1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV	140
224	2. Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV ² ,	
225	Experimental cut: $p_\ell^B > 1.0$ GeV	144
226	3. Phase-space region: $p_\ell^B > 2.1$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV	148
227	2. p_ℓ^B fit	152
228	1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV	152
229	2. Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV ² ,	
230	Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV ²	156
231	3. M_X fit	159

232	1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV	159
233	4. $M_X:q^2$ fit	162
234	1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV	162
235	5. $p_\ell^B:q^2$ fit	166
236	1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV	166
237	2. Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, Experimental cut:	
238	$p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV	170
239	G. K_L	174
240	H. Hadron ID	177
241	I. Fits with control regions	178
242	1. Fit 3	178
243	2. Fit 5	182
244	J. Low-lepton-momentum region	186
245	K. Fit with split lepton flavour	187
246	1. Electron	187
247	2. Muon	190
248	L. Continuum reweighting variables	192
249	1. Before applying corrections	192
250	2. After applying corrections	202
251	M. Track and cluster multiplicities	212
252	N. Low FEI cut studies	214
253	O. Strange quark fragmentation	218
254	1. γ_s	218
255	2. Fraction of kaons before/after hybridisation	218
256	P. Variable dependent efficiency	220
257	Q. VR_2 fits	223
258	1. Fit 3	223
259	2. Fit 5	227
260	R. Signal region distributions	231
261	1. Splits in lepton flavour	231
262	2. Fit 3 region: $E_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV	233
263	3. Fit 5 region: $E_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV	234
264	S. Additional fit corrections	235
265	1. Fit 3	235
266	2. Fit 5	237
267	References	239

1. INTRODUCTION

The discrepancy between inclusive and exclusive measurements of the CKM matrix element $|V_{ub}|$ has posed a longstanding puzzle. The major difficulty involved with the inclusive $|V_{ub}|$ measurement is the determination of the non-perturbative distribution function describing the internal Fermi motion of the b -quark. On the experimental aspect, the measurement of $B \rightarrow X_u \ell \nu$ is challenged by the high background of the CKM-favoured $B \rightarrow X_c \ell \nu$ decays. Belle II has unique advantage for the inclusive semileptonic decays with a clean experimental environment. As 3-body decays, semi-leptonic events can be characterised by the following three variables: the lepton momentum, p_ℓ (equivalent to the lepton energy E_ℓ ¹, in the case of massless leptons), the mass of the hadronic system, M_X and the invariant mass of the leptonic system $\ell - \nu$, q^2 .

This work aims to measure the partial branching fraction of charmless semileptonic $B \rightarrow X_u \ell \nu$ decays, where $\ell = e, \mu$ and the inclusive X_u includes the non-resonant contribution and the exclusive resonances, e.g. $\pi, \eta, \eta', \rho, \omega$. Subsequently, the inclusive $|V_{ub}|$ can be extracted based on the underlying theoretical decay rates. The branching fraction and $|V_{ub}|$ will be measured in different phase-space regions defined by cuts to the three kinematic variables p_ℓ^B , M_X and q^2 . This measurement will also provide separate results for testing the lepton flavour universality in (e, μ) and iso-spin asymmetry of (B^+, B^0) . Further measurements including the differential spectra of key kinematic variables, a ratio of inclusive $|V_{ub}|$ and $|V_{cb}|$, etc. are also planned.

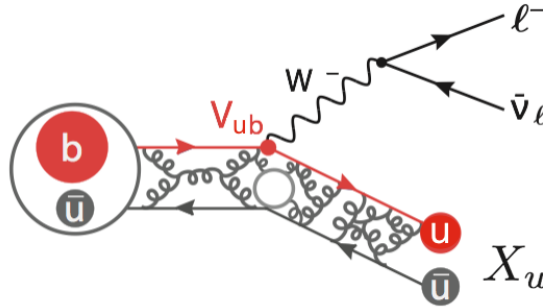


FIG. 1: Standard Model charmless semileptonic decay.

The analysis presented within this note is co-developed with the Weak Annihilation in charmless semi-leptonic decays measurement presented in BELLE2-NOTE-PH-2023-060 [2]. There is significant overlap in the data and Monte Carlo samples, the analysis selections, Monte Carlo corrections, and analysis methods between both analyses. The basf2 release light-2311-nebelung is used.

¹ These will be denoted p_ℓ^B , E_ℓ^B when taken in the B -meson rest frame.

2. MODELLING

1. Data

The analysis will be performed on the Long Shutdown 1 (LS1) data sample, corresponding to $364.558 \pm 0.021_{\text{stat}} \pm 2.297_{\text{syst}} \text{ fb}^{-1}$ collected on-resonance and a further $42.561 \pm 0.007_{\text{stat}} \pm 0.269_{\text{syst}} \text{ fb}^{-1}$ collected off-resonance.

2. Monte Carlo simulation

The signal and background are simulated using generic MC15 (`MC15ri_b`) which corresponds to 1 ab^{-1} of $q\bar{q}$ events and 2.8 ab^{-1} of $b\bar{b}$ events. Specific modes must also be added. All the branching fractions quoted below are up-to-date with the PDG2022 [3].

1. $B \rightarrow X_u \ell \nu$

The exclusive and inclusive $B \rightarrow X_u \ell \nu$ modes are taken from signal MC. For some of the decay modes, the branching fraction needs to be updated to a more up-to-date value (see Section 6.6). The considered modes with their associated branching fraction are listed in Tab. I.

Various theoretical models exist to describe the inclusive $B \rightarrow X_u \ell \nu$ mode. In the generic MC, the De Fazio - Neubert (DFN) model [4] is used with outdated parameters. Following Tab. IV in Ref. [5], the values of the non-perturbative parameters $\bar{\Lambda}$ and λ_1 are updated. Their values can then be converted to values of the b -quark mass and non-perturbative parameter a in the Kagan-Neubert renormalisation scheme. They are updated in the signal MC to: $m_b^{\text{KN}} = (4.66 \pm 0.04) \text{ GeV}$ and $a^{\text{KN}} = 1.3 \pm 0.5$. The uncertainties related to these parameters are taken into account in this analysis. The DFN model is used as the nominal model for our samples. We were initially advised to use the Bosch - Lange - Neubert - Paz (BLNP) model [6] but various issues were spotted. These are discussed in Section 2.2.3. However, as the modelling of inclusive $B \rightarrow X_u \ell \nu$ decays is known to be imperfect, the difference between the DFN and BLNP models is taken as an additional uncertainty (*c.f.* Section 6 for more details).

Resonant decays are described by form factors. We give here a quick overview of the models used for each exclusive mode but more details about the models and associated uncertainties are given in and Section 6.3.

π : Generated following the expansion of Bourrely, Caprini, and Lellouch (BCL) [7] to 8 terms using outdated parameters. Updated to the BCL expansion truncated at 5 terms of Tab. 57 of Ref.[8] via reweighting in q^2 .

ρ/ω : Generated following the BCL expansion with outdated parameters. Updated to parameter values of Tab. 4 of Ref.[9] via reweighting in q^2 .

$\eta^{(\prime)}$: Generated via ISGW2. Updated to the light-cone-sum-rules based model of Ref.[10] via reweighting in hadronic recoil, $w = \frac{m_B^2 + m_X^2 - q^2}{2m_B m_X}$.

331 The γ_S parameter in `PYTHIA8` controls the suppression of s quark production relative to u
332 or d production. This eventually impacts the number of K produced. Since the number of
333 $K^\pm$ and K^0 is used in the MVA classifier (see Section 5), this parameter can impact the
334 post-classifier signal efficiency. Samples with varied γ_S values were produced: one sample
335 with the `PYTHIA8` default value, $\gamma_S = 0.217$; one with a more accurate value taken from the
336 average of two measurements [11, 12], $\gamma_S = 0.300 \pm 0.09$ and one with the $+1\sigma$ variation
337 taken from the same average, $\gamma_S = 0.390$. $\gamma_S = 0.300$ is the nominal value.

338

339 For fragmentation and subsequent hadronisation of the X_u system in non-resonant
340 modes, `PYTHIA8` is used. No Belle II dedicated tune of `PYTHIA8` has been performed so
341 far. The known poor modelling of fragmentation is taken into account as an additional
342 uncertainty (see Section 6)

TABLE I: Assumed branching fractions, number of generated events and decfile for the $B \rightarrow X_u \ell \nu$ channels. For the inclusive channel, the first entry gives the nominal sample and the following give details on samples used to study systematic uncertainties.

Decay mode	$\mathcal{B} [\times 10^{-4}]$	Model	Dec File	Code	N Events $[\times 10^6]$
$B^- \rightarrow \pi^0 \ell \bar{\nu}$	0.780 ± 0.027	BCL	1290710003		0.188×50
$B^- \rightarrow \rho^0 \ell \bar{\nu}$	1.58 ± 0.11	BCL	1290710003		0.380×50
$B^- \rightarrow \eta \ell \bar{\nu}$	0.350 ± 0.040	DM	1290710003		0.091×50
$B^- \rightarrow \eta' \ell \bar{\nu}$	0.240 ± 0.070	DM	1290710003		0.055×50
$B^- \rightarrow \omega \ell \bar{\nu}$	1.19 ± 0.09	BCL	1290710003		0.286×50
$B^- \rightarrow X_u^0 \ell \bar{\nu}$	19.2 ± 2.4	DFN ($\gamma_s = 0.300$)	1290710022		30
$B^- \rightarrow X_u^0 \ell \bar{\nu}$	19.2 ± 2.4	DFN ($\gamma_s = 0.217$)	1290710012		10
$B^- \rightarrow X_u^0 \ell \bar{\nu}$	19.2 ± 2.4	DFN ($\gamma_s = 0.390$)	1290710032		10
$B^- \rightarrow X_u^0 \ell \bar{\nu}$	19.2 ± 2.4	BLNP ($\gamma_s = 0.300$)	1290710021		20
$\bar{B}^0 \rightarrow \pi^+ \ell \bar{\nu}$	1.50 ± 0.06	BCL	1190710003		0.330×50
$\bar{B}^0 \rightarrow \rho^+ \ell \bar{\nu}$	2.94 ± 0.21	BCL	1190710003		0.670×50
$\bar{B}^0 \rightarrow X_u^+ \ell \bar{\nu}$	17.6 ± 2.2	DFN ($\gamma_s = 0.300$)	1190710022		30
$\bar{B}^0 \rightarrow X_u^+ \ell \bar{\nu}$	17.6 ± 2.2	DFN ($\gamma_s = 0.217$)	1190710012		10
$\bar{B}^0 \rightarrow X_u^+ \ell \bar{\nu}$	17.6 ± 2.2	DFN ($\gamma_s = 0.390$)	1190710032		10
$\bar{B}^0 \rightarrow X_u^+ \ell \bar{\nu}$	17.6 ± 2.2	BLNP ($\gamma_s = 0.300$)	1190710022		20

343 2. Hybrid model

344 Models describing inclusive $B \rightarrow X_u \ell \nu$ decays cannot make predictions about resonant
345 decays. In the latter case quark-hadron duality breaks down and therefore form factor
346 models must be used. The models used in our samples are given in Tab. I. See also Section
347 6.3 for a discussion about form factors. For our simulation to cover all decays, inclusive
348 and exclusive contributions are combined with the *hybrid* approach [13]: the exclusive and
349 inclusive contributions are summed in three-dimensional bins of E_ℓ , M_X and q^2 ; in each
350 bin, the inclusive part is scaled down such that the total contribution matches the initial
351 inclusive contribution; this way, the total branching fraction matches the inclusive branching

fraction and the exclusive contribution is included. Therefore, the inclusive part is scaled
by the weights:

$$w_{ijk} = \frac{\Delta\mathcal{B}_{ijk}^{\text{inc}} - \Delta\mathcal{B}_{ijk}^{\text{exc}}}{\Delta\mathcal{B}_{ijk}^{\text{inc}}} \quad (1)$$

where $\Delta\mathcal{B}^{\text{inc}}$, $\Delta\mathcal{B}^{\text{exc}}$ are the partial branching fraction of the exclusive and inclusive components respectively and the indices i, j, k define a three-dimensional bin. These three-dimensional bins are defined by the combinations of the following one-dimensional bins:

$$\begin{aligned} q^2 &: [0, 2.5, 5, 7.5, 10, 12.5, 15, 20, 25] \text{ GeV}^2, \\ E_\ell^B &: [0, 0.5, 1, 1.25, 1.5, 1.75, 2, 2.25, 3] \text{ GeV}, \text{ and} \\ M_X &: [0, 1.4, 1.6, 1.8, 2, 2.5, 3, 3.5] \text{ GeV}. \end{aligned}$$

The hybrid reweighting is illustrated in Fig. 2.

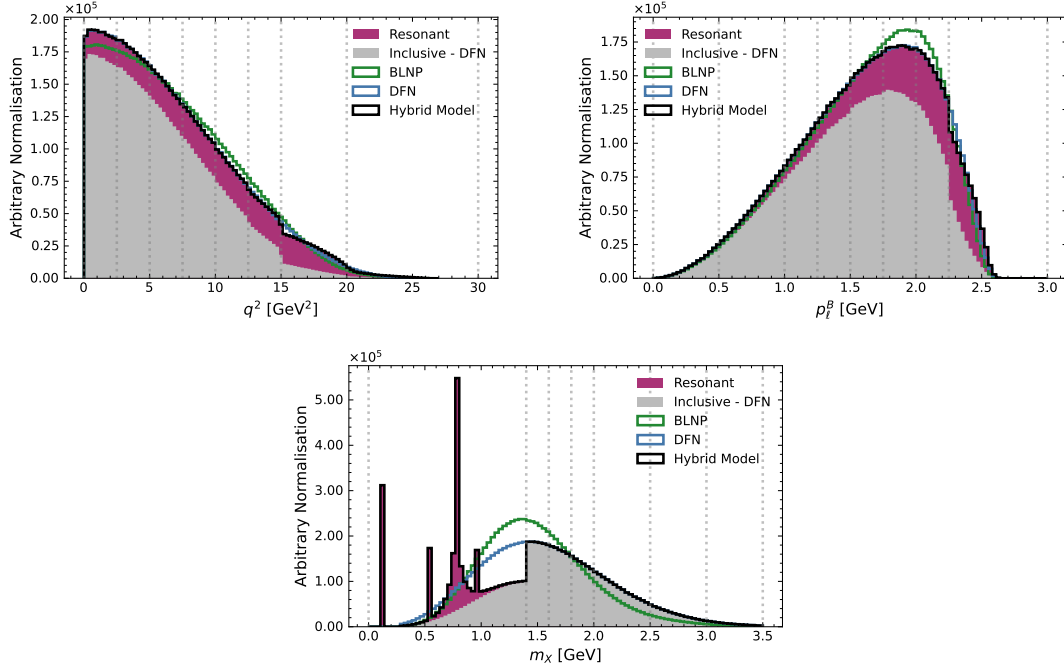


FIG. 2: Projections of the hybrid MC in M_X , E_ℓ^B and q^2 in the $B^+ \rightarrow X_u \ell \nu$ channel. The grey vertical dotted lines show the boundaries of the bins used for hybrid weights calculations from Equ. 1. A finer binning is used to illustrate the hybrid procedure.

3. BLNP model issues

First, many of the calculated hybrid weights were noticed to be negative, corresponding
to bins where the number of exclusive events exceeds the number of inclusive ones. These
weights are usually set to 0 which has the consequence of over-estimating the inclusive rate.
However, if the number of negative weights is limited they will have little to no effect. In

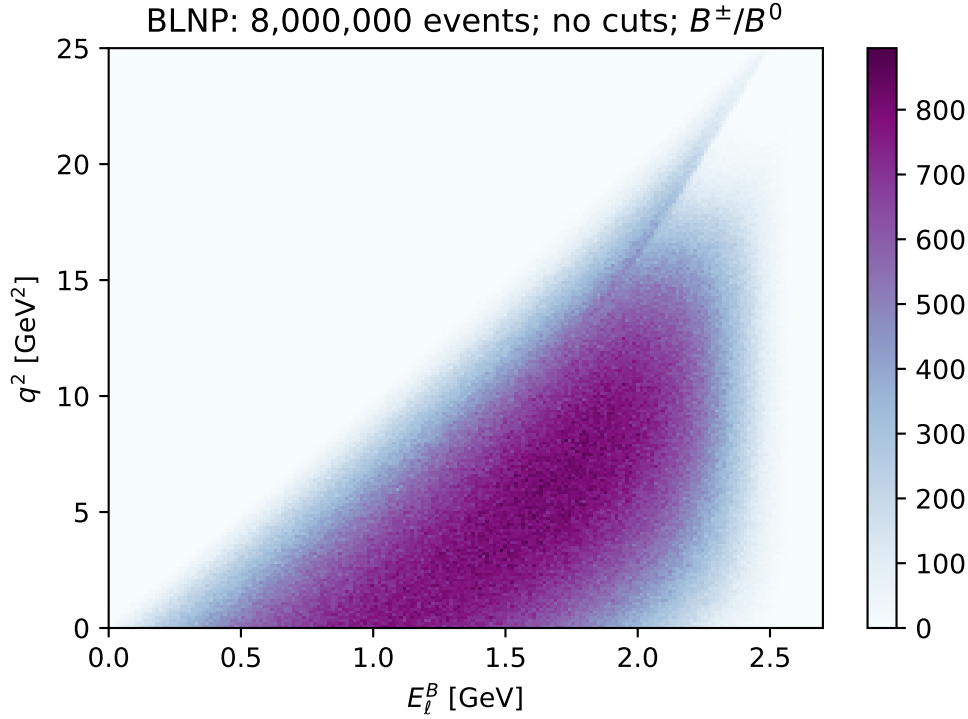


FIG. 3: Two-dimensional histogram of q^2 versus E_ℓ^B for 8 million events generated with the most up-to-date input values for the BLNP model in EvtGen.

360 this case their number was relatively large and they could've spoiled the hybrid modelling
 361 of $B \rightarrow X_u \ell \nu$. Moreover, as illustrated in Fig. 3 an unexpected spike of events was observed
 362 in 2D plots of E_ℓ^B , q^2 and M_X which doesn't appear in 1D plots. More details about this
 363 feature are discussed in Appendix A.

364 4. $B \rightarrow X_c \ell \nu$

365 Specific $B \rightarrow X_c \ell \nu$ decays are also replaced in the generic MC by dedicated signal MC. In
 366 particular, modes such as $B \rightarrow D_0(\rightarrow D\pi\pi)\ell\nu$ must be added to the samples in order to
 367 fill the gap between the total $B \rightarrow X_c \ell \nu$ branching fraction measured from the sum of all
 368 exclusive modes and the one measured from the inclusive mode (so far no inclusive predic-
 369 tions exist for $B \rightarrow X_c \ell \nu$ decays as they do for charmless decays). The nominal $B \rightarrow X_c \ell \nu$
 370 and gap modes are listed in Tab. II with associated branching fractions for B^+ and B^0 .

371

372 We give here a quick overview of the $B \rightarrow X_c \ell \nu$ form factor models:

- 373 • $B \rightarrow D \ell \nu$: in the generic MC these decays are modelled using the BGL form factor
 374 parametrisation [14]. $B \rightarrow D \ell \nu$ events are modelled using the $N = 3$ expansion (8
 375 parameters). The $B \rightarrow D \ell \nu$ decays are described using the leptonic system invariant
 376 mass q^2 (or alternatively the w recoil parameter)² and the angle between the direction

² $w = \frac{m_B^2 + m_X^2 - q^2}{2m_B m_X}$

of the lepton momentum in the W rest frame and the direction of the W momentum in the B rest frame, θ_ℓ . We update the modelling to the BLPRXP model [15] using the HAMMER package [16].

- $B \rightarrow D^* \ell \nu$: in the generic MC, modelled using the $(1, 1, 2)$ expansion of the BGL parametrisation (6 parameters). As D^* mesons decay typically to $D\pi/\gamma$, to describe them one needs to consider in addition of q^2 and θ_ℓ the following variables: the angle between the direction of the D meson momentum in the D^* rest frame and the direction of the D^* momentum in the B rest frame, θ_V and the angle between the planes defined by the leptonic system and the D^* system, χ . Also updated to the BLPRXP model.
- $B \rightarrow D^{**} \ell \nu$: modelled using the BLR form factor model (previously called LLSW). There are 3 parameters for D_0 and D'_1 and 4 parameters for D_1 and D_2 . The values of these parameters were updated following Ref. [17]. An issue was spotted with the modelling of the two broad D^{**} channels, D_0 and D'_1 . Because of their very large width, in some events the generated D^{**} mass is much larger than the nominal one. It was observed that these events lead to an unphysical enhancement of the $w \approx 1$ region. Therefore, events with a mass more than 3 (resp. 2.5) times higher than the nominal D_0 (resp. D'_1) mass are rejected. After this cut is applied, all the remaining events are up-scaled accordingly. The scale factor is calculated separately for each D^{**} decay channel including the gap modes with EvtGen truth samples.
- $B^- \rightarrow D_s^{(*)+} K^- \ell \bar{\nu}$: taken from the generic MC, where they are modelled with PHSP, for lack of a better model. We expect these decays to have a negligible impact, as their branching fraction is low, $\sim 3 \times 10^{-4}$ [18]
- $B \rightarrow D_{\text{Gap}}^{**} \ell \nu$: There remains a significant gap between the sum of branching fractions of all measured channels and the measured branching fractions of $B \rightarrow X_c \ell \nu$. The missing branching fraction is filled with final states of the type $\bar{B} \rightarrow D^{(*)} \pi \pi \ell \nu$ and $\bar{B} \rightarrow D^{(*)} \eta \ell \nu$. Within the generic MC these channels are simulated via PHSP. As these are four and five body decays this gives an extremely soft lepton momentum spectrum. We replace the events from the generic MC with events simulated through intermediate resonances, using the two broad D^{**} resonances (D_0^* and D'_1) as the intermediate resonances. It is important to note that D_0^* and D'_1 are only known to decay to $D\pi$ and $D^*\pi$ respectively. Some of the decays used for the gap modes could be physically allowed [19] but they have never been observed. Throughout this analysis, the gap modes are treated as regular $B \rightarrow D^{**} \ell \nu$ decays (which is mostly relevant for the form factor reweighting *c.f.* Section 6.3.2). The generic MC additionally contains some $\bar{B} \rightarrow D^{(*)} \pi \ell \bar{\nu}$ events. These are removed as their measured branching fractions are saturated by production via the D^{**} resonances. In Ref. [20] an unphysical enhancement at $w = 1$ was observed in the broad D^{**} decays, caused by events generated several widths away from the nominal mass. We adopt the procedure of Ref. [20] and add a generator level selection, requiring events to be, at most 3 (2.5) widths above the nominal D_0 (D'_1) mass. The weight of the remaining events is increased to compensate for the removed events. The unphysical feature, and the post selection spectrum for an example channel can be seen in Fig. 4.

Decay mode	$[\times 10^{-2}]$	Model	Dec File	Code	N Events $[\times 10^6]$
$B^- \rightarrow D \ell \bar{\nu}$	2.27 ± 0.06	BLPRXP		Generic	-
$B^- \rightarrow D^* \ell \bar{\nu}$	5.27 ± 0.12	BLPRXP		Generic	-
$B^- \rightarrow D^1 \ell \bar{\nu}$	0.64 ± 0.10	BLR		Generic	-
$B^- \rightarrow D_0^* \ell \bar{\nu}$	0.13 ± 0.19	BLR		Generic	-
$B^- \rightarrow D_1' \ell \bar{\nu}$	0.28 ± 0.04	BLR		Generic	-
$B^- \rightarrow D_2^* \ell \bar{\nu}$	0.32 ± 0.03	BLR		Generic	-
$B^- \rightarrow D_s K \ell \bar{\nu}$	0.030 ± 0.013	PHSP		Generic	-
$B^- \rightarrow D_s^* K \ell \bar{\nu}$	0.029 ± 0.019	PHSP		Generic	-
$B^- \rightarrow D_0^*(\rightarrow D \eta) \ell \bar{\nu}$	0.90 ± 0.90	BLR	1296708001		10
$B^- \rightarrow D_1'(\rightarrow D^* \eta) \ell \bar{\nu}$	0.90 ± 0.90	BLR	1296708000		10
$B^- \rightarrow D_1'(\rightarrow D \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.07 \pm 0.09)$	BLR	1296700001		10
$B^- \rightarrow D_0^*(\rightarrow D \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.07 \pm 0.09)$	BLR	1296700003		10
$B^- \rightarrow D_1'(\rightarrow D^* \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.22 \pm 0.10)$	BLR	1296700002		10
$B^- \rightarrow D_0^*(\rightarrow D^* \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.22 \pm 0.10)$	BLR	1296700004		10
$\bar{B}^0 \rightarrow D \ell \bar{\nu}$	2.11 ± 0.06	BLPRXP		Generic	-
$\bar{B}^0 \rightarrow D^* \ell \bar{\nu}$	4.90 ± 0.11	BLPRXP		Generic	-
$\bar{B}^0 \rightarrow D^1 \ell \bar{\nu}$	0.59 ± 0.10	BLR		Generic	-
$\bar{B}^0 \rightarrow D_0^* \ell \bar{\nu}$	0.12 ± 0.18	BLR		Generic	-
$\bar{B}^0 \rightarrow D_1' \ell \bar{\nu}$	0.26 ± 0.04	BLR		Generic	-
$\bar{B}^0 \rightarrow D_2^* \ell \bar{\nu}$	0.30 ± 0.03	BLR		Generic	-
$\bar{B}^0 \rightarrow D_0^*(\rightarrow D \eta) \ell \bar{\nu}$	0.86 ± 0.86	BLR	1196708001		10
$\bar{B}^0 \rightarrow D_1'(\rightarrow D^* \eta) \ell \bar{\nu}$	0.86 ± 0.86	BLR	1196708000		10
$\bar{B}^0 \rightarrow D_1'(\rightarrow D \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.07 \pm 0.08)$	BLR	1196700001		10
$\bar{B}^0 \rightarrow D_0^*(\rightarrow D \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.07 \pm 0.08)$	BLR	1196700003		10
$\bar{B}^0 \rightarrow D_1'(\rightarrow D^* \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.20 \pm 0.10)$	BLR	1196700002		10
$\bar{B}^0 \rightarrow D_0^*(\rightarrow D^* \pi \pi) \ell \bar{\nu}$	$0.5 \times (0.20 \pm 0.10)$	BLR	1196700004		10

TABLE II: Assumed branching fractions, models, number of generated events and decay files for the channels. The gap channels to $D^{(*)}\pi\pi$ are taken to be fully correlated across both intermediate D^{**} resonances.

5. Continuum

We use the generic continuum samples: 1 ab^{-1} of generic on-resonance events and 50 fb^{-1} of off-resonance. $e^+e^- \rightarrow q\bar{q}$ processes are simulated using KKMC [21]. As the modelling of continuum is known to be poor, we perform a data-driven reweighting using the off-resonance sample. This is described in section 4.

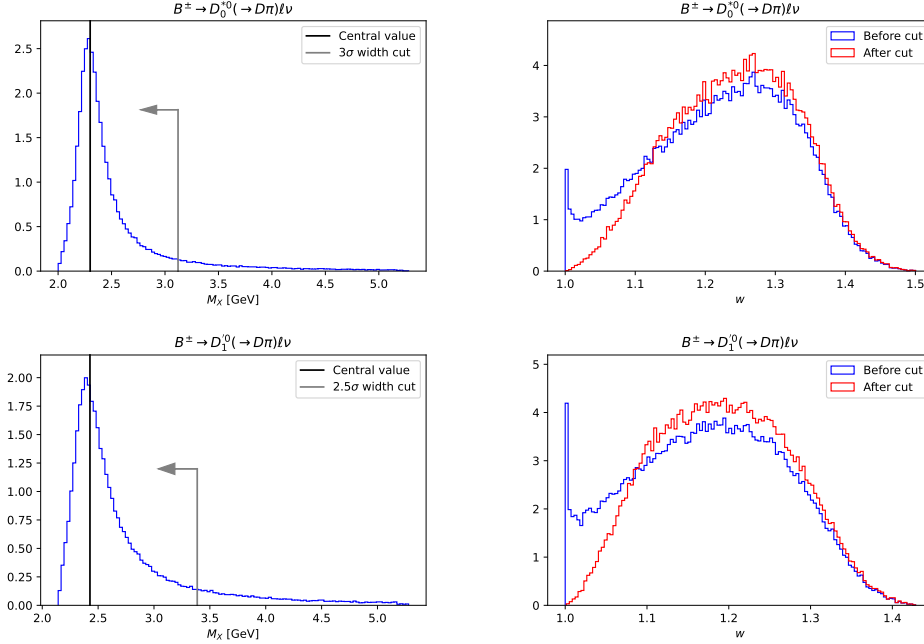


FIG. 4: Two examples of the broad D^{**} mass cuts. On the left-hand side the generator level mass distributions are shown together with the corresponding mass nominal values as well as the maximum allowed mass. On the right-hand side, the generator level w distributions are shown before and after the cut and rescaling are applied. The unphysical spike at $w \approx 1$ is removed thanks to the mass cut.

6. Other

For completeness we list here the remaining event categories, which originate primarily from the generic MC. These are:

a. Secondaries Events in which we have selected a lepton which is not a primary child of the B -meson. This includes leptons originating from semileptonic decays of D mesons, and leptons originating from leptonic decays of taus. In the case of the tau channels, the leptons are expected to be low momentum and thus lie outside the region of interest. Semileptonic decays of the D are further discussed in section 2.3.

b. Fakes Events in which we have misidentified a hadron as the signal lepton. These events are taken from the generic MC.

3. Charm Decay Modelling

Inclusive studies at Belle and Belle II have reported poor data to MC agreement in quantities related to the hadronic system (m_X , m_{miss}^2 , q^2 , ...) [20, 22–25]. We therefore reweight the decay of the charm mesons to the most up to date values, and more importantly, attempt to capture the uncertainty of these branching fractions. The full list of decays and the assumed branching fractions are attached to this note on the document server.

As we cannot re-generate all events, we reweight the signal side charm decays within the existing MC, expanding on the procedure of Ref. [26]. This is achieved by finding the

443 matching MC particle to the selected lepton candidate, searching up the decay tree until
 444 finding a B meson, then recursively searching down the decay tree for up to four $D^{(*)}$ mesons
 445 and storing their decays. In this way we additionally aim to minimise any impact on the
 446 tag side as the corrections to the FEI efficiency were calculated without this update of the
 447 charm decay modelling.

448

449 The branching fractions are updated to the latest PDG2023 versions (for some decays
 450 we must refer back to 2008 PDG as subsequent versions no longer list all sub-modes of
 451 $D^0 \rightarrow K\pi\pi\pi$). There are several caveats to the reweighting procedure that should be
 452 noted:

- 453 • We can only include channels that were listed in the decay file with a non-zero branch-
 454 ing fraction. This means there are channels (although usually they are very rare
 455 channels) in the PDG which are not reproduced in the decay file.
- 456 • When a decay has been measured through a decay channel of an intermediate resonance
 457 (ex. $D^0 \rightarrow K^- a_1^+ (a_1^+ \rightarrow \rho^0 \pi^+)$), then the branching fraction is corrected assuming
 458 the branching fraction of the intermediate resonance used in the decay file ($(a_1^+ \rightarrow$
 459 $\rho^0 \pi^+) = 0.30875$). No uncertainty is associated to this branching fraction.
- 460 • If a channel has been measured through multiple intermediate resonances, we try to
 461 infer which was previously used in the decay file and use the same.
- 462 • If only a limit is available, the branching fraction is set to the limit with 100% uncer-
 463 tainty.
- 464 • Uncertainties are taken to be fully uncorrelated. This can lead to significantly in-
 465 flated uncertainties when a poorly known resonance contribution is subtracted from
 466 an inclusive rate.
- 467 • The reweighting procedure only considers the branching fraction. It is not feasible
 468 to reweight the decay models, as either no better model exists, or a large amount of
 469 additional information would need to be stored per event. A significant portion of
 470 events is modelled via PHSP and thus likely to still be poorly modelled even with
 471 corrected branching fractions.
- 472 • The $D^0 \rightarrow K^- a_1^+$, as measured through $a_1 \rightarrow \rho^0 \pi^+$, rate has increased several per-
 473 centage points from 7.8% to 14%. Combined with all other measured channels at their
 474 respective central values, and zeroing out the unmeasured $D^0 \rightarrow K_{S/L} \pi^+ \pi^- \pi^0 \pi^0$,
 475 the total D^0 branching fraction is $\sim 102\%$. As the branching fractions of the a_1 are
 476 not well known, we decrease the $D^0 \rightarrow K^- a_1^+$ branching fraction to $11.94 \pm 2.3\%$, such
 477 that the total D^0 branching fraction correctly sums to unity. The difference to the
 478 nominal branching fraction is added as an additional uncertainty on this channel.
- 479 • For the D^+ we face the opposite problem, with the measured branching fractions cov-
 480 ering approximately 98% of the total decay rate. We therefore increase the branching
 481 fraction of the unmeasured channels ($D^+ \rightarrow \eta 5\pi$) by a factor of ~ 2.5 over what is
 482 currently in the decay file. The branching fractions of these channels are given a 100%
 483 uncertainty. This gives these channels branching fractions a factor of ~ 2 larger than
 484 the equivalent $\eta 4\pi$ or $\eta 3\pi$ channels, which appears somewhat unphysical. For lack of

other channels with which to fill the gap we nevertheless adopt these large branching fractions.

Table III presents a summary of the final state particles produced in comparison with the inclusive measurements available in the PDG [18], and the official decay table. The table is created by recursively expanding all unstable intermediate states, as outlined above no uncertainty is associated to the branching fractions of the intermediate resonances. Neutral pions are considered stable, if they are allowed to decay, the electron rate increases by approximately 1 percentage point.

Curiously, the updated decay file has largely shifted the inclusive particle production rates further from the measured values than the decay file used for MC15 production. The inclusive neutral kaon production rate in the D^0 channel is especially concerning, at a $\sim 2.3\sigma$ deviation between the PDG value and the updated decay file. As the updated values are the most up-to-date branching fractions available, we nevertheless adopt them.

Final State	D^0 [%]			D^+ [%]		
	PDG	Dec File	Updated Dec File	PDG	Dec File	Updated Dec File
$K_S^0 X$	-	20.44	18.48 ± 1.46	-	30.39	32.01 ± 0.78
$K_L^0 X$	-	20.12	18.50 ± 1.46	-	30.32	31.95 ± 0.78
$(K_S^0 \text{ or } K_L^0) X$	47.00 ± 4.00	40.07	36.46 ± 2.45	61.00 ± 5.00	59.70	63.32 ± 1.27
$K^+ X$	3.40 ± 0.40	3.36	3.24 ± 0.27	5.90 ± 0.80	5.60	4.40 ± 0.15
$K^- X$	54.70 ± 2.80	55.85	59.84 ± 4.33	25.70 ± 1.40	27.66	27.57 ± 1.20
$(K^+ \text{ or } K^-) X$	-	56.47	60.45 ± 4.33	-	31.41	30.23 ± 1.20
$\mu^+ X$	6.80 ± 0.60	6.27	6.96 ± 1.34	17.60 ± 3.20	16.26	15.27 ± 0.25
$e^+ X$	6.49 ± 0.11	6.54	7.20 ± 1.32	16.07 ± 0.30	16.17	16.11 ± 0.82

TABLE III: Comparison of the inclusive particle production, as listed in the PDG, the MC15 generic dec file, and the updated branching fractions used within this study.

3. EVENT SELECTION AND RECONSTRUCTION

We use the hadronic FEI to reconstruct a tag B meson, which is paired with a lepton candidate from the remaining tracks. All remaining tracks and clusters are inclusively summed to form the rest of event (ROE). The ROE is used to represent the hadronic system, X .

1. Tag Side

The tag candidate is reconstructed by the FEI in one of 32 (36) charged (neutral) hadronic B channels. As we use the official FEI calibration channels, we adopt the common tag side selection as defined within the FEI calibration studies. The selections are:

- A beam energy difference, $\Delta E = E_B^* - E_{\text{beam}}^*$, of $-0.15 < \Delta E < 0.10$ GeV.
- A beam constrained mass, $m_{\text{bc}} = \sqrt{E_{\text{beam}}^* - p_B^{*2}}$, of $m_{\text{bc}} > 5.24$ GeV. This loose selection provides a useful sideband to the calibration studies. After the selection of the best candidate on the tag side, we tighten this to $m_{\text{bc}} > 5.27$ GeV.
- An FEI signal probability, $\mathcal{P}_{\text{FEI}} > 0.01$.
- A selection on the continuum suppression variable $\cos \theta_{\mathcal{T}_B \mathcal{T}_O}^{\text{cM}}$, the cosine of the angle between the thrust of the tag B meson and the rest of event, of $\cos \theta_{\mathcal{T}_B \mathcal{T}_O} < 0.9$, to reduce the continuum contamination at a cost of approximately 10% of the signal. The details of the rest of event selections used to calculate $\cos \theta_{\mathcal{T}_B \mathcal{T}_O}^{\text{cM}}$ are given in section 3.1.1.

After these selections, we perform a tag-side best candidate selection, selecting the reconstructed B meson with the highest $\mathcal{P}_{\text{FEI}}$. We then tighten the m_{bc} requirement to 5.27 GeV. For off-resonance events we apply the selection to a scaled beam-constrained mass to account for the energy difference between on-resonance and off-resonance events,

$$m_{\text{bc}}^{\text{corr.}} = \sqrt{\frac{10.58^2}{4} - \left(\frac{10.58}{10.52}\right)^2 p_{B_{\text{tag}}}^{*2}}, \quad (2)$$

where $p_{B_{\text{tag}}}^*$ is the centre-of-mass frame four momentum of the tag B candidate.

1. Continuum Suppression Masks

To calculate $\cos \theta_{\mathcal{T}_B \mathcal{T}_O}^{\text{cM}}$, we must define which particles in the rest of event are considered ‘good’, and enter the event-shape calculations. We adopt the official selection used in calibrating the FEI. These selections are only used for the definition of $\cos \theta_{\mathcal{T}_B \mathcal{T}_O}$ within this selection and are marked by the superscript cM. The selections are:

- Tracks:
 - originate close to the interaction point with $dr < 2$ cm and $|dz| < 4$ cm.

- Have a minimum transverse momentum of $p_t > 0.2 \text{ GeV}$.
- Have momentum within the CDC acceptance passing `thetaInCDCAcceptance`.

- Clusters:

- Have a minimum energy of 80, 30, or 60 MeV respectively in the forward, barrel, and backwards regions of the ECL.
- Clusters from the edge of the ECL are rejected, requiring $0.2967 < \theta_{\text{cl}} < 2.6180 \text{ rad}$.
- Clusters must be recorded within 200 ns either side of t_0 , consistent with originating from photons created in the event.

For the remainder of this note, all event shape variables are calculated using the nominal rest of event selection defined in Sec. 3.2.1.

2. Signal Side

Our signal consists of a lepton, an invisible neutrino and finally the ROE which is used as the inclusive hadronic system X . Even though we cannot detect the neutrino, we can use the known initial state energy to calculate the missing energy carried by the neutrino.

1. Rest Of Event

The ROE is used to get kinematic information about the hadronic system without explicitly reconstructing it. Various sets of selections (called masks) can be used for the ROE reconstruction. We use a mask without any cuts as well as a mask based on the selections described in Ref. [22] for reference. The nominal selections were optimised they are listed below. The optimisation is discussed and illustrated in Appendix A in Ref. [2].

- Tracks:

- Originate close to the interaction point with $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$.
- Have momentum within the CDC acceptance, passing `thetaInCDCAcceptance` ($17^\circ < \theta < 150^\circ$), and must have been seen in the CDC: `nCDCHits` ≥ 2 .
- Polar angle: $0.31 < \theta < 2.61$.
- Pass the MVA based duplicate curl track rejection: `isCurl==0`.

- Clusters:

- Have a minimum energy of 70, 60, or 70 MeV in the forward, barrel, and backward regions of the ECL respectively.
- Clusters must be recorded within 70 ns of t_0 , consistent with photons created in the event: `abs(clusterTiming) < 70`.
- Fake photons are suppressed using an MVA based classifier: `fakePhotonSuppression > 0.5`.

Events with lepton candidates in the ROE (based on the criteria defined below for leptons) are rejected as they are indicative of $B \rightarrow X_c \ell \nu$ events and they have a worse resolution than events without ROE leptons.

2. Slow π

The slow charged and neutral π are treated carefully as they are used to reject $B \rightarrow D^* \ell \nu$ events in the MVA classifier (cf. Section 5). D^* mesons mostly decay to a D and a π : approximately two-thirds of the $D^{*\pm}$ decay to $D^0 \pi^\pm$ and the remaining third to $D^\pm \pi^0$; D^{*0} likewise decay to $D^0 \pi^0$ at a rate of about two-thirds, with the remainder being $D^0 \gamma$ ($D^{*0} \rightarrow D^\pm \pi^\mp$ is not possible as the sum of the $D^\pm$ and $\pi^\pm$ masses exceeds the D^{*0} mass). The D^* and D masses are very close and therefore the π produced has a very low momentum. These π are called slow π and their properties can be used to reject $B \rightarrow D^* \ell \nu$ events. The corrections and uncertainties associated to slow π are explained in Section 6.1. Slow π have a momentum between 50 and 200 MeV. The efficiency of these π is calibrated in independent studies (Ref. [27] and [28] for $\pi^\pm$ and π^0 respectively). For the quoted calibration factors to be used in our data sets, the exact corresponding selections need to be applied to π . Slow charged π are identified as tracks with momentum in the [50; 200] MeV range. The selections for π^0 are summarised in Tab. IV. They are reconstructed from a pair of photons³.

	Selections on MC	Additional selections on data
	eff40 list	
γ	<code> clusterTiming < 200 ns</code> <code>fakePhotonSuppression > 0.1</code>	apply <code>correctEnergyBias</code>
π^0	<code> dM < 0.015 GeV</code> mass constrained <code>kFit</code>	

TABLE IV: Selections applied on π^0 . The selections of the eff40 list can be found in Ref. [29]

3. High-momentum $\pi^\pm$

An uncertainty must be associated to the hadronisation of the X_u system by PYTHIA8. For that purpose, the $\pi^\pm$ multiplicity is used (see Section 6.1.4 for a precise description of the procedure applied to extract the uncertainty). We only consider $\pi^\pm$ candidates covered by the $\pi^\pm$ identification and $K^\pm$ to $\pi^\pm$ fake rate correction tables. The correction factors are given in bins of lab-frame momentum p and polar angle θ . The momentum is taken between $0.3 < p < 4.5$ GeV and the polar angle is taken between $0.3 < \theta < 2.6$ rad. To specifically identify $\pi^\pm$, a `pionID > 0.6` cut is applied. Both for $\pi^\pm$ and $K^\pm$, the neural network identification scores were considered but it was noticed that the fraction of fake hadrons is in general higher than for the likelihood-based scores (see Appendix H).

³ The branching fraction $\pi^0 \rightarrow \gamma\gamma$ is known to be about 99%.

591 4. Kaons

592 D mesons decay very often to a final state containing a kaon. As D mesons are always part
 593 of the decay chain of D^{**} and D^* mesons, the presence of kaons in an event can be used
 594 to tag this event as a B decay to a charm meson. In particular, charged and neutral kaon
 595 multiplicities are used in our $B \rightarrow X_c \ell \nu$ MVA classifier (*c.f.* Section 5).

596
 597 Charged kaons $K^\pm$ are simply reconstructed as charged tracks with a kaon identifica-
 598 tion probability `kaonID` required to be greater than 0.6. Similarly to high-momentum $\pi^\pm$,
 599 only $K^\pm$ covered by the correction tables are considered.

600
 601 The K_S^0 are reconstructed from two oppositely charged tracks on which a pion mass hypoth-
 602 esis is applied. As the K_S^0 is long lived we do not require the tracks to pass the selection de-
 603 scribed in Section 3.2.1. The K_S^0 candidate is required to have an invariant mass between 0.47
 604 and 0.53 GeV, and have a momentum vector consistent with the vector between the interac-
 605 tion point and its vertex, requiring `cosAngleBetweenMomentumAndVertexVector` > 0.998 .

606 5. Signal Lepton

607 The signal lepton can either be an electron or a muon. Tracks are required to have a high
 608 likelihood of being an electron or a muon. To identify electrons, the track is required to have
 609 an identification BDT score `pidChargedBDTScore` > 0.9 [30]. To identify muons, the track
 610 is required to have a likelihood identification score `muonID_noSVD` > 0.9 . Other track quality
 611 cuts are applied to the signal leptons: `thetaInCDCAcceptance` ($17^\circ < \theta < 150^\circ$), `nCDCHits`
 612 > 0 , $dr < 1$, and $|dz| < 2$. Furthermore, tracks with a lab frame momentum lower than 0.3
 613 GeV are rejected. These selections are summarised in Tab. V. The signal electron also need
 614 to be corrected for Bremsstrahlung with the `correctBremsBelle` module and an additional
 615 cut of `protonID` < 0.99 is applied on electrons in order to reject protons.

Lepton	$e^\pm$	$\mu^\pm$
Identification	<code>pidChargedBDTScore</code> > 0.9	<code>muonID_noSVD</code> > 0.9
Selections	$ \Delta r < 1$ cm $ \Delta z < 3$ cm <code>thetaInCDCAcceptance</code> ($17^\circ < \theta < 150^\circ$) <code>nCDCHits</code> > 0 ^a $p_{\text{lab}} > 0.3$ GeV	

^a According to the following page: <https://confluence.desy.de/pages/viewpage.action?pageId=349455022>,
`nCDCHits` > 20 is not used for lepton ID. However, it is recommended to use it for hadrons.

TABLE V: Summary of selections applied on tracks to identify signal leptons.

616 After having been pre-selected, the signal lepton candidates are combined with the selected
 617 best B_{tag} . If there are multiple lepton candidates, the event is rejected as this is indicative
 618 of secondary semileptonic decays, identifying the event as $B \rightarrow X_c \ell \nu$. The signal lepton
 619 multiplicity is illustrated in Figure 5. Events with lepton candidates in the ROE are also
 620 rejected for the same reason. We additionally define a side-band selection, consisting of

events in which the lepton candidate is vetoed by the J/ψ or photon conversion vetoes (see below) to study these contributions.

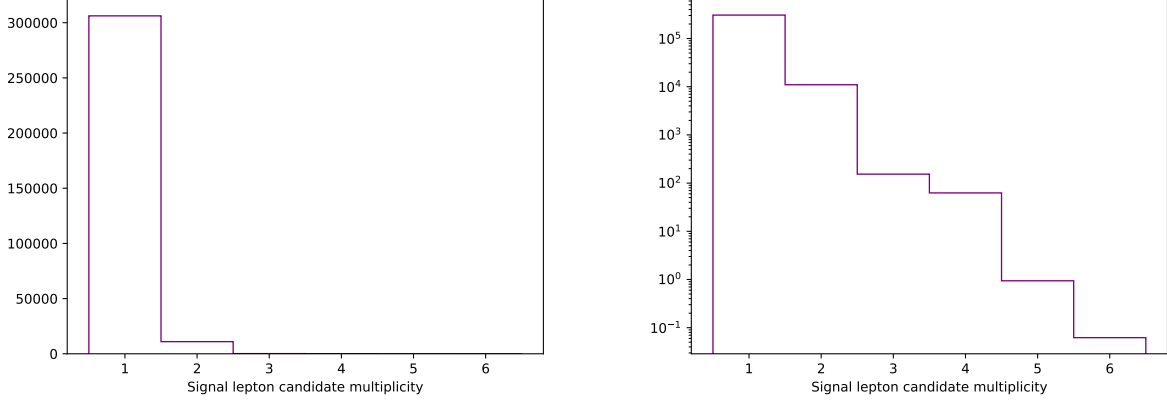


FIG. 5: Signal lepton candidate multiplicity before rejecting events with multiple leptons in linear scale (left) and log scale (right).

6. J/ψ and Photon Conversion Vetoes

In addition to the channels outlined in section 2.2 true leptons may arise from the decay of the J/ψ or, in the case of electrons, from photon conversion in the detector materials. In both cases, the two-body nature of the decay allows for efficient vetoing of these contributions.

We combine the lepton candidate with all oppositely charged tracks in the rest-of-event, and select the pair with an invariant mass closest to the mass of the J/ψ or 0 GeV (for photon conversion) where the oppositely charged track is assigned the same mass hypothesis as the signal lepton. For the J/ψ veto we require the oppositely charged track to pass the selection outlined in Section 3.2.1, for the photon conversion veto such a selection is not required as we expect a significant fraction of conversions to happen at the beam pipe, and applying strict dr and $|dz|$ requirements may miss such conversion events. Tracks used with an electron hypothesis undergo the same bremsstrahlung recovery procedure as the signal lepton. No particle identification selections are made on the second track to remain inclusive. We veto any candidate with:

- $m_{e^+e^-} < 0.050$ GeV, where the threshold has been selected to capture 95.45% of the true photon conversion tracks.
- $3.043 < m_{e^+e^-} < 3.129$ GeV, where the thresholds have been selected to capture 95.45% of true J/ψ candidates above the nominal J/ψ mass and 68.27% below the nominal J/ψ mass. The coverage in the lower mass direction has been reduced due to the long and broad tail caused by bremsstrahlung.
- $3.072 < m_{\mu^+\mu^-} < 3.122$ GeV, where the upper threshold has been selected to capture 95.45% of true J/ψ candidates above the nominal J/ψ mass. The mass peak has a

small broad ‘base’ towards the negative direction. To capture only the peak, and not this broad base, we symmetrise the upper threshold about the nominal J/ψ mass.

The distribution of the selected candidates in the dilepton invariant masses is shown in Fig. 6.

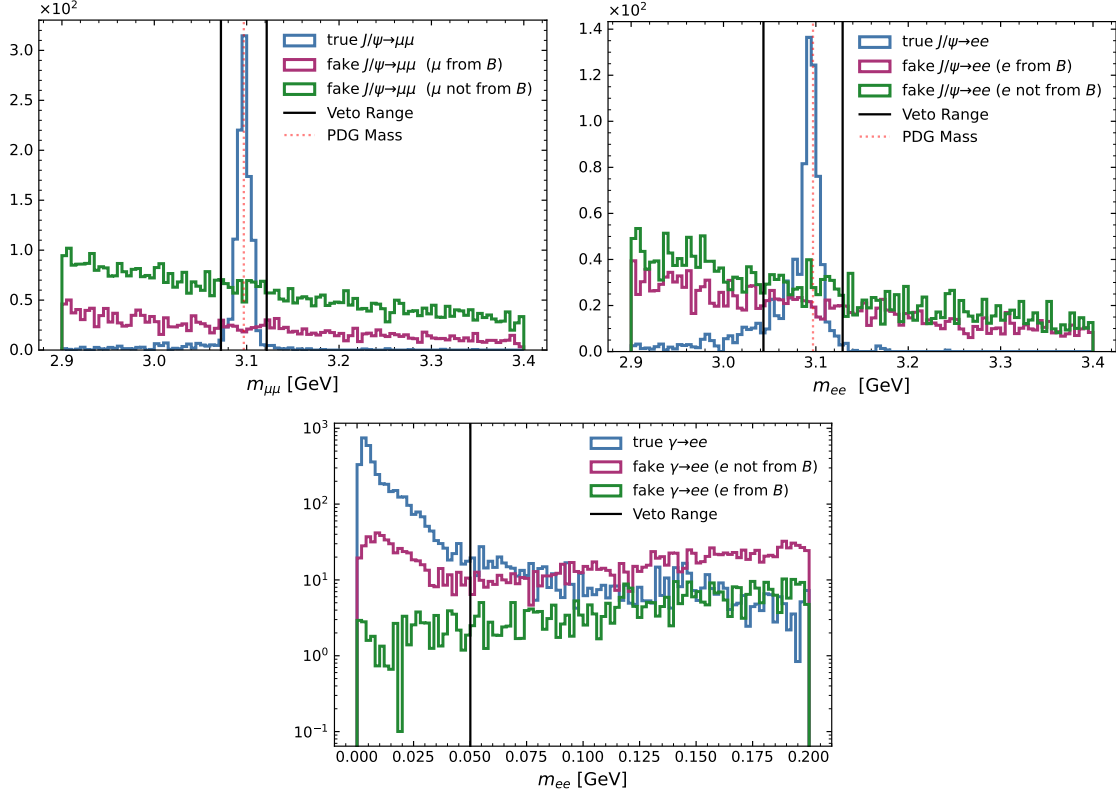


FIG. 6: J/ψ vetoes (*left*: muon, *right*: electron) and photon conversion veto (*bottom*).

3. Other selections

Cuts on the main variables of interest, p_ℓ^B , M_X and q^2 , are very effective to reject background. These cuts are of high importance for theory predictions. A minimal p_ℓ^B cut will always be added as it cuts away very large portions of continuum background as well as background coming from secondary and fake leptons. The low p_ℓ^B region of phase-space is sparsely populated with signal events. This selection is set to $p_\ell^B > 1$ GeV for now but lower cuts will be explored. Additional cuts on M_X and q^2 will be also be considered. They will be mentioned whenever relevant (see Section 8).

4. CONTINUUM MODELLING

Despite ongoing efforts to improve the simulation of the continuum processes, recent studies still show significant disagreement between continuum MC and data. Therefore, we perform a data-driven reweighting in two steps, correcting the normalisation and shape of the continuum sample.

To improve the normalisation, we compare the number of off-resonance data events in the training sample (defined below) to the expected number of off-resonance events from the MC using the training sample defined below. This is done independently for the charged and neutral B_{tag} channels, since the rate at which false B mesons are reconstructed by the FEI may differ between both channels. We find correction factors of 0.874 and 0.938 for the charged and mixed channels.

To improve the shape of the continuum sample, we follow the approach proposed in Ref. [31], and used in the recent Belle $B \rightarrow \mu\nu$ search [32]. This approach is detailed in Ref. [33]. Our primary goal is to improve the modelling of the event-shape variables, as these are subsequently used to train a continuum suppression classifier. We therefore train a classifier on these variables, and a few selected other variables, to distinguish off-resonance data from off-resonance MC. This classifier can then be applied to on-resonance MC, assigning a weight of

$$w_{\text{CW}} = \frac{\mathcal{P}_{\text{CW}}}{1 - \mathcal{P}_{\text{CW}}}, \quad (3)$$

where $\mathcal{P}_{\text{CW}}$ is the classifier response, which is larger as the event is more data-like. The ratio $\mathcal{P}_{\text{CW}}/(1 - \mathcal{P}_{\text{CW}})$ can be interpreted as an estimate of the likelihood ratio $\mathcal{L}^{\text{Data}}/\mathcal{L}^{\text{MC}}$, where $\mathcal{L}^{\text{Data}}, \mathcal{L}^{\text{MC}}$ are the likelihoods of the continuum event to be from data and simulation respectively.

1. Input Variables

As input variables, we include the standard event-shape variables. The full set of variables is:

- Thrust based variables:
 - The magnitude of the thrust of the tag B , $|\mathcal{T}_B|$, and rest of event particles $|\mathcal{T}_O|$.
 - the cosine of the angle between the two thrust axes, $\cos \theta_{\mathcal{T}_B \mathcal{T}_O}$.
 - the cosine of the angle between the thrust of the tag B and the z -axis, $\cos \theta_{\mathcal{T}_B z}$.
- CLEO Cones [34]:
 - The fraction of total momentum flow in nested cones of 10° about the thrust axes: $\mathcal{C}_1, \dots, \mathcal{C}_9$.
- (Kakuno-Super-)Fox-Wolfram (KSFW) moments [35, 36]:
 - The normalised ratio of the second to zeroeth Fox-Wolfram moment R_2 .

- 695 – The KSFW moments: $\mathcal{H}_{00}^{\text{so}}, \mathcal{H}_{01}^{\text{so}}, \mathcal{H}_{02}^{\text{so}}, \mathcal{H}_{03}^{\text{so}}, \mathcal{H}_{04}^{\text{so}}, \mathcal{H}_{10}^{\text{so}}, \mathcal{H}_{12}^{\text{so}}, \mathcal{H}_{14}^{\text{so}}, \mathcal{H}_{20}^{\text{so}}, \mathcal{H}_{22}^{\text{so}}, \mathcal{H}_{24}^{\text{so}},$
696 $\mathcal{H}_0^{\text{oo}}, \mathcal{H}_1^{\text{oo}}, \mathcal{H}_2^{\text{oo}}, \mathcal{H}_3^{\text{oo}}, \mathcal{H}_4^{\text{oo}}.$
- 697 – The missing mass squared, $m_{\text{miss,CS}}^2$ and transverse energy, $E_{\text{t,CS}}.$

698 1. Classifier Training

699 We reconstruct 59687 data and 89142 MC events in the off-resonance sample. These are
700 split into three categories:

- 701 • training (40%), used directly to train the sample and derive the normalisation correc-
702 tion.
- 703 • validation (10%), to prevent the growth of unnecessarily complex models by termi-
704 nating training when performance on this sample, independent of the training sample,
705 plateaus for 20 training iterations.
- 706 • testing (50%), used to check classifier performance. As this sample is not used in
707 training the classifier, it remains statistically independent and can be used later to
708 further constrain the continuum.

709 As this is a simple classification task, we use a boosted decision tree (BDT) as the classi-
710 fier, via the `xgboost` package. Hyperparameters optimisation is performed via a Bayesian
711 optimisation process as implemented in Ref. [37, 38] running for 500 trials, optimising for
712 the best loss on the validation sample. We take a quasi-deep learning approach to this
713 optimisation, floating almost all hyperparameters in the process. The complete set of hy-
714 perparameters along with a brief description, their allowed ranges, and optimised values are
715 given in Tab. VI.

716 We adopt an exponentially decaying learning rate (l) to allow the model to quickly learn to
717 separate the samples while allowing later estimators to fine-tune the separation. To prevent
718 a vanishing learning rate, we enforce a minimum l_{min} . The learning rate for tree n is thus
719 given by

$$l(n) = \max(l_0(1 - r)^n, l_{\text{min}}), \quad (4)$$

721 where l_0, l_{min} and the decay rate r are hyper-parameters that are optimised.
722 Figure 7 provides the loss curve (binary cross-entropy) on the training and validation sam-
723 ples and the evolution of the loss for each trial of the hyperparameter optimisation. The
724 optimisation is able to rapidly find the region of minimum loss, with the majority of itera-
725 tions fine-tuning within this region.

726 For the interpretation of the ratio $\mathcal{P}_{\text{CW}}/(1 - \mathcal{P}_{\text{CW}})$ as a ratio of likelihoods to be valid,
727 the classifier must be well calibrated, with its score

Parameter	Value	Range	Description
Max Depth (d)	5	[1, 11]	Maximum depth of each decision tree. Note the maximum number of trees permitted to the model is dependent on the depth of each tree. If <i>lossguide</i> is adopted as the grow policy we instead limit the tree to 2^d nodes.
Initial Learning Rate (l_0)	0.124	[0.01, 1.0]	Initial learning rate of model. See text for more details.
Minimum Learning Rate ($l_{\min}$)	0.034	[0.001, 0.1]	Minimum learning rate of model. See text for more details.
Learning Rate Decay (r)	0.0026	$[10^{-6}, 0.05]$	Decay of the learning rate. See text for more details.
Alpha (α)	0.078	$[10^{-5}, 10^4]$	Minimum reduction in loss required to further split a node of the tree. Can help to prevent over-fitting.
Lambda (λ)	53	[0.05, 1000]	L1 regularisation term on the weights. Can help to prevent over-fitting.
Gamma (γ)	0.007	[0.001, 2]	L2 regularisation term on the weights. Can help to prevent over-fitting.
Subsample	0.895	[0.5, 1.0]	Fraction of training events sampled used to train each tree. Can help to prevent over-fitting.
Column Subsample	0.559	[0.5, 1.0]	Fraction of variables sampled used in training each tree. Can help to prevent over-fitting by forcing the model to rely on a broader range of variables.
Minimum Child Weight	5	-	Minimum number of training events required in each node after splitting a sample. Can prevent over-fitting by preventing extremely targeted selections.
Grow Policy	<i>depthwise</i>	[<i>depthwise</i> , <i>lossguide</i>]	Whether new nodes are added to a tree in order of closest to the root node or in order of maximum reduction in loss. If <i>lossguide</i> is selected rather than limiting the maximum depth of the tree we limit the maximum number of nodes of each tree.
Tree Method	<i>gpu_hist</i>	-	The method used by XGBoost to train the trees. The <i>hist</i> methods pre-bin continuous features into histograms and base their node splitting on the bin indices of the histograms. This strategy is also employed by <i>FastBDT</i> [39].
nBins	512	-	Number of bins used to bin continuous features.

TABLE VI: Hyperparameters of the XGBoost model. For parameters tuned by Bayesian optimisation, the range investigated by the optimiser is provided in the “Range” column. All parameters not listed are left at their default value as per version 1.7.6 of XGBoost.

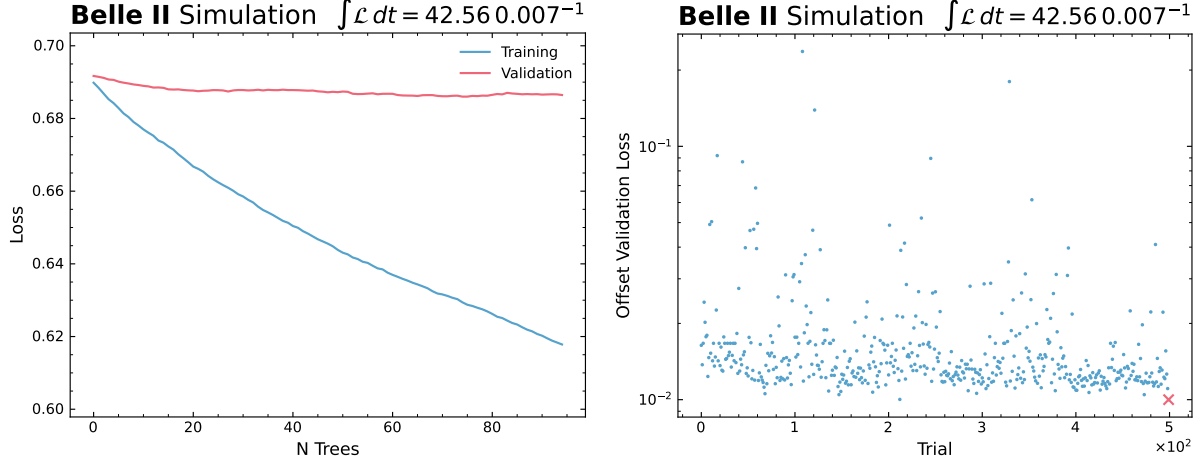


FIG. 7: *Left*: Validation and training loss for the selected classifier. The early stopping terminates the model growth after 182 trees. *Right*: Final validation loss of the individual trials of the hyper-parameter optimisation. The selected classifier is shown as a pink cross. To better show the variations near the minimum loss, all validation losses have been offset such that the minimum is at 0.01.

2. Variable Importance

We estimate the importance of each variable to the classifier via their Shapley values [40], which provide the average marginal contribution of a feature to the output, as implemented in the SHAP package of Refs. [41, 42]. In this context, a higher importance indicates a larger disagreement between the data and MC in these variables. The importances, as measured on the validation sample, are presented in Fig. 8.

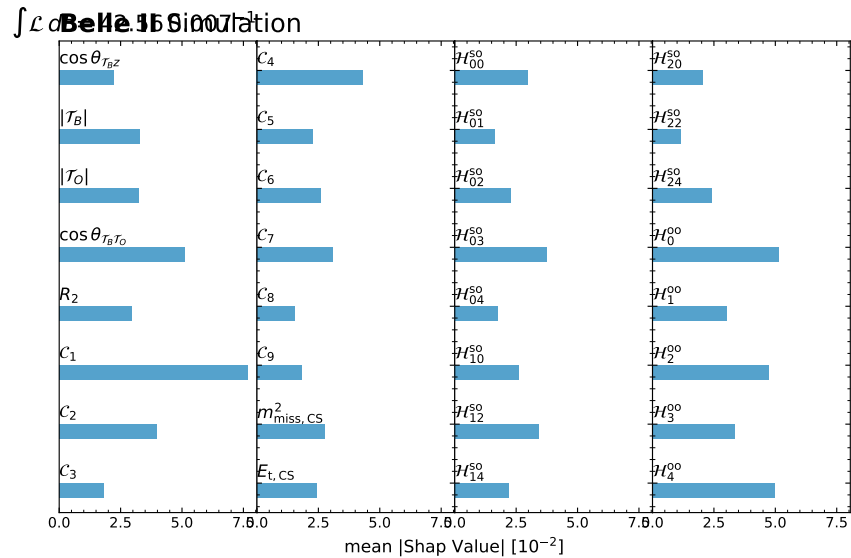


FIG. 8: Variable importance for the data-driven continuum reweighting.

735 3. Uncertainty

736 Two sources of systematic uncertainty are assigned to the reweighting procedure. Firstly,
 737 the full size of the normalisation shift is taken as an uncertainty. Secondly, we propagate the
 738 statistical uncertainty of the training sample by repeating the training 200 times, varying
 739 the weight of each event by drawing from a Poisson distribution of mean 1.

740 4. Results

741 Figure 9 shows the output of the nominal classifier. Figure 10 gives an example of the
 742 pre-reweighting and post-reweighting distributions in R_2 . Figure 11 shows a data-MC com-
 743 parison of the three main kinematic variables in the off-resonance sample. For a comparison
 744 of all the input variables before and after the reweighting see Appendix L.

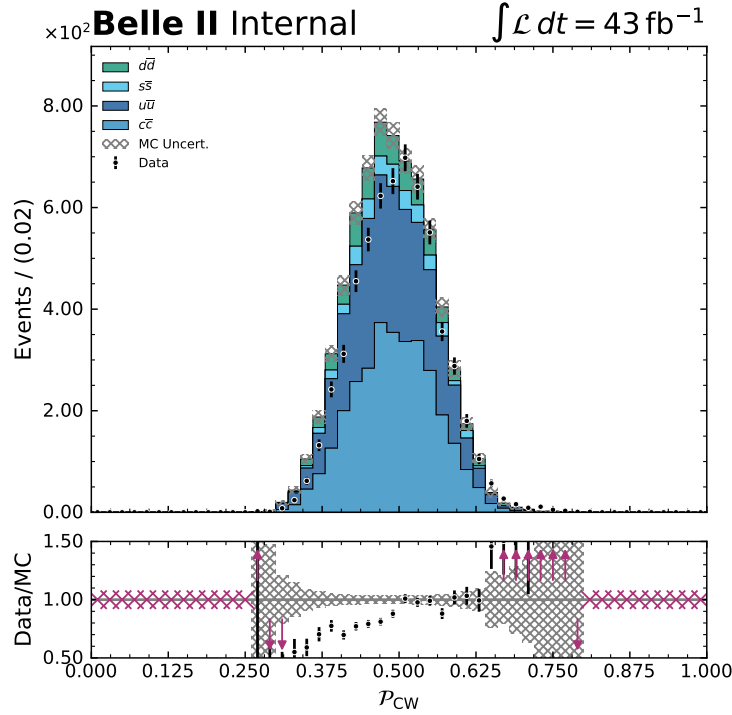


FIG. 9: Distribution of the nominal classifier response for the classifier trained to distinguish data from Monte Carlo. All MC uncertainties except the continuum reweighting ones are shown here.

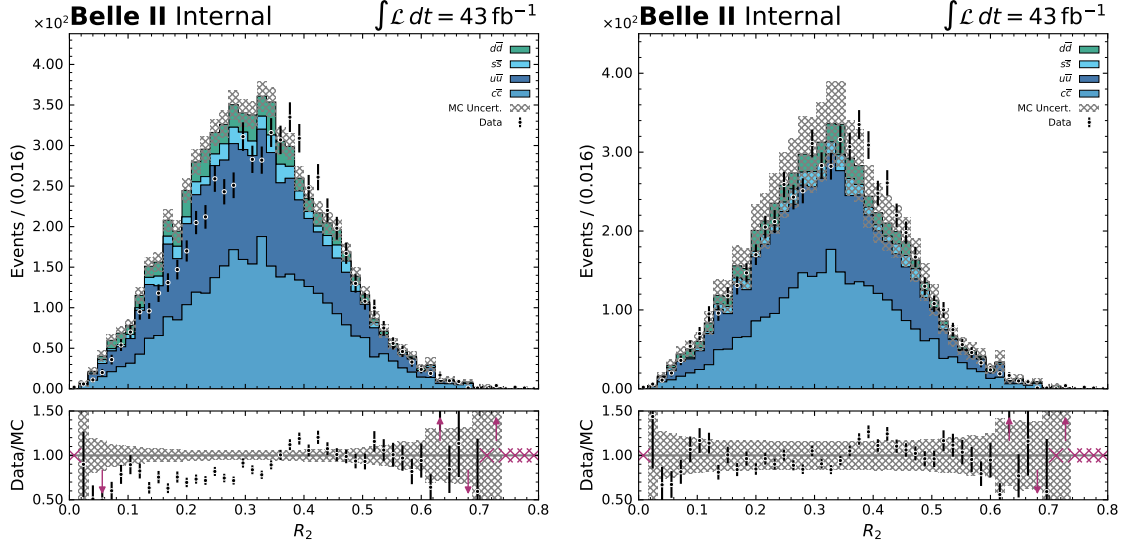


FIG. 10: Distribution of R_2 before and after the data-driven reweighting. All MC uncertainties except the continuum reweighting ones are shown on the left. On the right these are also included.

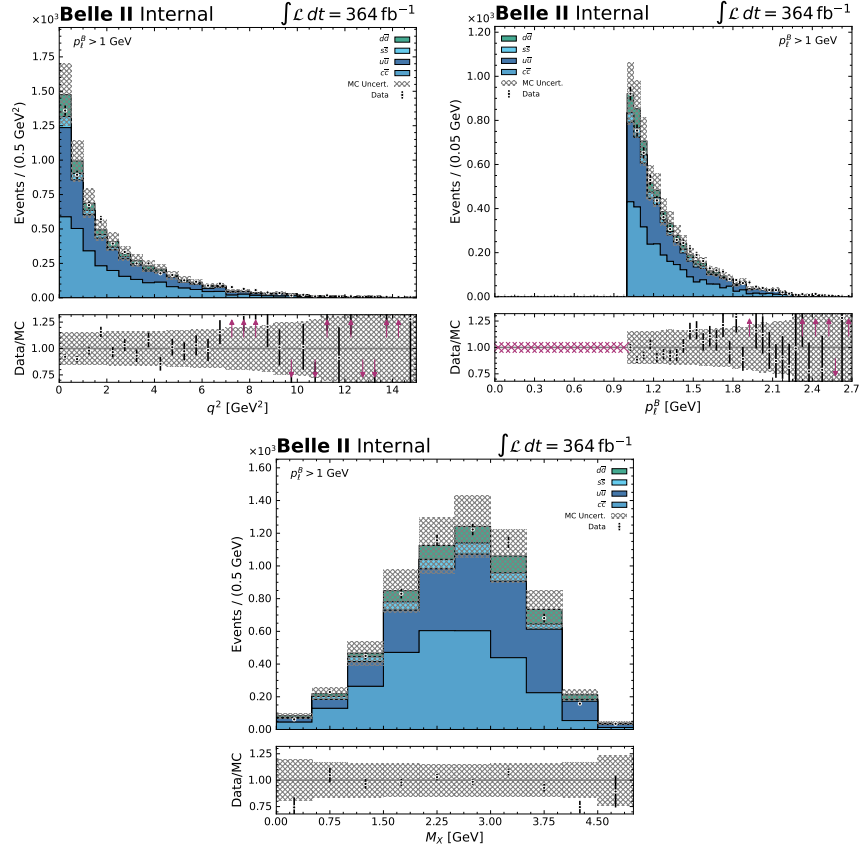


FIG. 11: Distribution of q^2 , p_ℓ^B and M_X after the data-driven reweighting. All MC uncertainties including the continuum reweighting ones are shown here.

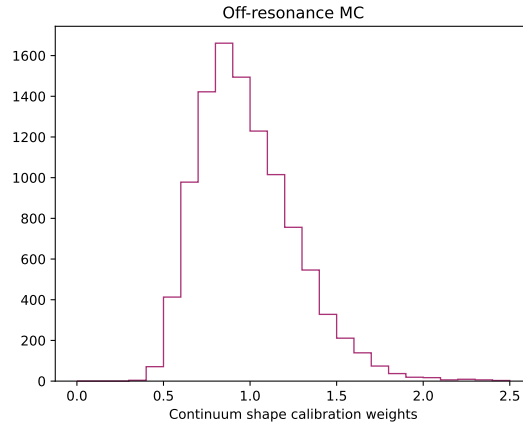


FIG. 12: Distribution of continuum calibration weights in the nominal preselected sample.

745 5. MULTIVARIATE SELECTION

746 After an initial preselection, two Multivariate (MVA) classifiers are applied to the data in
 747 order to further distinguish between signal and background events. Signal corresponds to
 748 $B \rightarrow X_u \ell \nu$ events. The first classifier trains against continuum events and the second one
 749 against $B \rightarrow X_c \ell \nu$ events. In both cases, a neural network (NN) from the `tensorflow`
 750 package is trained. The hyperparameters are optimised using the `optuna` package. For
 751 both classifiers, 5% of the MC $X_u \ell \nu$ signal and 20% of the respective MC background were
 752 randomly selected to form a training sample. The rest forms the test sample. Both the
 753 training sample and test sample are rescaled to match the initial total weight of the MC
 754 sample.

755 1. Classifiers and training

756 1. Continuum suppression

757 The variables used to reject continuum are listed in Section 4.1. The corresponding distri-
 758 butions are shown in Appendix B. Based on the observed data-MC agreement the following
 759 14 variables are chosen:

- 760 • $\mathcal{C}_2$
- 761 • $\cos \theta_{\mathcal{T}_B \mathcal{T}_O}$
- 762 • $\cos \theta_{\mathcal{T}_B z}$
- 763 • $\mathcal{H}_1^{\text{oo}}$
- 764 • $\mathcal{H}_3^{\text{oo}}$
- 765 • $\mathcal{H}_4^{\text{oo}}$
- 766 • $\mathcal{H}_{01}^{\text{so}}$
- 767 • $\mathcal{H}_{02}^{\text{so}}$
- 768 • $\mathcal{H}_{03}^{\text{so}}$
- 769 • $\mathcal{H}_{04}^{\text{so}}$
- 770 • $\mathcal{H}_{12}^{\text{so}}$
- 771 • $\mathcal{H}_{14}^{\text{so}}$
- 772 • $\mathcal{H}_{20}^{\text{so}}$
- 773 • N_{tr}^{CS}

Parameter	Value	Range	Description
N Layers	2	[2, 4]	Number of layers in the feed forward network.
Layer Size	[256, 128]	{8, 16, 32, 64, 128, 256}	Number of neurons in each layer of the network. The number is independently selected for each layer and is restricted to a power of two.
Activation Function	Swish	-	We use the swish [43] activation function of the form $f(x) = \frac{x}{1+\exp(-x)}$. Tests with ReLU showed similar performance, indicating the network is largely agnostic to the specific activation function.
Optimiser	AdamW	-	We adopt the AdamW [44] optimiser which computes adaptive learning rates for each parameter and has demonstrated rapid convergence in few epochs. Tests using the SGD optimiser converged on similar models.
Learning Rate	$9.3 \cdot 10^{-3}$	$[10^{-5}, 10^{-2}]$	Learning rate of the model. Governs the rate at of gradient descent.
Weight Decay	$9.6 \cdot 10^{-4}$	$[10^{-8}, 10^{-3}]$	Strength of the weight decay regularisation. Helps to avoid over-fitting by penalising overly complex models.
Dropout Rate	0.25		We add a dropout layer after each dense layer. This parameter governs the rate at which the dropout layers randomly drop a fraction of input connections. The remaining connections are re-scaled to compensate. Dropout is used to reduce overtraining of the classifier so its value is fixed.
Label Smoothing	$8.8 \cdot 10^{-5}$	$[10^{-5}, 0.05]$	Slightly smooths the labels by changing them from binary, 1 or 0 for signal and background, to $1 - s$ and s . This helps to prevent over-fitting, and prevents issues with many events being assigned to exactly 1 or 0 due to floating point precision.
Batch Size	4096	{1024, 2048, 4096}	Size of a minibatch for each epoch of training. The batches are large to allow for accurate calculation of the DisCo loss term for each batch.

TABLE VII: Hyperparameters of the continuum suppression neural network. All hyperparameters not mentioned are left at their default values as of **tensorflow v2.17.0**.

774 The optimal cut on this classifier is chosen according to the maximal significance defined as:
775 $S/\sqrt{S+B}$, where S is the number of $B \rightarrow X_u \ell \nu$ events and B the number of continuum
776 events that pass the selection. The values of the hyperparameters for the optimal classifier
777 are shown in Tab. VII.

778

779 The performance of the continuum suppression classifier is illustrated in Figures 13, 14 and
780 15. We evaluate overtraining from the corresponding Receiver Operating Characteristic
781 (ROC) curve as shown in Fig. 16.

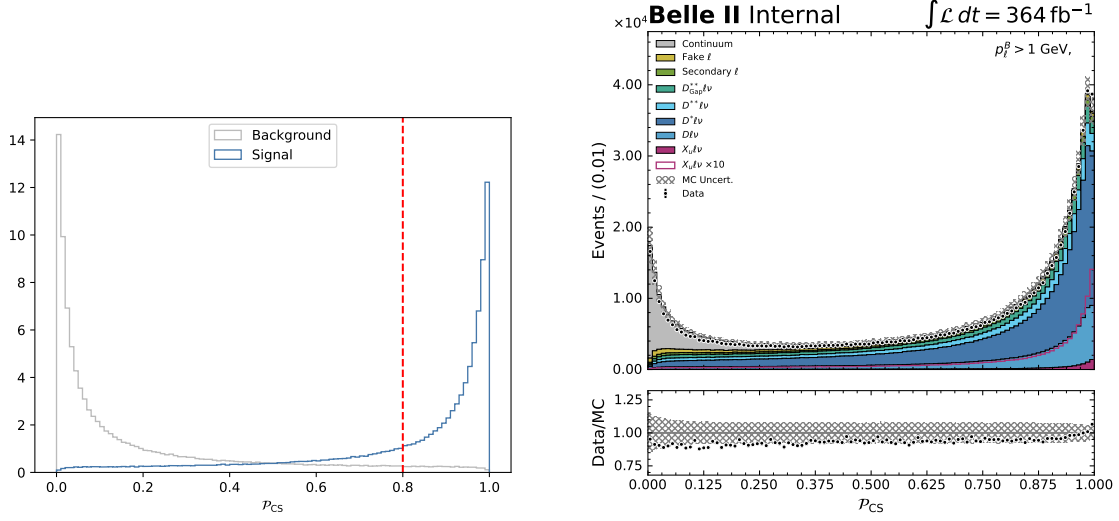


FIG. 13: The continuum suppression score. On the left the score is shown as signal vs background. After optimisation, the cut score is equal to 0.80. The plot on the right shows the good data-MC agreement of the score.

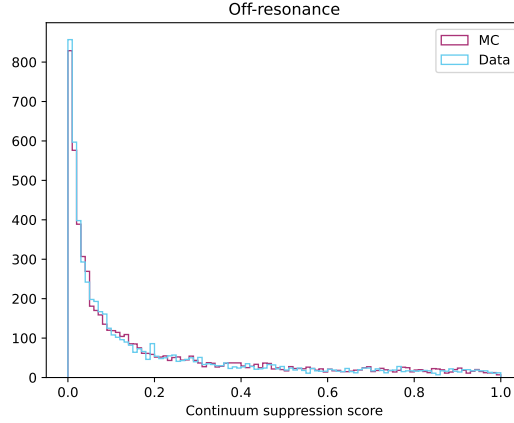


FIG. 14: The continuum suppression score of the MC and data off-resonance samples.

2. $B \rightarrow X_c \ell \nu$ background

The variables used in the second MVA classifier trained against $B \rightarrow X_c \ell \nu$ events are listed in Tab. VIII. This classifier is trained and applied after adding the continuum suppression selection discussed above. In addition, a kaon veto is applied as K are characteristic of charm meson decays. The cut on K multiplicities is discussed in Section 7.

The MVA cut minimising the value of the partial $B \rightarrow X_u \ell \nu$ branching fraction total uncertainty in an Asimov fit is chosen as the optimal MVA cut. The extraction of the partial branching fraction and its associated uncertainty is discussed in later Sections. The optimal MVA cut is obtained from a nominal fit. We illustrate this classifier's performance using a cut obtained from a fit to q^2 in a region of phase-space where $p_\ell^B > 1.0$ GeV. The optimal

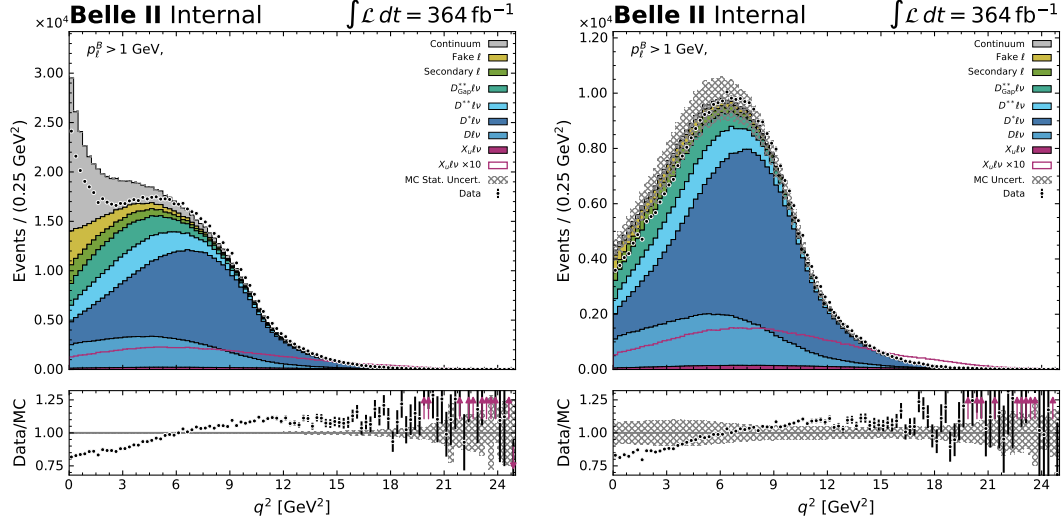


FIG. 15: Distribution of q^2 before (left) and after (right) applying the optimal continuum suppression selection.

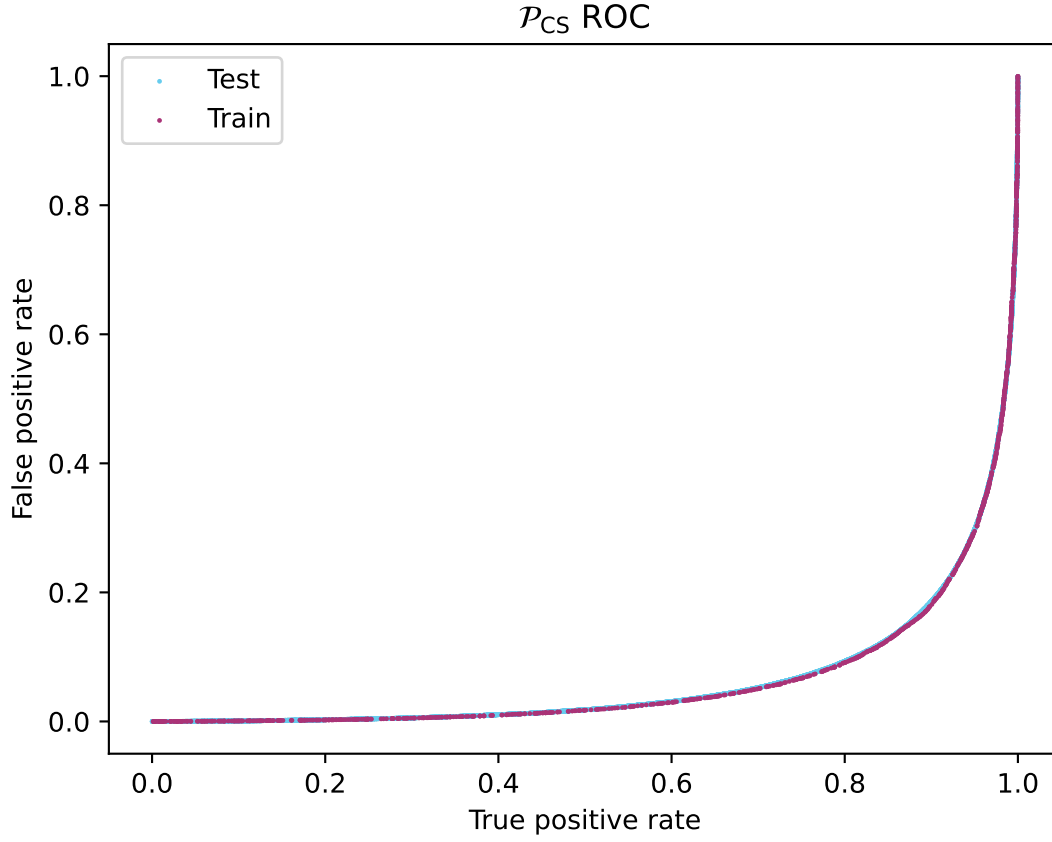


FIG. 16: Receiver Operating Characteristic curve for the continuum suppression MVA classifier.

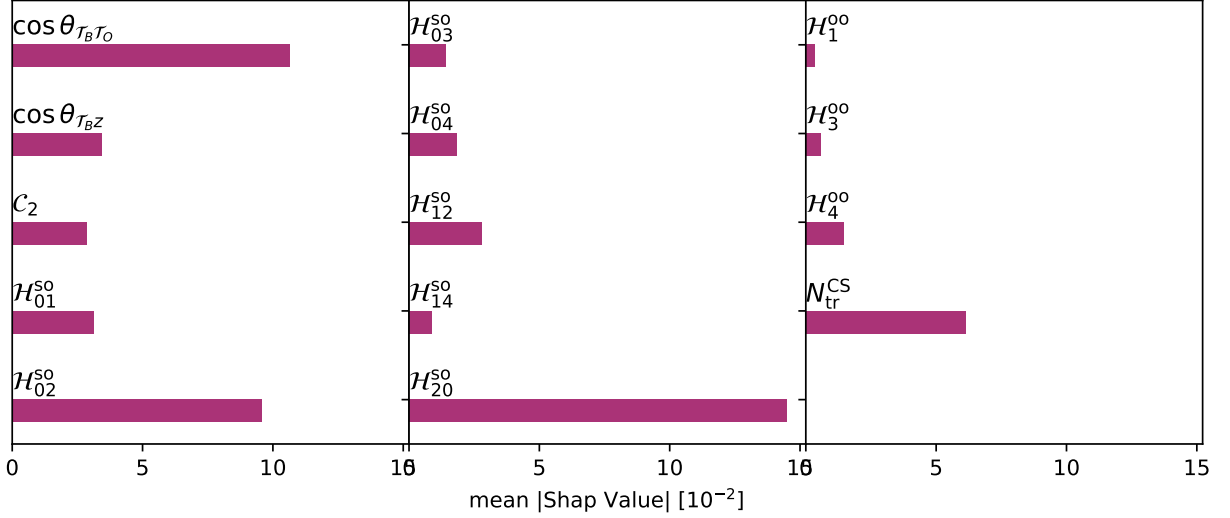
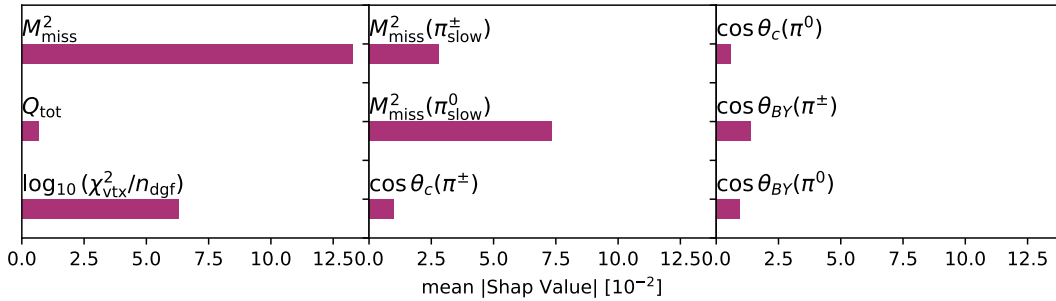
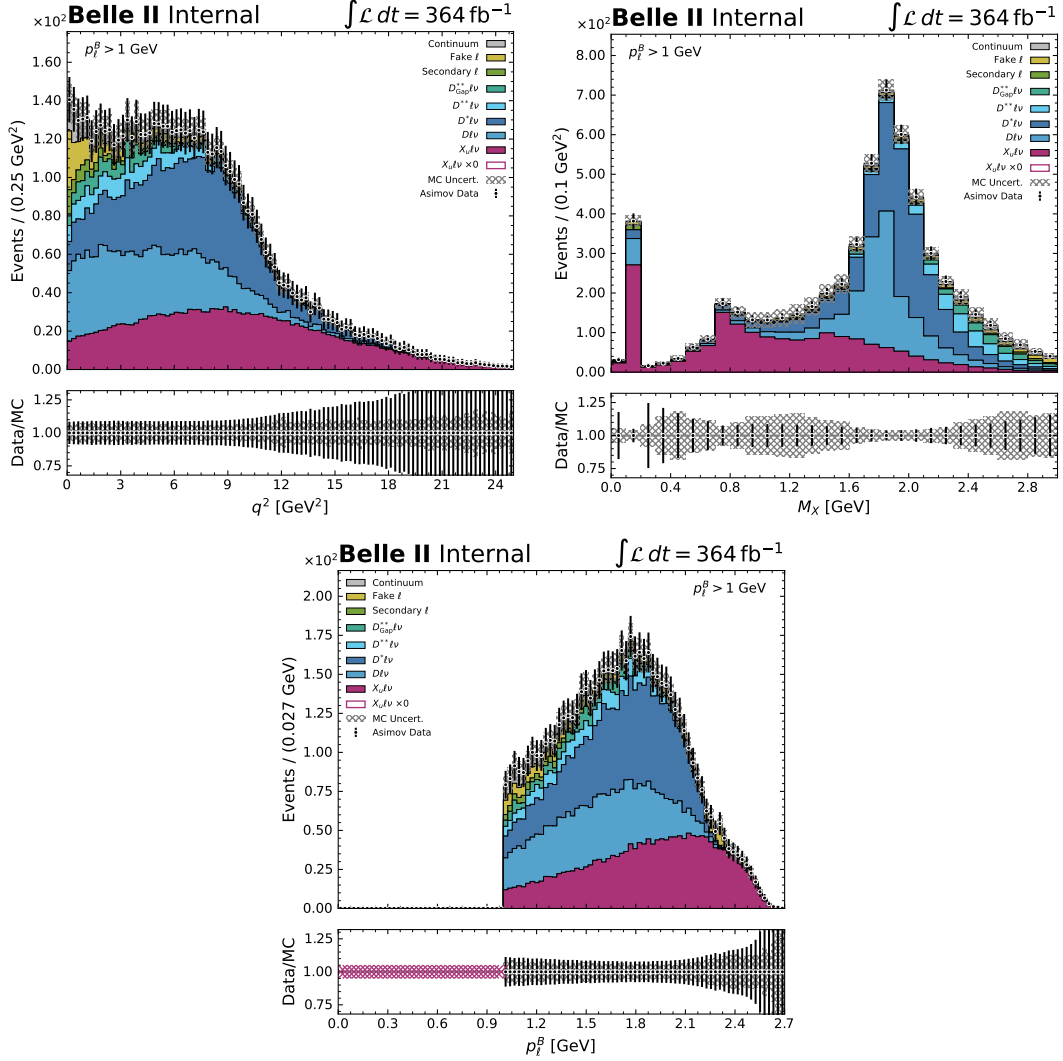


FIG. 17: Input feature importances (quantified using the **shap** values) for the continuum suppression MVA classifier.

Notation	Definition
M_{miss}^2	Missing mass squared of the semi-leptonic event
$M_{\text{miss}}^2(\pi_{\text{slow}}^\pm)$	Invariant mass squared of the system recoiling against the reconstructed slow $\pi^\pm$ coming from a D^*
$M_{\text{miss}}^2(\pi_{\text{slow}}^0)$	Invariant mass squared of the system recoiling against the reconstructed slow π^0 coming from a D^*
$\log_{10}(\chi_{\text{vtx}}^2/n_{\text{dof}})$	χ^2 of the ROE vertex fit
$\cos \theta_c(\pi^\pm)$	Cosine of the angle between the D^* reconstructed from the slow $\pi^\pm$ and the lepton flight directions in the CMS frame
$\cos \theta_c(\pi^0)$	Cosine of the angle between the D^* reconstructed from the slow π^0 and the lepton flight directions in the CMS frame
$\cos \theta_{BY}(\pi^\pm)$	Cosine of the angle in CMS frame between the B reconstructed from the slow $\pi^\pm$ and a nominal B particle
$\cos \theta_{BY}(\pi^0)$	Cosine of the angle in CMS frame between the B reconstructed from the slow π^0 and a nominal B particle
Q_{tot}	Total charge of event

TABLE VIII: $B \rightarrow X_c \ell \nu$ MVA input features

cut is equal to 0.87. The classifier's performance is shown in Fig. 18 and its score in Fig. 20. Its ROC curve is shown in Fig. 21 and its loss curve is shown in Fig. 22. We also display the importance of each input feature in Fig. 19. The importance is calculated with the **shap** package using the Kernel method. The values of the optimised classifier hyper-parameters are shown in Tab. IX.



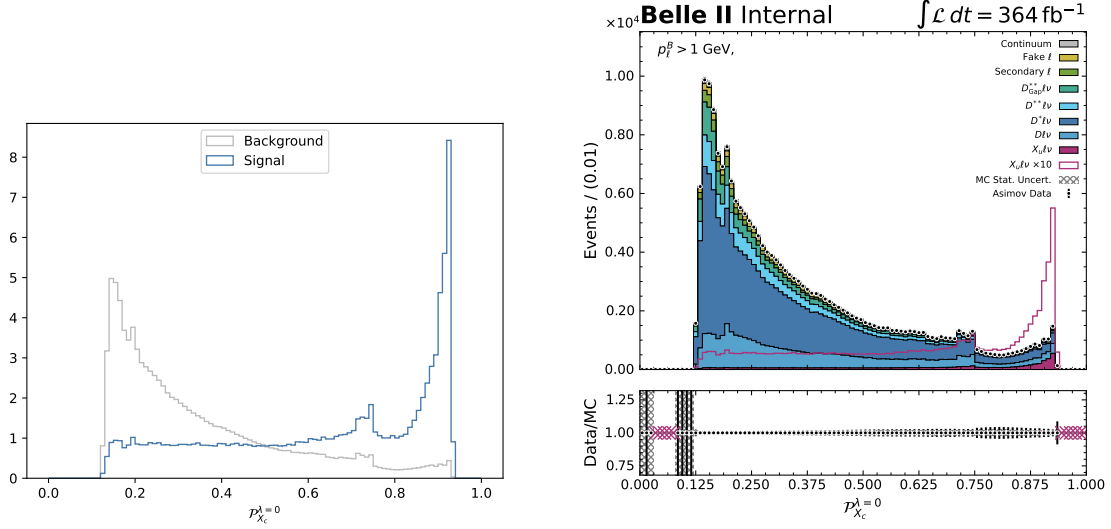


FIG. 20: The $B \rightarrow X_c \ell \nu$ suppression score. On the left, the yields of each distribution are normalised to be visible on the same scale. On the right, the composition of the score is broken down. The optimal cut is found to be 0.87.

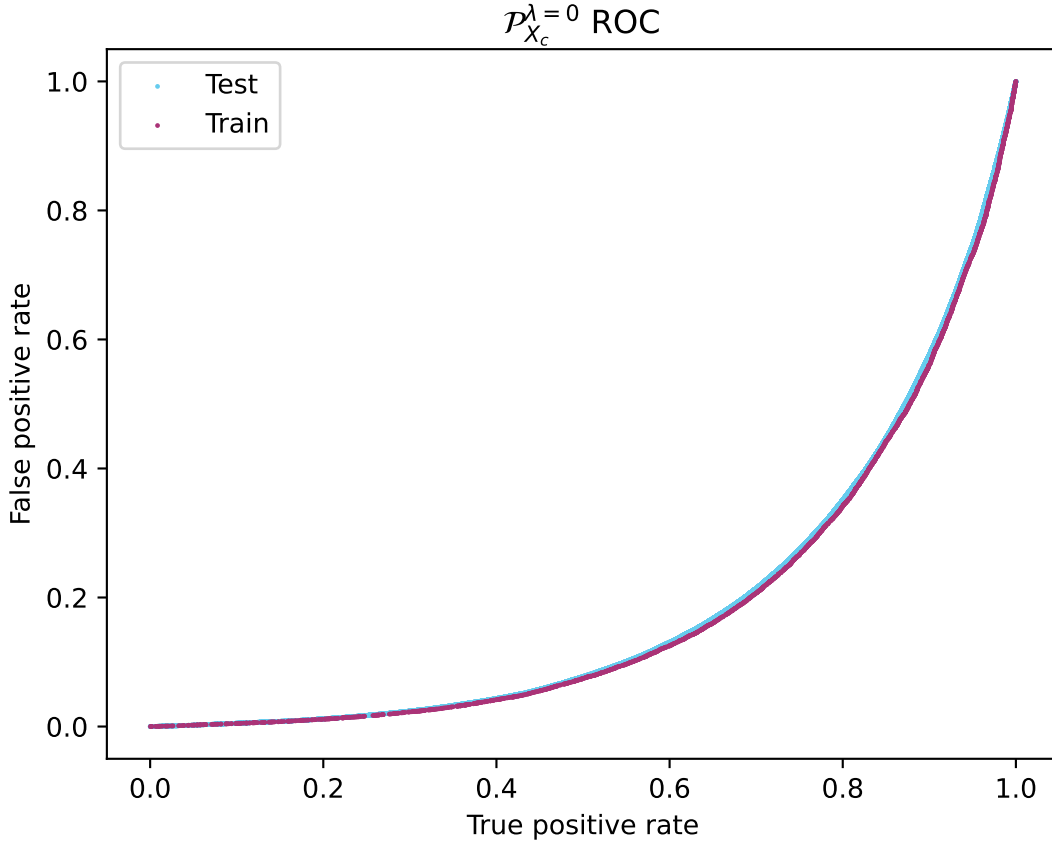


FIG. 21: ROC curve of the optimised $B \rightarrow X_c \ell \nu$ suppression MVA classifier.

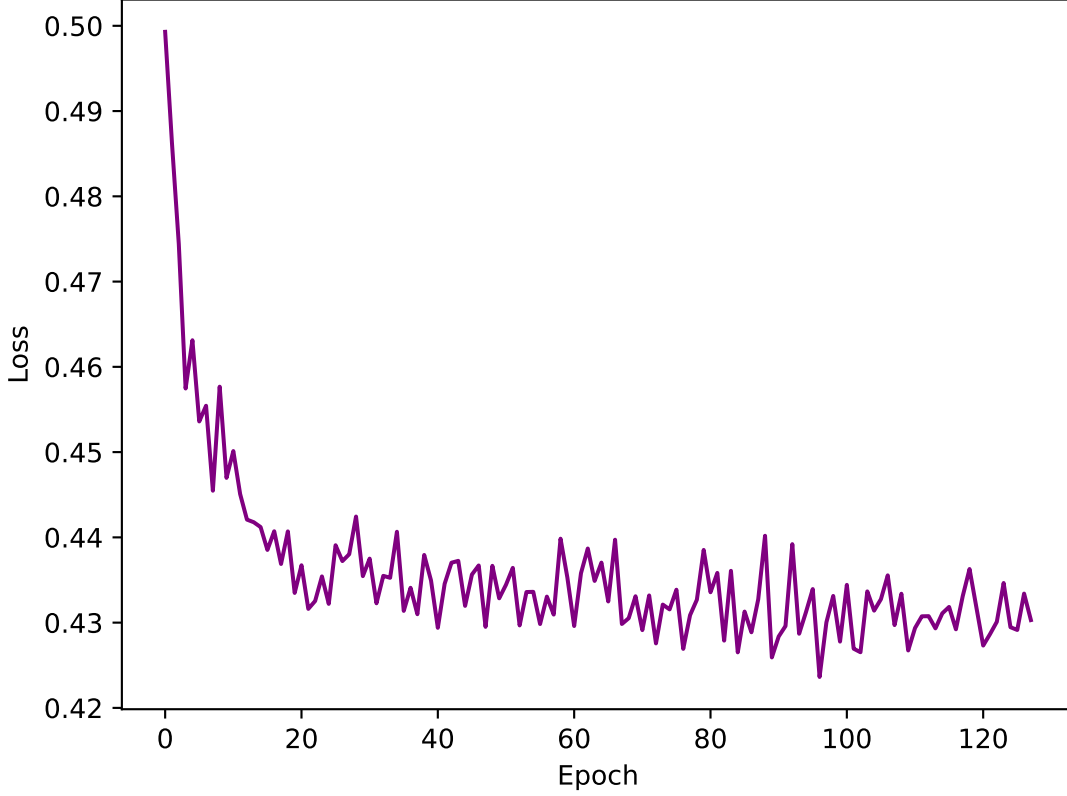


FIG. 22: Training loss epoch by epoch of the $B \rightarrow X_c \ell \nu$ suppression MVA classifier. The training of the neural network is stopped after 128 epochs.

800 The output score of the $B \rightarrow X_c \ell \nu$ classifier (Fig. 20) exhibits a non-smooth behaviour
801 which could point to overtraining. This behaviour was investigated and it was concluded
802 that the spikes that appear in the score distribution correspond to particularities of the
803 input features. To isolate the effects of the input features, the classifier was retrained using
804 the nominal hyper-parameter values given in Tab. IX with a subset of the input features.
805 The following observations were made:

- 806 • Classifier trained using only the 6 slow pion variables from Tab. VIII: for all events for
807 which a slow pion hasn't been found and reconstructed these variables are not defined.
808 For these events, a common value is given to the slow pion variables for all events. The
809 value is chosen to be far away from the distribution of the given variable for events
810 where it's defined. The output score for these events is therefore always equal to 0.5
811 as the classifier cannot use them. This results in a large spike at 0.5 in the output
812 score distribution.
- 813 • Classifier trained using the slow pion variables and M_{miss}^2 : a large spike towards large
814 scores (0.8 - 0.9) is observed. When isolating this spike in the distribution it was
815 observed that it corresponds to a tight cut of M_{miss}^2 around 0. M_{miss}^2 being the most
816 important variable in the classifier, it is expected that a tight cut around the nominal

Parameter	Value	Range	Description
N Layers	4	[2, 4]	Number of layers in the feed forward network.
Layer Size	[64, 32, 128, 128]	{8, 16, 32, 64, 128, 256}	Number of neurons in each layer of the network. The number is independently selected for each layer and is restricted to a power of two.
Activation Function	Swish	-	We use the swish [43] activation function of the form $f(x) = \frac{x}{1+\exp(-x)}$. Tests with ReLU showed similar performance, indicating the network is largely agnostic to the specific activation function.
Optimiser	AdamW	-	We adopt the AdamW [44] optimiser which computes adaptive learning rates for each parameter and has demonstrated rapid convergence in few epochs. Tests using the SGD optimiser converged on similar models.
Learning Rate	$3.30 \cdot 10^{-3}$	$[10^{-5}, 10^{-2}]$	Learning rate of the model. Governs the rate at of gradient descent.
Weight Decay	$1.15 \cdot 10^{-5}$	$[10^{-8}, 10^{-3}]$	Strength of the weight decay regularisation. Helps to avoid over-fitting by penalising overly complex models.
Dropout Rate	0.40	-	We add a dropout layer after each dense layer. This parameter governs the rate at which the dropout layers randomly drop a fraction of input connections. The remaining connections are re-scaled to compensate. Dropout is used to reduce overtraining of the classifier so its value is fixed.
Label Smoothing (s)	$7.05 \cdot 10^{-3}$	$[10^{-5}, 0.05]$	Slightly smooths the labels by changing them from binary, 1 or 0 for signal and background, to $1 - s$ and s . This helps to prevent over-fitting, and prevents issues with many events being assigned to exactly 1 or 0 due to floating point precision.
Batch Size	4096	{1024, 2048, 4096}	Size of a minibatch for each epoch of training. The batches are large to allow for accurate calculation of the DisCo loss term for each batch.
L1/L2 Regularisers	0.01, 0.01	-	L1 and L2 regularisations are penalty terms to the loss function. They reduce the complexity of the model which helps avoiding overtraining.

TABLE IX: Hyperparameters of the $B \rightarrow X_c \ell \nu$ suppression neural network. All hyperparameters not mentioned are left at their default values as of **tensorflow v2.11.0**. The classifier is trained over 128 epochs.

neutrino mass indeed isolates signal events. This is illustrated in Figure 24.

- Classifier trained using the slow pion variables and the total event charge: a large spike of events appears in the distribution. When isolated it was observed that it corresponds to values of Q_{tot} equal to 0.

The combination of all the above mentioned effects results in the non smooth distribution

822 observed in Fig. 20.

823 2. MVA-induced model dependence

824 As the MVA classifiers we use are trained on simulated data, the output score used to select
825 events depends on the modelling of the MC. To avoid introducing model dependence various
826 strategies have been developed for MVA classifiers (adversarial networks [45], DisCo [46]...).
827 The continuum is corrected using off-resonance data (see Section 4) and relies on well-
828 modelled variables. The model dependence induced by the continuum suppression classifier
829 is therefore expected to have a very limited impact. However, as the $B \rightarrow X_c \ell \nu$ classifier
830 relies on the modelling of both $B \rightarrow X_c \ell \nu$ and $B \rightarrow X_u \ell \nu$ events, it is preferable to avoid
831 any model dependence. To evaluate this effect, we monitor the correlation between the
832 classifier output score and each of the three kinematic variables p_ℓ^B , M_X and q^2 . This is
833 shown in Fig. 23 where no strong correlations are observed between the three variables and
834 the classifier score. As the proportion of signal to background is not uniform in the three
835 kinematical variables, it is expected that more events with a higher or lower score will be
836 present in particular regions of phase-space. The DisCo algorithm was implemented and
837 tested in order to limit the correlation between the score and the kinematical variables.
838 However, its implementation leads to unexpected behaviour of the classifier and its effect
839 appeared to be limited in our case so it was decided to not use it. The superscript $\lambda = 0$
840 appearing in the score label indicates that the DisCo algorithm was not used during the
841 neural network training.

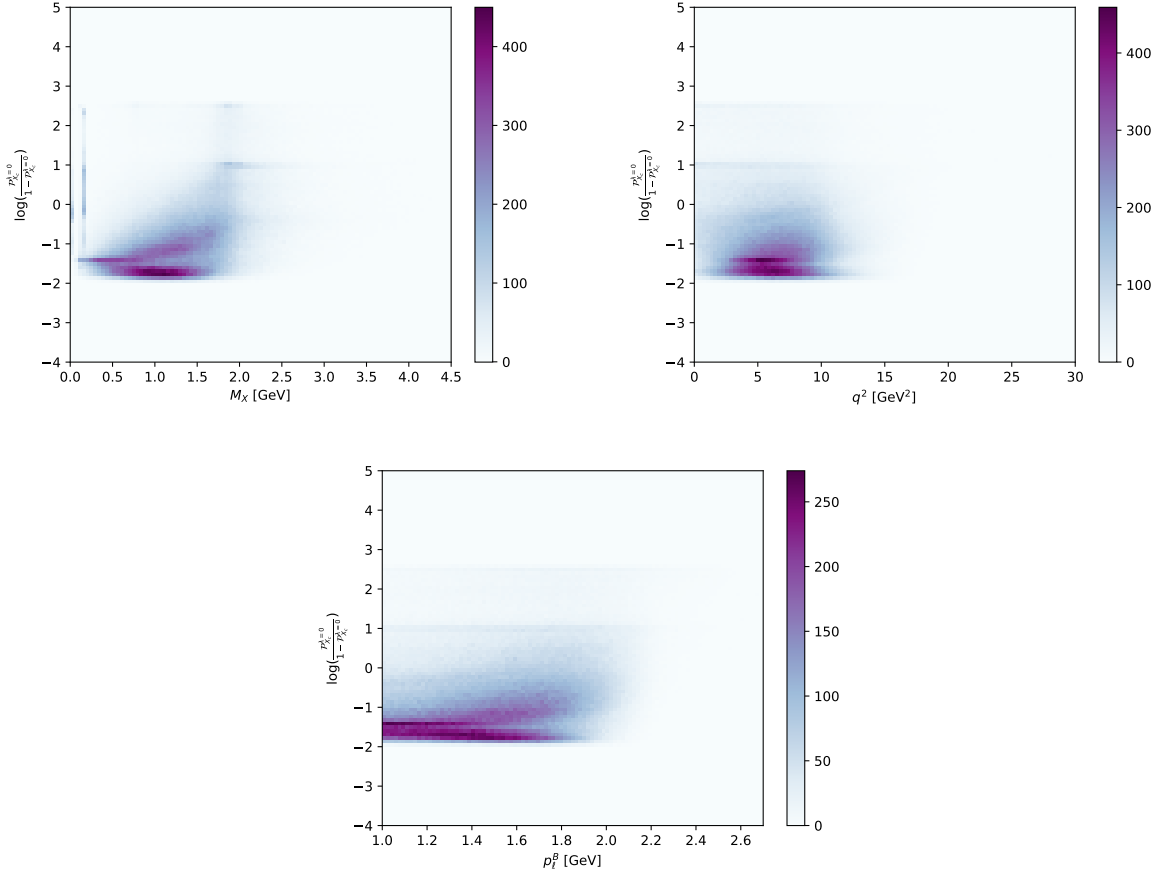


FIG. 23: Two-dimensional distributions of the $B \rightarrow X_c \ell \nu$ MVA score versus the three kinematical variables p_ℓ^B , M_X and q^2 .

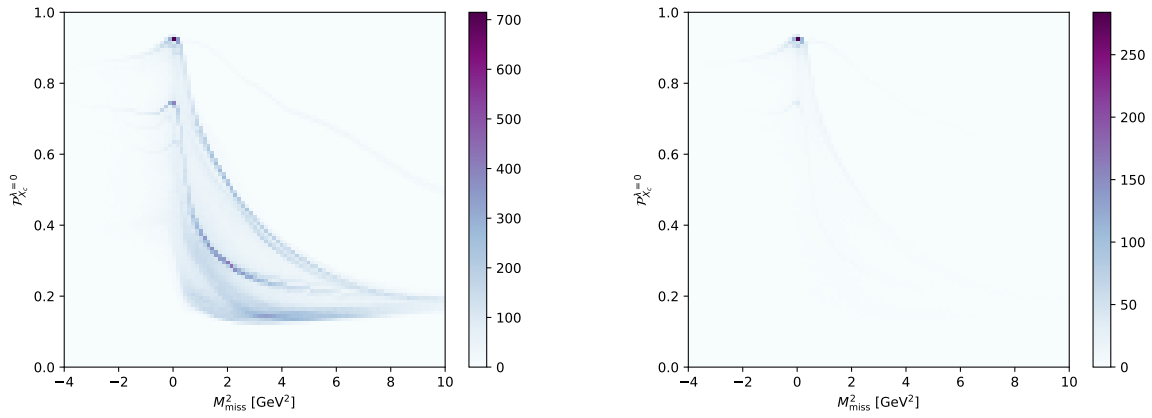


FIG. 24: Two-dimensional distributions of the $B \rightarrow X_c \ell \nu$ MVA score versus M_{miss}^2 for all events (left) and for signal events (right).

6. CORRECTIONS AND SYSTEMATICS

1. Slow π efficiency

We carefully treat slow π coming from $B \rightarrow D^* \ell \nu$ events as their properties are used in the MVA selection. They can be rejected by the ROE mask but they need to be taken into account in the slow π efficiency reweighting. Charged and neutral π are treated separately. The selections applied on slow π are given in Section 3.

All $\pi^\pm$ reconstructed in the ROE and $\pi^\pm$ reconstructed from a D^* decay and not caught by the ROE mask are assigned a correction factor. Each $\pi^\pm$ corresponds to a track so the correction factor assigned to these pions corresponds to a track correction. The track correction defined in Section 6.2 is only applied on tracks with a momentum higher than 200 MeV. As π^0 don't leave tracks, only the ones used in a D^* reconstruction and caught by the ROE mask are corrected.

The efficiencies are given in three bins of momentum between 50 and 200 MeV. The correction factors are summarised in Tab. X and are taken from Ref. [27] and [28] for $\pi^\pm$ and π^0 respectively. The efficiency in the 200 - 400 MeV bin is assumed to be 1 and is therefore used as a normalisation for the three low momentum bins. For each π of a given type in an event and a given bin the corresponding efficiency scale factor is applied to the event.

High momentum π^0 ($p > 200$ MeV) are also reconstructed using a different channel than the slow π^0 . Because of the higher statistics in this momentum region, the extracted correction factors are deemed more reliable. Therefore, the correction factor obtained for $200 \text{ MeV} < p < 400 \text{ MeV}$ is used as an additional normalisation factor for slow π^0 . This number is equal to 0.818 and is directly multiplied to the correction factors.

$\pi^\pm$			
Bins (GeV)	[0.05, 0.12]	[0.12, 0.16]	[0.16, 0.2]
Correction factors	$0.996 \pm 0.020 \pm 0.013 \pm 0.003$	$0.990 \pm 0.015 \pm 0.013 \pm 0.003$	$0.987 \pm 0.017 \pm 0.013 \pm 0.003$
π^0			
Bins (GeV)	[0.05, 0.10]	[0.10, 0.15]	[0.15, 0.2]
Correction factors	$1.027 \pm 0.103 \pm 0.054$	$0.963 \pm 0.063 \pm 0.024$	$1.033 \pm 0.031 \pm 0.033$

TABLE X: Slow π efficiency correction factors. An additional factor of 0.818 is applied on the π^0 factors. The uncertainties are respectively the uncorrelated statistical, correlated statistical and systematic uncertainties for $\pi^\pm$ and (uncorrelated) statistical and systematic for π^0 . Refer to the main text for details.

Three types of uncertainties are quoted: the uncorrelated and correlated statistical errors (because of the normalisation to the [200; 400] MeV bin) and the systematical error. The total covariance matrix is calculated as follows:

$$C = \begin{bmatrix} \sigma_{\text{tot}, 1}^2 & \sigma_{\text{corr}, 1}\sigma_{\text{corr}, 2} + \sigma_{\text{syst}, 1}\sigma_{\text{syst}, 2} & \sigma_{\text{corr}, 1}\sigma_{\text{corr}, 3} + \sigma_{\text{syst}, 1}\sigma_{\text{syst}, 3} \\ \sigma_{\text{corr}, 2}\sigma_{\text{corr}, 1} + \sigma_{\text{syst}, 2}\sigma_{\text{syst}, 1} & \sigma_{\text{tot}, 2}^2 & \sigma_{\text{corr}, 2}\sigma_{\text{corr}, 3} + \sigma_{\text{syst}, 2}\sigma_{\text{syst}, 3} \\ \sigma_{\text{corr}, 3}\sigma_{\text{corr}, 1} + \sigma_{\text{syst}, 3}\sigma_{\text{syst}, 1} & \sigma_{\text{corr}, 3}\sigma_{\text{corr}, 2} + \sigma_{\text{syst}, 3}\sigma_{\text{syst}, 2} & \sigma_{\text{tot}, 3}^2 \end{bmatrix}, \quad (5)$$

where the diagonal elements are:

$$\sigma_{\text{tot}, i}^2 = \sigma_{\text{uncorr}, i}^2 + \sigma_{\text{corr}, i}^2 + \sigma_{\text{syst}, i}^2. \quad (6)$$

If no correlated error is used (as is the case for π^0), it can simply be set to 0. The covariance matrix is then decomposed as:

$$C = V \text{diag}(\vec{\lambda}) V^{-1}, \quad (7)$$

where V is the matrix of eigenvectors ($\vec{e}_i$) of C and $\text{diag}(\vec{\lambda})$ the diagonal matrix of eigenvalues (λ_i). The $\pm 1\sigma$ eigendirection variations are computed from the central vector $\vec{x}_{\text{central}}$ as:

$$\vec{x}_i^{\pm} = \vec{x}_{\text{central}} \pm \sqrt{\lambda_i} \vec{e}_i, \quad (8)$$

where $\vec{x}_i^{\pm}$ are the varied vectors.

2. Tracking efficiency

The tracking efficiency uncertainty is applied as a percentage per track. The current correlation is 0.24%/track. It's applied on tracks with momentum higher than 200 MeV.

3. Form factors

Semileptonic resonant B decays are described by form factors (FF). For each decay, different FF models exist. One specific model for each decay is implemented in the `basf2` EvtGen framework. The form factor parametrisation used in the decay files must be updated for some of the $b \rightarrow u$ and $b \rightarrow c$ semi-leptonic decays. We update to a more recent model and/or more up-to-date parameter values. In addition, an uncertainty is associated to each form factor parameter. Because these parameters are correlated with each other, the systematic uncertainty associated to each decay is determined as follows (similar method as for the slow π , see Section 6.1):

- Combine the individual errors and the total correlation matrix to obtain a covariance matrix
- Compute the eigenvalues and eigenvectors of this matrix
- Vary each parameter by $\pm 1\sigma$ in the eigendirection

For n form factor parameters, this gives a set of n up/down variations. For each variation, new hybrid weights are calculated and the difference with the nominal distribution is taken as a systematic uncertainty.

896 1. $B \rightarrow X_u \ell \nu$ form factors

897 For the $B \rightarrow X_u \ell \nu$ decays, the **eFFORT** [47] package was used for the form factor reweighting.

898

899 The form factor models were already discussed in Section 2.2. We give here a short overview
900 of the model update for each type of resonant decay.

- 901 • $B \rightarrow \pi \ell \nu$: the BCL model is used but a more up-to-date set of parameter values is
902 given in Tab. 57 of Ref. [8]. The truncation of the BCL expansion is different and
903 therefore 5 parameters are used instead of 8.
- 904 • $B \rightarrow \rho/\omega \ell \nu$: the BCL expansion is used but the values are updated to the ones found
905 in Ref. [9]. The model has 11 parameters. The $B \rightarrow \omega \ell \nu$ normalised differential decay
906 rate as a function of q^2 is illustrated in Fig. 25 together with the impact of form factor
907 parameter variations.
- 908 • $B \rightarrow \eta/\eta' \ell \nu$: in the decay files the ISGW2 model is used. It's considered to be out-
909 of-date and therefore the model is updated to the more recent Light Cone Sum Rules
910 (LCSR) QCD model from Ref. [10]. This change is illustrated in Fig. 26 where the
911 normalised differential decay rate as a function of the recoil parameter w is plotted.
912 ISGW2 does not have any parameter, whereas the new model has two parameters.

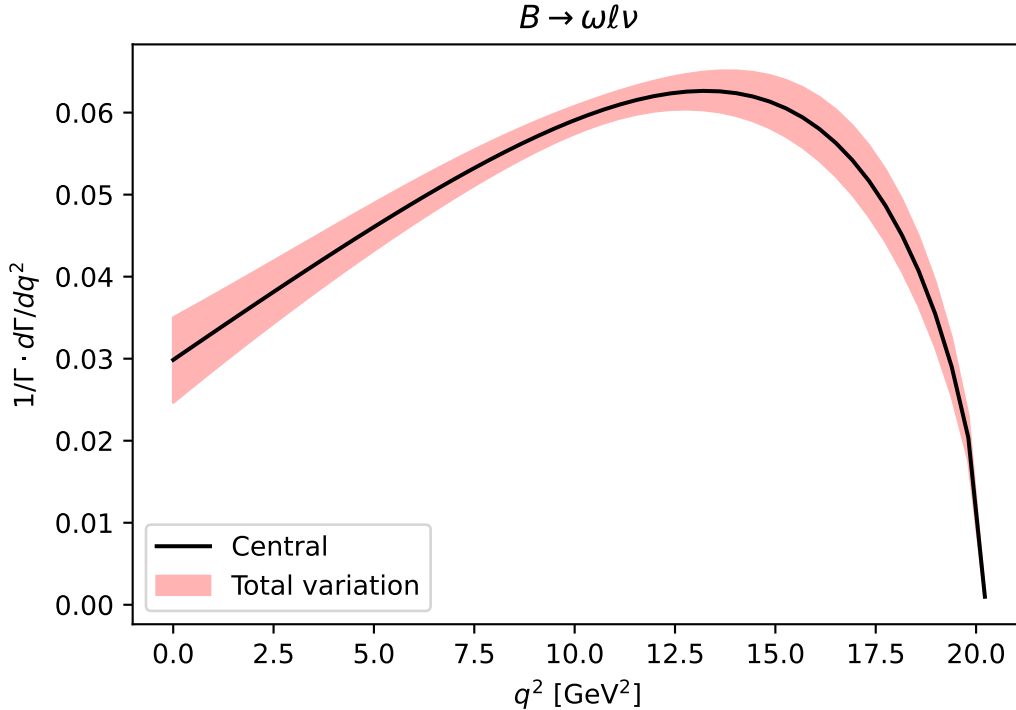


FIG. 25: The central $B \rightarrow \omega \ell \nu$ normalised differential decay rate is illustrated here (black line). The red band corresponds to the total impact of the 11 up/down $B \rightarrow \omega$ form factor parameter variations.

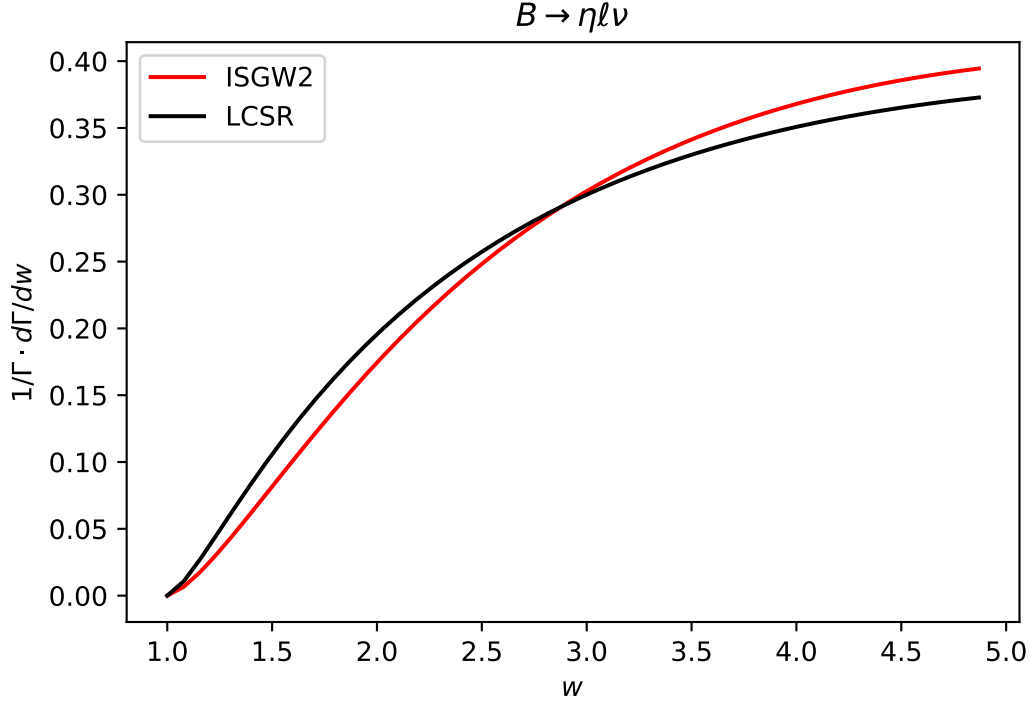


FIG. 26: The central normalised $B \rightarrow \eta \ell \nu$ differential decay rate as a function of w . The decay rate is shown for both the ISGW2 (red) and the more up-to-date LCSR (black) $B \rightarrow \eta$ form factor models.

913 2. $B \rightarrow X_c \ell \nu$ form factors

914 For $B \rightarrow X_c \ell \nu$ form factors, the **HAMMER** framework [16] was used.

- 915 • $B \rightarrow D \ell \nu$: the BGL [14] parametrisation is used to model these decays in the decay
 916 file ($N = 3$ expansion, 8 parameters). However, the theoretical calculation on which
 917 the **basf2** implementation is based was criticised by theorists in Ref. [48]. Therefore,
 918 a new **HAMMER** class correcting these errors was used. These mistakes have a negligible
 919 impact on the form factor calculation and they do not invalidate the parameter values
 920 used for the decay file. Nevertheless, they are fixed for completion⁴. The $B \rightarrow D$
 921 form factor modelling was updated to the BLPRXP model [15]. This model has the
 922 advantage to fit both D and D^* data which makes it more accurate.
- 923 • $B \rightarrow D^* \ell \nu$: in **basf2**, these decays are modelled using the BGL model with the
 924 $(1, 1, 2)$ expansion. They are updated to the BLPRXP model. The difference
 925 between the two models as well as the variations associated to the BLPRXP model
 926 are illustrated in Fig. 27.
- 927 • $B \rightarrow D^{**} \ell \nu$: the name D^{**} is not a single state but covers all four states D_0^* , D_1 , D_1^*
 928 and D_2^* . For $B \rightarrow D^{**} \ell \nu$ decays, only the BLR model exists. The two broad states

⁴ These changes are discussed for example in https://gitlab.desy.de/belle2/software/basf2/-/merge_requests/1631 where a new EvtGen class was created.

D_0^* and D_1^* are not described by the same parameters as the narrower D_1 and D_2^* (3 parameters for the former, 4 for the latter). The values of these parameters are updated to the ones found in Tab. 5 in [17].

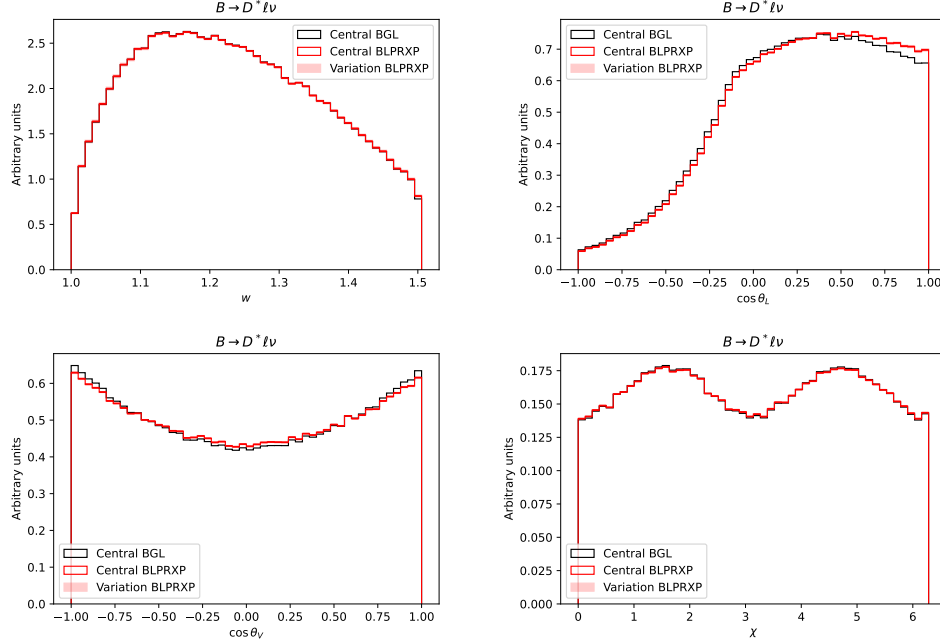


FIG. 27: The four main variables w , $\cos \theta_L$, $\cos \theta_V$ and χ used to describe $B \rightarrow D^* \ell \nu$ decays are shown here for the two form factor models described in the text. For the model that we use in our samples (BLPRXP), the total variation due to the 9 parameters uncertainties is also shown.

4. Charged particle identification

We correct for charged-particle identification efficiencies using the PIDVar package. This includes the following:

- $\ell^\pm \rightarrow \ell^\pm$ efficiency corrections. Taken from the official v0b tables.
- $\pi^\pm \rightarrow \ell^\pm$ fake rate corrections. Taken from the official v0b tables.
- $K^\pm \rightarrow \ell^\pm$ fake rate corrections. Taken from the official v0b tables.
- $p^\pm \rightarrow \ell^\pm$ fake rate corrections. Estimated with the systematics framework. Includes the protonID veto requirement.
- $K^\pm \rightarrow K^\pm$ efficiency corrections. Estimated with the systematics framework.
- $\pi^\pm \rightarrow K^\pm$ fake rate corrections. Estimated with the systematics framework.

942 The correction tables provide weights binned in track charge, lab frame momentum, and
 943 polar angle. In each bin a correction weight is applied, $\eta_k = \epsilon^{\text{Data}}/\epsilon^{\text{MC}}$, where k gives
 944 the bin index, and $\epsilon^{\text{Data}}, \epsilon^{\text{MC}}$ are the measured identification efficiencies in data and MC
 945 control samples. For each bin the statistical and systematic uncertainties of this ratio are
 946 also available. Using PIDVar we draw 200 variations of the $\vec{\eta}$ values using multivariate
 947 gaussians, about the nominal values, where the statistical uncertainties are taken to be fully
 948 uncorrelated, and the systematic (within a correction table) to be fully correlated.

949

950 In a small number of cases, mostly $K \rightarrow \ell$ fake rates, the tracks fall outside the cover-
 951 age of the correction tables. When this occurs, the correction factor is assumed to be 1
 952 with an uncertainty that combines the average of the total uncertainties in all measured
 953 bins, and the average difference of the measured correction factors from unity in quadrature.

954

955 When used to correct veto efficiencies, as when searching the rest of event for charged
 956 kaon or lepton candidates, rather than a binary veto of events in MC, we apply correction
 957 weights (w_k) as

$$w_k = 1 - \eta_k, \quad (9)$$

958 such that if the identification in data is less efficient than in MC, the MC events are kept
 959 with a small weight. If identification in data is more efficient than in MC the MC events
 960 are assigned a negative weight. This occurs rarely, such that when performing the binned
 961 fit, all bins have positive yield expectation values.

962 5. FEI tagging

963 For Belle II, a novel algorithm, the Full Event Interpretation [49] (FEI), was developed to
 964 reconstruct exclusive tag-candidates. The algorithm relies on machine learning to identify
 965 plausible B -meson decay-chains and uses hadronic and semi-leptonic tag-side decays to do
 966 so. In order to apply the tag-side reconstruction algorithm in a physics measurement it is
 967 important to quantify and correct for potential mismatches in the tag-side efficiency between
 968 data and simulation. The calibration factors are taken from Ref. [50]. They are computed
 969 from a combination of the signal-side $B \rightarrow X\ell\nu$ and $B \rightarrow D^{(*)}\pi$ channel calibrations. The
 970 correction factors are split by B meson charge. One calibration factor per tag-side channel
 971 is given (the least common channels were in fact grouped and one factor is given for all of
 972 them). There are respectively 11 and 12 factors for the neutral B and charged B modes.
 973 Furthermore, different factors were extracted for the two most common FEI probability
 974 $\mathcal{P}_{\text{FEI}}$ cuts, 0.01 and 0.001. As mentioned in Section 3.1, we use $\mathcal{P}_{\text{FEI}} > 0.01$.

975

976 The variations are calculated using the same method as for the slow π and form factor
 977 variations (see Sections 6.1 and 6.3 respectively). The covariance matrix is diagonalised and
 978 the eigenvariations are calculated. As the FEI calibration factors and covariance matrices
 979 are separated for charged and neutral B mesons, the variations are also separated.

980 6. Branching fractions

981 The branching fractions for B and D decays have to be updated. For B decays they are
 982 simply changed to more up-to-date values. The associated uncertainty needs to be taken

into account. For each variation, new hybrid weights are computed and the difference with the nominal distribution is taken as an uncertainty. There is therefore one up/down variation per branching fraction.

D decays are discussed in details in Section 2.3. For these decays, the branching fraction is varied according to its associated uncertainty and the difference with the nominal distribution is taken as a systematic uncertainty.

7. Hybrid parameter variations

The DFN model is chosen as our nominal model for the description of $B \rightarrow X_u \ell \nu$ events. It requires two input parameters for their modelling with EvtGen, m_b and a (*cf.* Section 2.2.1). These parameters are correlated and their uncertainties yield two different sets of up/down variations. The hybrid weights are recalculated for each one of these eigenvariations and the difference with the nominal distribution is taken as a systematic uncertainty. The parameters and their values are given in Tab. XI. Their values in two dimensions are shown in Fig. 28.

	m_b	a
Central	4.658	1.328
Variation 1	4.698	1.021
Variation 2	4.618	1.659
Variation 3	4.649	1.042
Variation 4	4.667	1.734

TABLE XI: DFN parameters central values and up/down correlated variations

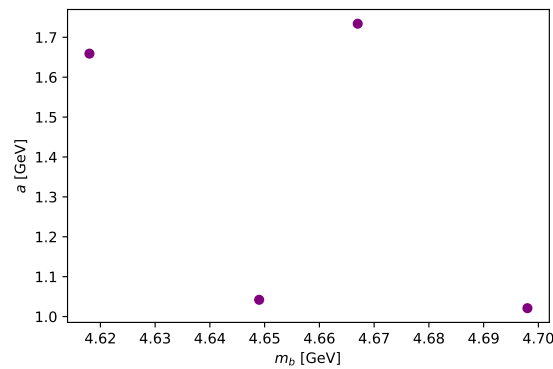


FIG. 28: Illustration of the four variations of the DFN parameters on a 2D plane.

8. Inclusive model differences

Discrepancies are observed between different inclusive models. Therefore, an additional uncertainty is associated to the difference between the BLNP and DFN models. The hybrid

weights are recalculated for the nominal DFN model and the difference with the nominal BLNP distribution is taken as a systematic uncertainty.

9. K_S^0 efficiency

Following the recommendations of the performance group, we assign a weight of 0.55% per cm of flight distance from the interaction point to K_S^0 candidates. The full correction is taken as a systematic uncertainty.

10. $s\bar{s}$ fragmentation

In Section 2.2, the variation of the PYTHIA parameter controlling the production of s quarks was discussed. The difference between the nominal distribution and those obtained with the varied γ_S parameter is taken as a systematic uncertainty.

11. MC statistics

The number of events in Monte Carlo samples being finite, the associated statistical uncertainty needs to be taken into account. It is simply calculated as:

$$\sqrt{\sum_i w_i^2}, \quad (10)$$

w_i being all the weights applied to each event.

12. $f^{\pm/0}$

We vary the fraction of $\Upsilon(4S) \rightarrow B^+B^- / \Upsilon(4S) \rightarrow B^0\bar{B}^0$ according to the value obtained in [51]: $f_{+-/00} = 51.1 \pm 1.1\%$.

13. Continuum calibration

The full size of the normalisation corrections discussed in Sec. 4, are assigned as systematic uncertainties. The corrections for the charged and neutral tagged samples are taken to be uncorrelated. Uncertainties on the BDT based shape correction arising from the limited statistics of the off-resonance data and simulated samples are propagated by training 200 additional classifiers on bootstrapped samples of the off-resonance data and MC.

14. X_u hadronisation modelling

The hadronisation modelling of the non-resonant $X_u\ell\nu$ component is performed by PYTHIA8. However, its parameters have yet to be tuned in Belle II and therefore a method must be devised to estimate the uncertainty related to the hadronisation modelling. It was attempted

to extract this uncertainty by changing the values of a limited number of parameters which could impact the X_u modelling in a similar way to the γ_S uncertainty described above. Modifying the values of this small subset of `PYTHIA8` parameters had a very limited effect on tested distributions and therefore it was concluded that a specific large-scale study should be performed. The tuning of `PYTHIA8` parameters is currently ongoing in Belle II.

The alternative method relies on the $\pi^\pm$ multiplicity in the ROE and can only be performed with signal region data. $\pi^\pm$ are selected with a hadron ID cut of 0.6. The nominal fit results are projected onto the $\pi^\pm$ multiplicity using the method described in Appendix E4. The non-resonant $X_u\ell\nu$ component is scaled in bins of the generator-level $\pi^\pm$ multiplicity such that the post-fit $\pi^\pm$ multiplicity in MC matches exactly the data distribution. The resulting weights are applied on the nominal sample and the nominal fit is repeated. The difference between the two $B \rightarrow X_u\ell\nu$ branching fraction central values is taken as an additional uncertainty. This procedure will be completed after unblinding of the signal region. We use the $E_\ell^B:q^2$ $\text{CR}_{K,\text{low}}\text{-}\text{CR}_{K,\text{high}}$ fit described in Section 8.10.2 to check the pre and postfit agreement of the $\pi^\pm$ multiplicity in the $\text{CR}_{K,\text{high}}$ region. If resonant $B \rightarrow X_c\ell\nu$ decays are properly modelled, then the assumption that the postfit disagreement is due only to the non-resonant $B \rightarrow X_u\ell\nu$ component holds. The $\pi^\pm$ multiplicity is shown in Figure 29. The projection method is described in Appendix E4.

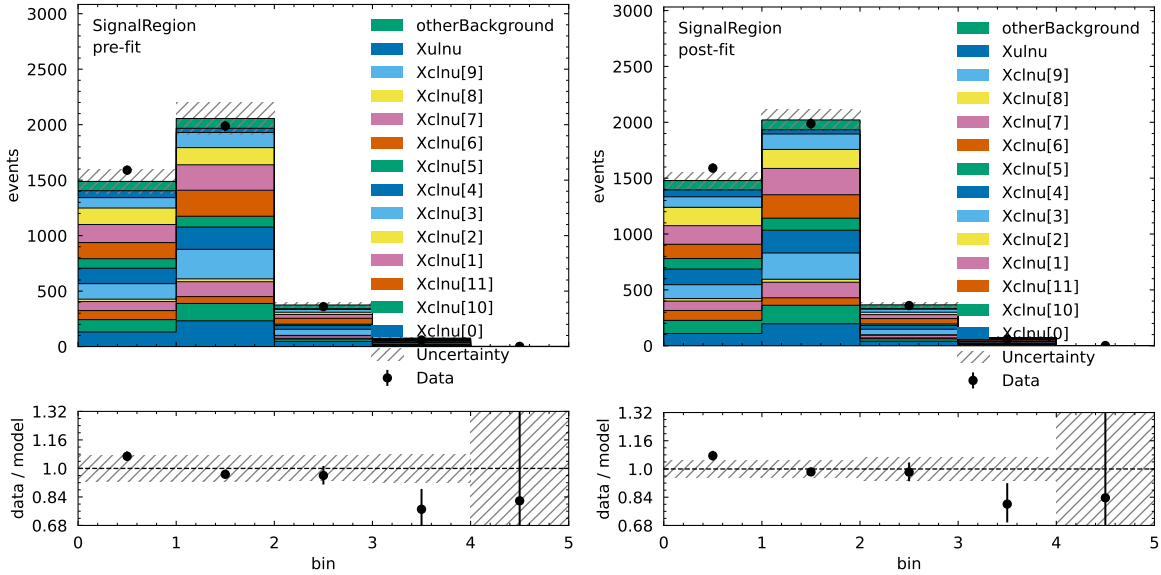


FIG. 29: Pre and postfit $\pi^\pm$ multiplicities in $\text{CR}_{K,\text{high}}$

15. Uncertainties related to the fitting procedures

The signal extraction procedure is detailed in Section 8. Additional uncertainties are assigned to the measured BR based on the corrections described in Section 9.3.

1051 7. DATA-MC AGREEMENT

1052 In this Section we compare data and MC distributions in different control regions. The pre-
 1053 selections presented in Section 3 are applied on all samples. We define 5 different control
 1054 regions where the data can be unblinded safely as the signal is overwhelmed by different
 1055 sources of background. The signal region is defined by selecting events without K and with
 1056 an MVA score greater than 0.87. By applying or not and inverting the K and MVA score
 1057 selections, one can define various control regions. A cut on $p_\ell^B > 1.0$ GeV is always applied
 1058 on all events. As described in Section 8, other selections on kinematical variables are also
 1059 applied to probe regions with higher signal purity. All the plots and results discussed in this
 1060 Section are for the nominal case where only the p_ℓ^B cut is applied. The control regions are
 1061 defined as follows:

- 1062 • Control region 1 (CR1): no other selections than the pre-selections defined in Section 3
 1063 are applied (Fig. 30)
- 1064 • $B \rightarrow X\ell\nu$ control region (CR $_{X\ell\nu}$): the continuum suppression classifier is applied
 1065 (Fig. 31); in this region, semi-leptonic decays to any hadronic system are enhanced
 1066 ($X = X_u, X_c$)
- 1067 • Wrong sign lepton region (CR $_{\text{WSLR}}$): charged B events where the tag-side B and the
 1068 lepton have the same charge (neutral B -mesons are not considered because of mixing);
 1069 the continuum suppression classifier is applied (Fig. 32)
- 1070 • Kaon, low score region (CR $_{K,\text{low}}$): events are required to have at least one K and a
 1071 $B \rightarrow X_c\ell\nu$ MVA classifier score lower than 0.60; the continuum suppression classifier
 1072 is applied (Fig. 33)
- 1073 • Kaon, high score region (CR $_{K,\text{high}}$): events are required to have at least one K and a
 1074 $B \rightarrow X_c\ell\nu$ MVA classifier score higher than 0.87; the continuum suppression classifier
 1075 is applied (Fig. 34)
- 1076 • No kaon, low score region (CR $_{0,\text{low}}$): events are required to have no K and a $B \rightarrow X_c\ell\nu$
 1077 MVA classifier score lower than 0.60; the continuum suppression classifier is applied
 1078 (Fig. 35)

1079 In this analysis it is not straightforward to find a control mode to validate the efficiency of
 1080 the main selections defining the signal region. In fact, the most similar decays to our signal
 1081 $B \rightarrow X_u\ell\nu$ decays are the $B \rightarrow X_c\ell\nu$ decays. We can therefore use a $B \rightarrow X_c\ell\nu$ -enhanced
 1082 region to estimate the selection efficiencies of $B \rightarrow X_u\ell\nu$ decays. It is important to note that
 1083 this estimate remains fairly qualitative. Charm and charmless semileptonic B decays exhibit
 1084 important differences in particular in their modelling and kinematics. $B \rightarrow X_u\ell\nu$ events
 1085 tend to have softer hadronic systems and harder leptonic systems. Their modelling partly
 1086 relies on specific models of the non-resonant contribution. We quote below the efficiencies
 1087 and their associated total uncertainty for three important cuts: the continuum suppression
 1088 selection on CR1 events, the K veto on CR $_{X\ell\nu}$ events and the $X_c\ell\nu$ suppression MVA cut
 1089 on CR $_{X\ell\nu}$ events with at least one K . The SR efficiency will be checked in data as part of
 1090 our unblinding procedure (see Section 9). The total signal efficiency in MC as a function of
 1091 the three main kinematical variables is shown in Appendix P.

- 1092 • Continuum suppression
 - 1093 – Data: $(48.9 \pm 0.1)\%$
 - 1094 – MC: $(47.6 \pm 1.0)\%$
 - 1095 – Compatible within 1.3σ
- 1096 • K veto
 - 1097 – Data: 30.3%
 - 1098 – MC: 29.7%
 - 1099 – The exact uncertainty which factors in correlations is challenging to derive for
 - 1100 technical reasons but the total MC uncertainty is expected to be about 0.5 – 1%
 - 1101 and therefore the efficiencies would be compatible within about 1σ
- 1102 • $B \rightarrow X_c \ell \nu$ suppression MVA
 - 1103 – Data: $(2.95 \pm 0.05)\%$
 - 1104 – MC: $(3.23 \pm 0.12)\%$
 - 1105 – Compatible within 2.2σ

1106 The conclusion of this test is fairly qualitative but given the large difference in modelling
 1107 between $X_u \ell \nu$ and $X_c \ell \nu$ events, this is the best check that can be done. The disagreement
 1108 between MC and data in this test is corrected using $X_c \ell \nu$ events as detailed later in this
 1109 Section. Furthermore, the pre-fit uncertainty on $X_u \ell \nu$ events is expected to be larger than
 1110 that of $X_c \ell \nu$ events because of the important modelling uncertainties that are added for the
 1111 signal mode. It is therefore expected that any discrepancy in the signal efficiency between
 1112 MC and data is well covered by the total uncertainty.

1113 1. Kinematic distributions

1114 In Figures 30 to 35 plots of p_ℓ^B , M_X and q^2 are shown in the control regions defined above.
 1115 In addition, we show the same distributions in the signal region without data (see Fig. 36).
 1116 In the plots, 'MC Uncert.' indicates that all sources of uncertainties are considered in
 1117 the uncertainty bands whereas 'MC Stat. Uncert.' indicates that only the MC statistical
 1118 uncertainty is considered.

1119 2. Sample compositions

1120 For each sample, in Tables XII and XIII, the composition in terms of signal and background
 1121 components and the $B \rightarrow X_c \ell \nu$ composition are given respectively.

1122 3. Data-MC disagreement

1123 A large disagreement is observed in all three variables in the control regions discussed above.
 1124 The exact origin of this disagreement is not yet known but we give in the next Section an
 1125 outline of our current understanding of the data-MC disagreement as well as a potential
 1126 solution to it.

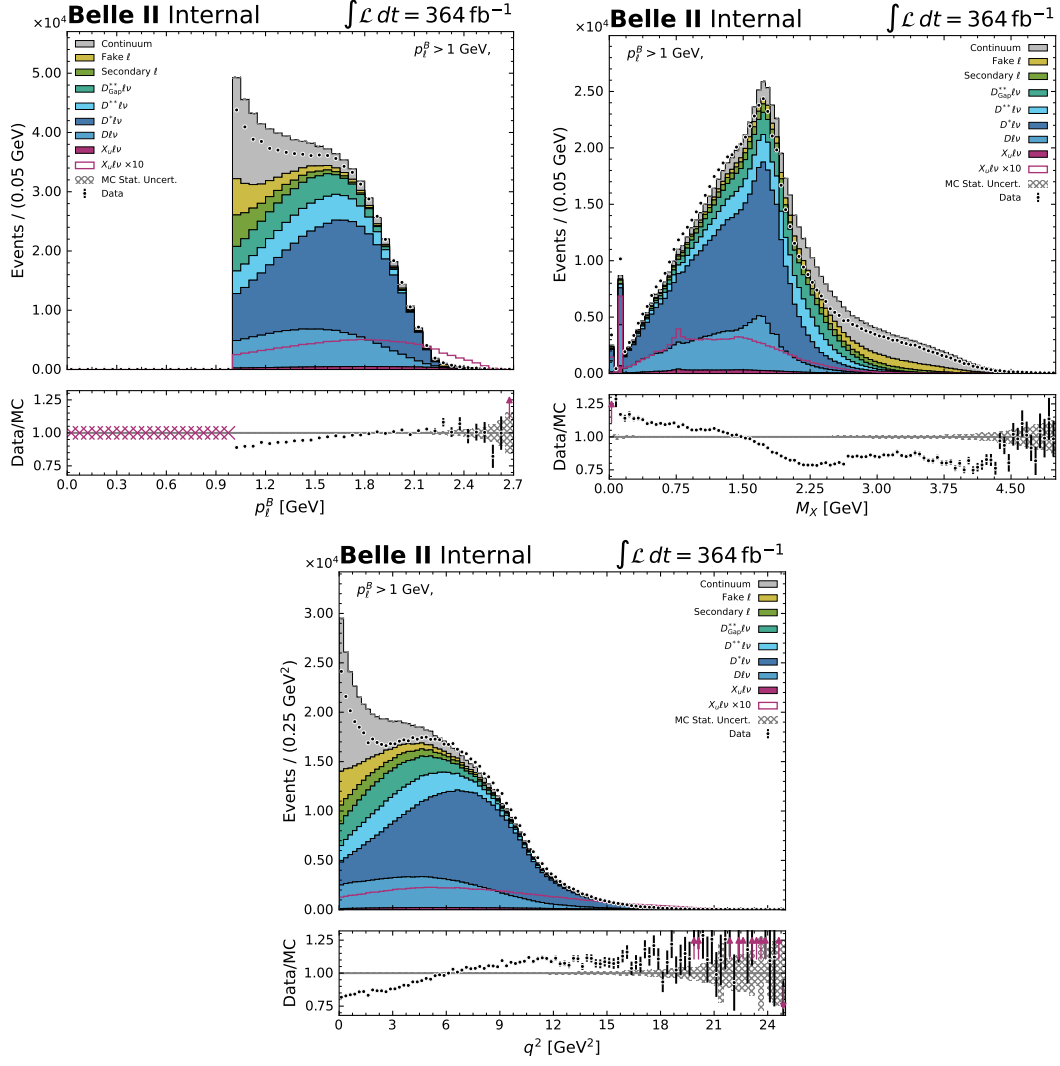


FIG. 30: Data-MC comparison of p_ℓ^B , M_X and q^2 in control region 1. The uncertainty band includes only the MC statistical error.

	$CR_{K,low}$	$CR_{K,high}$	$CR_{0,low}$	Signal
$B \rightarrow X_u \ell \nu$	$0.46 \pm 0.01\%$	$2.55 \pm 0.04\%$	$1.49 \pm 0.01\%$	$29.5 \pm 0.2\%$
$B \rightarrow X_c \ell \nu$	$93.2 \pm 0.2\%$	$92.2 \pm 0.8\%$	$90.9 \pm 0.1\%$	$61.5 \pm 0.4\%$
Secondary leptons	$3.19 \pm 0.02\%$	$1.60 \pm 0.06\%$	$3.30 \pm 0.01\%$	$1.70 \pm 0.05\%$
Fake leptons	$2.27 \pm 0.01\%$	$2.65 \pm 0.08\%$	$2.74 \pm 0.01\%$	$3.98 \pm 0.08\%$
Continuum	$0.88 \pm 0.02\%$	$0.99 \pm 0.10\%$	$1.56 \pm 0.02\%$	$3.30 \pm 0.14\%$

TABLE XII: Composition of various background regions and the signal region. The statistical error is shown.

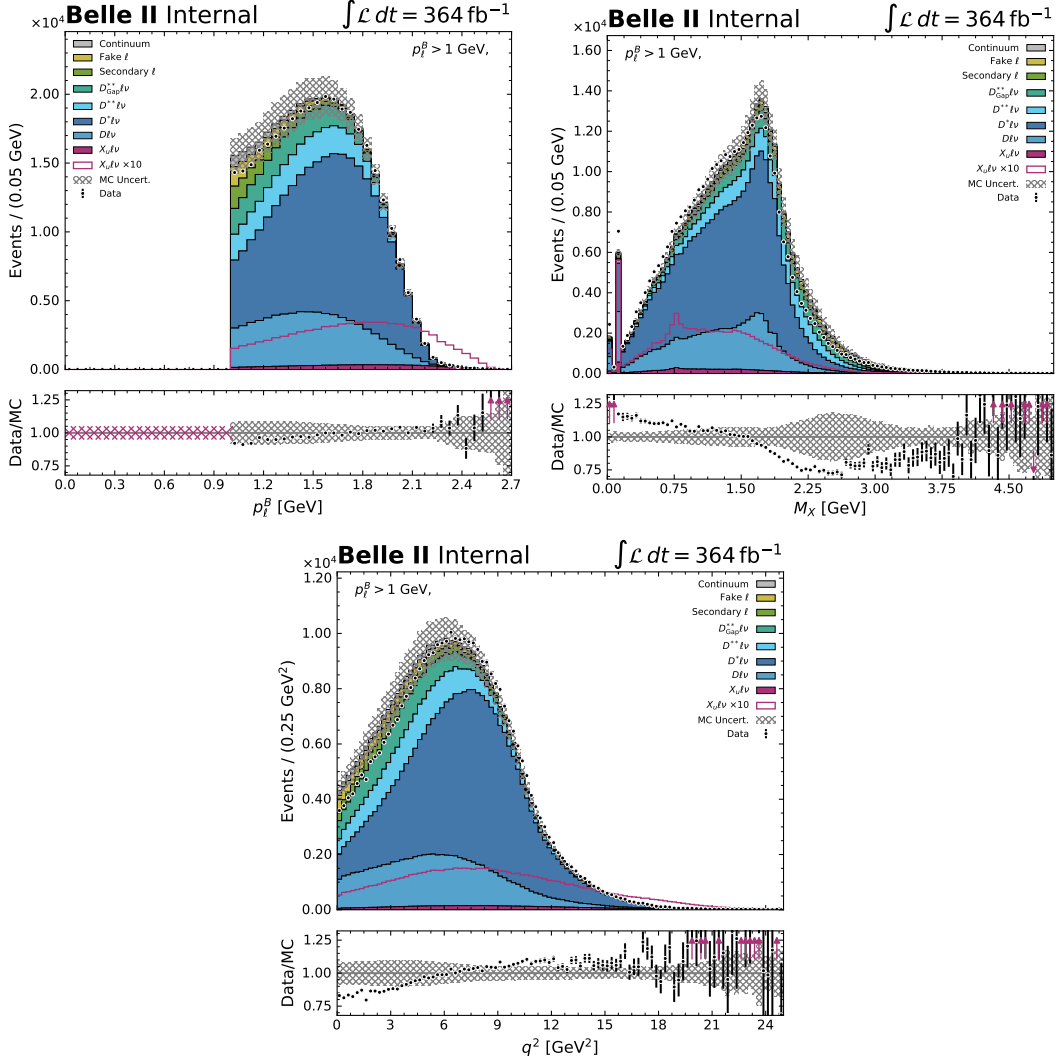


FIG. 31: Data-MC comparison of p_ℓ^B , M_X and q^2 in the $B \rightarrow X\ell\nu$ control region.

	$\text{CR}_{K,\text{low}}$	$\text{CR}_{K,\text{high}}$	$\text{CR}_{0,\text{low}}$	Signal
$B \rightarrow D\ell\nu$	$17.33 \pm 0.05\%$	$40.4 \pm 0.4\%$	$19.1 \pm 0.04\%$	$37.3 \pm 0.4\%$
$B \rightarrow D^*\ell\nu$	$61.21 \pm 0.12\%$	$47.3 \pm 0.5\%$	$58.7 \pm 0.09\%$	$49.2 \pm 0.5\%$
$B \rightarrow D^{**}\ell\nu$	$9.66 \pm 0.03\%$	$6.15 \pm 0.13\%$	$10.0 \pm 0.03\%$	$6.85 \pm 0.14\%$
$B \rightarrow D_{\text{Gap}}^{**}\ell\nu$	$11.8 \pm 0.10\%$	$6.10 \pm 0.40\%$	$12.2 \pm 0.07\%$	$6.62 \pm 0.35\%$

TABLE XIII: Composition of the $X_c\ell\nu$ component in various background regions and the signal region. The statistical error is shown.

1. Origin of the data-MC disagreement

Similar data-MC differences were already observed in past inclusive measurements in Belle II, in particular in Ref. [52]. We divide our MC templates in three categories and discuss them below: lepton backgrounds (fake and secondary leptons), continuum and $B \rightarrow X\ell\nu$.

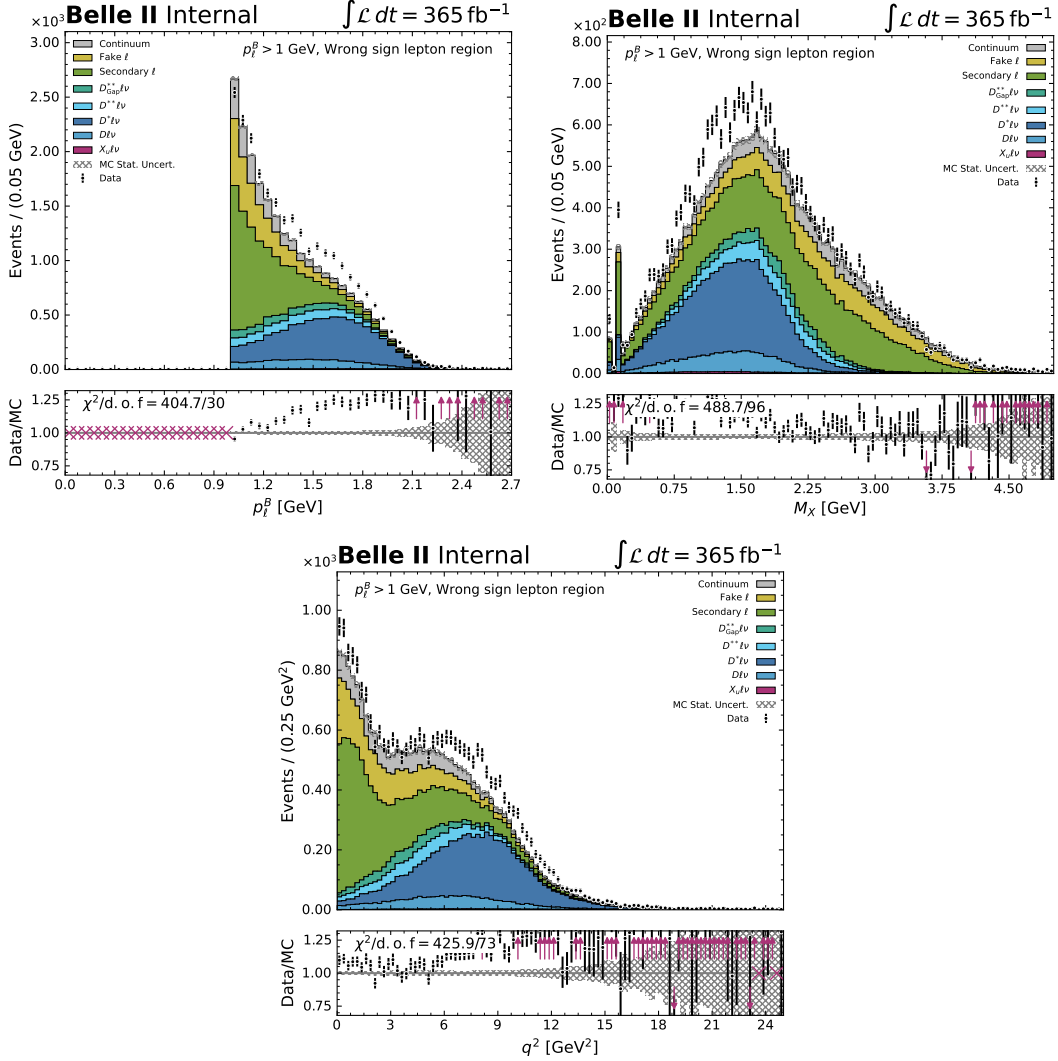


FIG. 32: Data-MC comparison of p_ℓ^B , M_X and q^2 in the wrong sign lepton control region.

1131

1132 The fake and secondary lepton components correspond to a very small portion of the events
 1133 in the $X\ell\nu$ control region and are thus not expected to be the main cause of disagreement.
 1134 However, the lepton background contributions are increased in the wrong sign lepton region
 1135 region compared to the $X\ell\nu$ control region. Similar trends as in the $X\ell\nu$ control region can
 1136 be seen so the fake and secondary leptons could also contribute to the data-MC disagreement.

1137

1138 The continuum is in general known to be poorly modelled. However, as described in
 1139 Section 4, the continuum is rescaled using off-resonance data. After these corrections, in
 1140 the off-resonance region, the data-MC agreement is much improved. However, the on- and
 1141 off-resonance samples present small differences. In particular, as illustrated in Fig. 38,
 1142 shape differences can be observed in the variables used for the continuum calibration. As
 1143 the scaling factors are extracted using off-resonance MC and data and then applied to on-
 1144 resonance samples, some disagreement between MC and data is expected in continuum in
 1145 the on-resonance region. Nevertheless, the disagreement coming from continuum modelling

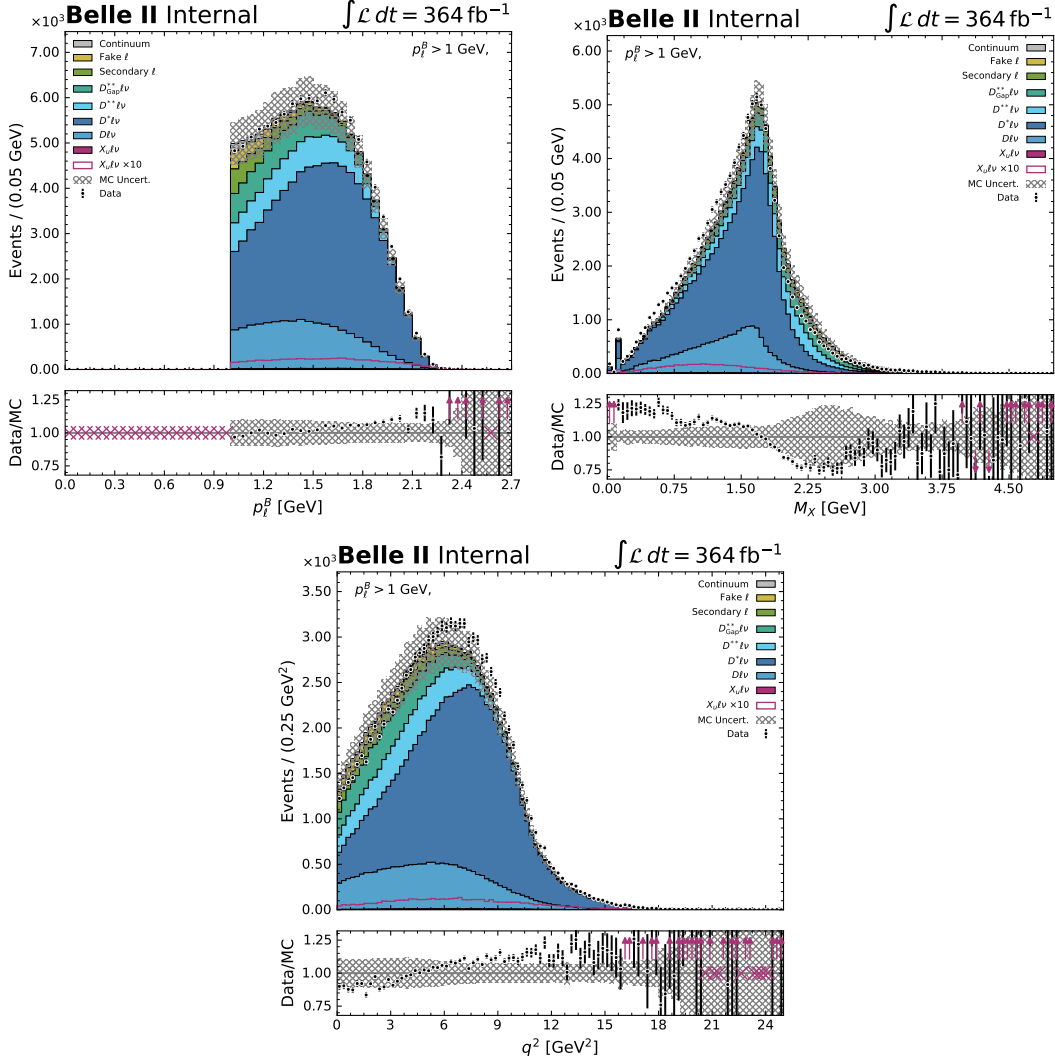


FIG. 33: Data-MC comparison of p_ℓ^B , M_X and q^2 in the $\text{CR}_{K,low}$.

can hardly explain entirely the large data-MC differences observed in the $X\ell\nu$ control region.

Therefore, the data-MC disagreement is expected to be largely explained by the $B \rightarrow X\ell\nu$ modelling. From Fig. 31, it can be seen that the $B \rightarrow X_u\ell\nu$ component contribution is too small in this region but that most events are $B \rightarrow X_c\ell\nu$ events. As described in Section 22, the inclusive $B \rightarrow X_c\ell\nu$ component is modelled as a sum of exclusive charm meson decays and gap modes. As the latter are not physical but are added to fill the gap between sum of exclusive and total inclusive, they are not expected to perfectly model the non-resonant part of $B \rightarrow X_c\ell\nu$. Large uncertainties are associated to this contribution to compensate for its imperfect modelling. Another potential cause for the poor $B \rightarrow X_c\ell\nu$ modelling are D meson decays. In particular, hadronic decays to four or more particles are modelled using phase-space modelling instead of decay-specific models. As D^* and D^{**} mesons always eventually decay to a D meson, these decays appear in effectively every $B \rightarrow X_c\ell\nu$ event and they could spoil the modelling of this component. Another type of

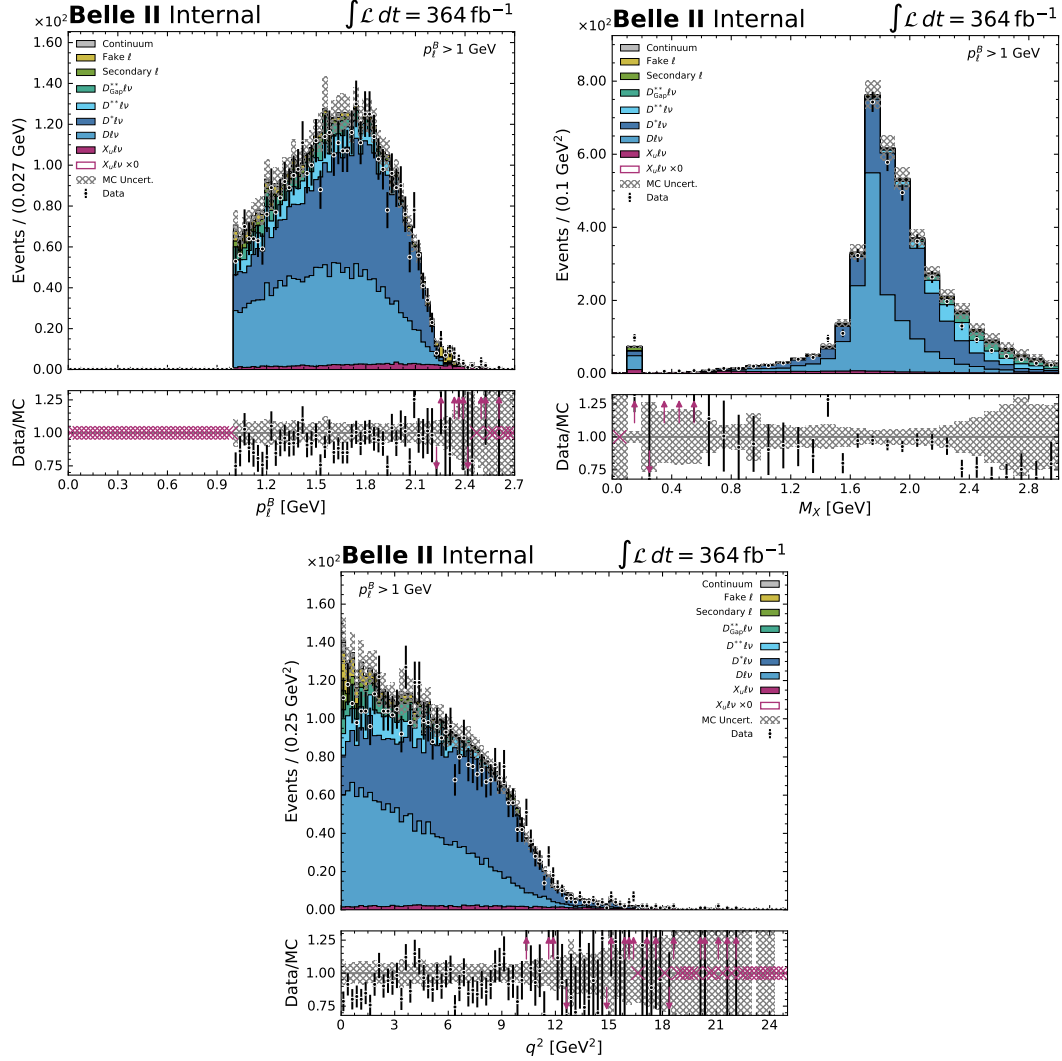


FIG. 34: Data-MC comparison of p_ℓ^B , M_X and q^2 in the $\text{CR}_{K,\text{high}}$.

1161 D decays that are known to be badly modelled are decays to K_L^0 . In Ref. [53], an overall
 1162 correction factor of 30% for these decays was derived from data. To test how much this
 1163 component could impact our modelling we use the CR1 and we split it in the following
 1164 three templates: continuum, B decays with $D \rightarrow K_L^0 X$ decays and B decays without
 1165 $D \rightarrow K_L^0 X$ decays, X representing any generic system. We correct the template with K_L^0
 1166 by 30% and we observe no improvement in the data-MC comparison (cf. Fig. 39). These
 1167 decays do not represent a large portion of all events and therefore a simple normalisation of
 1168 this component could hardly cover the gap between data and MC that is currently observed.
 1169

1170 To conclude this discussion, it is likely that the difference between data and MC can be
 1171 explained by the cumulative effect of a few mismodelled components rather than a unique
 1172 template. Nevertheless, it appears that the $B \rightarrow X_\ell \ell \nu$ poor modelling is the largest cause
 1173 of the disagreement. In the next Section, we describe a method to correct this component.

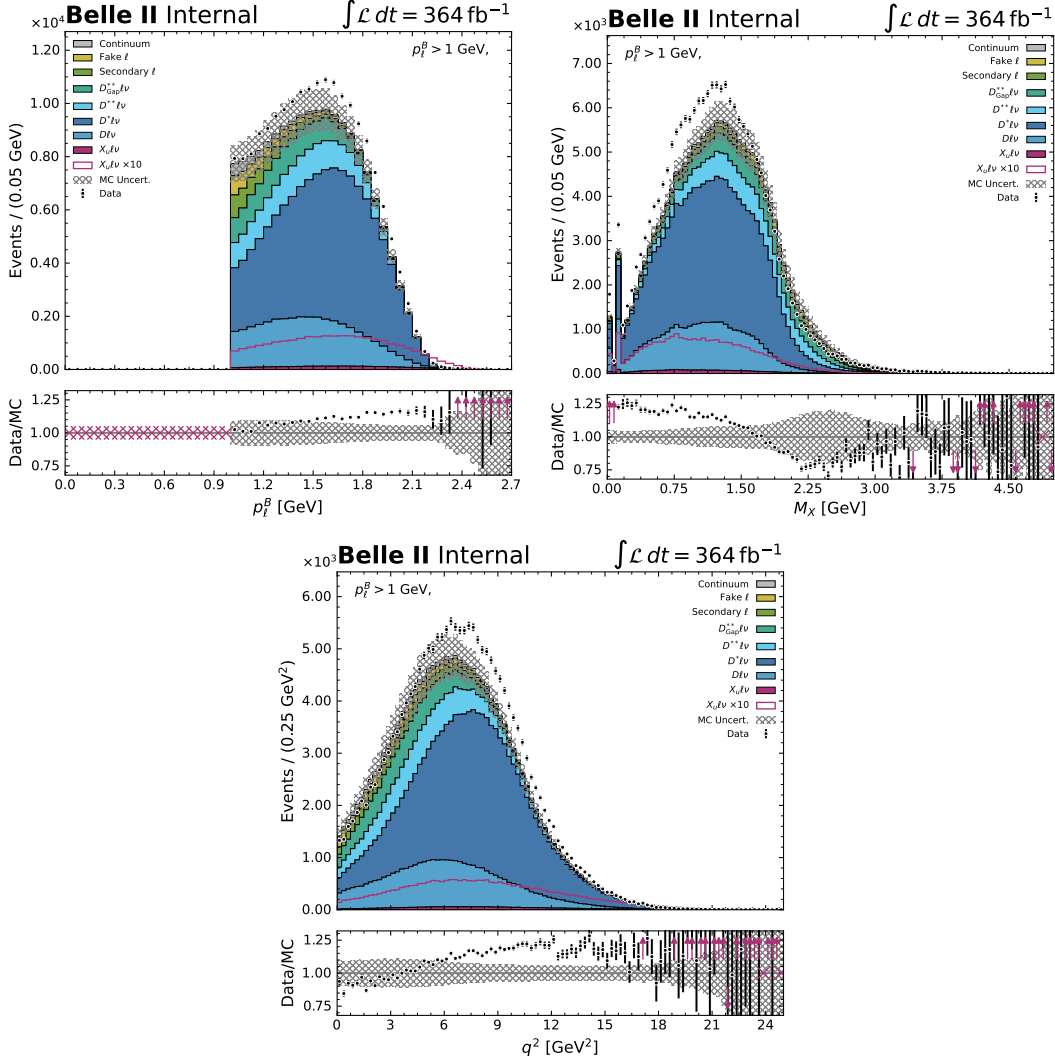


FIG. 35: Data-MC comparison of p_ℓ^B , M_X and q^2 in the $\text{CR}_{0,\text{low}}$.

2. Normalisation mismodelling

From the various plots shown in this Section, it is clear that the shape of the $B \rightarrow X_c \ell \nu$ component is not well modelled. However, the total normalisation difference between MC and data is also widely different between different control regions. This can be explained by the $X_c \ell \nu$ MVA score and K multiplicity distributions in Figures 40 and 41. Depending on the selection on these variables the normalisation is expected to shift. The normalisations in the different control regions defined above are summarised in Table XIV.

The control regions built based on the K and MVA score cut are illustrated in Figure 42. We therefore need to correct for both effects. When using the `pyhf` package, fits where both the normalisation and the shape of a component are free-floated at the same time yield very unpredictable and unstable results. We therefore choose to split the normalisation and shape corrections in two parts. The later is corrected in a simultaneous fit of the signal region and a control region as detailed later.

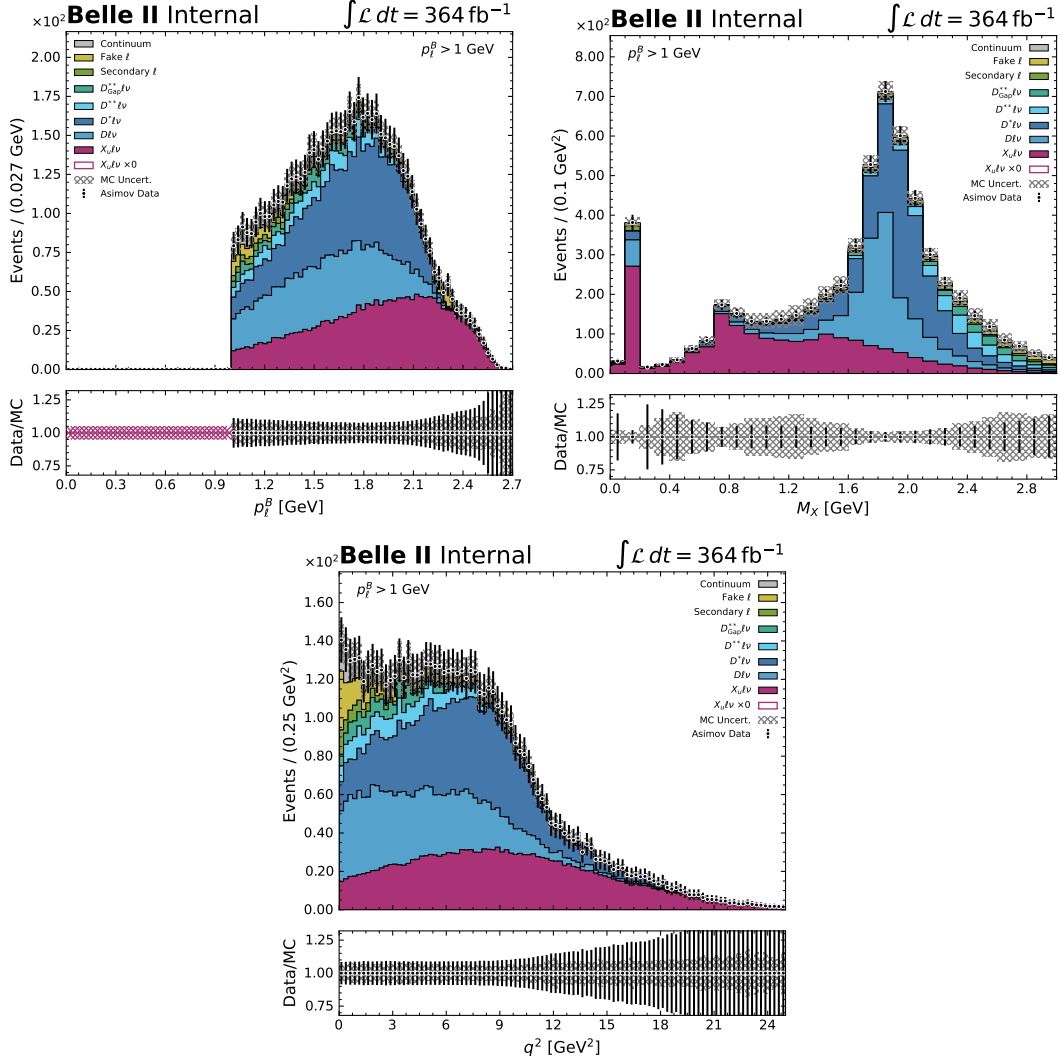


FIG. 36: p_ℓ^B , M_X and q^2 in the signal region.

Region	Data/MC ratio
$B \rightarrow X \ell \nu$	0.986 ± 0.002
$\text{CR}_{K,\text{low}}$	1.039 ± 0.004
$\text{CR}_{K,\text{high}}$	0.946 ± 0.018
$\text{CR}_{0,\text{low}}$	1.113 ± 0.003

TABLE XIV: $B \rightarrow X_c \ell \nu$ data-MC ratio for the different control regions described in this Section for a kinematical selection $p_\ell^B > 1.0$ GeV. The uncertainty is statistical.

1188

1189 To derive a normalisation correction for the signal region we use what is commonly re-
 1190 ferred to as the ABCD method. We split the phase-space into four separate regions based
 1191 on the K multiplicity and the $B \rightarrow X_c \ell \nu$ MVA score which are two weakly correlated
 1192 variables. The signal and control regions were defined above and Figure 42 illustrates them

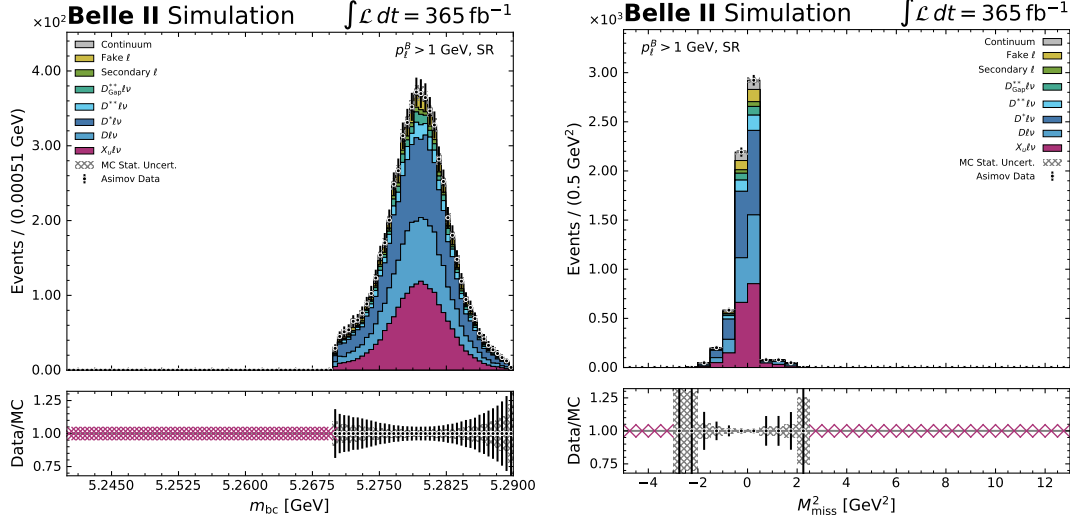


FIG. 37: M_{bc} and M_{miss}^2 in the signal region.

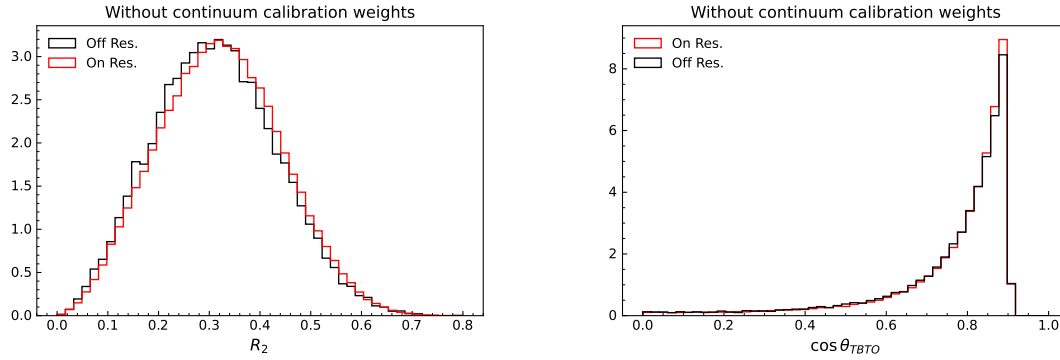


FIG. 38: Comparison of MC continuum on- and off-resonance samples using R_2 and $\cos \theta_{TBTO}$. As shape differences can be seen, it is expected that despite the continuum correction being applied to on-resonance samples, the data and MC distributions in this region will exhibit some disagreement.

in a 2D phase-space. Since the trend in the MVA score as seen in Figure 40 appears to be very similar when the K veto is applied or not, we assume that the signal region data/MC ratio can be well approximated by using the three control regions $CR_{K,low}$, $CR_{K,high}$ and $CR_{0,low}$. In the following when talking about data/MC ratio we only refer to the ratio for $B \rightarrow X_c \ell \nu$ events. When computing these ratios, all other components are removed from MC and the MC-estimated yields are also subtracted from data: $(\text{data} - \text{non-}X_c) / X_c$. The data/MC ratio R_{SR} in the SR is then equal to:

$$R_{SR} = \frac{R_{0,low} \cdot R_{K,high}}{R_{K,low}}, \quad (11)$$

This assumption can be tested by using the two validation regions defined by an MVA score between 0.60 and 0.87. Using $R_{VR2} = R_3 \cdot R_{VR1} / R_1$, when only a kinematical cut $p_\ell^B > 1.0$ GeV is applied, we find a value of $R_{VR2} = 1.07 \pm 0.01$ where the error is the

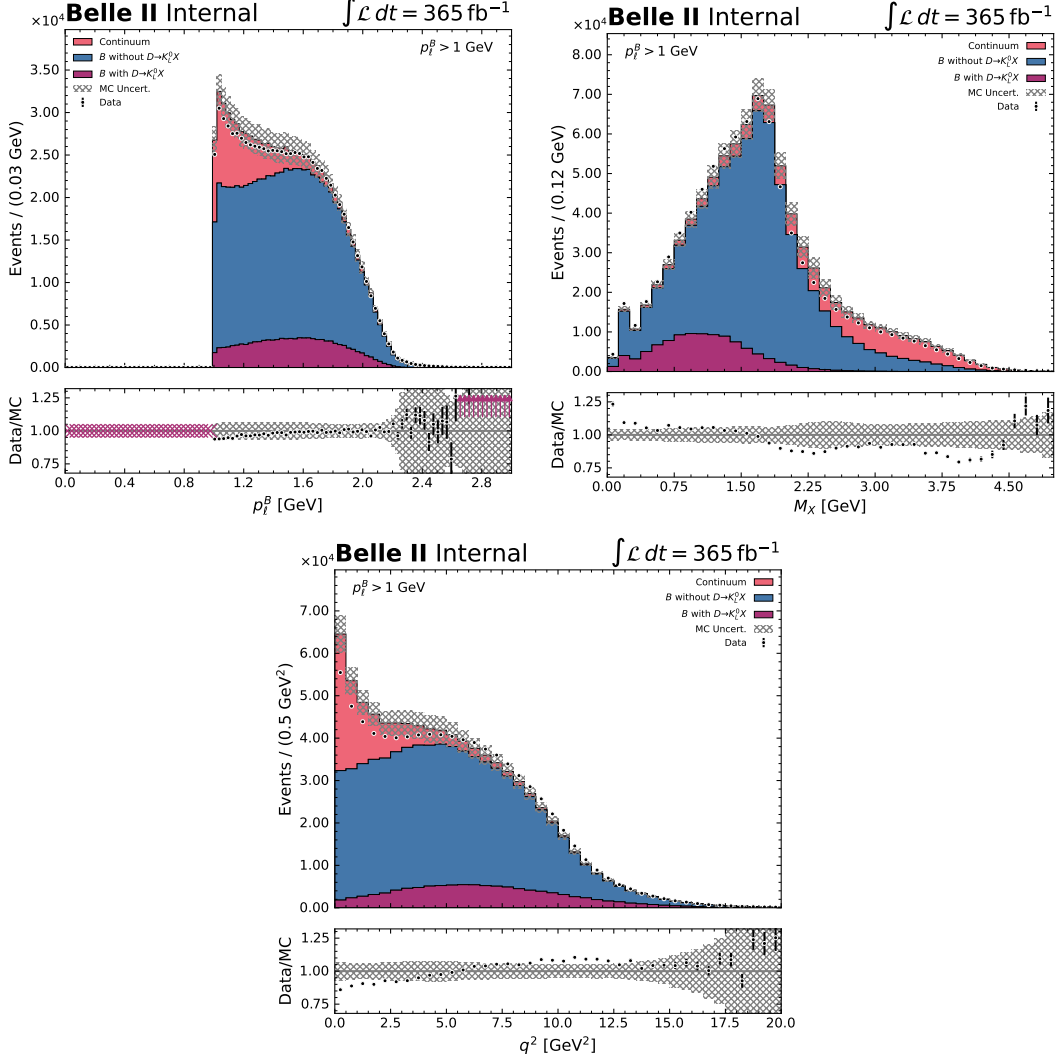


FIG. 39: The three main kinematic variables in CR1 after rescaling by 30% the B decay events containing a $D \rightarrow K_L^0 X$ decay.

1203 statistical uncertainty. The value obtained directly from using MC and data in this region
 1204 is $R_{VR2} = 1.03 \pm 0.01$ (stat.). The gap between the two values is used as an additional
 1205 uncertainty on the SR data/MC. An additional uncertainty is associated to the fact that the
 1206 non- $X_c \ell \nu$ components also have a non-perfect modelling. The ratio R_{SR} is computed again
 1207 by varying the non- $X_c \ell \nu$ yields by 50%. The difference with respect to the nominal value of
 1208 R_{SR} is used as an additional uncertainty. Finally, we find: $R_{SR} = 1.01 \pm 0.02 \pm 0.06 \pm 0.05 =$
 1209 1.01 ± 0.08 where the first uncertainty is statistical, the second comes from the validation
 1210 region estimate and the third one from the non- $X_c \ell \nu$ yield variation. As justified in the
 1211 next Section, in the fit we use $CR_{0,low}$. Its normalisation correction is directly taken from
 1212 data and MC. An uncertainty is also derived by varying the non- $X_c \ell \nu$ component yields
 1213 giving: $R_{0,low} = 1.11 \pm 0.01(\text{stat.}) \pm 0.05(\text{additional uncert.}) = 1.11 \pm 0.05$. When extracting
 1214 the signal branching fraction, we also probe regions where the cuts $M_X < 1.7$ GeV and/or
 1215 $q^2 > 8$ GeV² are applied. We repeat the procedure described above for these regions and
 1216 the SR and $CR_{0,low}$ X_c data/MC ratios are summarised in Table XV.

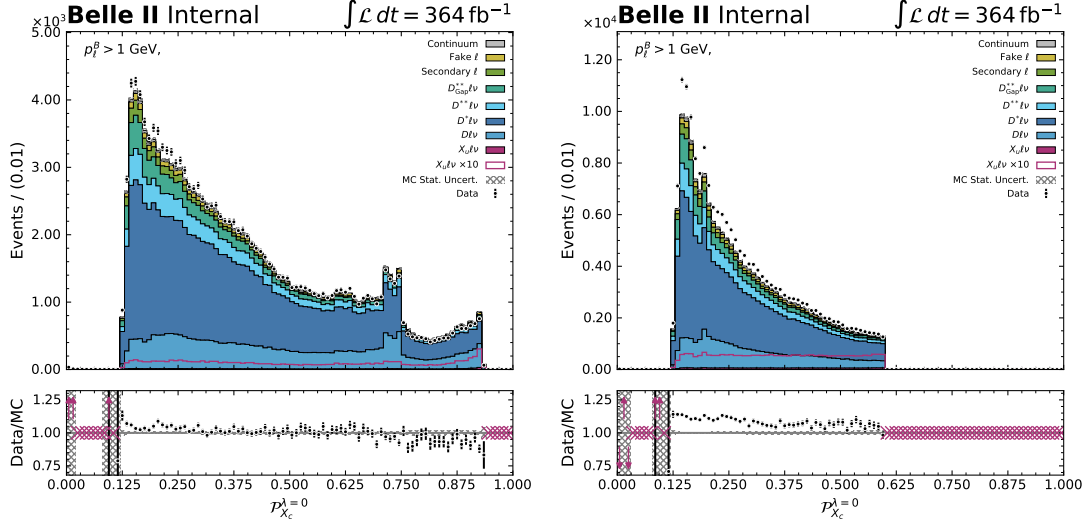


FIG. 40: The $B \rightarrow X_c \ell \nu$ suppression score after applying continuum suppression is shown when requiring the presence of at least one K (left) and after applying the K veto (right). The score is cut at 0.60 for the later plot to avoid unblinding the signal region.

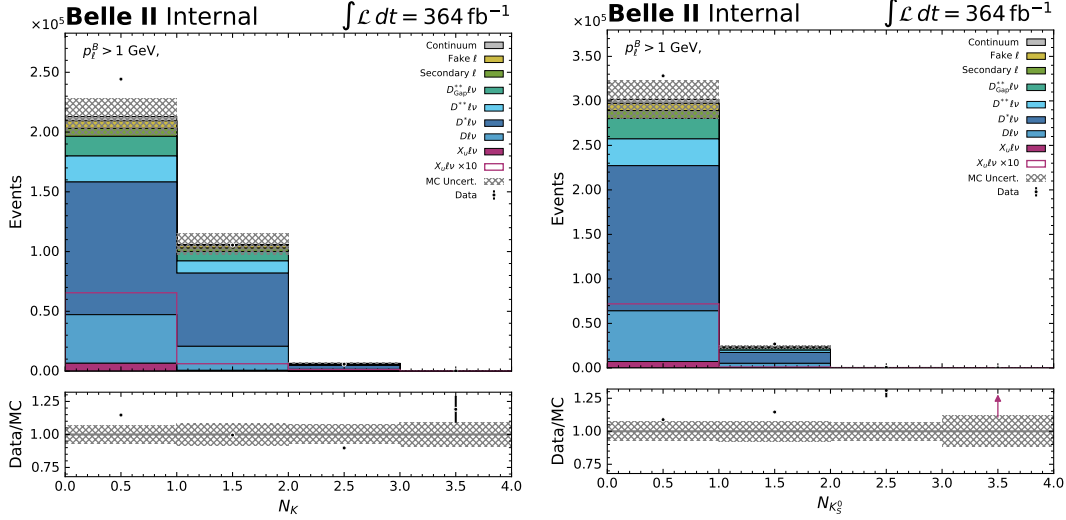


FIG. 41: The charged (left) and neutral (right) K multiplicities after applying continuum suppression are shown.

Kinematic selection	CR _{0,low}	SR
$p_\ell^B > 1.0 \text{ GeV}$	1.11 ± 0.05	1.01 ± 0.08
$p_\ell^B > 1.0 \text{ GeV}, M_X < 1.7 \text{ GeV}$	1.18 ± 0.04	1.07 ± 0.13
$p_\ell^B > 1.0 \text{ GeV}, M_X < 1.7 \text{ GeV}, q^2 > 8 \text{ GeV}^2$	1.24 ± 0.03	1.09 ± 0.25

TABLE XV: CR and SR $X_c \ell \nu$ normalisation factors for different kinematic selections.

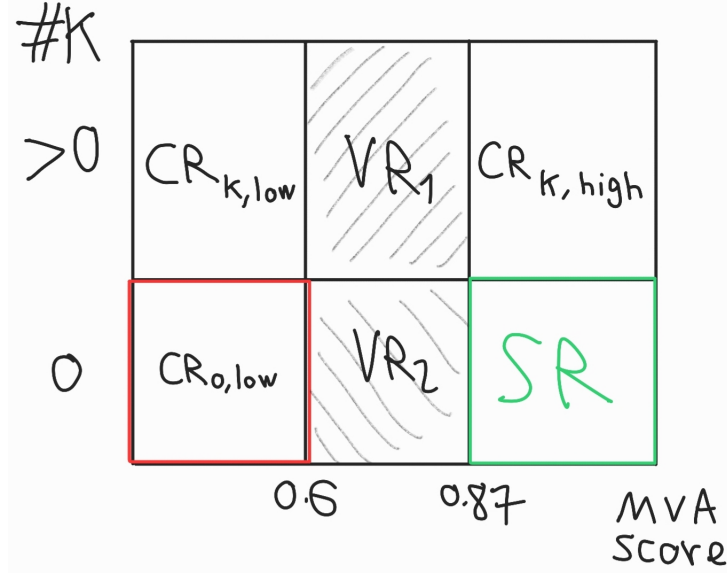


FIG. 42: Illustration of the CR and SR definitions based on the K multiplicity and the X_c MVA score. The SR is highlighted in green. The CR used in the fit setup is highlighted in red (see Section 7.3.3 for details on the choice of this region). The regions VR_1 and VR_2 are used as validation regions for the ABCD method used to estimate the SR $X_{c\ell\nu}$ component normalisation (see Section 7.3.2 for details about this method).

1217 3. Shape mismodelling

1218 Even though the $X_{c\ell\nu}$ normalisation doesn't translate from one region to the other, the
 1219 shape of the data/MC ratio appears to be similar in all regions as illustrated in Figure 43.
 1220 We would like to use a control region orthogonal to the signal region in order to correct the
 1222 $B \rightarrow X_{c\ell\nu}$ component. A few different contributions could explain the mismodelling of this
 1223 component both on the level of detector modelling and the level of event generation. We
 1224 therefore correct this component as a whole.

1225

1226 The shape correction factors are extracted together with the $B \rightarrow X_{u\ell\nu}$ branching fraction
 1227 in a simultaneous fit of the signal and control regions. Binned free-floating normalisation
 1228 factors are assigned to the $B \rightarrow X_{c\ell\nu}$ component in order to let the fit correct it. The
 1229 SR $B \rightarrow X_{c\ell\nu}$ component is normalised to the value of R_{SR} that we found above and
 1230 the $B \rightarrow X_{c\ell\nu}$ MC yields in the CR are directly normalised to data. With this strategy,
 1231 all correlations between nuisance parameters are properly handled and the uncertainty on
 1232 the extracted branching fraction incorporates the $B \rightarrow X_{c\ell\nu}$ shape correction. A nuisance
 1233 parameter is added for the X_c normalisation in each of the two regions based on the central
 1234 values and errors found. More details about the fitting setup are given in the next Section.

1235

1236 $CR_{0,low}$ is orthogonal to the signal region and is mostly composed of $B \rightarrow X_{c\ell\nu}$ events.
 1237 The yields of these events are large enough in this region to obtain statistically significant
 1238 $X_{c\ell\nu}$ shape factors. Using $CR_{K,high}$ was also considered but the compositions and shapes
 1239 of this region are much more similar to the signal region than for $CR_{0,low}$ and therefore
 1240 the correction factor extraction would rely heavily on the signal region which we want to

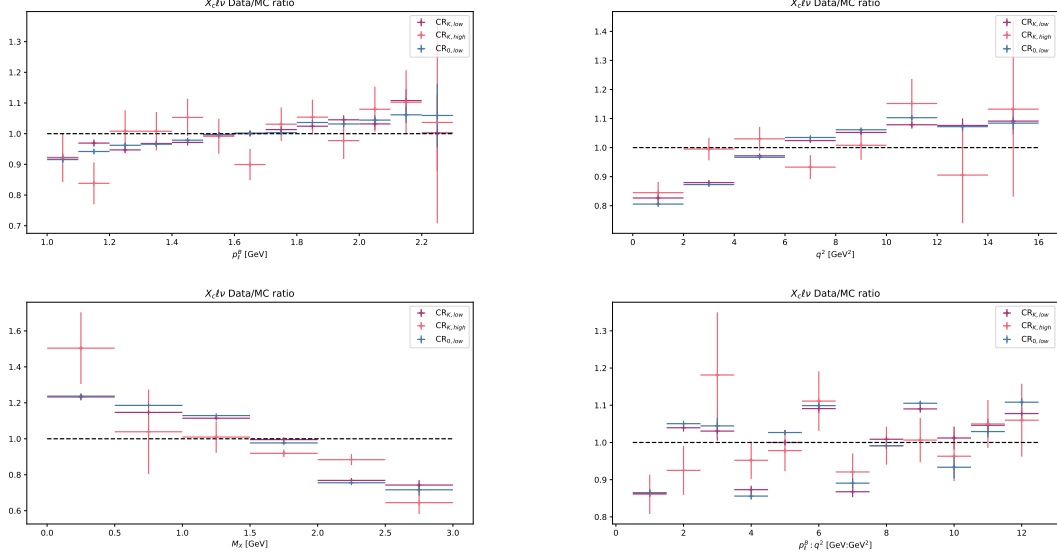


FIG. 43: The normalised data/MC ratio is shown here for the three control regions $CR_{K,low}$, $CR_{0,low}$ and $CR_{K,high}$ defined above. The shape of the disagreement appears to be similar within uncertainties for all regions.

1241 avoid. Nevertheless, as the $X_c \ell \nu$ component is corrected as a whole, an uncertainty on the
 1242 $X_c \ell \nu$ composition difference between the K and signal regions (*c.f.* Tables XIII) needs to
 1243 be considered. This is also discussed in the next Section.

8. SIGNAL EXTRACTION

To extract signal yields, the `pyhf` and `cabinetry` packages are used to extract the $B \rightarrow X_u \ell \nu$ branching fraction from data distributions. We perform a binned template fit using MC distributions. As discussed in the previous Section, we simultaneously fit the signal region and the kaon region in order to constrain the $B \rightarrow X_c \ell \nu$ shape using data and automatically correct for mismodelling of this component in the signal region. The MC yield is split into four templates including the signal template. The templates are discussed in Section 8.2. The signal strength μ is extracted as a Parameter of Interest (POI) and nuisance parameters (NP) based on the aforementioned systematic uncertainties are added to the fit. This section details the fitting procedure.

Different variables can be used to extract the signal. As we are considering a semi-leptonic decay, the three variables M_X , q^2 and p_ℓ^B are used. Two-dimensional fits are also considered. In addition to the preselection and MVA cuts described previously, different phase-space selections applied on generator level and different experimental selections on p_ℓ^B , M_X and q^2 are also applied. These selections directly impact the theoretical prediction for the triple differential decay rate $d\Gamma/dM_X dq^2 dp_\ell^B$ as they reduce the phase-space region from which the $B \rightarrow X_u \ell \nu$ signal is extracted. This is also discussed in Section 8.2.

1. pyhf

The `pyhf` package is widely used in HEP as a fitting tool. It is based on the `HistFactory` package. An outline of its functioning is given here. A general Poisson probability model can be written as follows:

$$\mathcal{P}(n_b|\mu) = L(\mu) \propto \prod_{b \in \text{bins}} \text{Pois}(n_b | \mu \nu_b^{\text{sig}} + \nu_b^{\text{bkg}}) \quad (12)$$

where $L(\mu)$ is the likelihood function, b is the bin index, μ is the signal strength, n_b is the data histogram and ν_b are the signal and background MC histograms.

In order to include multiple samples as well as various normalisations and variations, the likelihood template needs to be generalised. Most importantly, multiplicative and additive modifiers are added to the definition of the event rates ν_b . Each modifier corresponds to a nuisance parameter. For each modifier a constraint term is added to limit the rate modification. The modifiers can influence both the total normalisation and the shape of the considered distribution. The exact statistical model and the event rates as defined in `pyhf` are given in Appendix E. The parameters assigned to each template are summarised in Section 8.3.

2. Fit templates, free-floating parameters and phase-space regions

Historically, various cuts on p_ℓ^B , M_X and q^2 have been applied in $B \rightarrow X_u \ell \nu$ measurements as they reduce background contributions. For instance, we always apply a requirement that the lepton momentum is greater than 1.0 GeV as the low lepton momentum region is mainly populated with continuum and lepton backgrounds. Moreover, a selection on the hadronic

mass at $M_X < 1.7$ GeV cuts away most $B \rightarrow X_c \ell \nu$ processes as the lightest charm meson, the D meson, has a mass around 1.86 GeV and is very narrow. The inclusive $B \rightarrow X_u \ell \nu$ models take these selections into account and predictions for the decay rate are calculated in specific phase-space regions. In order to test these predictions we apply various phase-space cuts on the generator level on signal events in the distributions we fit. Signal events are therefore split in two templates: events in the phase-space region (signal-in component) and events outside of the phase space region (signal-out component). The former template is our nominal signal template and the signal strength is extracted from it. The background is split into a $B \rightarrow X_c \ell \nu$ template and a template including all other background contributions (consisting of continuum as well as fake and secondary leptons). Additionally, we apply different cuts on the experimental level in order to reject more or less background. In certain cases, very few signal events pass the experimental cuts but not the phase-space cuts. This is the case for instance when the p_ℓ^B cut at 1.0 GeV is applied on both levels because of the good resolution of this variable. In these cases the signal contribution outside of the phase-space is neglected and a single template is defined for the signal. In other cases the signal-out component is small but cannot be neglected. In these fits, the same free-floating normalisation is assigned to the signal-in and signal-out templates. More details about the fit parameters are given in the following sections.

1300

1301 We consider the following phase-space cuts:

- 1302 • $p_\ell^B > 1.0$ GeV: always applied as the equivalent experimental cut is already applied
1303 on our samples
- 1304 • $M_X < 1.7$ GeV: the equivalent experimental cut is known to reduce $B \rightarrow X_c \ell \nu$
1305 background so this phase-space cut is also probed
- 1306 • $q^2 > 8$ GeV²: the equivalent experimental cut rejects $B \rightarrow X_c \ell \nu$ background; this cut
1307 could also reduce the sensitivity to specific theoretical components

1308 The acceptance of each phase-space region is calculated from MC as the ratio of number
1309 of generated events before and after applying the phase-space cuts. The acceptance for
1310 different phase-space regions is given in Tab. XVI.

Phase space cuts	Acceptance $\varepsilon_{\Delta\mathcal{B}}$
$E_\ell^B > 1.0$ GeV	86.7%
$E_\ell^B > 1.0$ GeV $M_X < 1.7$ GeV	56.5%
$E_\ell^B > 1.0$ GeV $M_X < 1.7$ GeV $q^2 > 8$ GeV ²	31.5%

TABLE XVI: Phase-space acceptance for different selections.

1311

1312

1313

1314 In the following Sections, the phase-space region that is probed and the experimental
1315 cuts that are applied are clearly mentioned.

1316 3. Fit parameters

1317 In our fit, the following `pyhf` modifiers are implemented:

- 1318 • Normalisation uncertainty (**normsys**): modifier to take into account overall normali-
1319 sation uncertainties on the considered distribution. This modifier enters the fit as the
1320 up and down variations of the overall normalisation factor associated to the param-
1321 eter (the central value of the normalisation factor being equal to 1). As the other-
1322 backgrounds template is small in the fitted distributions, a **normsys** modifier is also
1323 attached to it in order to constrain it and limit fit instabilities (Gaussian constraint
1324 with a width of 50% around the nominal yields). As explained Section 7.3.3, the
1325 $X_c \ell \nu$ normalisation is corrected in the SR and CR and the associated uncertainty is
1326 implemented in the fit as a Gaussian constrained normalisation uncertainty.
- 1327 • Correlated shape (**histosys**): modifier to take into account shape variations on the
1328 considered distribution. This modifier enters the fit as the binned sample up and down
1329 rate variations associated to the parameter.
- 1330 • MC statistical uncertainty (**statererror**): specific modifier for the MC statistical un-
1331 certainty. The uncertainty per bin is estimated using a set of bin-wise constrained
1332 scale factors.
- 1333 • Unconstrained normalisation (**normfactor**): this modifier is used for the strength of
1334 the signal templates (signal-in and signal-out when it is not neglected). It scales the
1335 sample rate by a free-floating parameter. The signal template strength is the POI and
1336 eventually allows the computation of the $B \rightarrow X_u \ell \nu$ branching fraction. The signal
1337 strength is only assigned to the $B \rightarrow X_u \ell \nu$ component in the signal region which keeps
1338 the (small) signal component fixed in the control region.
- 1339 • Data-driven shape (**shapefactor**): free bin-wise normalisation factors. This modifier
1340 is used to fit the normalisation and shape of the $B \rightarrow X_c \ell \nu$ template as each bin is
1341 normalised separately. It is assigned to both regions in the fit.

1342 In our fit, every nuisance parameter except for the MC statistical uncertainty is split into a
1343 **normsys** and a **histosys** contribution. An uncertainty is associated to $\mathcal{B}(B \rightarrow X_u \ell \nu)$ and
1344 this is incorporated in the fit. However, this is also the number we want to extract from
1345 the fit. Therefore, the associated nuisance parameter is allowed to modify the shape of the
1346 considered distribution but not its overall normalisation.

1347
1348 As explained in the next Section, for certain systematic uncertainty sources with a large
1349 number of associated nuisance parameters, this number is reduced such that 99% of the
1350 associated variance is covered. The number of bins is also different for different fits and
1351 therefore the number of nuisance parameters can slightly vary from one fit to another (dif-
1352 ferent fitted variable and different phase-space region).

1353
1354 As mentioned above, in some of the fits, the signal-out template is completely neglected as
1355 its yields are extremely small relatively to the the signal-in template. In other fits, the yields
1356 are small but not negligible and their shape is similar to the background template shapes
1357 and in particular the $B \rightarrow X_c \ell \nu$ shape. This is illustrated in Fig. 44 for a phase-space

1358 region defined by the cuts $p_\ell^B > 1.0$ GeV and $M_X < 1.7$ GeV. Therefore, the normalisa-
 1359 tion of this template can correlate very strongly to the other three normalisations and its
 1360 resulting uncertainty and bias can be very large. In these cases the two $X_u \ell \nu$ templates are
 1361 kept separate but the same free-floating normalisation μ^{X_u} is assigned to both templates.
 1362 Finally, in other fits, the signal-out component is large enough to be treated as a completely
 1363 separate template and is therefore assigned its own independent free-floating normalisation.
 1364 The parameters assigned to each of the four templates are summarised in Table XVII.
 1365 Besides, in fits with a very tight phase-space region (when applying a tight M_X or q^2 cut
 1366 for example), the yields of "other backgrounds" events become very small compared to the
 1367 semi-leptonic event yields. Floating this component therefore renders the fit unstable and
 1368 we fix its normalisation (all NPs related to this component are still allowed to float its
 1369 normalisation and shape).

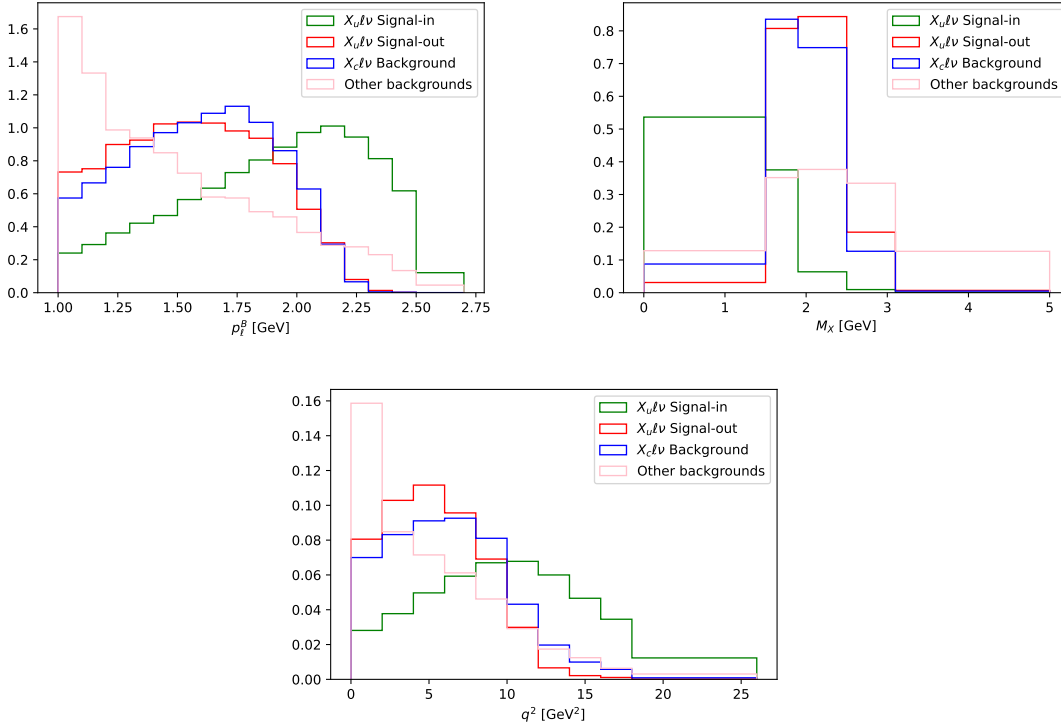


FIG. 44: p_ℓ^B , M_X and q^2 split into the four templates defined for the fit. The signal component is split in signal-in and signal-out templates using the cuts $p_\ell^B > 1.0$ GeV and $M_X < 1.7$ GeV applied on generator level. Each histogram is normalised to compare the shapes.

1370

1371

1372

1374 There are in total $N + 2$ free-floating parameters where N is the number of bins in the con-
 1375 sidered fit. If the signal-out component is ignored there are $N + 1$ free-floating parameters.

1376 1. Implementation of systematic uncertainties within the fit

1377 a. Two-sided variation

1378

	Signal region	Control region
$X_u\ell\nu$ -in	1 free-floating normalisation factor	-
$X_u\ell\nu$ -out	Neglected, same normalisation factor as $X_u\ell\nu$ -in or independent normalisation factor	-
$X_c\ell\nu$	N free-floating normalisation factors (+1 Gaussian constrained normalisation NP)	N free-floating normalisation factors (+1 Gaussian constrained normalisation NP)
Other backgrounds	Gaussian constrained normalisation NP (with width of 50% around nominal yields) or fixed when yields are too low	Gaussian constrained normalisation NP (with width of 50% around nominal yields) or fixed when yields are too low

TABLE XVII: Summary of fit parameters used for each of the four templates (for N bins).

The majority of systematics we consider are what we label *two-sided* variations. These correspond to systematic uncertainties such as branching fractions of the $B \rightarrow X\ell\nu$ channels, form factors or slow π calibration factors in which we have auxiliary measurements from which we can take $\pm\sigma$ variations. These variations are propagated through the reconstruction by reweighting events to generate two sets of varied MC templates. When the fit pulls on the nuisance parameters, the templates are morphed in the direction of the varied templates. When several parameters are correlated, the variations are calculated from the covariance matrix and the corresponding eigenvectors and eigenvalues as explained in Section 6. For each pair of up/down variations, one nuisance parameter is added to the fit.

b. One-sided variation

Systematics in which we have only a one-sided variation, such as switching the nominal inclusive $B \rightarrow X_u\ell\nu$ model (DFN) to the alternate (BLNP), are challenging to deal with, as there is no natural choice for the distribution of the prior for the nuisance parameter. Within this analysis the choice of the inclusive $B \rightarrow X_u\ell\nu$ model is the most impactful nuisance parameter. As the modelling of $B \rightarrow X_u\ell\nu$ is in general not well known, we symmetrise such one-sided variations by mirroring about the nominal shape to create two-sided variations.

c. Toy variation

Several uncertainties, such as those associated to the statistical uncertainty on BDT based continuum shape corrections, or particle identification corrections, are propagated through the analysis as N sets of toy weights. To capture the bin-to-bin correlations of these uncertainties within the `pyhf` framework, we construct the covariance matrix of the binned MC samples from the N sets of weights. The covariance matrix is constructed in a flattened global binning that assigns a unique bin index to the combination of channel (only one channel is considered for now but splitting the B charge or the lepton flavour is also possible), template and fitted variable(s). The covariance matrix is diagonalised and its eigenvectors/eigenvalues are calculated as described earlier.

To capture the full covariance matrix, the number of variations, and hence nuisance param-

eters in the fit, would be equivalent to the number of bins in the flattened global binning. Most of these nuisance parameters would be negligible. To avoid adding these, we truncate the number of components, taking the largest n components such that the sum of their eigenvalues exceeds 99% of the sum of all eigenvalues, and thus covers 99% of the variance.

1418

We additionally consider a special case of the two-sided variations in which we build a covariance matrix from a large number of two-sided variations and follow the procedure above to reduce the number of nuisance parameters. This is used to reduce the number of nuisance parameters associated with the modelling of the charm decay from 234 to ~ 10 and to reduce the number of nuisance parameters associated with the FEI calibration from a total of 23 to one for the B^0 channel and two for the $B^\pm$ channel.

1425 4. Symmetrising

Most systematics that are considered are symmetric about the nominal template in the upward and downward varied templates. However, to avoid introducing biases in the fit, the variations that are known to be asymmetric are symmetrised. These are mainly form factor variations and inclusive model parameter variations. The asymmetry is tested via:

$$A = \max_i \frac{(H_i - H_i^-) - (H_i^+ - H_i)}{H_i}, \quad (13)$$

where i runs over the flattened global bin index, H_i is the nominal yield in the bin, and $H_i^\pm$ are the yields of the up and down varied templates in the bin. If the asymmetry is found to be small $A < 0.005$, we average the distance between the nominal and varied templates. If the asymmetry is found to be large, we take the maximum envelope of the varied template in each bin.

1435 5. List of fits

The list of fits we perform is given below. We present results for all considered variables only for Fit 1 to illustrate them but subsequent fits are repeated only for the variable yielding the most precise and most stable result. Compared to the previous iteration of the analysis, fits 2 and 6 were removed. In these setups, the $X_u \ell \nu$ -out component was assigned the same normalisation parameter as the $X_u \ell \nu$ -in component even though its yields weren't negligible. This makes the fit output much more dependent on the modelling of the signal template in the considered phase-space region. Fit 4 is kept for illustration purposes but this setup leads to biased results as is shown in Appendix F.

1444

1445

1446 • Fit 1

- 1447 – Signal phase-space selection: $p_\ell^B > 1.0$ GeV
- 1448 – Experimental cut: $p_\ell^B > 1.0$ GeV
- 1449 – Signal-out: $\approx 0.1\%$ of all signal events so it is neglected

- 1450 – Other backgrounds: floated normalisation within a Gaussian constraint of 50%
- 1451 of the nominal yields
- 1452 – Fitted variables: p_ℓ^B , q^2 , M_X , 2D $M_X:q^2$, 2D $p_\ell^B:q^2$
- 1453 • Fit 3
- 1454 – Signal phase-space selection: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV
- 1455 – Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV
- 1456 – Signal-out: $\approx 2.5\%$ of all signal events so it is assigned the same free-floating
- 1457 normalisation as the signal-in component
- 1458 – Other backgrounds: fixed normalisation
- 1459 – Fitted variables: 2D $p_\ell^B:q^2$
- 1460 • Fit 4
- 1461 – Signal phase-space selection: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV²
- 1462 – Experimental cut: $p_\ell^B > 1.0$ GeV
- 1463 – Signal-out: $\approx 46\%$ of all signal events so it is assigned an independent free-floating
- 1464 normalisation
- 1465 – Other backgrounds: floated normalisation within a Gaussian constraint of 50%
- 1466 of the nominal yields
- 1467 – Fitted variables: q^2
- 1468 • Fit 5
- 1469 – Signal phase-space selection: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV²
- 1470 – Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV²
- 1471 – Signal-out: $\approx 4.5\%$ of all signal events so it is assigned the same free-floating
- 1472 normalisation as the signal-in component
- 1473 – Other backgrounds: fixed normalisation
- 1474 – Fitted variables: p_ℓ^B
- 1475 • Fit 7
- 1476 – Signal phase-space selection: $p_\ell^B > 2.1$ GeV
- 1477 – Experimental cut: $p_\ell^B > 1.0$ GeV
- 1478 – Signal-out: $\approx 68\%$ of all signal events so it is assigned an independent free-floating
- 1479 normalisation
- 1480 – Other backgrounds: floated normalisation within a Gaussian constraint of 50%
- 1481 of the nominal yields
- 1482 – Fitted variables: q^2

6. $\Delta\mathcal{B}(B \rightarrow X_u\ell\nu)$ measurement

To extract $\Delta\mathcal{B}(B \rightarrow X_u\ell\nu)$, we use an MC sample on which all selections discussed in Sections 3 and 5 are applied. From the fit presented above, the signal strength μ is extracted. Given the input $\mathcal{B}(B \rightarrow X_u\ell\nu)$, one can therefore obtain the measured branching fraction. The MC $B \rightarrow X_u\ell\nu$ events were generated with the branching ratio values of $\mathcal{B}(B^\pm \rightarrow X_u\ell\nu) = (1.92 \pm 0.24) \cdot 10^{-3}$ and $\mathcal{B}(B^0 \rightarrow X_u\ell\nu) = (1.76 \pm 0.22) \cdot 10^{-3}$. The input value used for the total full phase-space branching fraction is extracted from generator level samples as: $\mathcal{B}(B \rightarrow X_u\ell\nu) = N_{X_u\ell\nu}^{\text{gen}}/(2N_{B\bar{B}}) = (1.82 \pm 0.03) \cdot 10^{-3}$. Where the uncertainty comes from the uncertainty on the number of $B\bar{B}$ pairs $N_{B\bar{B}} = (387.1 \pm 5.6) \cdot 10^6$ (the statistical uncertainty on the number of $X_u\ell\nu$ events generated is negligible). Cuts on the phase-space reduce the theoretical acceptance $\varepsilon_{\Delta\mathcal{B}}$ as discussed in Section 8.2. The predicted branching fraction in each phase-space region is therefore scaled by the corresponding acceptance.

1. Asimov test

Before unblinding the data we first perform an Asimov fit⁵. An Asimov fit consists of using the MC distribution itself as pseudo-data in both regions used in the fit. When performing the fit, one would then expect to recover exactly the initial settings *i.e.* every single free-floating and nuisance parameter should be equal to its initial value. In the Asimov test, a signal strength of $\mu = 1.0$ is obtained as expected. The statistical uncertainty is obtained from 1,000 toy fits where all nuisance parameters are fixed to their initial value. The resulting distribution of signal strengths is fitted to a Gaussian whose width corresponds to the statistical uncertainty. The systematic uncertainty is then derived by subtracting in quadrature the statistical uncertainty from the total uncertainty.

We want to perform fits for different variables in different phase-space regions and with different experimental cuts improving the signal to background ratio. A lepton momentum cut at 1.0 GeV is applied by default on all samples. The results from all these fits are shown in Tables XVIII to XXII. The optimal $B \rightarrow X_c\ell\nu$ MVA cut is chosen as the cut that minimises the value of the total uncertainty in fit 1 with a phase-space and experimental cuts of $p_\ell^B > 1.0$ GeV (*cf.* Section 5). The value of the cut was found to be equal to 0.87. In addition, for this same fit, we show the pre- and post-fit distributions in Fig. 45. Even though each systematic source is symmetrised according to the procedure described above, the total uncertainty can still be asymmetric and it is computed as such. Nevertheless, if the asymmetry is small, we quote the larger uncertainty as symmetric. Fits using $M_X(q^2)$ are not performed with an experimental $M_X(q^2)$ cut at 1.7 GeV (8 GeV²). The binning chosen to fit each variable is shown in Tab. XXIII.

The separate impact of every nuisance parameter can be obtained directly with `pyhf`. The impact of a nuisance parameter is defined as the difference between the nominal POI post-fit value and its value from a fit where the nuisance parameter is held constant at its central value plus or minus its associated uncertainty. The 30 parameters with the largest impact from the q^2 fit in the region $p_\ell^B > 1.0$ GeV are displayed in Fig. 46. The ranking

⁵ The name Asimov comes from Isaac Asimov's short story *Franchise* where one person is selected to represent the entire electorate during the United States presidential election.

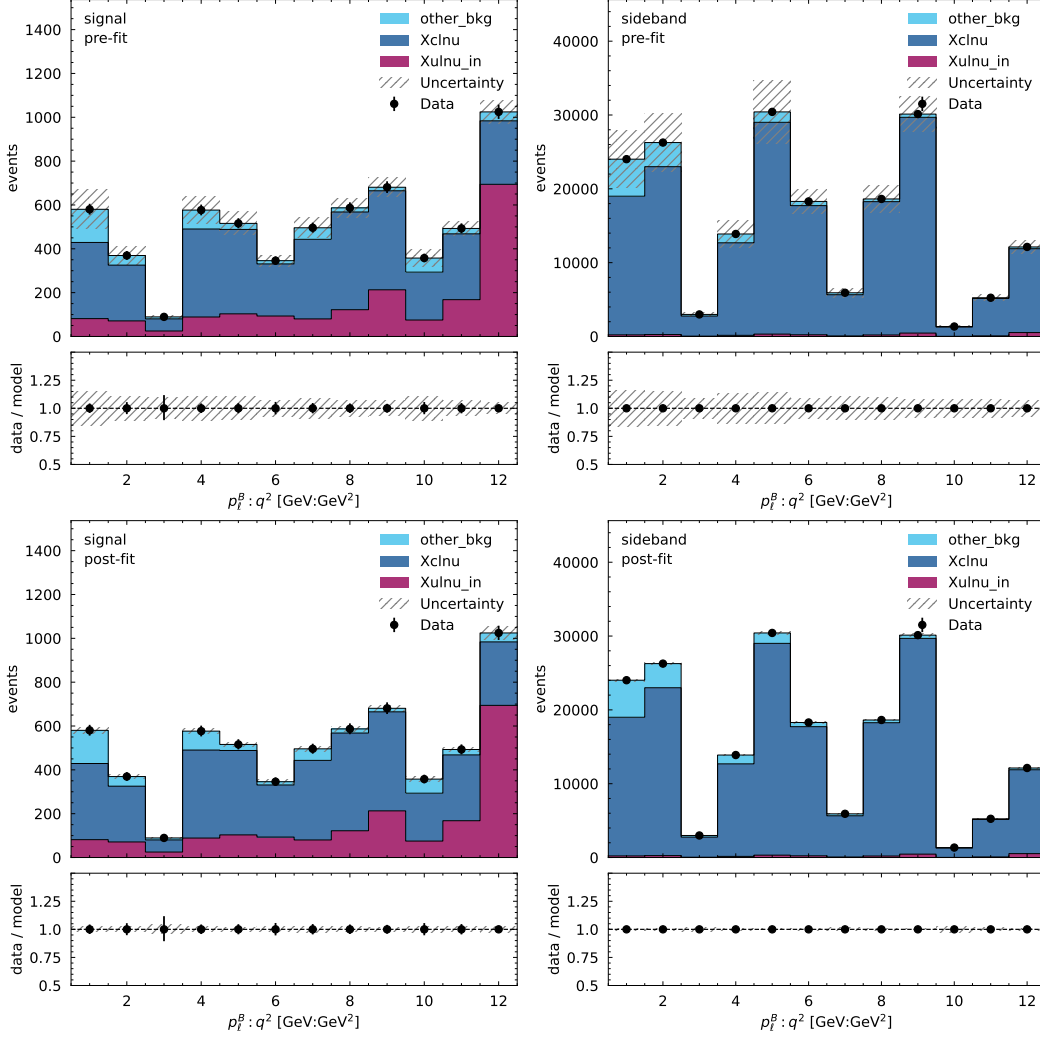


FIG. 45: Pre-fit (top) and post-fit (bottom) $p_\ell^B:q^2$ distributions for a simultaneous fit of the signal region (left) and $\text{CR}_{0,\text{low}}$ (right) using Asimov data. The $X_{u\ell\nu}\text{-in}$ region is defined by a selection $p_\ell^B > 1.0$ GeV (fit 1).

Fitted variable	$\Delta\mathcal{B}(B \rightarrow X_u\ell\nu) \times 10^3$
p_ℓ^B	$1.58^{+0.18}_{-0.16}$
q^2	$1.58 \pm 0.13 = 1.58 \pm 0.08 \pm 0.10$
M_X	$1.58^{+0.21}_{-0.18}$
$M_X:q^2$	1.58 ± 0.13
$p_\ell^B:q^2$	$1.58 \pm 0.13 = 1.58 \pm 0.08 \pm 0.10$

TABLE XVIII: Partial branching fractions obtained from various Asimov fits for a phase space selection of $p_\ell^B > 1.0$ GeV and an experimental selection of $p_\ell^B > 1.0$ (Fit 1). The uncertainties given for the branching fraction errors are respectively the statistical and systematic errors.

Fitted variable	$\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) \times 10^3$
$p_\ell^B : q^2$	$1.03 \pm 0.08 = 1.03 \pm 0.06 \pm 0.06$

TABLE XIX: Partial branching fractions obtained from various Asimov fits for a phase space selection of $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV and an experimental selection of $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV (Fit 3). The uncertainties given for the branching fraction are respectively the statistical and systematic errors.

Fitted variable	$\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) \times 10^3$
q^2	0.57 ± 0.05

TABLE XX: Partial branching fractions obtained from various Asimov fits for a phase space selection of $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV² and an experimental selection of $p_\ell^B > 1.0$ GeV (Fit 4). The uncertainties given for the branching fraction are respectively the statistical and systematic errors.

Fitted variable	$\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) \times 10^3$
p_ℓ^B	$0.57 \pm 0.06 = 0.57 \pm 0.03 \pm 0.05$

TABLE XXI: Partial branching fractions obtained from various Asimov fits for a phase space selection of $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV² and an experimental selection of $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV² (Fit 5). The uncertainties given for the branching fraction are respectively the statistical and systematic errors.

Fitted variable	$\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) \times 10^3$
q^2	$0.35 \pm 0.05 = 0.35 \pm 0.04 \pm 0.03$

TABLE XXII: Partial branching fractions obtained from various Asimov fits for a phase space selection of $p_\ell^B > 2.1$ GeV and an experimental selection of $p_\ell^B > 1.0$ GeV (Fit 7). The uncertainties given for the branching fraction are respectively the statistical and systematic errors.

Variable	Binning
p_ℓ^B	12 bins: 11 bins of width 0.1 GeV in [1.0, 2.1] GeV and [2.1, 2.7] GeV (fit 1) 10 bins: [1.0, 1.4, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.7] GeV (fit 5)
q^2	10 bins: 9 bins of width 2 GeV ² in [0, 18] GeV ² and [18, 26] GeV ²
M_X	4 bins: [0, 1.5, 1.9, 2.5, 5.0] GeV
$M_X : q^2$	[0, 1.5] GeV $\times$ [0, 2, 4, 6, 8, 10, 12, 14, 26] GeV ² [1.5, 1.9] GeV $\times$ [0, 2, 4, 6, 26] GeV ² [1.9, 2.5] GeV $\times$ [0, 2, 4, 26] GeV ² [2.5, 5.0] GeV $\times$ [0, 2, 26] GeV ²
$p_\ell^B : q^2$	12 bins: [1.0, 1.3, 1.6, 1.9, 2.7] GeV $\times$ [0, 4, 8, 26] GeV ²

TABLE XXIII: Binning of variables used in the fits described in this Section.

of the most impactful nuisance parameters for every fit is shown in Appendix F. The uncertainty of the DFN parameters has the largest impact on this fit (particularly the first eigenvariation). For each nuisance parameter, the pull, which is defined below in Section 87, is also shown on Fig. 46.

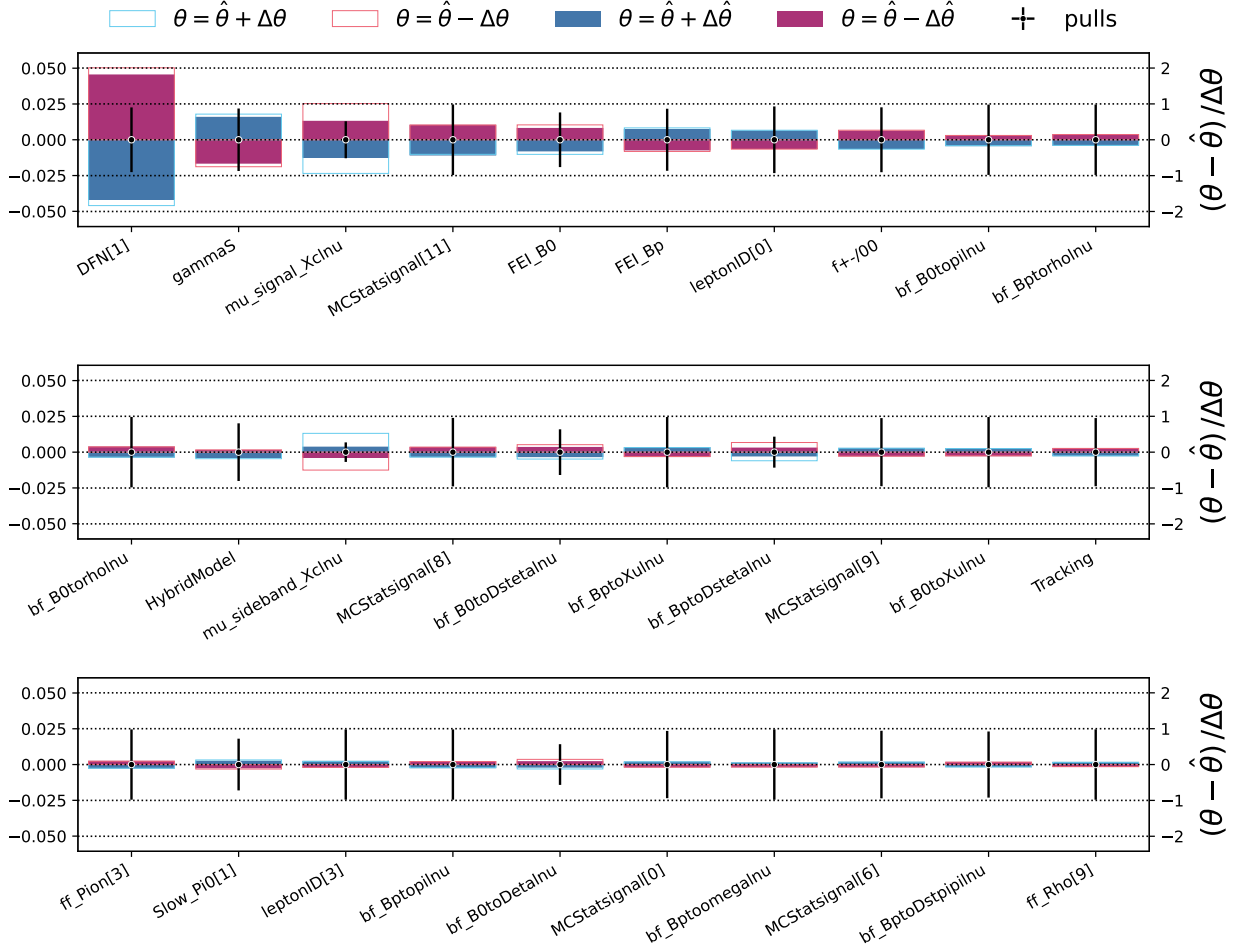


FIG. 46: Nuisance parameter impact ranking for the 30 most important sources for fit 1. The post-fit (resp. pre-fit) impact is shown as a full (resp. empty) box. The blue (resp. purple) band corresponds to the POI impact of the parameter when it is held to its nominal value plus (resp. minus) its associated uncertainty. More details are given in the text.

Fit 4 and 7 are dropped as a non-negligible bias of the signal-out normalisation is observed as shown in Appendix F. Based on the various checks discussed below and the expected precision, we propose to quote the following four fits in a publication with the nominal fit in **bold**:

- **Fit 1 with $p_\ell^B:q^2$**
- Fit 3 with $p_\ell^B:q^2$
- Fit 5 with p_ℓ^B

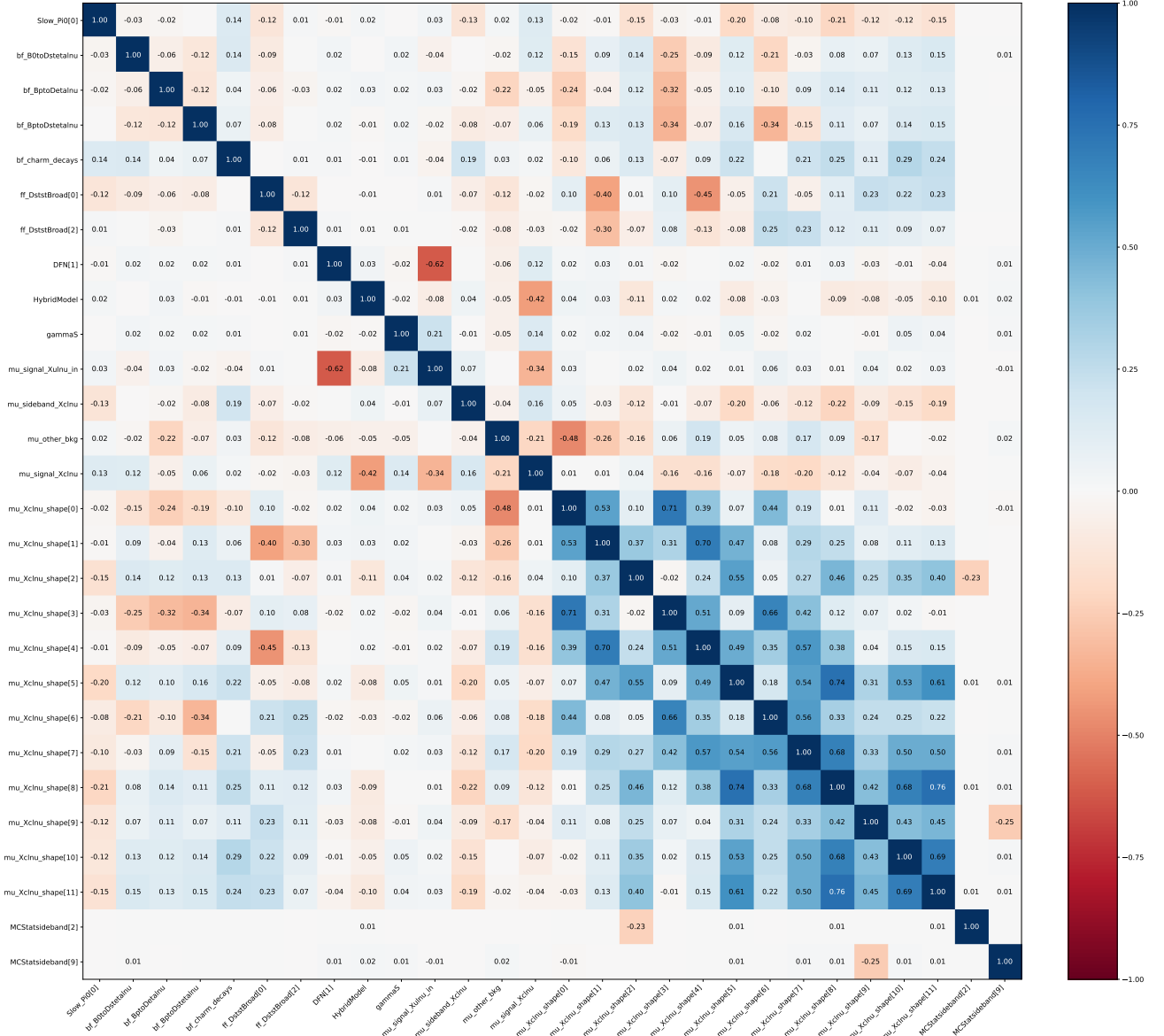


FIG. 47: Correlation matrix for fit 1 ($p_\ell^B:q^2$ fit with $p_\ell^B > 1.0$ GeV cut for phase-space and experimental selection). Only rows/columns where a value of 0.20 or larger is found are shown.

2. Systematics breakdown

We break down the systematic uncertainty contributions by producing a set of 1000 toys whose data yields are fixed to the Asimov yields and whose auxiliary data is fixed to the initial value for all systematics except the one we evaluate the contribution of. Each toy is then fitted and the obtained uncertainty is taken as the systematic source contribution to the total uncertainty. The breakdown for all fits is summarised in Table XXIV. The DFN model parameter uncertainty is the largest contribution for all four fits discussed in this Section.

Systematics source	Systematics uncertainty (%)		
	Fit 1	Fit 3	Fit 5
DFN parameters	4.7	4.2	5.8
DFN $\rightarrow$ BLNP	0.14	0.44	1.6
γ_S	1.6	1.2	1.8
$B \rightarrow \pi \ell \nu$ form factors	0.30	0.32	0.33
$B \rightarrow \rho \ell \nu$ form factors	0.27	0.30	0.24
$B \rightarrow \omega \ell \nu$ form factors	0.10	0.11	0.11
$B \rightarrow \eta^{(\prime)} \ell \nu$ form factors	0.01	0.01	0.02
$B^\pm \rightarrow X_u \ell \nu$ branching fractions	0.95	0.70	0.49
$B^0 \rightarrow X_u \ell \nu$ branching fractions	0.56	0.53	0.53
$B \rightarrow D_{\text{Broad}}^{**}$ form factors	0.45	0.23	0.03
$B \rightarrow D_{\text{Narrow}}^{**}$ form factors	0.06	0.02	0.01
$B \rightarrow D/D^* \ell \nu$ form factors	0.02	0.04	0.02
$B^\pm \rightarrow X_c \ell \nu$ branching fractions	0.73	0.56	0.06
$B^0 \rightarrow X_c \ell \nu$ branching fractions	0.66	0.22	0.09
D decay branching fractions	0.15	0.28	0.23
SR $X_c \ell \nu$ normalisation	1.5	2.3	3.8
CR $X_c \ell \nu$ normalisation	1.0	0.76	0.59
Other backgrounds normalisation	0.19	N/A	N/A
$N_{B\bar{B}}$	1.6	1.6	1.6
FEI	1.3	1.4	1.4
Slow π	0.40	0.25	0.20
ℓ ID	0.70	0.67	0.68
$f^{\pm/00}$	0.64	0.66	0.72
Continuum reweighting	0.12	0.15	0.18
Tracking	0.27	0.25	0.29
K_S^0 efficiency	0.05	0.06	0.04
$K^\pm$ ID	0.01	0.02	0.02
MC Statistics	1.2	1.1	0.85
Additional composition uncertainty	2.0	2.9	0.06
Total systematic	6.4	6.5	7.8
Statistical	5.1	5.0	4.6
Total	8.2	8.2	9.1

TABLE XXIV: Systematic uncertainty breakdown for all fits. The systematic sources are ranked in descending order of their contribution in fit 1.

1547 7. Fitter validation

1548 The pull of a parameter entering a fit is defined as:

$$\begin{cases} \frac{\mu_{init}-\mu}{\sigma_+}, & \text{if } \mu \leq \mu_{init} \\ \frac{\mu-\mu_{init}}{\sigma_-}, & \text{if } \mu > \mu_{init} \end{cases} \quad (14)$$

1549 where μ_{init} is the initial (true) value of the parameter, μ its post-fit value and σ_+/σ_- are
 1550 the amplitudes of associated positive/negative post-fit uncertainties. The pull is expected
 1551 to be equal to 0 for every parameter in an Asimov fit. This is confirmed in Fig. 48 for all
 1552 nuisance parameters for the p_ℓ^B fit in the region $p_\ell^B > 1.0$ GeV.

1553 1. Toy fit

1554 We want to test the fit set-up on a large number of randomly generated toy data sets. These
 1555 toys are generated from Poisson variations of the input MC yields. In addition, for each
 1556 nuisance parameter the initial value of the auxiliary data is Gaussian-varied. For each of
 1557 these toys, the fit is performed, the signal strength μ is extracted and its pull from the initial
 1558 value is calculated as defined in Equ. 14. For a sufficiently large number of toys, the pulls
 1559 are expected to be distributed as a Gaussian normal distribution. Any deviation from this
 1560 distribution would indicate problems in the fit, such as a bias on the POI. In Fig. 49, the
 1561 pull distribution obtained from the $p_\ell^B:q^2$ fit in the phase-space region $p_\ell^B > 1.0$ GeV for
 1562 5,000 toys is shown together with the normal distribution fit. Toy distributions from every
 1563 fit are shown in Appendix F.

1564 2. Linearity check

1565 In order to further asses the stability of the fit, we perform a linearity check. In this test,
 1566 different sets of toys are produced by varying the initial POI value. Each toy is fitted and
 1567 for each set of toys the mean value of the signal strength is extracted assuming a Gaussian
 1568 distribution. This value is expected to be close to the initial value. We test this for 11 initial
 1569 equidistant values between 0.5 and 1.5. In Fig. 50, we show the output value of the signal
 1570 strength as a function of its initial value for the $p_\ell^B:q^2$ fit in the phase-space region $p_\ell^B > 1.0$
 1571 GeV. The overall behaviour is linear and therefore this fit is expected to be stable. Linearity
 1572 check plots of every fit are shown in Appendix F. For some fits (in particular the M_X fits),
 1573 if the initial value of the signal strength given in the linearity test is small *i.e.* 0.5 – 0.7, the
 1574 output value is slightly overestimated. The M_X fits in general are expected to lead to large
 1575 uncertainties and we will therefore not investigate this behaviour further.

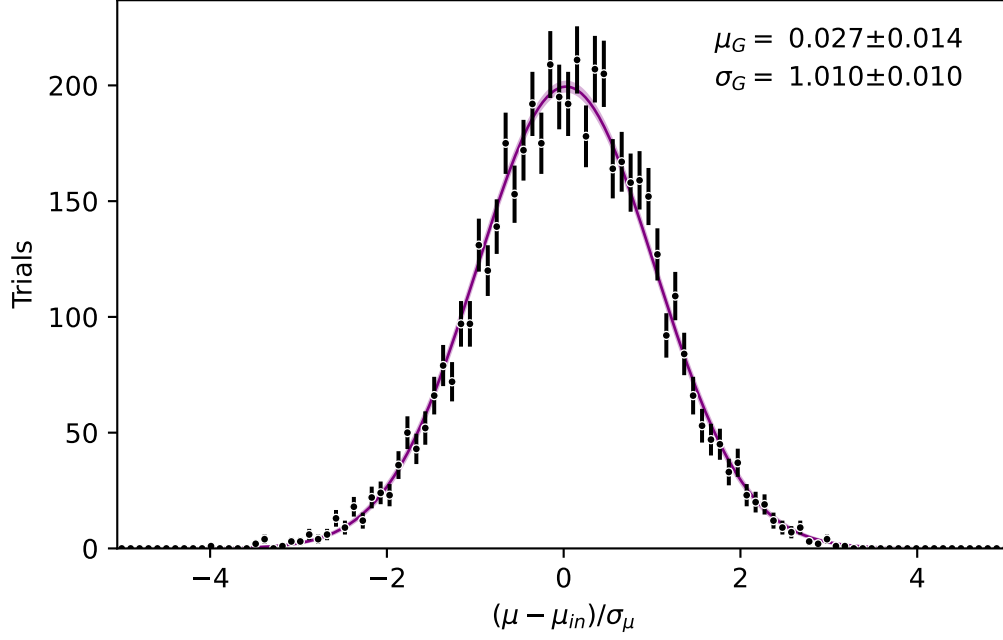


FIG. 49: Distribution of the signal strength pulls from 5,000 toys for fit 1. The distribution is fitted to a normal Gaussian distribution. The fitted values of the mean and variance of the normal distribution are displayed as well.

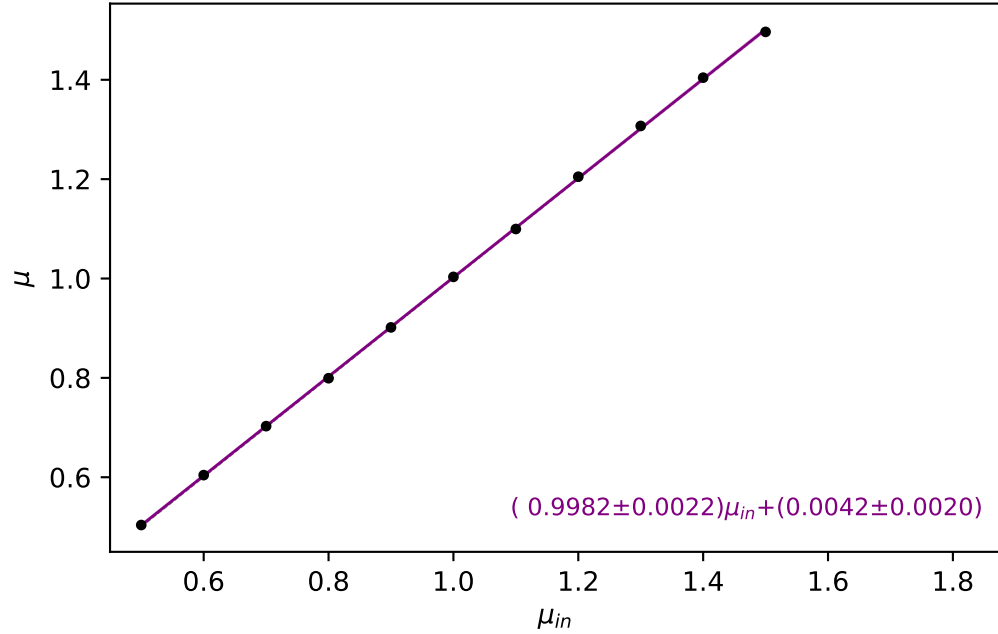


FIG. 50: Linearity check for fit 1. For each point, 1,000 toys were produced

1576 8. $B \rightarrow X_c \ell \nu$ factors

1577 In the Asimov fit discussed above, all $B \rightarrow X_c \ell \nu$ factors are equal to 1 as expected. However,
 1578 as the data-MC disagreement is large in the control region as illustrated in Fig. 35, the as-
 1579 sociated uncertainties are expected to be different from what is obtained from the Asimov fit.

1580

1581 In order to get a more realistic expectation of the $B \rightarrow X_c \ell \nu$ shape factor values and
 1582 their associated uncertainties, we perform a fit using data in the control region and pseudo-
 1583 data in the signal region. In order to create realistic pseudo-data yields, we use bin-by-bin
 1584 data-MC ratio factors obtained from $\text{CR}_{K,high}$. These factors are then multiplied to the MC
 1585 $B \rightarrow X_c \ell \nu$ component yields in the signal region. The obtained signal region pseudo-data
 1586 yields are therefore obtained by summing up the signal region MC yields of the signal and
 1587 other-backgrounds components together with the $X_c \ell \nu$ MC yields scaled by the data-MC
 1588 ratio obtained from the control region. The resulting pre-fit data and MC distributions are
 1589 shown in Fig. 51.

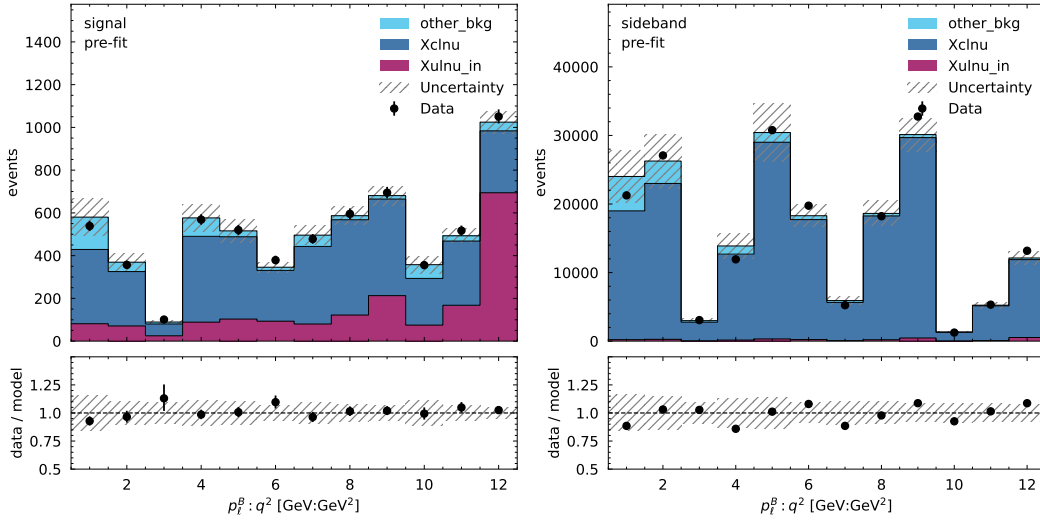


FIG. 51: Pre-fit $p_\ell^B : q^2$ distributions in the signal (left) and control (right) regions. Data is used in the control region and pseudo-data in the signal region.

1590 The $B \rightarrow X_c \ell \nu$ shape is corrected in the fit as one single component. However, its com-
 1591 position is different between the signal and control region as can be seen from Table XIII.
 1592 Because of these differences, the shape factors extracted in the simultaneous fit don't nec-
 1593 essarily translate perfectly from one region to the other as every charm component could
 1594 induce different sources of data-MC disagreement. We consider an additional uncertainty to
 1595 cover these differences. The procedure is described in this Section. Effectively, the bin-by-bin
 1596 composition of the $X_c \ell \nu$ component as well as the shapes of each individual sub-component
 1597 are different between the signal region and $\text{CR}_{0,low}$ which makes it extremely tedious to
 1598 properly estimate an uncertainty. Nevertheless, we assume that the total uncertainty ob-
 1599 tained from the fit and the additional composition systematic cover most of uncertainty
 1600 related to the differences between the signal and control region $B \rightarrow X_c \ell \nu$ component.

1601

1602 In order to estimate the uncertainty related to the compositions, we perform two fits
 1603 and compare the resulting $B \rightarrow X_u \ell \nu$ branching fraction. In the first fit, data is used in the

control region and pseudo-data in the signal region as described above. In the second fit, we use the same MC and data yields in the signal region and the same data yields in the control region as in the first fit. However, in the control region, the $X_c\ell\nu$ sub-components are scaled such that the composition matches the signal region composition. A comparison of the shape of the $X_c\ell\nu$ component with and without this scaling is shown in Figure 52. A comparison of the resulting pre-fit scaled $p_\ell^B:q^2$ distributions are shown in Fig. 53. In order to illustrate the procedure, we compare below the results from these two fits performed with the phase-space and experimental selections defined for Fit 1 ($p_\ell^B > 1.0$ GeV for both selections, *c.f.* Section 8.5). After unblinding, we plan to repeat this procedure for every fit discussed above using data both in the signal and control region. The $B \rightarrow X_c\ell\nu$ shape factors obtained from the two fits together with their uncertainties are shown in Fig. 54. The $\mathcal{B}(B \rightarrow X_u\ell\nu)$ obtained are also compared in Tab. XXV. The difference can then be added as an additional uncertainty (which assumes no correlation between this uncertainty and the total uncertainty). The test will be repeated with data in the signal region.

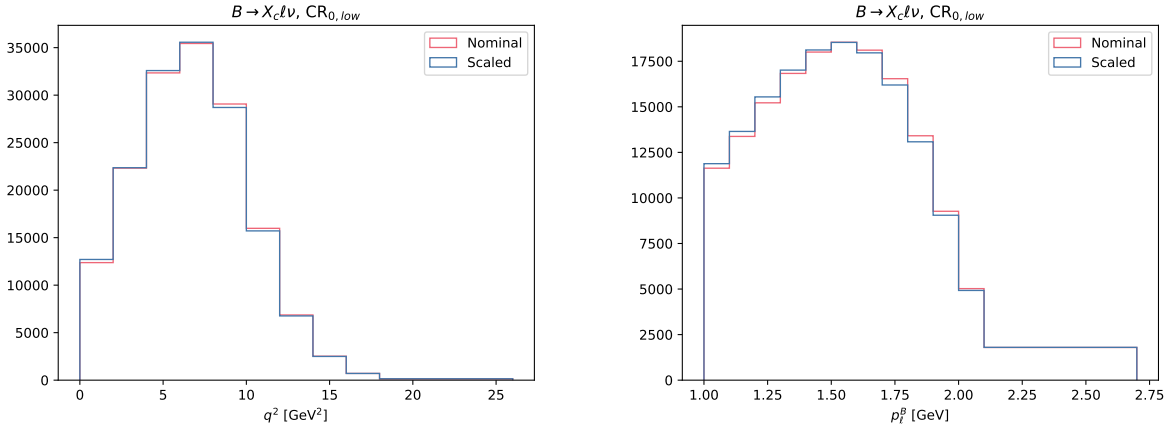


FIG. 52: Comparison of the q^2 (left) and p_ℓ^B (right) distributions in $\text{CR}_{0,low}$ of $X_c\ell\nu$ events before and after the $X_c\ell\nu$ sub-component composition scaling described in the text.

Variable	$p_\ell^B:q^2$
Unscaled $X_c\ell\nu$	1.52 ± 0.13
Scaled $X_c\ell\nu$	1.49 ± 0.13

TABLE XXV: Branching fractions obtained from the $p_\ell^B:q^2$ fit in the $p_\ell^B > 1.0$ GeV region using pseudo-data. The branching fractions obtained with the scaled and unscaled $X_c\ell\nu$ component are compared. See text for more details.

9. $|V_{ub}|$ measurement

Once the partial branching fraction $\Delta\mathcal{B}(B \rightarrow X_u\ell\nu)$ has been measured, one can derive the amplitude of V_{ub} via:

$$|V_{ub}| = \sqrt{\frac{\Delta\mathcal{B}(B \rightarrow X_u\ell\nu)}{\tau_B \cdot \Delta\Gamma(B \rightarrow X_u\ell\nu)}}, \quad (15)$$

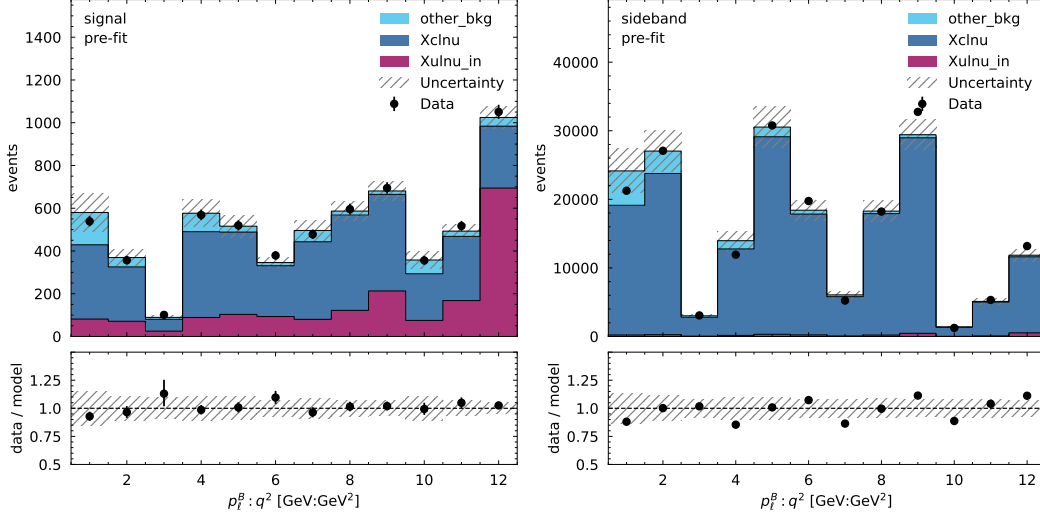


FIG. 53: Pre-fit $p_\ell^B : q^2$ distributions in the signal (left) and control (right) regions. Data is used in the control region and pseudo-data in the signal region. The $B \rightarrow X_c \ell \nu$ MC yields in the control region have been scaled as described in the text.

where $\tau_B = 1.579 \pm 0.003$ ps is the average charged/neutral B meson lifetime [3] and $\Delta\Gamma(B \rightarrow X_u \ell \nu)$ is the theoretical partial decay rate without $|V_{ub}|$.

1623

The decay rates for different models are extracted from previous measurements of $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu)$ and the most recent HFLAV averages of $|V_{ub}|$ (see Tables 94 and 95 in [54])⁶. Four inclusive models are considered: BLNP [6], GGOU [55], DGE [56] and ADFR [57]. The decay rates for the $p_\ell^B > 1.0$ GeV region are summarised in Tab. XXVI.

Phase-space region	BLNP	GGOU	DGE	ADFR
$p_\ell^B > 1.0$ GeV	$63.0^{+7.2}_{-5.6}$	$60.8^{+3.1}_{-2.6}$	$59.3^{+3.9}_{-3.3}$	$61.4^{+6.2}_{-5.4}$

TABLE XXVI: Partial decay rates $\Delta\Gamma$ for different models. Numbers are given in ps^{-1} .

Using the partial branching fraction obtained from the q^2 fit in the $p_\ell^B > 1.0$ GeV region (Tab. XVIII) and Equ. 15, we can therefore extract a value of $|V_{ub}|$ for each model quoted above. These are summarised in Tab. XXVII.

Phase-space region	BLNP	GGOU	DGE	ADFR
$p_\ell^B > 1.0$ GeV	$4.01 \pm 0.10 \pm 0.14^{+0.23}_{-0.18}$	$4.08 \pm 0.10 \pm 0.14^{+0.10}_{-0.09}$	$4.13 \pm 0.10 \pm 0.14^{+0.14}_{-0.12}$	$4.06 \pm 0.10 \pm 0.14^{+0.20}_{-0.18}$

TABLE XXVII: $|V_{ub}| \times 10^3$ for different models using the branching fraction extracted from a q^2 fit with the experimental and phase-space selections of Fit 1. The quoted uncertainties are statistical, systematic and theoretical respectively.

⁶ The numbers used to extracted the values of $|V_{ub}|$ quoted here were taken from the most recent HFLAV paper which is yet to be published.

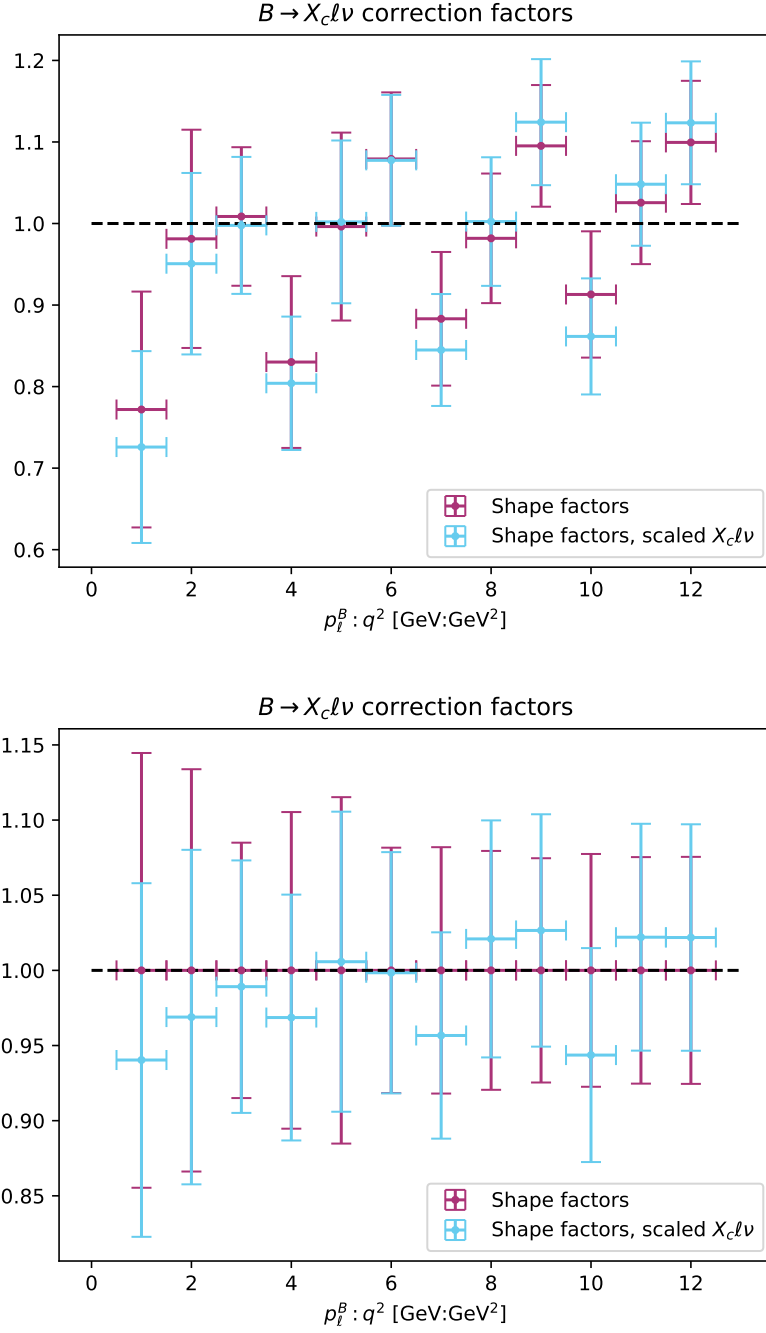


FIG. 54: The $B \rightarrow X_c \ell \nu$ shape factors obtained from the fits using pseudo-data. The cyan points correspond to the fit where the $B \rightarrow X_c \ell \nu$ yields have been scaled as described in the text. The bottom plot shows the same shape factors normalised to the factors obtained from the fit where $B \rightarrow X_c \ell \nu$ is left unscaled.

10. Fits with control regions

1. Fit with $CR_{0,low}$ split

We perform the fit procedure using $CR_{0,low}$ only. This region is split in two based on the $X_c\ell\nu$ MVA score: one with a score lower than 0.45 and one with a score between 0.45 and 0.60. We will call these two regions $CR'_{0,low}$ and VR'_2 respectively. The former is used as the control region in the fit and the latter as the signal region. We illustrate the test in this Section for fit 1. Plots for fit 3 and 5 are shown in Appendix I. Fit 7 uses the same kinematical cuts as fit 1 and only the signal phase-space selection is different. The $X_c\ell\nu$ background correction being unchanged, we do not repeat the test for fit 7.

The signal purity in $CR_{0,low}$ is about 1.50% as shown in Table XIII. The normalisation of the $X_c\ell\nu$ component in both regions is extracted exactly as described in Section 7.3.2 and the fit procedure is performed exactly as described in Section 8. Using data and MC, the normalisation in $CR'_{0,low}$ is found to be 1.11 ± 0.05 (stat.) and in VR'_2 , 1.06 ± 0.01 (stat.). Using the ABCD method, the normalisation in VR'_2 is found to be 1.08 ± 0.04 where the error is the uncertainty covering all sources described in Section 7.3.2.

As illustrated in Figure 55, the fit captures well the $B \rightarrow X_c\ell\nu$ shape in both regions which is confirmed by the obtained p -value of 0.99. We compare the output $X_c\ell\nu$ scale factors with the $X_c\ell\nu$ data/MC ratios for regions $CR'_{0,low}$ and VR'_2 in Figure 56.

2. Fit with $CR_{K,low}$ and $CR_{K,high}$

We also repeat the fit procedure using the two control regions where the presence of at least one kaon is required. The goal of this test is to confirm that the shape of $X_c\ell\nu$ as corrected in the low MVA region can reliably correct the $X_c\ell\nu$ component in the high MVA score region. We use $CR_{K,low}$ as CR and $CR_{K,high}$ as SR. In this case, the $X_c\ell\nu$ normalisations are taken directly from MC. We respectively find 1.04 ± 0.04 and 0.95 ± 0.05 , where the total uncertainty is obtained from the statistical uncertainty and the non- $X_c\ell\nu$ component variation. The pre and post-fit distributions for $p_\ell^B:q^2$ with the fit 1 setup are illustrated in Figures 57. A p -value of 0.60 is obtained. The NP pulls of this fit are shown in Figure 58. The $B \rightarrow X_c\ell\nu$ shape factors extracted from the fit are compared with data/MC ratios in the two regions from the fit in Figure 59. The fit is projected onto q^2 and E_ℓ^B as shown in Figure 60. The $CR_{K,low}$ - $CR_{K,high}$ fit is projected onto $E_\ell^B:q^2$ with the $X_c\ell\nu$ template broken down in subcomponents in order to compare the pre and postfit $X_c\ell\nu$ compositions. The comparison is summarised in Table XXVIII.

$CR_{K,high}$	$B \rightarrow D\ell\nu$	$B \rightarrow D^*\ell\nu$	$B \rightarrow D^{**}\ell\nu$	$B \rightarrow D_{Gap}^{**}\ell\nu$
Prefit	40.4%	47.3%	6.2%	6.1%
Postfit	40.2%	49.2%	6.0%	4.6%

TABLE XXVIII: Composition changes slightly except for the gap modes which, as the NP pulls show, are reduced.

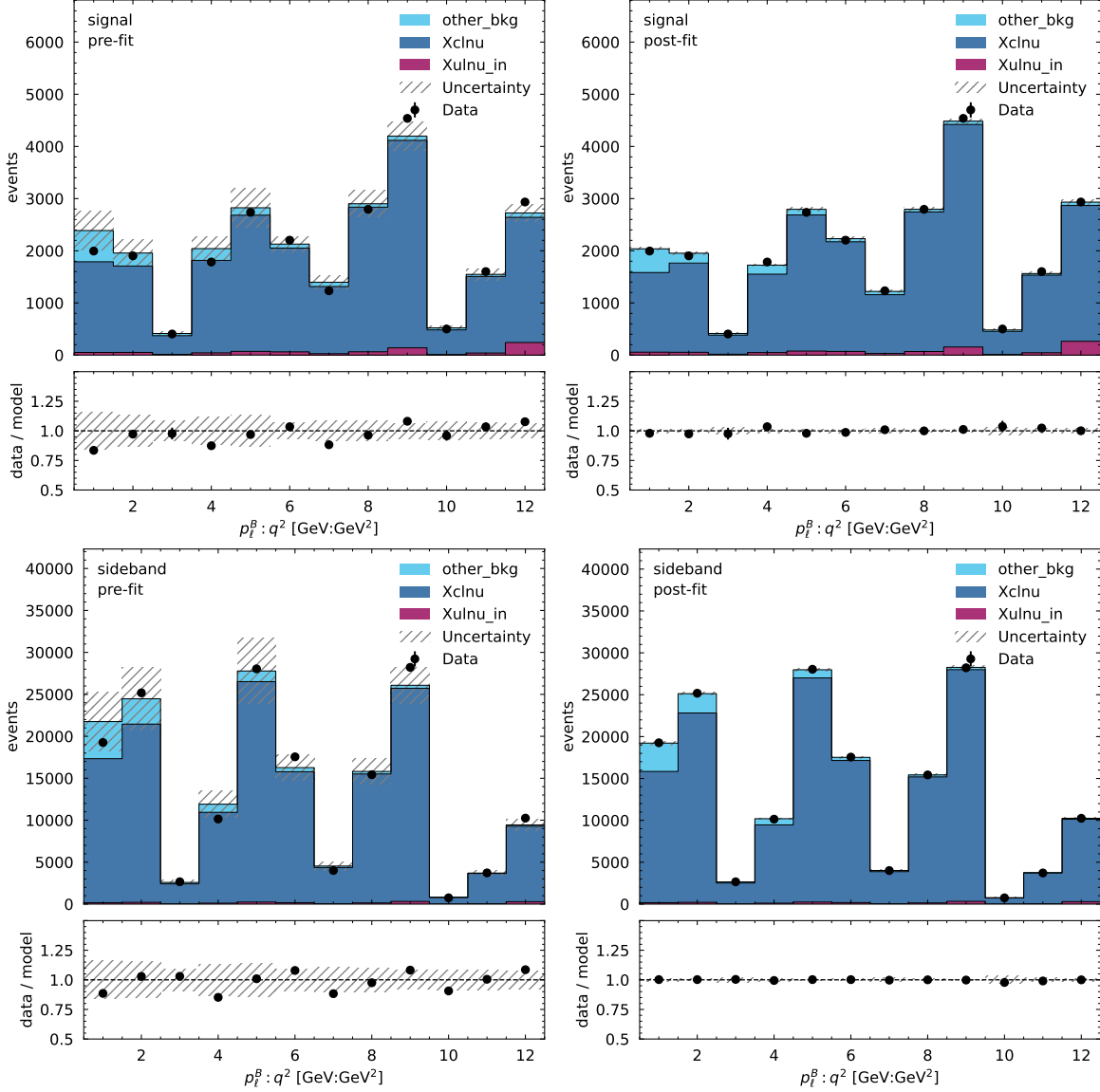


FIG. 55: Pre- (left) and post-fit (right) $p_\ell^B:q^2$ distributions in VR'_2 (top) and $\text{CR}'_{0,\text{low}}$ (bottom) for fit 1.

1671 3. Fit with $\text{CR}_{0,\text{low}}$ and $\text{CR}_{K,\text{high}}$

1672 We also repeat the fit procedure using the two control regions $\text{CR}_{0,\text{low}}$ as CR and $\text{CR}_{K,\text{high}}$
 1673 as SR. This represents a cross-check of the previous $\text{CR}_{K,\text{low}}\text{-CR}_{K,\text{high}}$ test. In this case,
 1674 the $X_c\ell\nu$ normalisations are taken directly from MC. We respectively find 1.11 ± 0.05 and
 1675 0.95 ± 0.05 , where the total uncertainty is obtained from the statistical uncertainty and the
 1676 non- $X_c\ell\nu$ component variation. The pre and post-fit distributions for $E_\ell^B:q^2$ with the fit 1
 1677 setup are illustrated in Figures 61. A p -value of 0.44 is obtained. The NP pulls of this fit
 1678 are shown in Figure 62.

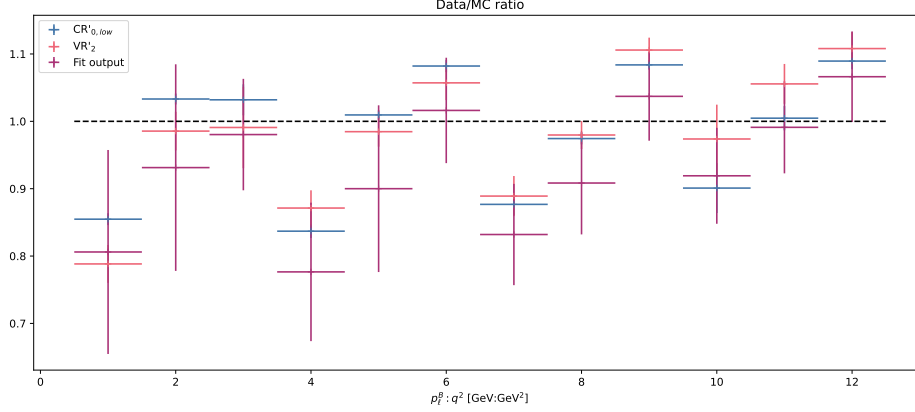


FIG. 56: Comparison between the normalised $X_c \ell \nu$ data/MC ratio for regions $\text{CR}'_{0,low}$ and VR'_2 and the fit output $X_c \ell \nu$ shape factors.

11. Split sample fits

With the current fit setup, one can split the signal and control region in order to perform a simultaneous fit of specific decay channels. In this Section, we discuss a simultaneous fit of the B^+ and B^0 channels as well as the $B \rightarrow e \ell \nu$ and $B \rightarrow \mu \ell \nu$ channels. We also discuss additional split fits used as cross-checks of our fit setup.

1. Fit with split B charge: $\Delta\mathcal{B}(B^0 \rightarrow X_u \ell \nu)$, $\Delta\mathcal{B}(B^\pm \rightarrow X_u \ell \nu)$

Before performing the signal extraction for the B^+ and B^0 channels split in two different channels, we must evaluate and, if necessary, correct the crossfeed between the charged and neutral B channels. This is of particular relevance in our fit setup as the crossfeed rate in $B \rightarrow X_c \ell \nu$ events has an impact on the normalisation and shape corrections of this component. The crossfeed for all events in $\text{CR}_{0,low}$ and SR is shown in Figure 63. The crossfeed for X_u events in the SR is shown in Figure 64. The crossfeed for signal events in the SR is relatively low and one could simply correct the extracted BR for B^+ and B^0 using the crossfeed rate of each B charge. However, the crossfeed rate for $B \rightarrow X_c \ell \nu$ events appears to be larger both in the CR and the SR. It also appears to be different between the two regions. As the normalisation and shape corrections are also extracted separately for each channel, one would need to carefully treat the crossfeed between charged and neutral B mesons. Given the complexity of properly correcting the crossfeed we have decided to not perform the measurement for split B charge.

2. Fit with split lepton flavour: $\Delta\mathcal{B}(B \rightarrow X_u e \nu)$, $\Delta\mathcal{B}(B \rightarrow X_u \mu \nu)$

The crossfeed between the electron and muon channels is negligible. We could extract the two separate branching fractions using the setup of fit 1. However, after further checking, it was noticed that the normalisation of the signal component in the electron channel is biased. Fixing the "other background" yields in the electron channel doesn't fix the bias so the e/μ

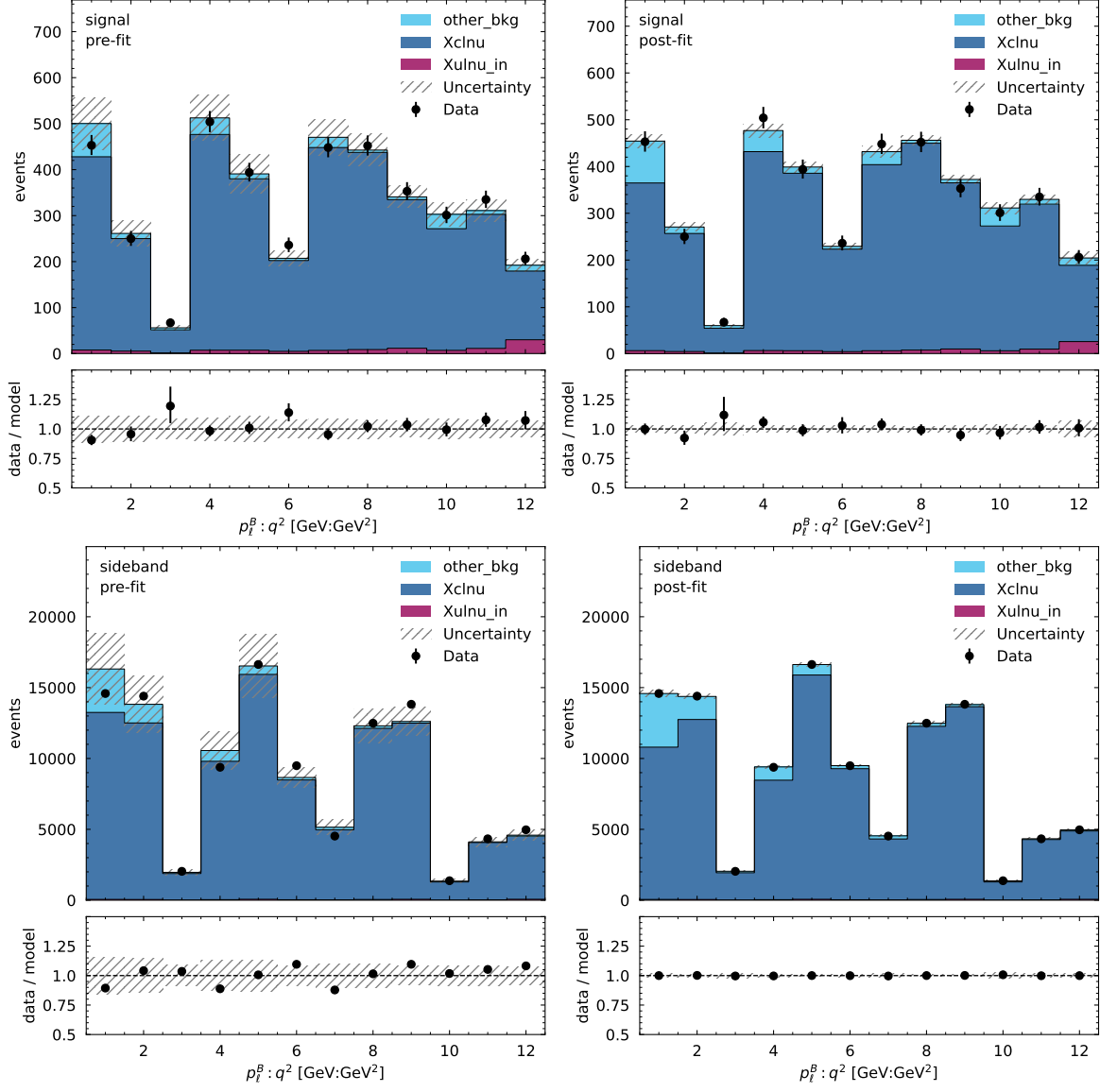


FIG. 57: Pre- (left) and post-fit (right) q^2 distributions in $CR_{K,high}$ (top) and $CR_{K,low}$ (bottom) for fit 1.

split is left as cross-check where each channel is fitted separately.

	e	μ
$CR_{0,low}$	1.13 ± 0.04	1.10 ± 0.07
SR	1.00 ± 0.08	1.03 ± 0.09

TABLE XXIX: $X_c \ell \nu$ normalisation factors for the e/μ split fit.

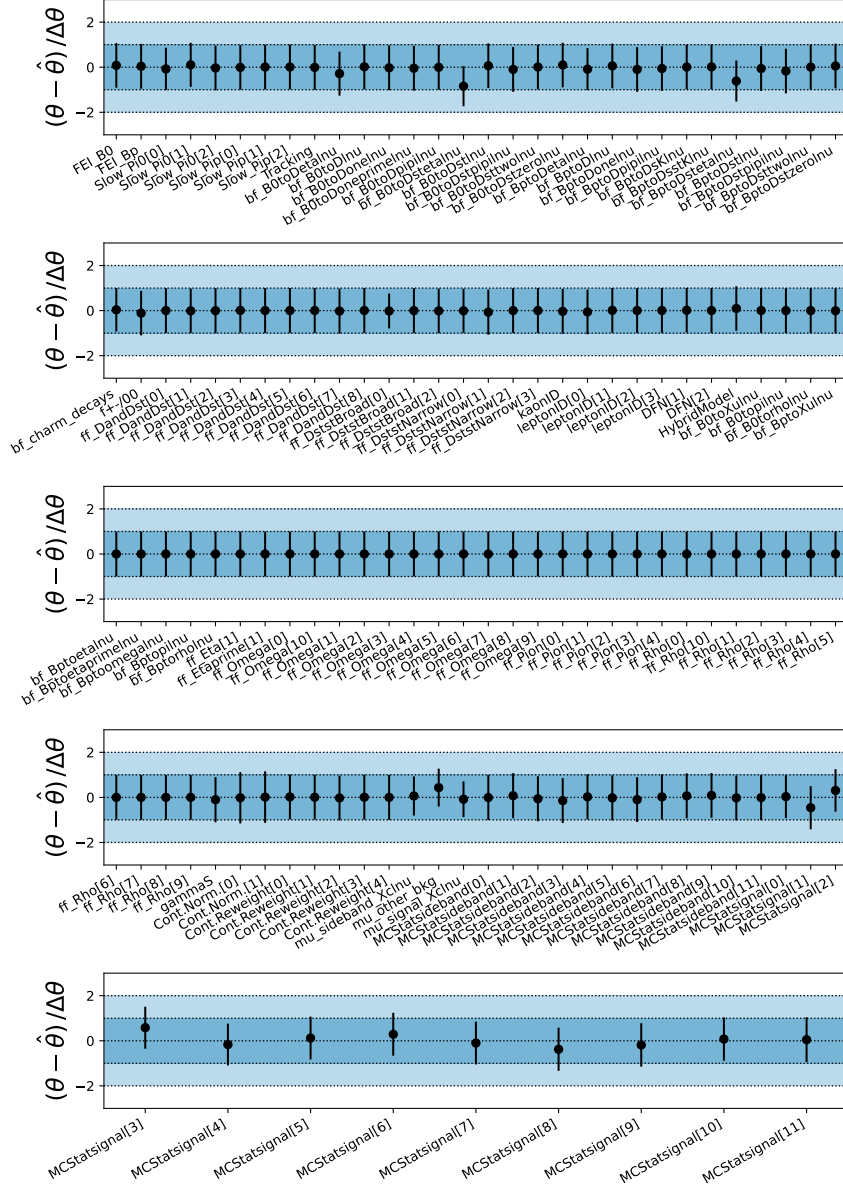


FIG. 58: Pull plot of the $\text{CR}_{K,low}\text{-CR}_{K,high}$ fit with $E_\ell^B:q^2$.

3. Additional split sample fits: split missing momentum θ angle and split lepton charge

Fit 1 is repeated with the SR and CR region split in two using the missing momentum polar angle, θ_{miss} . The cut splitting the two samples is set at $\pi/2$. We show the θ_{miss} distribution in SR and $\text{CR}_{0,low}$ in Figure 65. As this fit is meant as a cross-check rather than a physics result, each of the two channels is fitted separately. The $X_c\ell\nu$ component is normalised in the SR and CR separately following the ABCD method discussed in Section 7.3.2. We repeat the same split fits using the lepton charge (positive vs negative). We perform Asimov fits with these setups as well as the unblinded fit using regions $\text{CR}'_{0,low}$ and VR'_2 . The results of all fits look satisfying and we propose to perform these fits with an unblinded signal region during the box opening procedure in order to cross-check the stability of our fit setup.

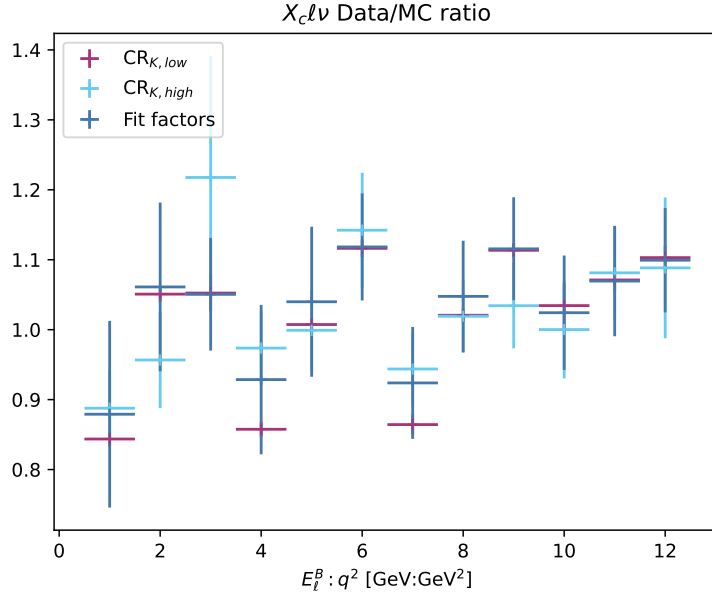


FIG. 59: $X_c \ell \nu$ shape factors as extracted from the $\text{CR}_{K,low}$ - $\text{CR}_{K,high}$ fit with $E_\ell^B : q^2$ compared to the data/MC ratios in these two regions. As expected, the shape factors match the $\text{CR}_{K,low}$ data/MC ratios.

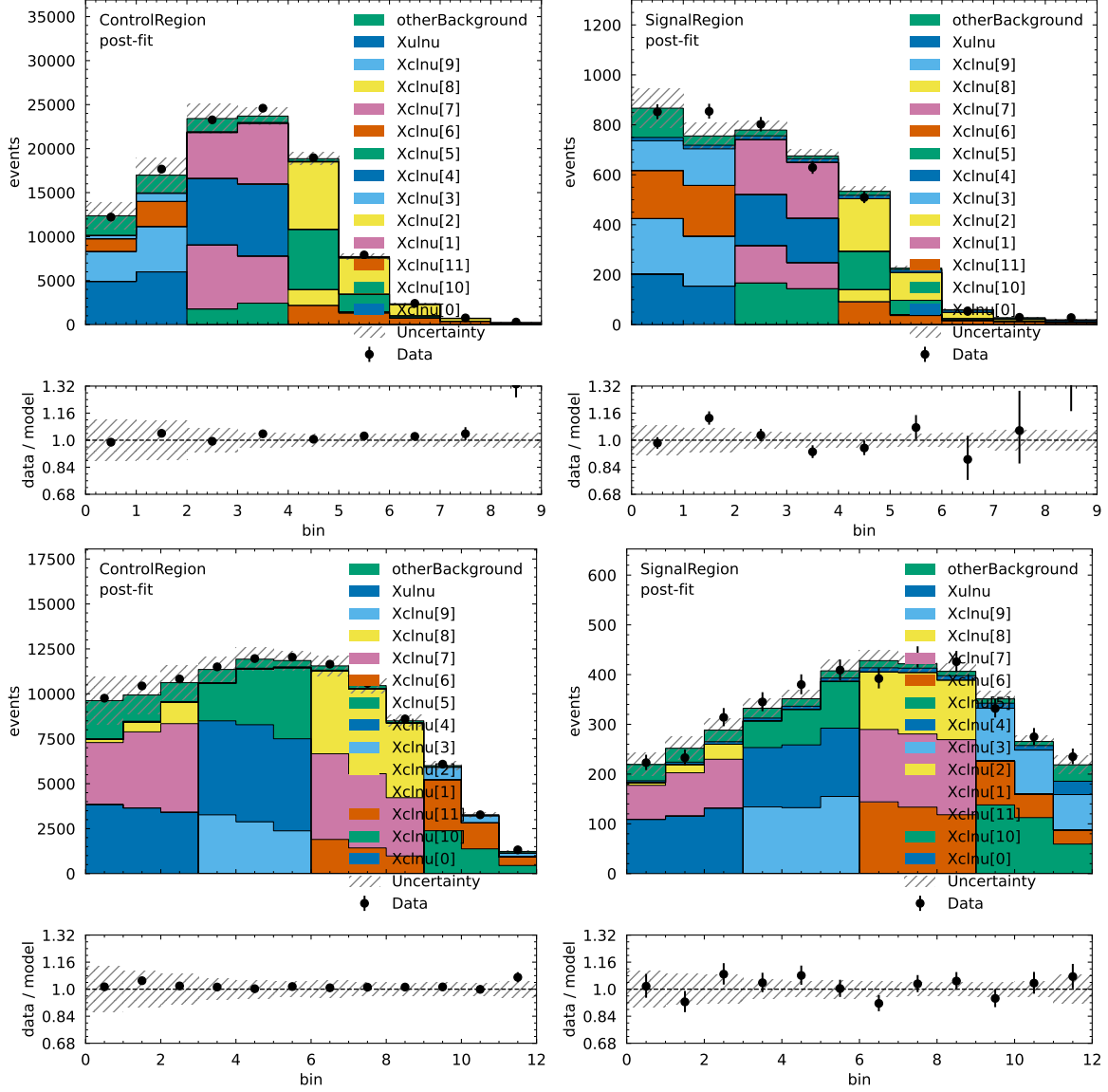


FIG. 60: Postfit projections of the $E_\ell^B:q^2$ $\text{CR}_{K,\text{low}}\text{-}\text{CR}_{K,\text{high}}$ fit on q^2 and E_ℓ^B . SignalRegion denotes $\text{CR}_{K,\text{high}}$ and ControlRegion $\text{CR}_{K,\text{low}}$.

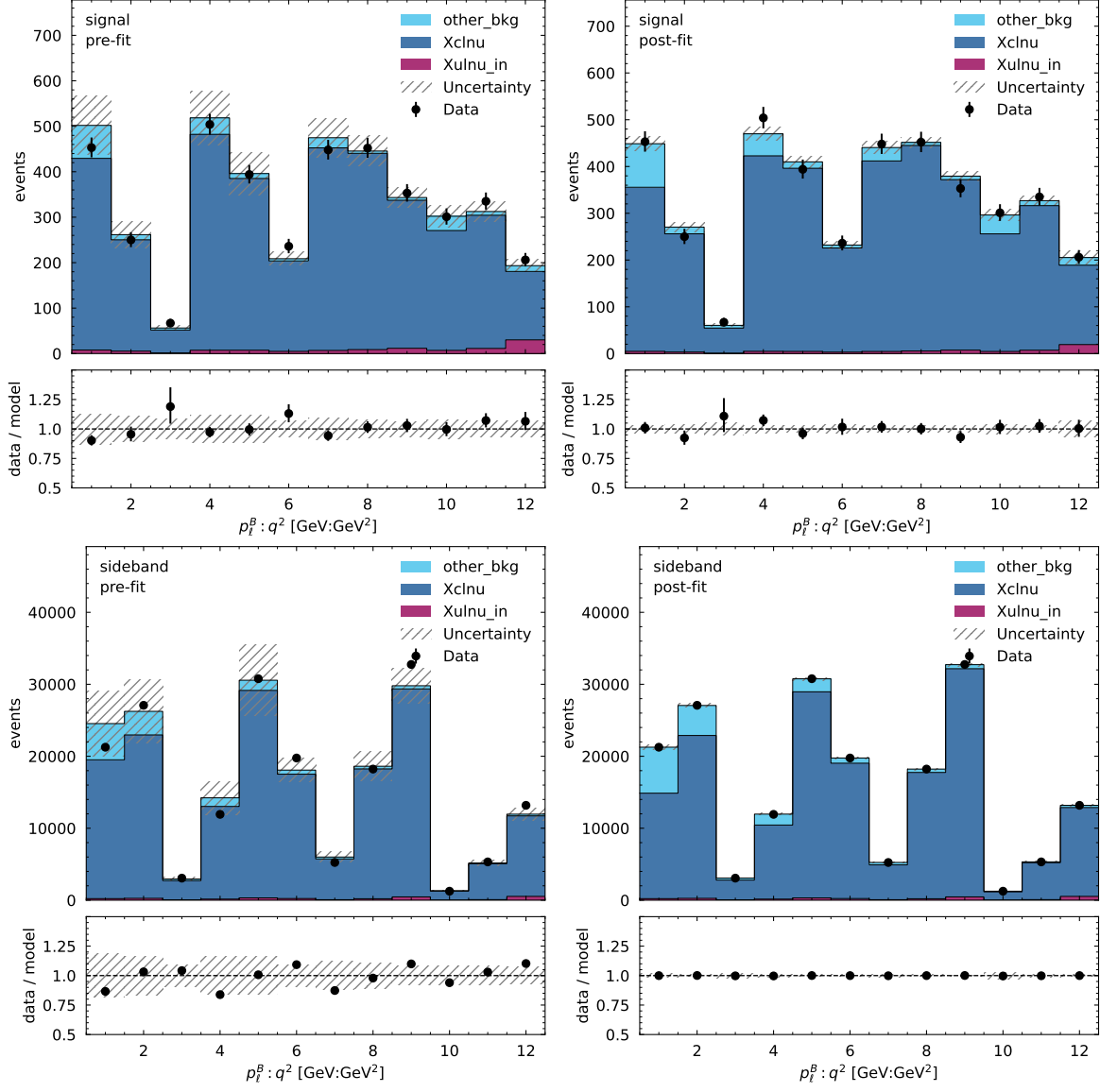
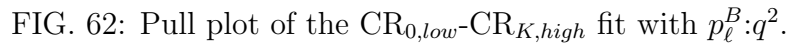


FIG. 61: Pre- (left) and post-fit (right) $E_\ell^B:q^2$ distributions in $CR_{K,high}$ (top) and $CR_{0,low}$ (bottom) for fit 1.



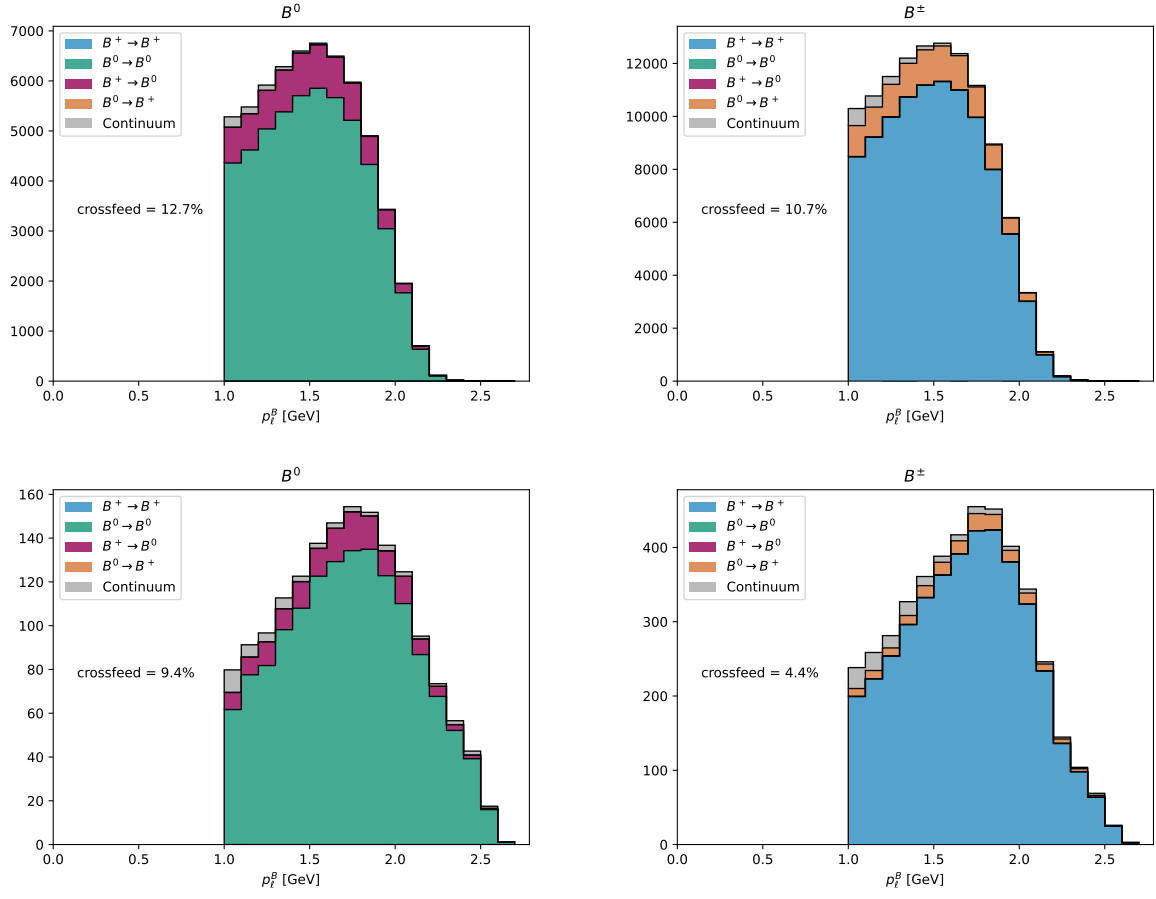


FIG. 63: p_ℓ^B in $\text{CR}_{0,low}$ (top) and SR (bottom) for the neutral (left) and charged (right) channels.

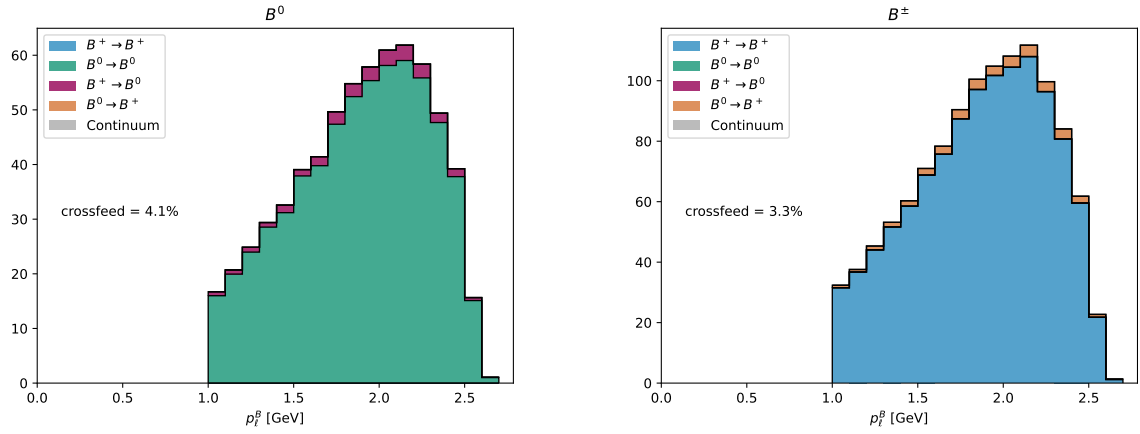


FIG. 64: p_ℓ^B for $B \rightarrow X_u \ell \nu$ events in the SR for the neutral (left) and charged (right) channels.

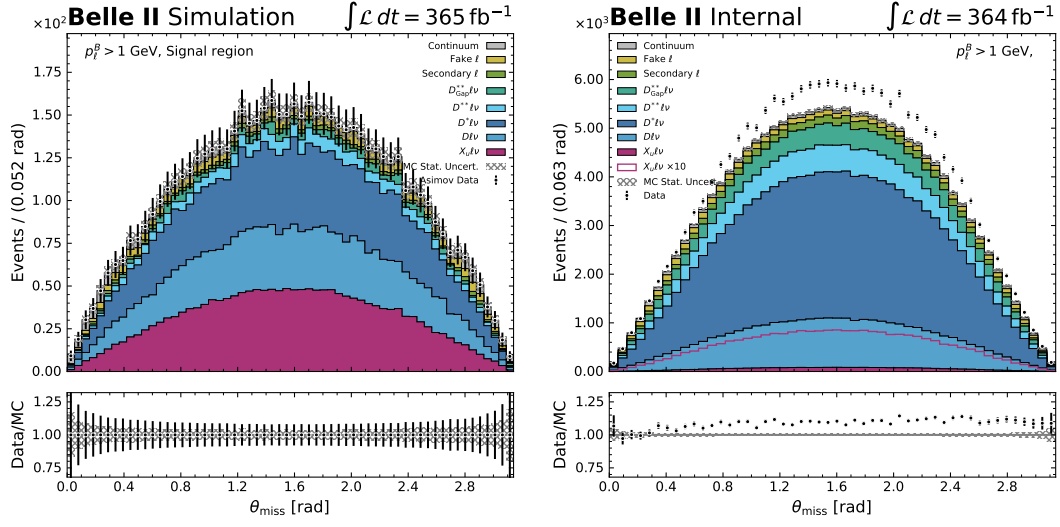


FIG. 65: θ_{miss} distribution in SR (left) and $\text{CR}_{0,\text{low}}$ (right)

9. BOX OPENING

1. Proposed unblinding strategy

- Perform all fits using $\text{CR}_{0,low}$ and VR_2 instead of SR in order to validate the background correction and signal extraction procedures with the validation region closest to the signal region
- Unbox the signal region, check the data-MC agreement on the following variables (combined and split e, μ modes): $E_\ell^B, q^2, M_X, M_{\text{miss}}^2$ and M_{bc} ; check the total efficiency of the SR in data as discussed in Section 7; check the normalisation of the signal-dominated region ($E_\ell^B > 2.1$ GeV) to make sure that the signal normalisation is consistent between MC and data within uncertainties; check the ratio of yields in the nominal sample and the low FEI cut sample in MC and data in the signal region
- Perform nominal fit (fit 1), keep branching fractions and $|V_{ub}|$ blinded, compare pre-fit and post-fit spectra and check output p -value, nuisance parameter pulls as well as total uncertainty; perform fit 1 separately in each split channel (θ_{miss} , lepton charge and flavour) and check the difference between each BR by adding an unknown factor to both results to keep them blinded; use post-fit charged π multiplicity to derive an additional uncertainty on the X_u system hadronisation
- Fully unblind BR and $|V_{ub}|$ obtained from the nominal fit. Perform fits 3 and 5 and all associated cross-checks.

2. Step 1: $\text{CR}_{0,low}$ - VR_2 fit

For all three fits, the $\text{CR}_{0,low}$ $X_c \ell \nu$ normalisation is extracted by directly comparing MC and data and the VR_2 normalisation is extracted via the ABCD method described in Section 7.3.2. The normalisation factors obtained for the three regions considered in our fits are summarised in Table XXX.

Kinematic selection	$\text{CR}_{0,low}$	VR_2
$p_\ell^B > 1.0$ GeV	1.11 ± 0.05	1.07 ± 0.07
$p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV	1.18 ± 0.04	1.15 ± 0.05
$p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV ²	1.24 ± 0.03	1.22 ± 0.07

TABLE XXX: $\text{CR}_{0,low}$ and VR_2 $X_c \ell \nu$ normalisation factors for different kinematic selections.

The pre and postfit distributions of fit 1 are shown in Figure 66. The pulls are shown in Figure 67. The normalised prefit $X_c \ell \nu$ data/MC ratios in $\text{CR}_{0,low}$ and VR_2 are compared to the fit $X_c \ell \nu$ shape factors in Figure 68. The pre and postfit projections on q^2, E_ℓ^B and M_X in the VR_2 region which is used as signal region in this setup are shown in Figure 69. For the three fits, we respectively obtain a p -value of 0.31, 0.35 and 0.53. The plots for fits 3 and 5 are shown in Appendix Q.

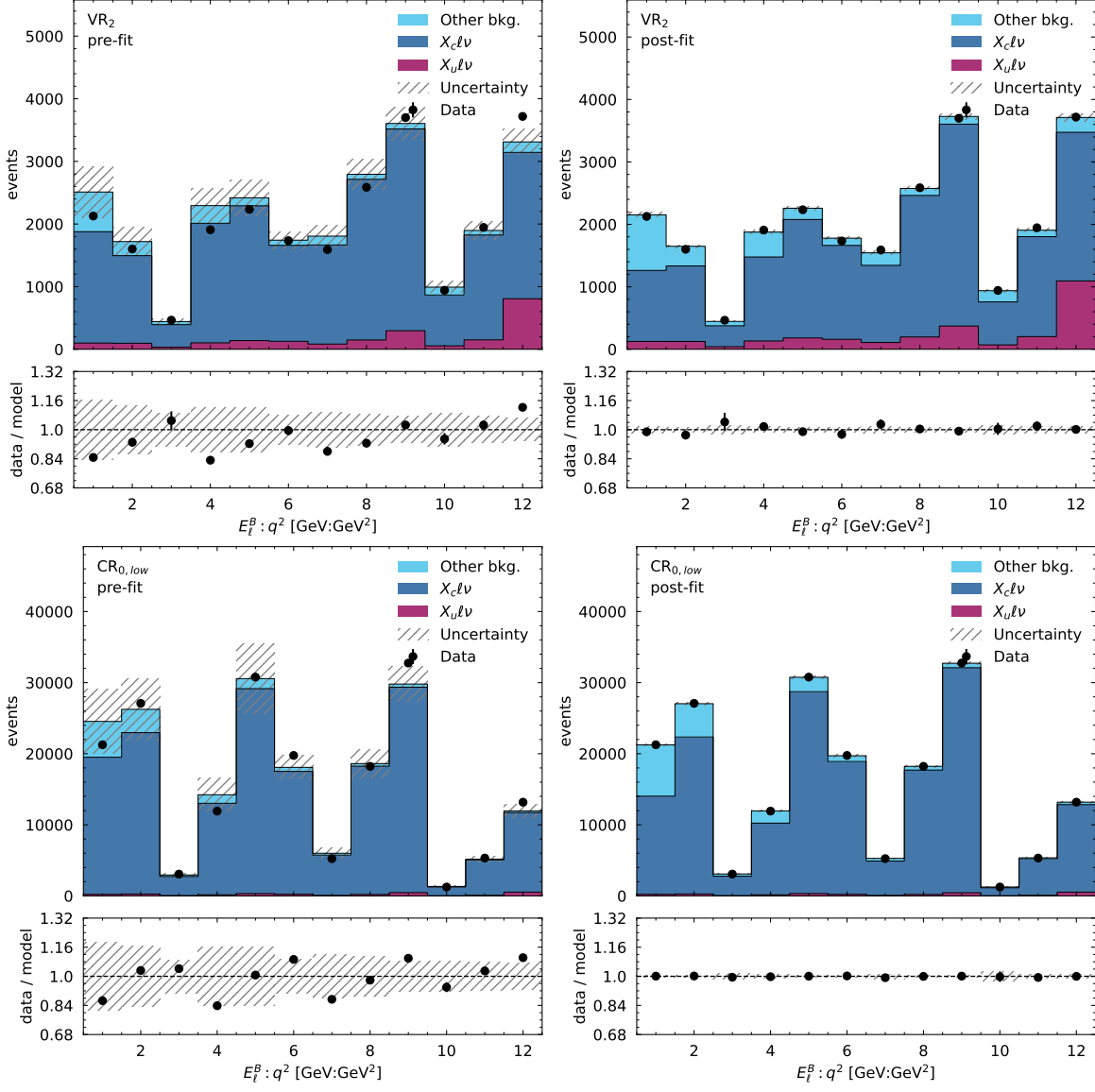


FIG. 66: Pre and postfit $E_\ell^B : q^2$ distributions of the $CR_{0,low}$ - VR_2 fit 1.

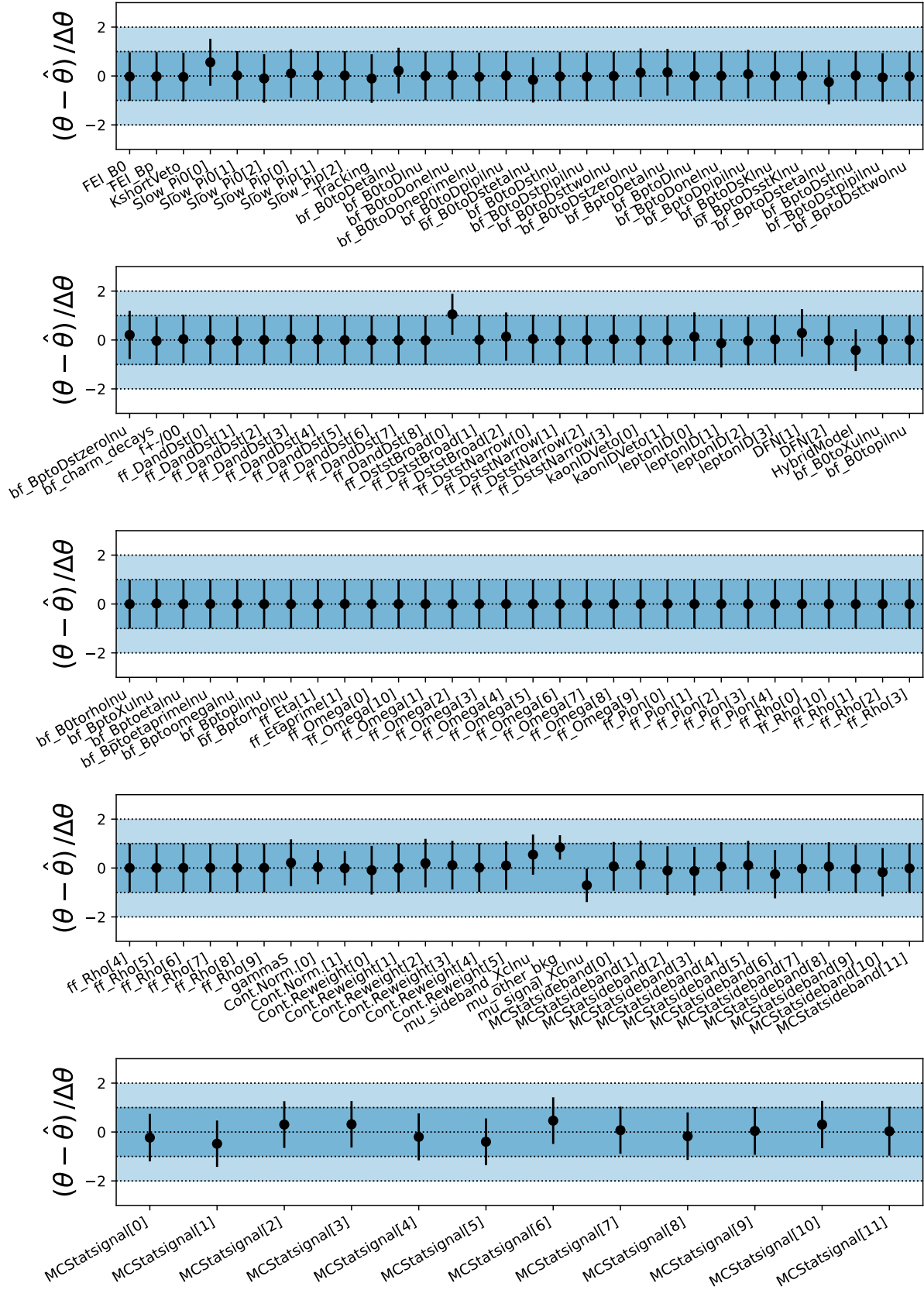


FIG. 67: $\text{CR}_{0,low}$ - VR_2 fit 1 nuisance parameter pulls

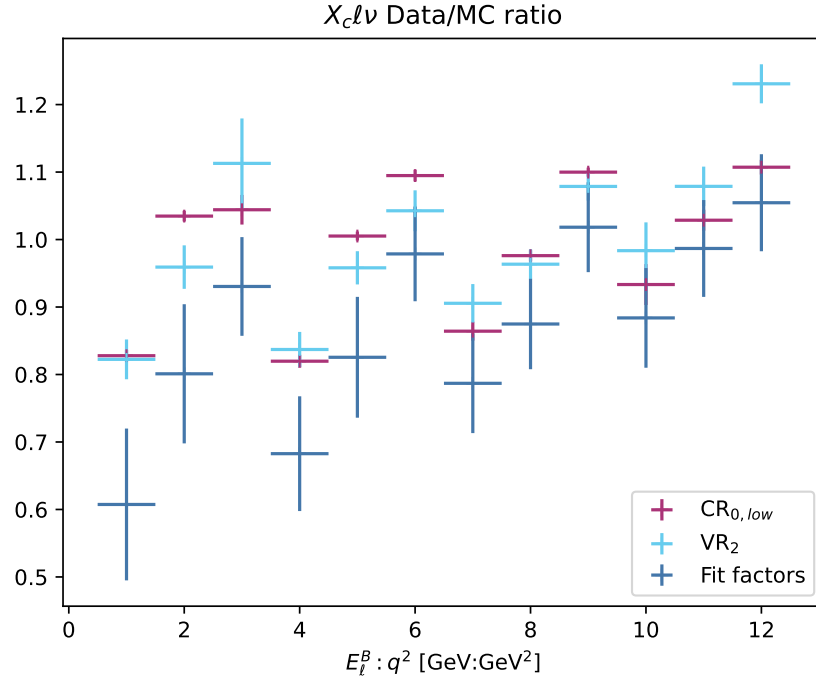


FIG. 68: Comparison of the normalised prefit (fit 1) $X_c \ell \nu$ data/MC ratios in $CR_{0,low}$ and VR_2 with statistical uncertainties and $X_c \ell \nu$ shape factors with full fit uncertainties.

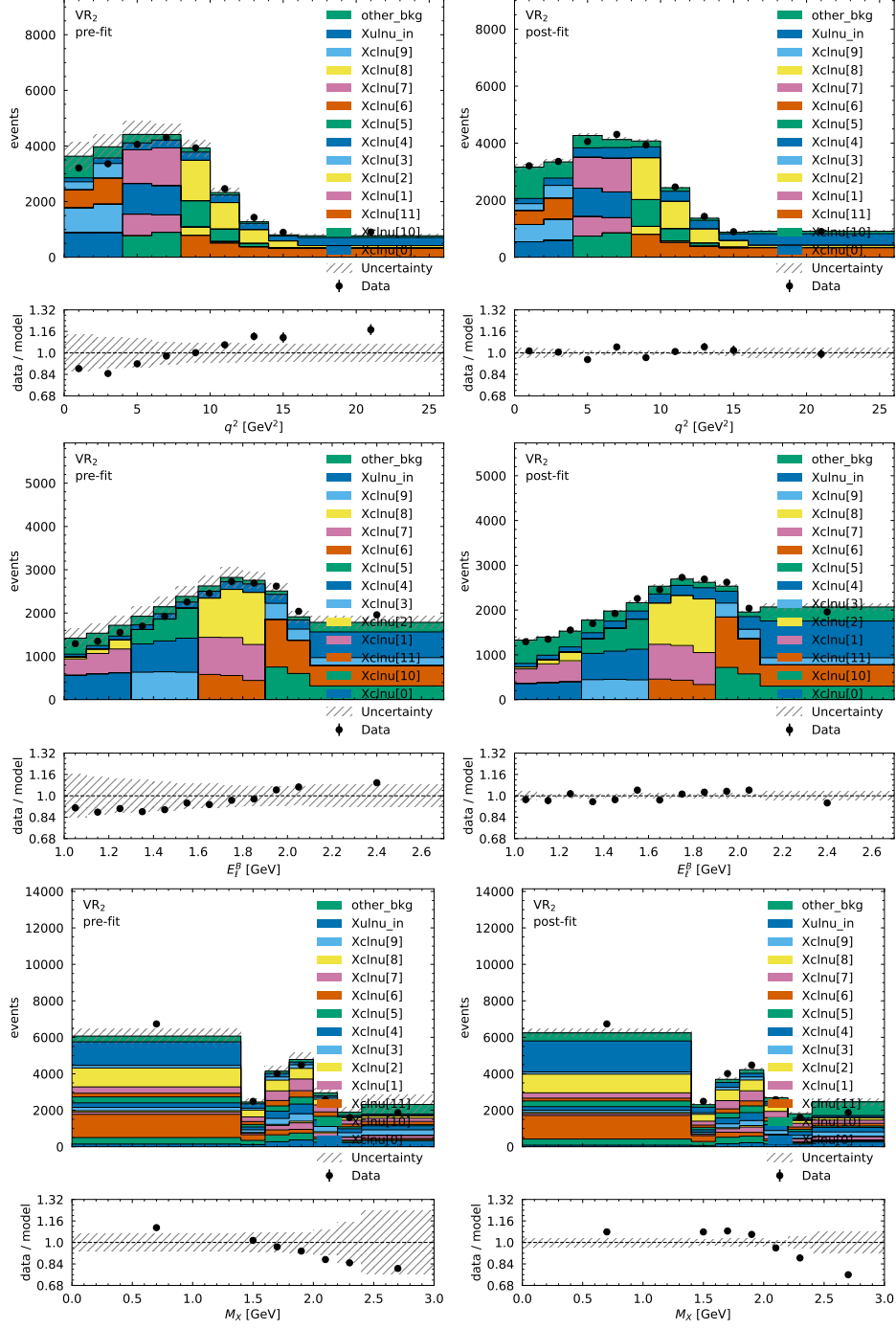


FIG. 69: Projections of the $E_\ell^B:q^2$ fit 1 on E_ℓ^B , q^2 and M_X .

1746 3. Step 2: signal region plots and first checks

1747 We show in Figure 70 distributions of E_ℓ^B , q^2 , M_X , $2D E_\ell^B:q^2$, M_{miss}^2 and M_{bc} in the signal
 1748 region. We add the signal region distributions split in lepton flavour and the distributions
 1749 in the regions considered for fit 3 and 5 in Appendix R.

1750

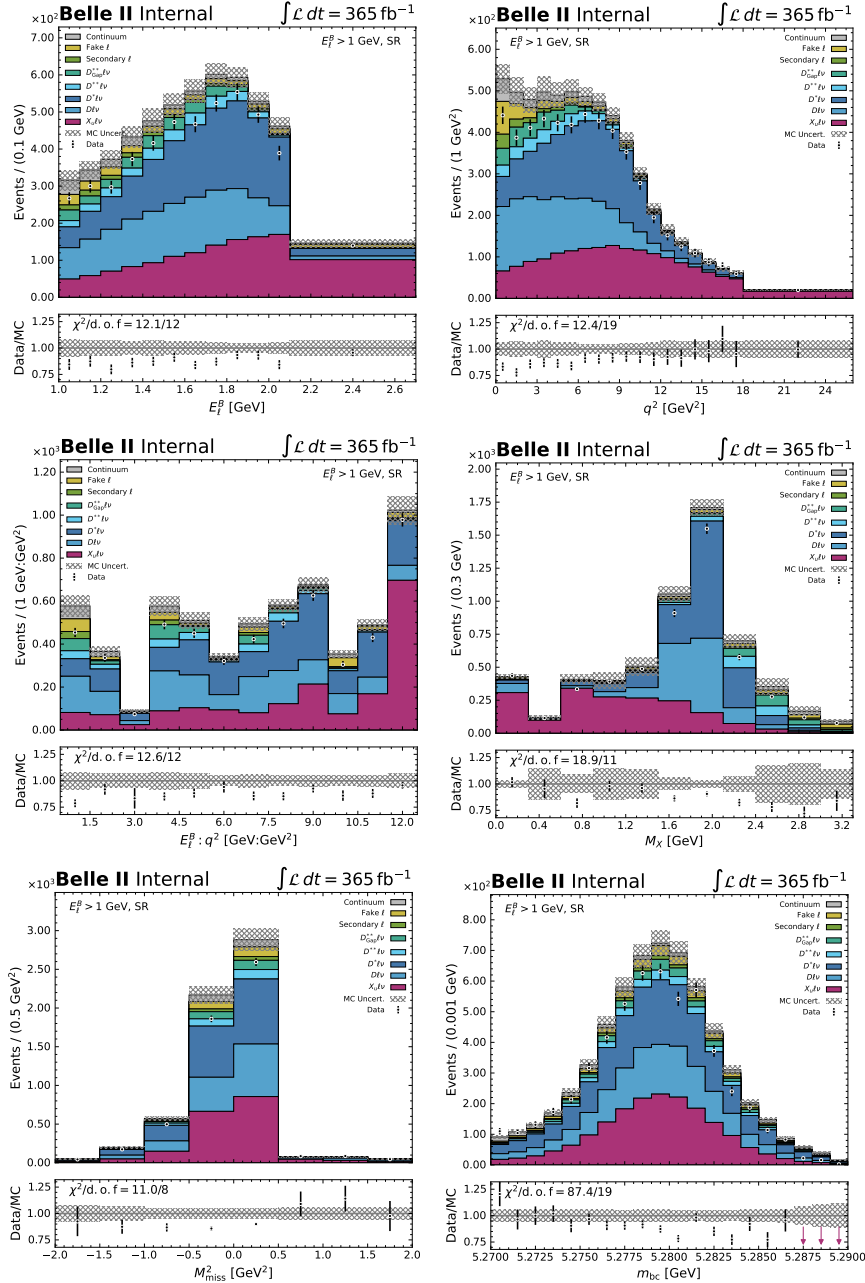


FIG. 70: Signal region distributions.

1751

1752

1753 As shown in Table XV, the SR normalisation is expected to be close to 1 which is clearly not
 1754 the case. A setup mimicking the nominal fit setup is created using the $\text{CR}_{K,\text{low}} - \text{CR}_{K,\text{high}}$
 1755 setup and injecting the SR MC $X_u \ell \nu$ yields in MC and data in $\text{CR}_{K,\text{high}}$. This setup allows

to control the normalisation of the signal injected in data while keeping the MC normalisation fixed and to control the normalisation of the $X_c\ell\nu$ template while keeping the data normalisation fixed. After careful investigation, two sources of bias in the fit were identified: first, the small differences in data/MC ratio shape between $\text{CR}_{K,low}$ ($\text{CR}_{0,low}$) and $\text{CR}_{K,high}$ (SR) which lead to an imperfect correction of the $X_c\ell\nu$ template in SR thus forcing the fit to pull on the signal normalisation to adjust the mismatch and second, the difference between the ABCD-extracted $X_c\ell\nu$ normalisation and the observed SR normalisation. A correction for the first bias is extracted by varying the normalisation of the signal injected in data in $\text{CR}_{K,high}$ choosing values in $[0.85, 0.90, 0.95, 1.00, 1.05, 1.10, 1.15]$. The output normalisations of these different tests are compared to the input normalisations. The resulting distribution shown in Figure 71 is linear and the fitted curve is used to correct the normalisation. The $B \rightarrow X_u\ell\nu$ branching fraction obtained from the fit will therefore be corrected by $f(x) = (x - 0.052)/0.920$.

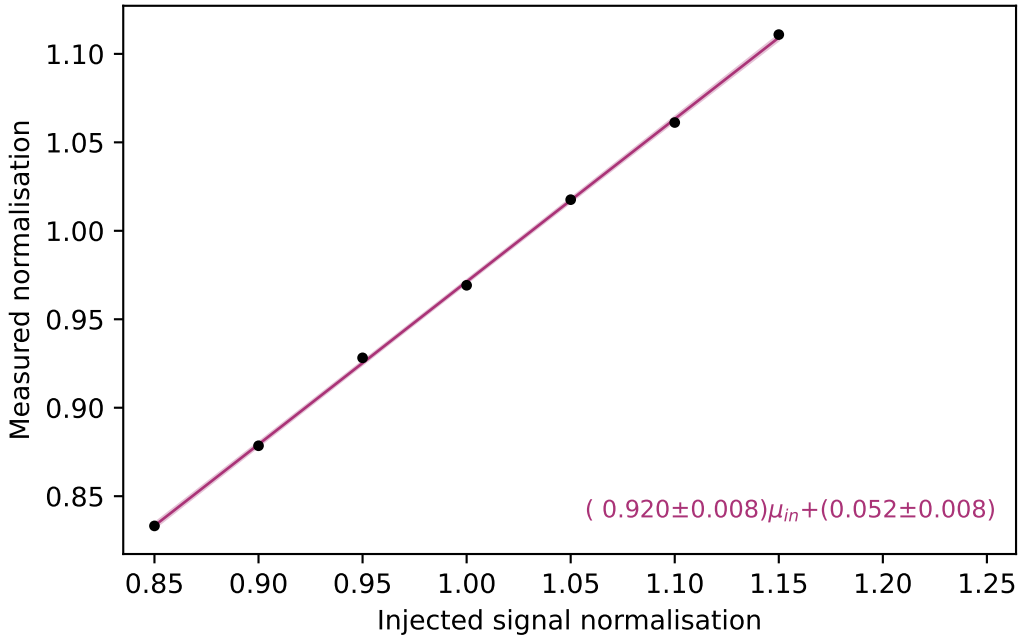


FIG. 71: Distribution of output vs input normalisations of the signal injected $\text{CR}_{K,low} - \text{CR}_{K,high}$ fit setup described in the text.

Since the correction factor is extracted from a specific setup, we need to derive an uncertainty related to the correction. For that, we repeat the signal-injected fit using 1,000 toys which are created by Gaussian-varying the data yields in $\text{CR}_{K,high}$ within their statistical uncertainties. This allows to test different configurations of the fit setup used to extract the bias correction instead of relying on a single setup. This test is repeated for different $X_u\ell\nu$ input normalisations in the same set of normalisations as above ($[0.85, \dots, 1.15]$) and for each test we plot the resulting distribution of 1,000 output normalisations as illustrated in Figure 72. The spread of the Gaussian distribution is between 0.022 and 0.023 for each input normalisation which corresponds to 2.0% to 2.5% relatively to the central value of the distribution. We use this spread to quantify the uncertainty related to the correction procedure and we therefore assign a flat 2.5% uncertainty.

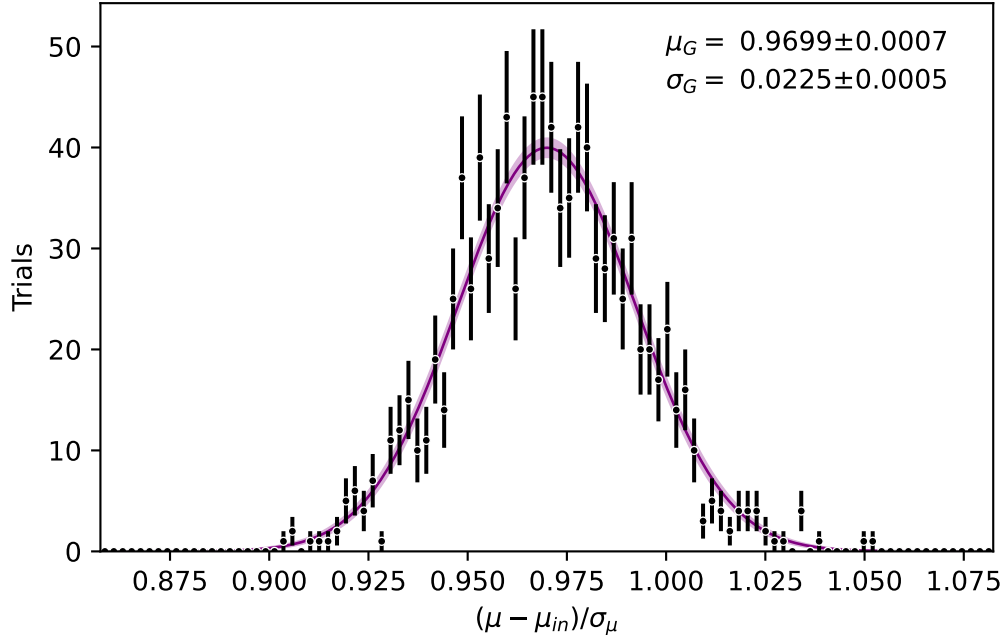


FIG. 72: Distribution of output normalisations with 1,000 toys created by varying the data yields in $\text{CR}_{K,\text{high}}$ within their statistical uncertainties. The normalisation of the signal injected in data is chosen to be 1.

1784

1785

1786 The correction for the bias coming from overestimating the $X_{c\ell\nu}$ normalisation is estimated
 1787 by keeping the input signal normalisation fixed 1 in MC and data and scaling the $X_{c\ell\nu}$
 1788 template in $\text{CR}_{K,\text{high}}$ in the signal-injected setup by different *overestimation factors* and
 1789 comparing them to the output signal normalisation. Again, the response is linear as shown
 1790 in Figure 73. It is however impossible to estimate the actual normalisation of the $X_{c\ell\nu}$
 1791 component in the true SR where the signal yields correspond to about a third of all events.
 1792 The data/MC ratio in the signal region is ~ 0.88 and we therefore derive a correction factor
 1793 using the linear response assuming an overestimation factor of 1.15. The resulting flat
 1794 correction factor is therefore $1/(-0.167 \times 1.15 + 1.166) \sim 1.026$. The full difference between
 1795 the corrected value and the non-corrected value is added as an uncertainty. We check that
 1796 this uncertainty is large enough to cover the true signal normalisation even if the true $X_{c\ell\nu}$
 1797 overestimation factor is very different from 1.15 as illustrated in Figure 74.

1799

1801 Finally, since we use the $\text{CR}_{K,\text{low}}$ and $\text{CR}_{K,\text{high}}$ regions to perform all these tests, we derive an
 1802 additional uncertainty to cover the small difference in $X_{c\ell\nu}$ composition between these two
 1803 control regions and $\text{CR}_{0,\text{low}}$ and SR. To do that, we repeat the signal-injected fit setup with
 1804 $X_{u\ell\nu}$ and $X_{c\ell\nu}$ normalisations fixed to 1 by rescaling the $X_{c\ell\nu}$ template subcomponents
 1805 in $\text{CR}_{K,\text{low}}$ (resp. $\text{CR}_{K,\text{high}}$) such that its composition matches the composition in $\text{CR}_{0,\text{low}}$
 1806 (resp. SR). The difference with the nominal setup normalisation is taken as an uncertainty
 1807 resulting in an additional 1.4% error. Such a procedure was used to derive an uncertainty
 1808 on the difference in SR and $\text{CR}_{0,\text{low}}$ $X_{c\ell\nu}$ compositions as described in Section 8.8. The
 1809 latter uncertainty is removed to avoid double-counting errors related to the composition of
 1810 $X_{c\ell\nu}$.

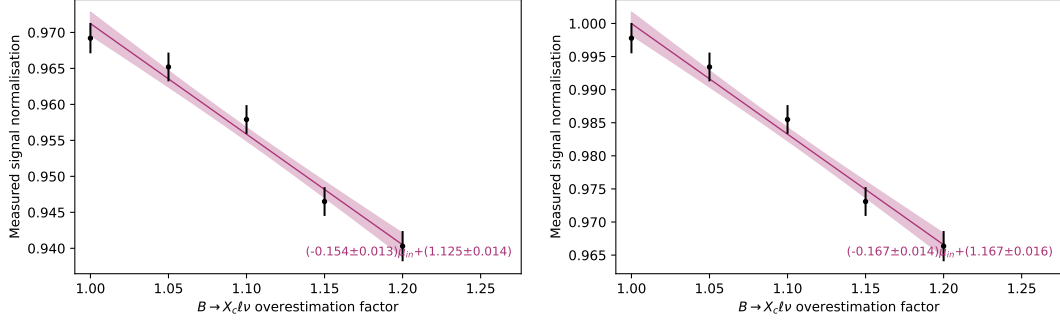


FIG. 73: Distribution of output signal normalisations vs $X_c \ell \nu$ overestimation factor in the signal injected $\text{CR}_{K,low} - \text{CR}_{K,high}$ fit setup described in the text. In the right-hand plot, the first source of bias described above has been corrected for. The normalisation of the injected signal is equal to 1.

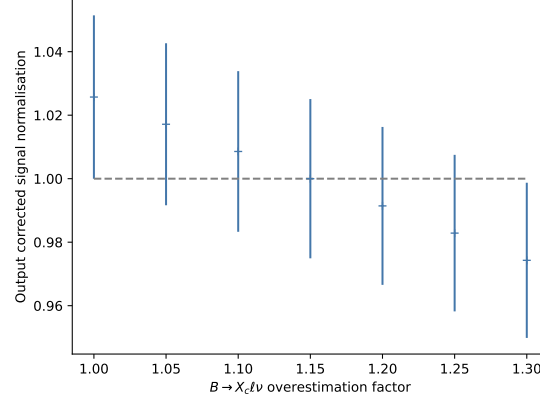


FIG. 74: Output signal normalisation corrected by the two corrections described in the text vs different $X_c \ell \nu$ overestimation factors. The uncertainty is the one added for the second source of bias.

1811

1812 The procedure described above is summarised as follows:

1813

- remove the composition uncertainty described in Section 8 8

1814

- correct for the bias coming from the data/MC ratio mismatch between CR and SR using $f(x) = (x - 0.052)/0.920$

1815

1816

- add a 2.6% uncertainty on the corrected normalisation based on the tests performed by varying the datasets used to extract the correction factor

1817

1818

- add a 1.4% uncertainty based on the small composition difference between $\text{CR}_{K,high}/\text{SR}$ and $\text{CR}_{K,low}/\text{CR}_{0,low}$

1819

1820

- correct for the bias coming from the overestimation of the $X_c \ell \nu$ normalisation in the SR using a flat correction factor of 1.026 and add the difference as an uncertainty

1821

1822 We give in the following an example with a measured normalisation of 0.952 and a total
1823 uncertainty of 8.0% (see Table XXIV):

- 1824 • Initial $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (1.504 \pm 0.120) \times 10^{-3}$
- 1825 • Apply first correction and add related uncertainty: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (1.546 \pm 0.136) \times$
1826 10^{-3}
- 1827 • Add composition uncertainty: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (1.546 \pm 0.138) \times 10^{-3}$
- 1828 • Correct for second bias and add uncertainty: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (1.588 \pm 0.148) \times 10^{-3}$

1829 This results in a total relative uncertainty of 9.3% to which only the X_u hadronisation mod-
1830 elling uncertainty needs to be added (see Section 6.14). For reference, the $X_u \ell \nu$ branching
1831 fraction measured by Belle had an uncertainty of 11.0%. We repeated the same procedure
1832 for fits 3 and 5 *i.e.* with additional cuts of $M_X < 1.7$ GeV and $M_X < 1.7$ GeV, $q^2 > 8$ GeV²
1833 respectively. All related plots are shown in Appendix S. The procedure for fit 3 outlines as
1834 follows:

- 1835 • remove the composition uncertainty described in Section 8.8
- 1836 • correct for the bias coming from the data/MC ratio mismatch between CR and SR
1837 using $f(x) = (x - 0.025)/0.988$
- 1838 • add a 2.3% uncertainty on the corrected normalisation based on the tests performed
1839 by varying the datasets used to extract the correction factor
- 1840 • add a 5.7% uncertainty based on the composition difference between $\text{CR}_{K,high}/\text{SR}$ and
1841 $\text{CR}_{K,low}/\text{CR}_{0,low}$
- 1842 • correct for the bias coming from the overestimation of the $X_c \ell \nu$ normalisation in the
1843 SR using a flat correction factor of 1.004 and add the difference as an uncertainty

1844 In general, the corrections and uncertainties added are smaller than for fit 1 owing to a
1845 larger signal purity and good signal modelling. However, the $X_c \ell \nu$ composition difference
1846 between $\text{CR}_{K,high}$ and SR is quite large as summarised in Table XXXI. Because of that,
1847 the composition uncertainty becomes the leading uncertainty bringing the total relative
1848 uncertainty from 7.7% to about 10% to be compared with an 8.7% uncertainty quoted by
1849 Belle for the same phase-space region. For fit 5, the correction procedure outlines as follows:

	$\text{CR}_{K,low}$	$\text{CR}_{K,high}$	$\text{CR}_{0,low}$	Signal
$B \rightarrow D \ell \nu$	22.05%	64.3%	21.5%	47.7%
$B \rightarrow D^* \ell \nu$	65.32%	33.6%	61.9%	47.3%
$B \rightarrow D^{**} \ell \nu$	6.43%	1.5%	8.0%	2.7%
$B \rightarrow D_{\text{Gap}}^{**} \ell \nu$	6.30%	0.6%	8.6%	2.3%

TABLE XXXI: Composition of the $X_c \ell \nu$ component in various background regions and the
signal region with an additional cut of $M_X < 1.7$ GeV.

1850

1851

- 1852 • remove the composition uncertainty described in Section 8.8

- correct for the bias coming from the data/MC ratio mismatch between CR and SR using $f(x) = (x - 0.050)/0.957$
- add a 1.8% uncertainty on the corrected normalisation based on the tests performed by varying the datasets used to extract the correction factor
- add a 0.2% uncertainty based on the composition difference between $CR_{K,high}/SR$ and $CR_{K,low}/CR_{0,low}$
- correct for the bias coming from the overestimation of the $X_c\ell\nu$ normalisation in the SR using a flat correction factor of 1.013 and add the difference as an uncertainty

In this case, the $X_c\ell\nu$ component also has a small impact. Besides, its composition doesn't impact the E_ℓ^B variable used for the fit. These additional uncertainties bring the total relative uncertainty from 9.1% to 9.9% to be compared with an uncertainty of 18% quoted by the Belle measurement in this phase-space region.

As part of the second unblinding step, we also need to compare the total efficiency of the SR selection in MC and data. As discussed above, the efficiency in MC doesn't match the data efficiency but the resulting bias is corrected. The data and MC distributions in the signal-dominated region is very similar which indicates a relatively good modelling of $B \rightarrow X_u\ell\nu$ processes. We finally check the ratio of yields in the nominal sample and the low FEI cut sample in MC and data in the signal region. The ratio is 0.648 ± 0.026 in MC and 0.649 ± 0.011 in data indicating a very good agreement.

4. Final unboxing step: perform the three nominal fits

We perform the 3 fits. The pre and post-fit distributions are shown in Figures 75 and 76 respectively. For fits 1, 3 and 5 we respectively find p -values of 0.79, 0.20 and 0.84 confirming the good agreement.

The pulls of all nuisance parameters are shown in Figures 77 to 79. In each fit, all pulls are within one standard deviation indicating that they are stable and yield reliable results. As expected, the $B \rightarrow X_c\ell\nu$ normalisation parameter in the signal region pulls significantly to adjust the overestimated normalisation of this component.

The event yields as extracted from fits 1, 3 and 5 are given in Table XXXII. In the two regions considered for fits 1 and 3, the signal yields are similar to those reported in the latest Belle measurement [58] while the background yields are approximately 30–45% lower, despite using a data set that is about half the size of the 711 fb^{-1} data set collected by the Belle detector.

Before applying all post-fit corrections and adding all post-fit uncertainties, the signal normalisations are

- fit 1: 0.926 ± 0.076 (8.2% relative uncertainty)
- fit 3: 0.928 ± 0.081 (8.7%)

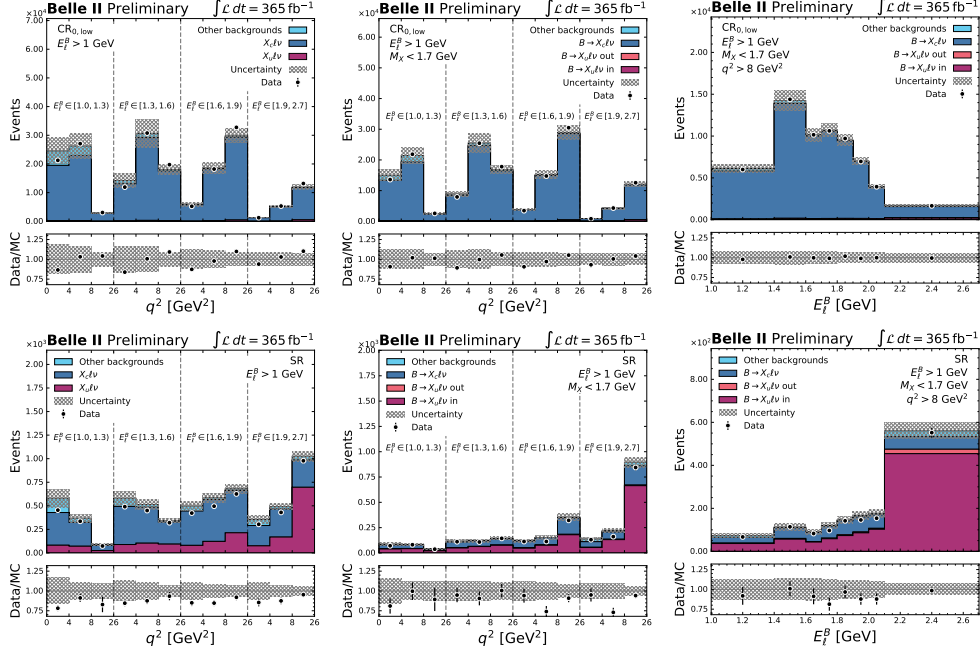


FIG. 75: The $E_\ell^B:q^2$ and E_ℓ^B distributions in data and simulated data in $\text{CR}_{0,\text{low}}$ (top) and the signal region (bottom) considered for fit 1 (left), 3 (middle) and 5 (right). The $E_\ell^B:q^2$ distribution is shown in bins of E_ℓ^B (in GeV) and q^2 as indicated in the figures. The $B \rightarrow X_c \ell \nu$ normalisation correction factors have been applied (see Table XV).

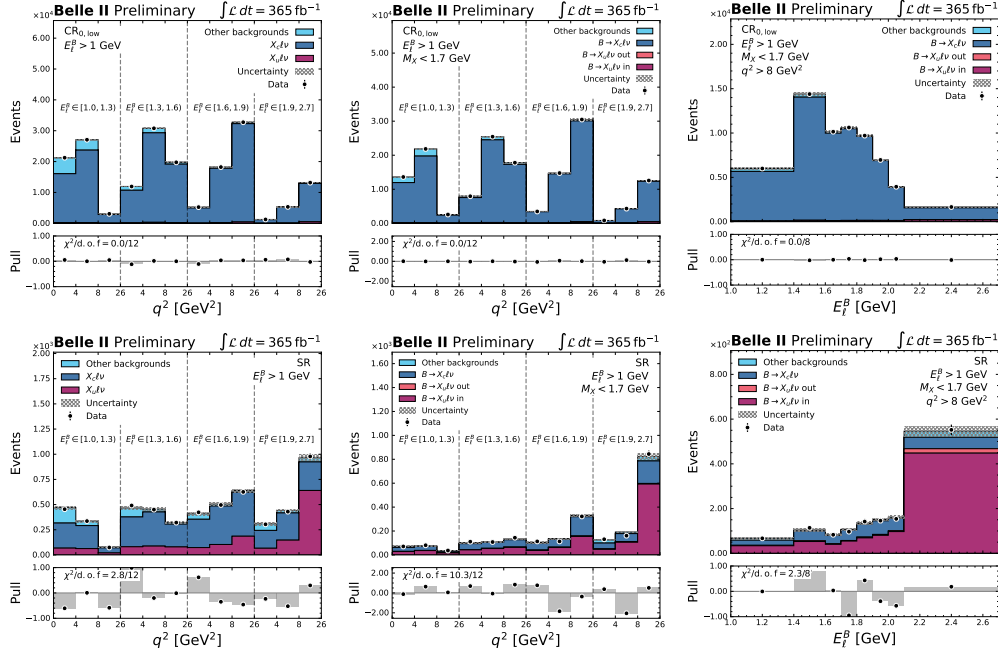


FIG. 76: Post-fit signal region distributions for fit 1 (left), 2 (middle) and 3 (right). The $E_\ell^B:q^2$ variable is flattened in bins of E_ℓ^B (in GeV) and q^2 .

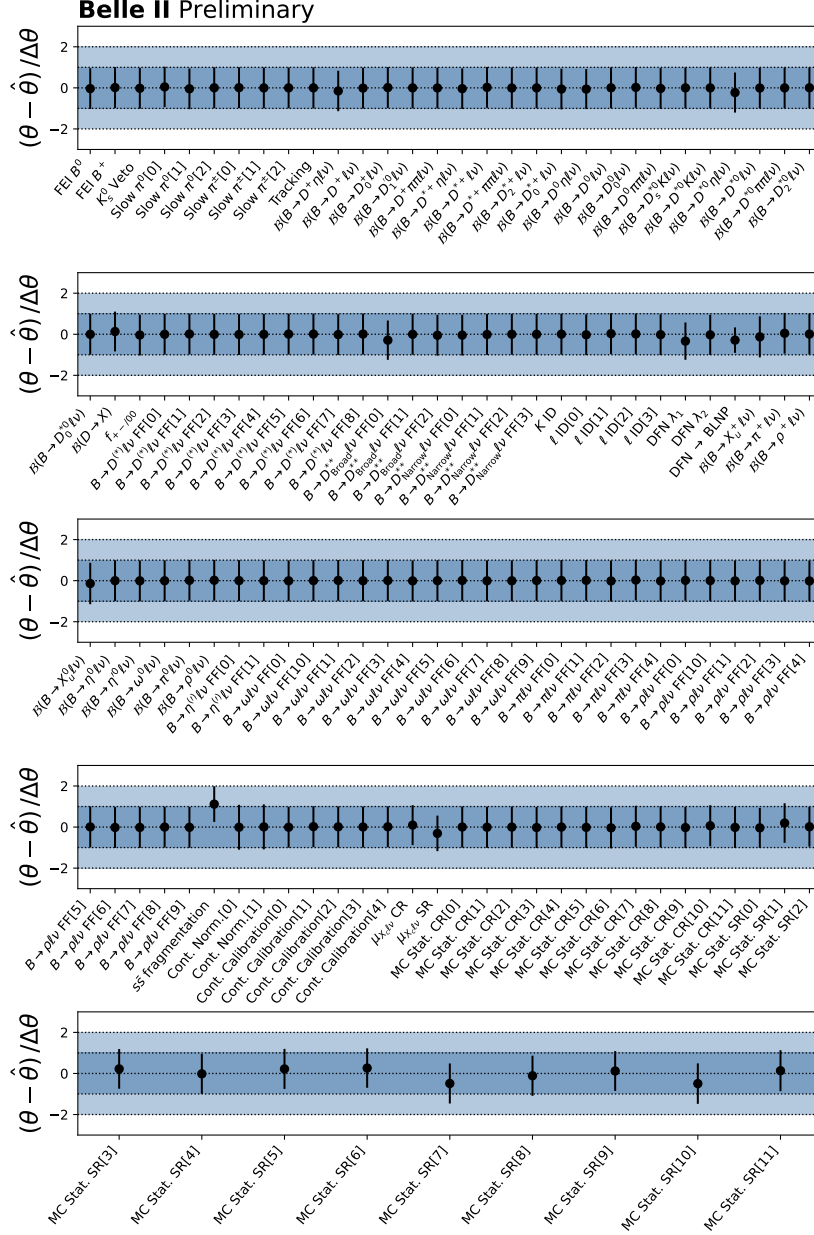


FIG. 78: Nuisance parameter pulls for fit 3. The acronyms FF, ID, Cont. and Norm. are used for form factors, identification, continuum and normalisation respectively. Sets of parameters related to the same source of uncertainty are indexed by the numbers in brackets.

resulting in the partial branching fractions

• fit 1: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (1.540 \pm 0.149) \times 10^{-3}$

• fit 3: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (0.945 \pm 0.112) \times 10^{-3}$

• fit 5: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (0.554 \pm 0.060) \times 10^{-3}$

which, when extrapolated to the full phase-space are

TABLE XXXII: Signal, background and data yields in the signal region as extracted from each fit. The error given for $B \rightarrow X_u \ell \nu$ and background yields is the total uncertainty as obtained from each fit.

Fit	$B \rightarrow X_u \ell \nu$ -in	$B \rightarrow X_u \ell \nu$ -out	Backgrounds	Data
Fit 1	1615 ± 122	–	3781 ± 216	5383
Fit 3	1243 ± 109	33 ± 5	962 ± 100	2236
Fit 5	881 ± 71	42 ± 4	434 ± 65	1355

method to properly evaluate the systematic correlation as any method ignores correlations between fit parameters which are sizeable in our case. These can be compared to the world average as given in the PDG [3]: $(1.92 \pm 0.21) \times 10^{-3}$ (10.9%). The 3 measured branching fractions agree within 0.5σ , 0.9σ and 0.6σ respectively with the world average. When split in statistical and systematic uncertainty, the three measured branching fractions read

- fit 1: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (1.540 \pm 0.083 \pm 0.122) \times 10^{-3}$
- fit 3: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (0.945 \pm 0.053 \pm 0.099) \times 10^{-3}$
- fit 5: $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu) = (0.554 \pm 0.025 \pm 0.054) \times 10^{-3}$

The systematics uncertainty breakdown is shown in Table XXXIII.

The value of $|V_{ub}|$ can be determined from a measured partial branching fraction $\Delta\mathcal{B}(B \rightarrow X_u \ell \nu)$ and partial decay rate $\Delta\Gamma(B \rightarrow X_u \ell \nu)$ using

$$|V_{ub}| = \sqrt{\frac{\Delta\mathcal{B}(B \rightarrow X_u \ell \nu)}{\tau_B \Delta\Gamma(B \rightarrow X_u \ell \nu)}}, \quad (16)$$

where $\tau_B = 1.578 \pm 0.003$ ps is the B meson lifetime [51]. Following Ref. [51], three different theoretical frameworks are used to extract values of $|V_{ub}|$:

- The BLNP model relies on complex theoretical technology to optimise the description of inclusive $B \rightarrow X_u \ell \nu$ decays in the shape-function region where the hadronic system energy is large while its mass is small. The model then smoothly interpolates between that region and the region where the HQE is valid. The BLNP framework provides decay rate predictions at next-to-leading order in α_s and second order in $1/m_b$ for arbitrary phase-space regions. The BLNP calculation relies on the shape function renormalisation scheme and uses as input $m_b^{\text{SF}} = 4.60 \pm 0.02$ GeV and $\mu_\pi^{2\text{SF}} = 0.18^{+0.05}_{-0.06}$ GeV².
- The Dressed Gluon Exponentiation (DGE) computation [56] relies on a resummation of the perturbation theory. It utilises the momentum scale hierarchy in the semileptonic decays without introducing a momentum cut-off. The DGE calculation is performed in the $\overline{\text{MS}}$ renormalisation scheme and uses $m_b^{\overline{\text{MS}}} = 4.21 \pm 0.04$ GeV.
- The model developed by Gambino, Giordano, Ossola and Uraltsev [59] doesn't focus on a particular region of phase space but instead separates between perturbative and

Uncertainty source	Relative uncertainty (%)		
	Fit 1	Fit 3	Fit 5
DFN parameters	4.4	4.5	5.7
DFN $\rightarrow$ BLNP	0.2	0.8	1.3
γ_S	1.7	2.1	2.1
$B \rightarrow \pi \ell \nu$ form factors	0.3	0.3	0.3
$B \rightarrow \rho \ell \nu$ form factors	0.3	0.3	0.2
$B \rightarrow \omega \ell \nu$ form factors	0.1	0.1	0.1
$B \rightarrow \eta^{(\prime)} \ell \nu$ form factors	< 0.1	< 0.1	< 0.1
$B^\pm \rightarrow X_u \ell \nu$ branching fractions	0.9	0.6	0.5
$B^0 \rightarrow X_u \ell \nu$ branching fractions	0.6	0.5	0.5
$B \rightarrow D_{\text{Broad}}^{**}$ form factors	0.5	0.1	0.2
$B \rightarrow D_{\text{Narrow}}^{**}$ form factors	0.1	< 0.1	< 0.1
$B \rightarrow D/D^* \ell \nu$ form factors	< 0.1	< 0.1	< 0.1
$B^\pm \rightarrow X_c \ell \nu$ branching fractions	0.7	0.5	0.2
$B^0 \rightarrow X_c \ell \nu$ branching fractions	0.6	0.2	0.1
D decay branching fractions	0.1	0.3	0.1
SR $X_c \ell \nu$ normalisation	1.6	3.5	3.4
CR $X_c \ell \nu$ normalisation	1.0	1.1	0.4
<i>Other backgrounds</i> normalisation	0.3	N/A	N/A
X_u fragmentation	0.3	4.4	3.9
$N_{\mathcal{R}(4S)}$	1.6	1.6	1.6
FEI	1.3	1.3	1.4
π_s	0.4	0.2	0.3
ℓ identification	0.7	0.7	0.6
$f^{\pm/00}$	0.6	0.7	0.6
Continuum calibration	0.2	0.2	0.2
Tracking	0.3	0.3	0.3
efficiency	0.1	0.1	< 0.1
$K^\pm$ ID	< 0.1	< 0.1	< 0.1
Simulated data statistics	1.1	1.1	0.8
Fit bias	2.6	2.3	1.8
Composition uncertainty	1.3	5.7	0.2
$B \rightarrow X_c \ell \nu$ overestimation correction	2.6	0.4	1.3
Total systematic	7.8	10.5	9.7
Statistical	5.4	5.6	4.5
Total	9.5	11.9	10.7

TABLE XXXIII: Systematic uncertainty breakdown for all fits. The systematic sources are ranked in descending order of their contribution in fit 1.

ture function describing the decay rate, the b -quark motion is parametrised as a single function for which different two-parameter functional forms are tested. The GGOU calculation is carried out in the kinetic renormalisation scheme and uses $m_b^{\text{kin}} = 4.57 \pm 0.01$ GeV and $\mu_\pi^{2\text{kin}} = 0.45 \pm 0.04$ GeV².

The values of $|V_{ub}|$ obtained from each model and each signal extraction fit using Equation 16 are summarised in Table XXXIV.

		Relative uncertainty (%)		
		Fit 1	Fit 3	Fit 5
Phase-space region		$E_\ell^B > 1$ GeV	$M_X < 1.7$ GeV $E_\ell^B > 1$ GeV	$q^2 > 8$ GeV ² $M_X < 1.7$ GeV $E_\ell^B > 1$ GeV
Fit variable		$E_\ell^B : q^2$	$E_\ell^B : q^2$	E_ℓ^B
Inclusive $B \rightarrow X_u \ell \nu$ model				
$\Delta\Gamma$ (ps ⁻¹)	BLNP	$63.0^{+6.4}_{-5.0}$	$47.4^{+5.2}_{-4.3}$	$24.6^{+3.2}_{-2.4}$
	DGE	$59.4^{+3.3}_{-2.8}$	$42.9^{+4.7}_{-3.9}$	$25.0^{+2.2}_{-1.7}$
	GGOU	$60.8^{+2.8}_{-2.0}$	$47.6^{+3.3}_{-3.0}$	$25.2^{+2.9}_{-2.0}$
$ V_{ub} \times 10^3$	BLNP	$3.94 \pm 0.11 \pm 0.16^{+0.20}_{-0.16}$	$3.56 \pm 0.10 \pm 0.19^{+0.20}_{-0.16}$	$3.78 \pm 0.09 \pm 0.18^{+0.25}_{-0.18}$
	DGE	$4.05 \pm 0.11 \pm 0.16^{+0.11}_{-0.09}$	$3.74 \pm 0.10 \pm 0.20^{+0.21}_{-0.17}$	$3.75 \pm 0.08 \pm 0.18^{+0.16}_{-0.13}$
	GGOU	$4.01 \pm 0.11 \pm 0.16^{+0.09}_{-0.07}$	$3.55 \pm 0.10 \pm 0.19^{+0.12}_{-0.11}$	$3.74 \pm 0.08 \pm 0.18^{+0.21}_{-0.15}$

TABLE XXXIV: Values of the $B \rightarrow X_u \ell \nu$ decay rate and $|V_{ub}|$ obtained from three different theoretical predictions in three separate phase-space regions.

The values of $|V_{ub}|$ extracted from the broadest phase-space region are compared to the latest inclusive and exclusive HFLAV averages [51] in Figure 80.

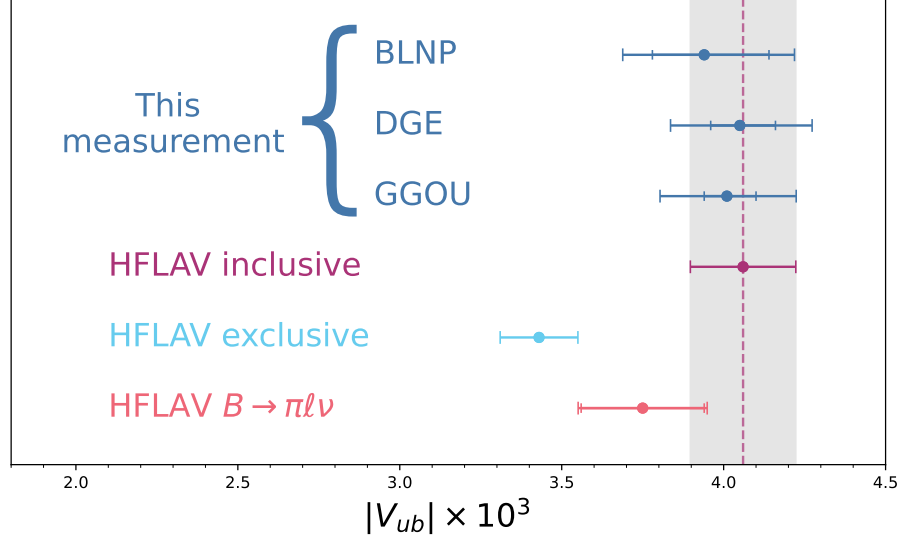


FIG. 80: Comparison between the three values of $|V_{ub}|$ obtained from fit 1 (blue), the inclusive (purple), exclusive (cyan) and $B \rightarrow \pi \ell \nu$ (pink) averages quoted in the latest HFLAV report [51]. The outer error bars represent the total uncertainty and for the three measured values and the $B \rightarrow \pi \ell \nu$ average, the inner error bars represent the theoretical uncertainty.

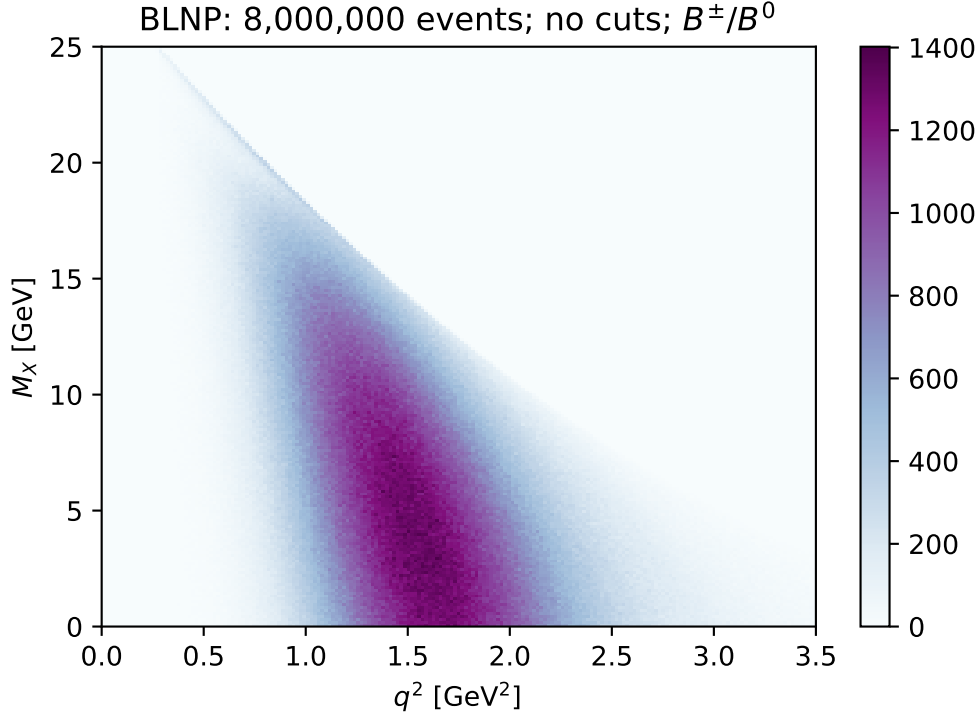


FIG. 81: Two-dimensional histogram of M_X versus q^2 for 8 million events generated with the most up-to-date input values for the BLNP model in EvtGen.

1959 A possible explanation for the unexpected spike observed in the generation of events with
1960 the BLNP model is given here.

1961

1962 In the BLNP model [6], a Soft Colinear Effective Theory (SCET) expansion is performed
1963 on the triple differential decay rate $d\Gamma/dq^2 dp_\ell^B dM_X$ in the so-called shape function region.
1964 We define the variables

$$P_+ = E_X - |\vec{p}_X|, \quad P_- = E_X + |\vec{p}_X|, \quad (17)$$

1965 where E_X and $|\vec{p}_X|$ are respectively the energy and 3-momentum magnitude of the hadronic
1966 system. In the shape function region, P_- is much larger than P_+ and therefore an expansion
1967 in powers of $P_+/(P_- - P_+)$ is performed. However, outside of this specific region, P_- and P_+
1968 can be of the same order which when combined with other factors can yield contributions of
1969 $\mathcal{O}(1)$ which are not suppressed by powers of $P_+/(P_- - P_+)$. Therefore, in regions of phase-
1970 space where $P_- \sim P_+$, an enhancement of events is observed. This seems to be confirmed
1971 when one isolates this particular region. In Fig. 83, the same plot as in Fig. 3 is presented
1972 with an additional selection of $\frac{1}{\sqrt{2}}(P_- - P_+) < 0.4 \text{ GeV}$. Only the problematic events survive
1973 this selection.

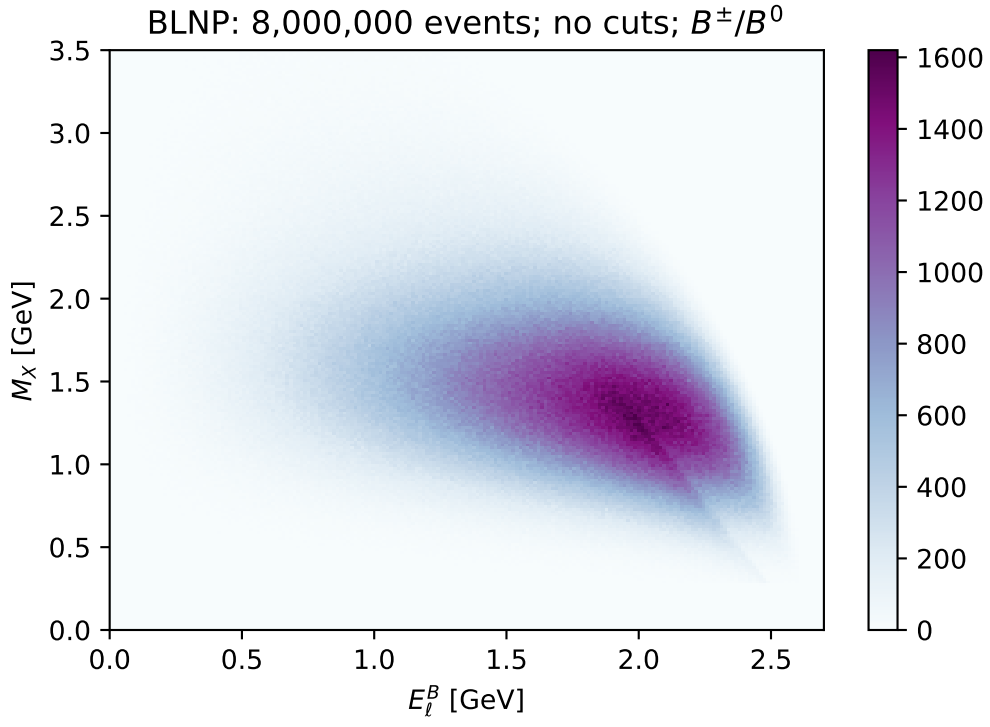


FIG. 82: Two-dimensional histogram of M_X versus E_l^B for 8 million events generated with the most up-to-date input values for the BLNP model in EvtGen.

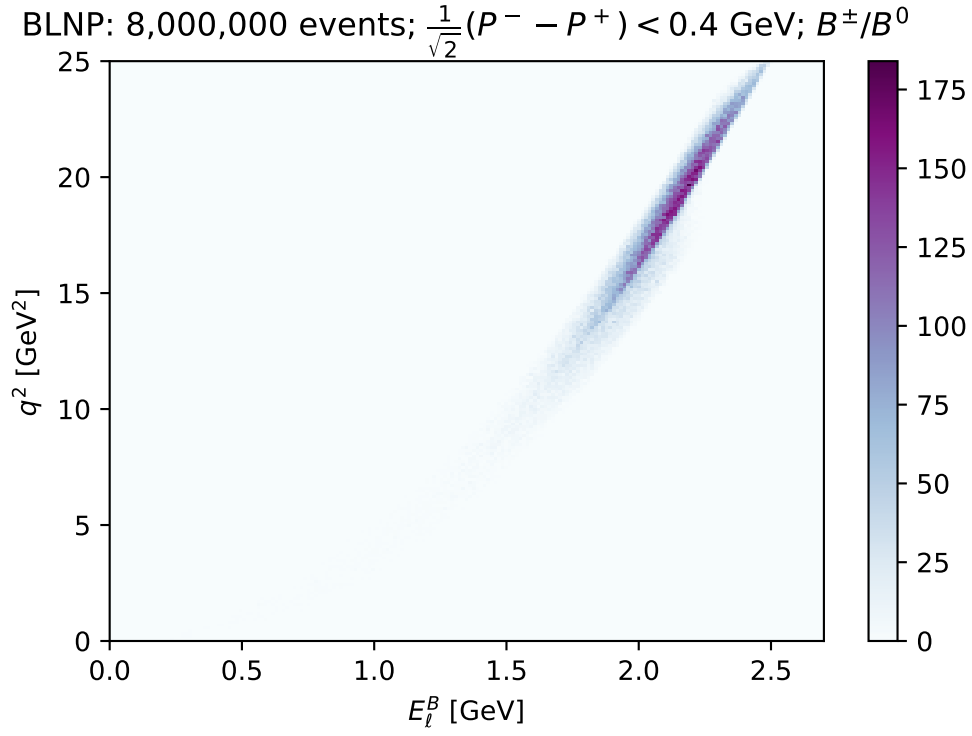
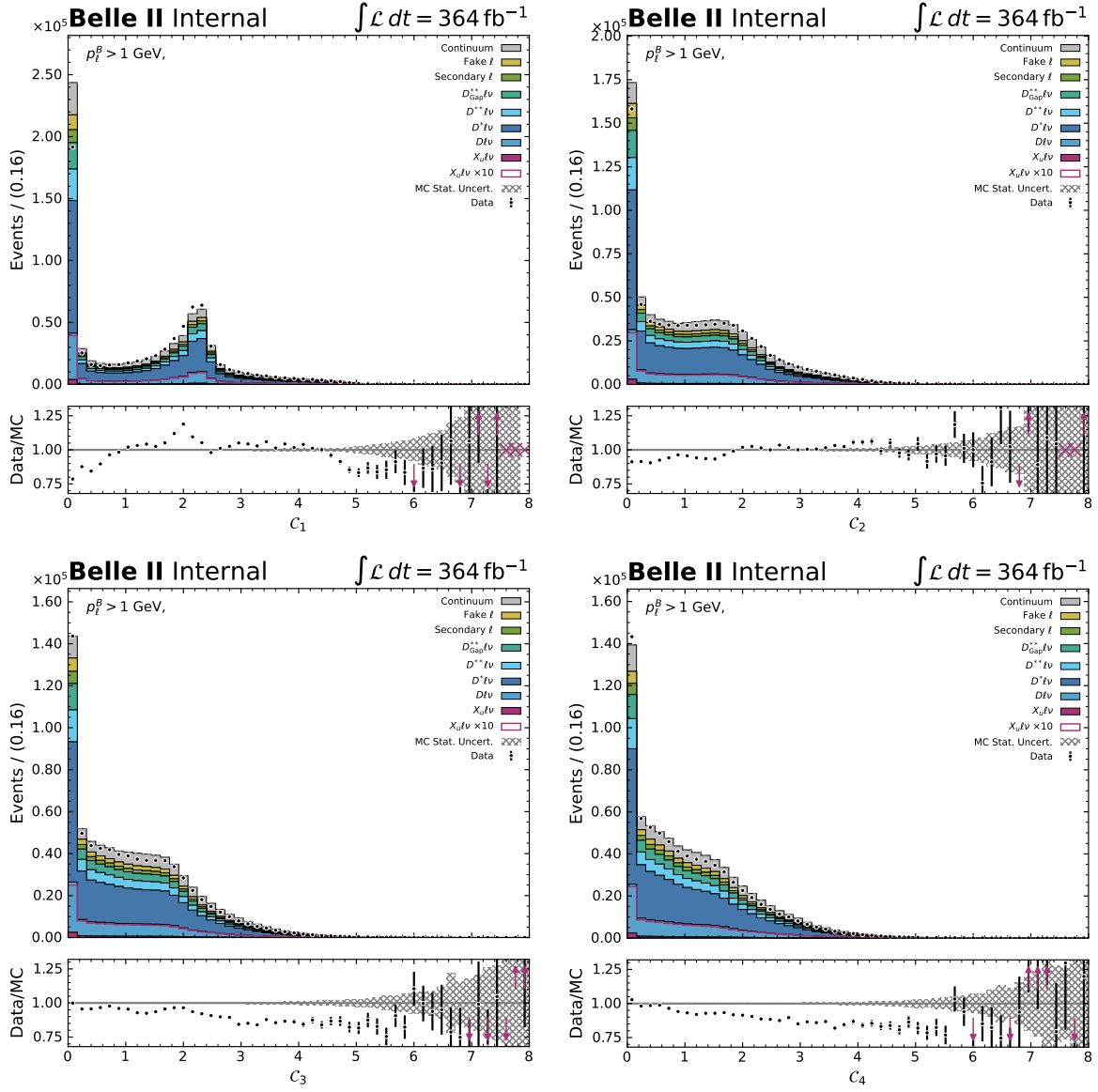
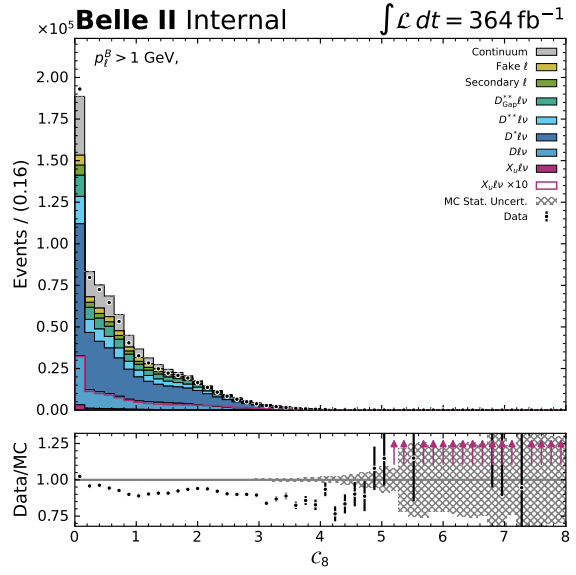
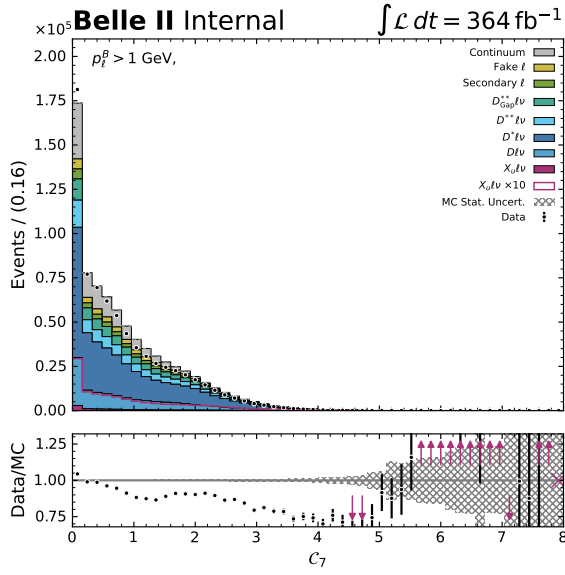
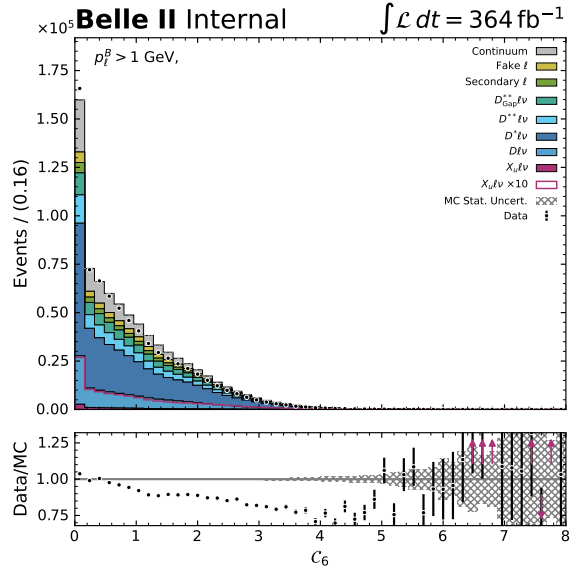
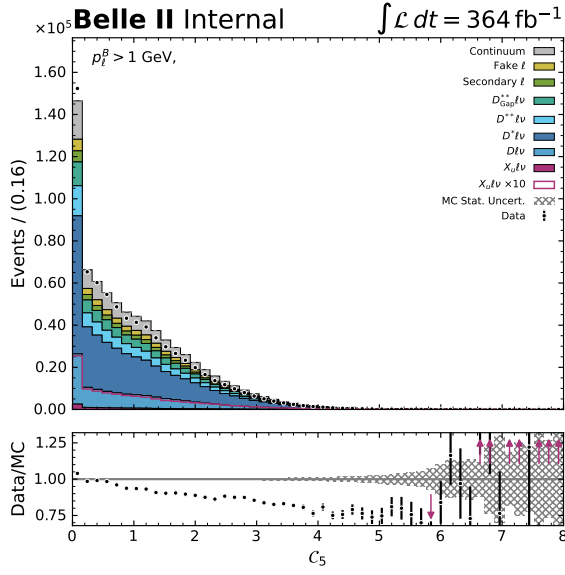


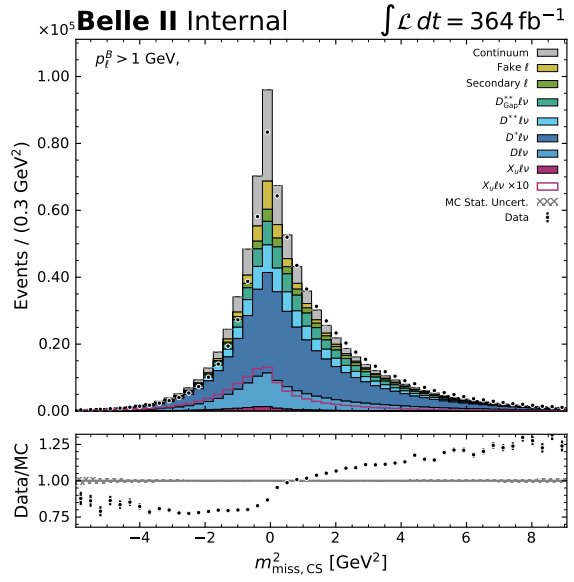
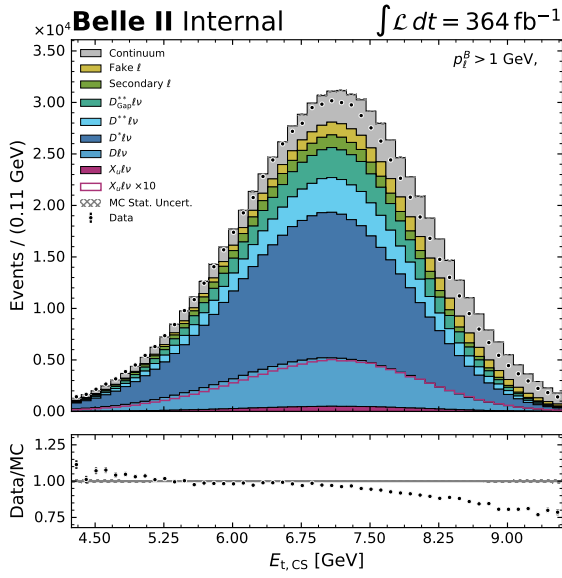
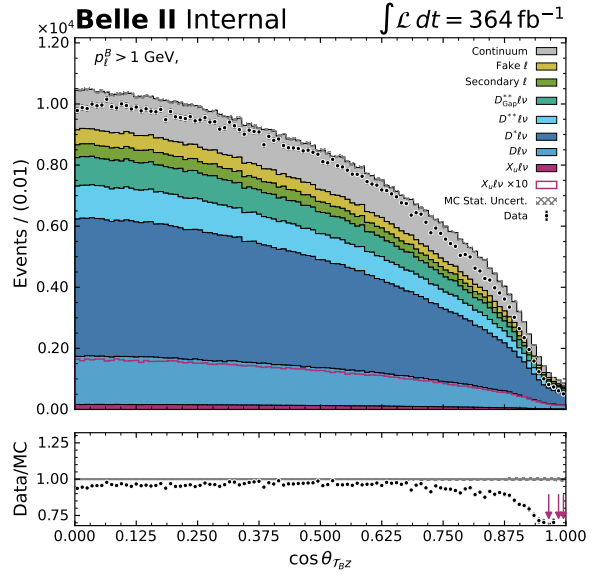
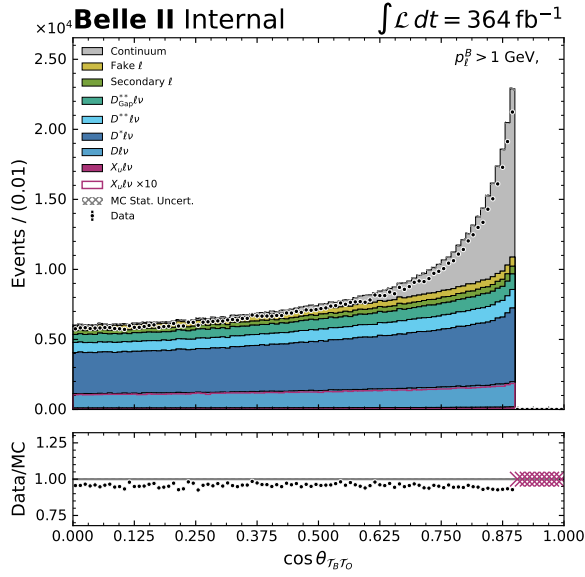
FIG. 83: Two-dimensional histogram of q^2 versus E_l^B for 8 million events generated with the most up-to-date input values for the BLNP model in EvtGen. An additional cut of $\frac{1}{\sqrt{2}}(P_- - P_+) < 0.4 \text{ GeV}$ is added.

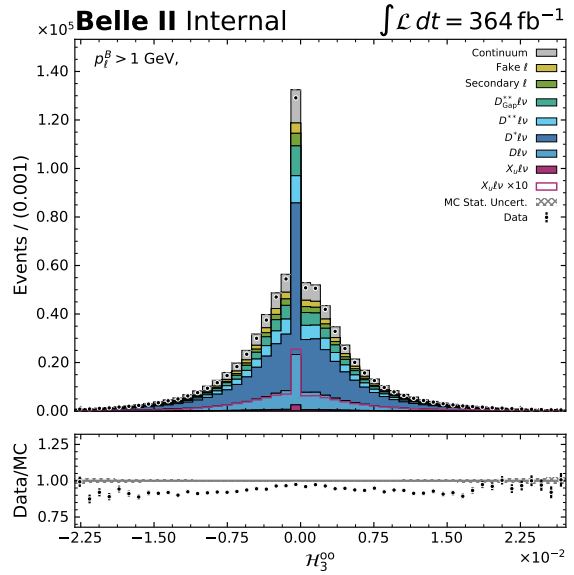
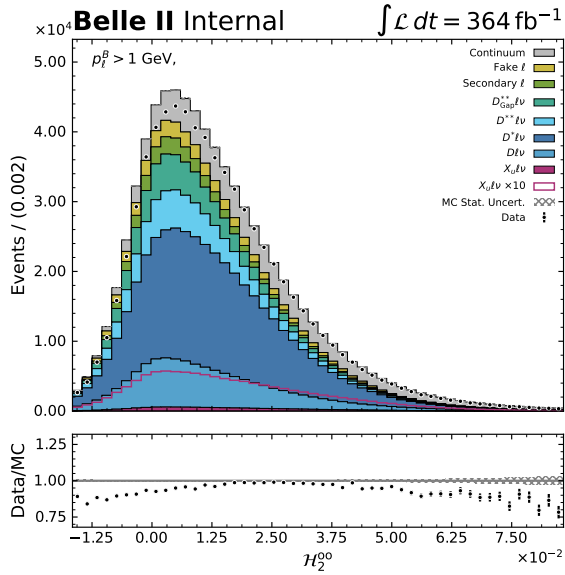
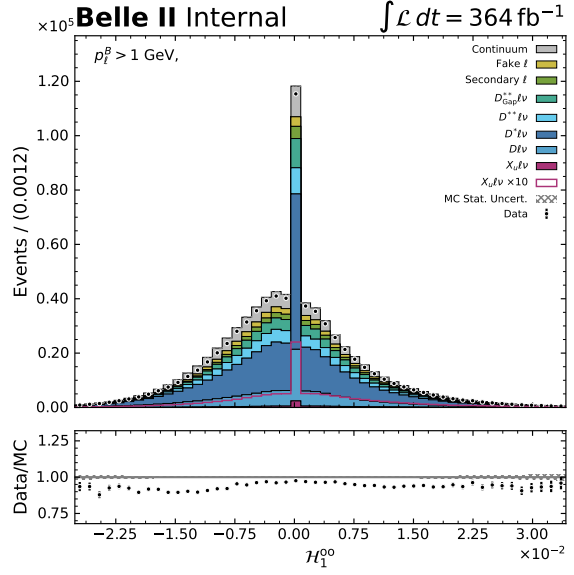
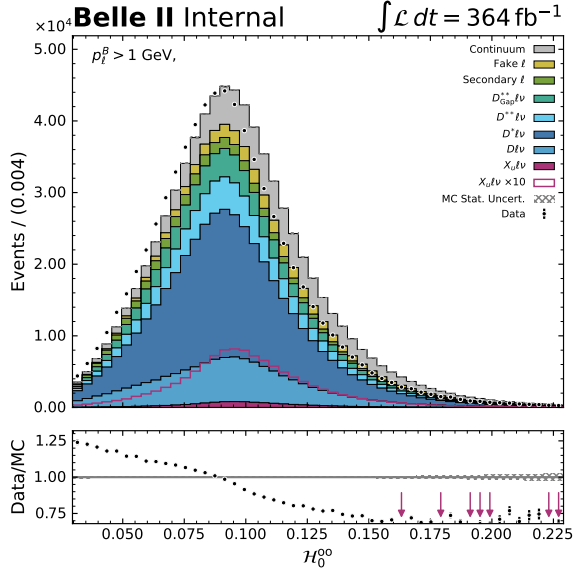
1974 B. CONTINUUM SUPPRESSION MVA INPUT FEATURES

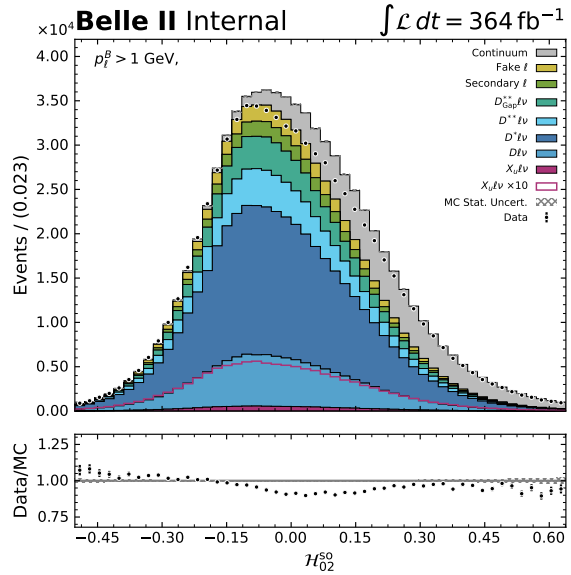
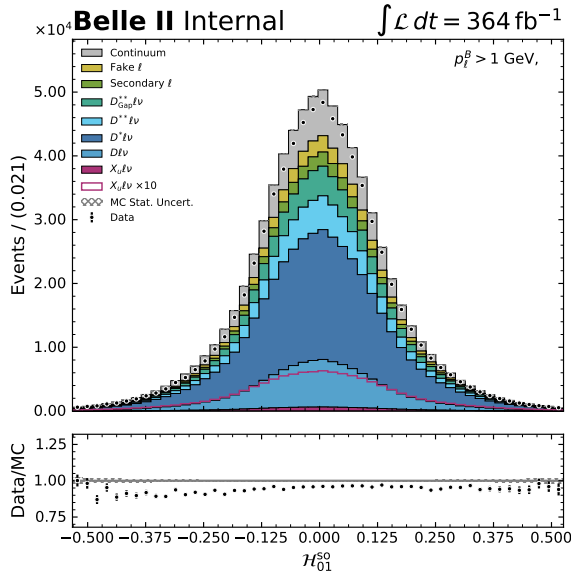
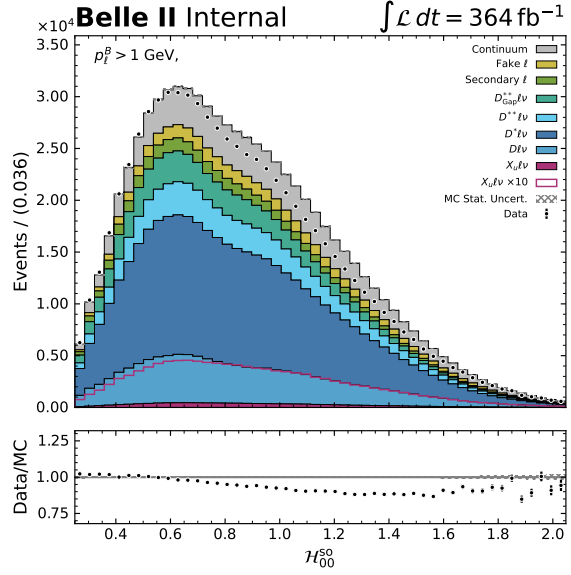
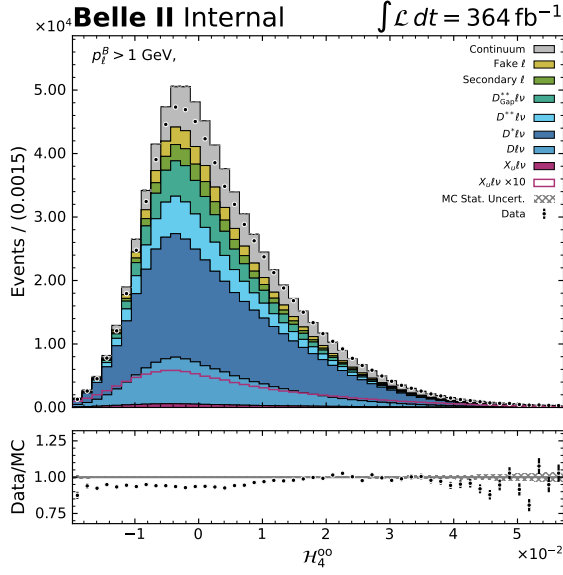
1975 The continuum suppression variables after reconstruction and preselection are shown here.

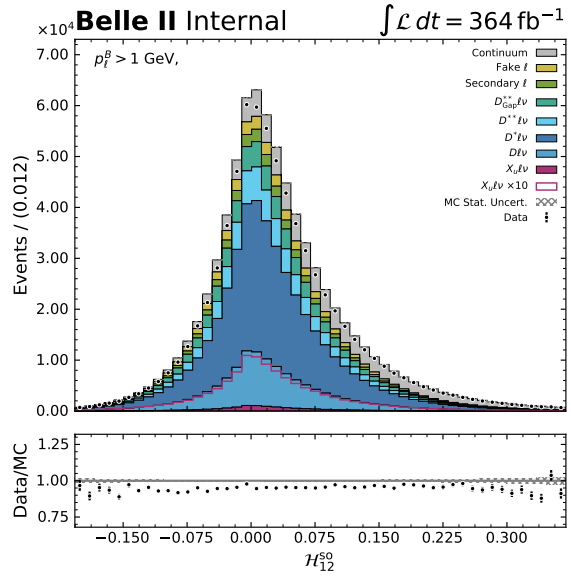
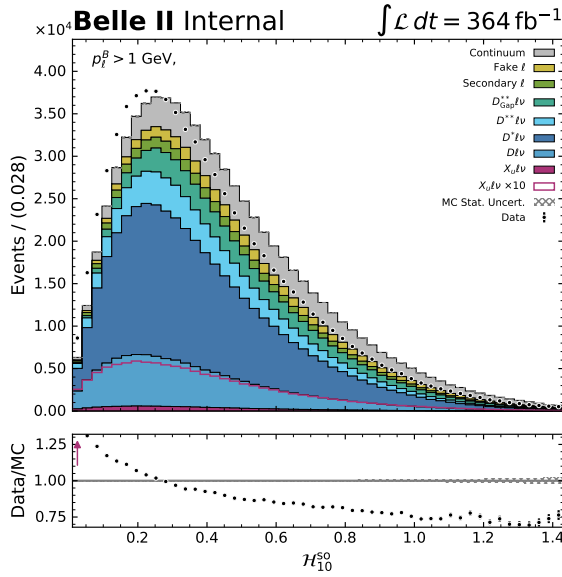
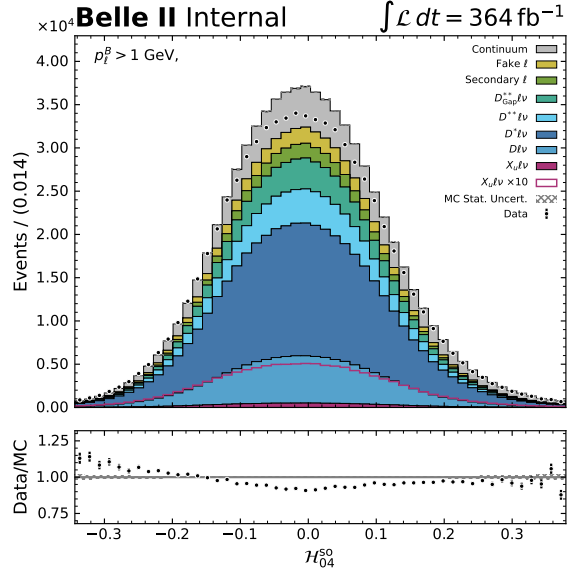
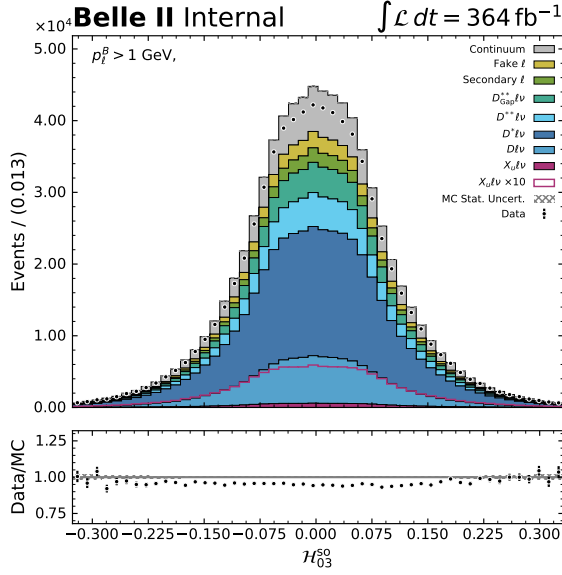


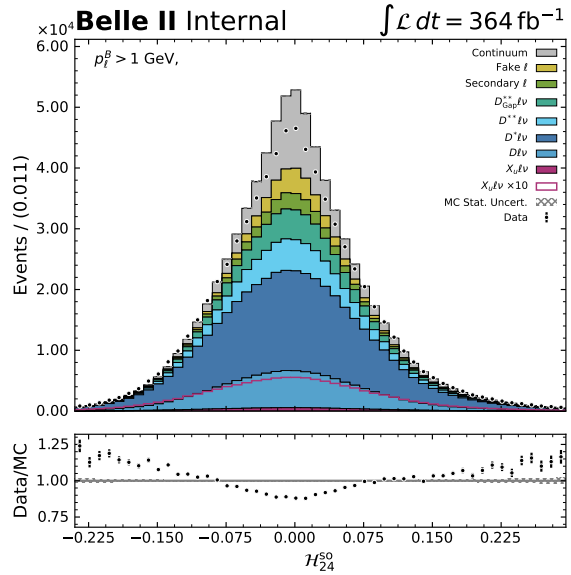
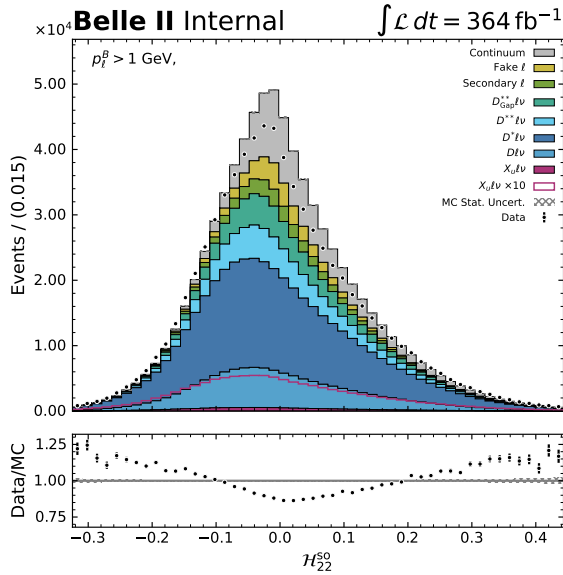
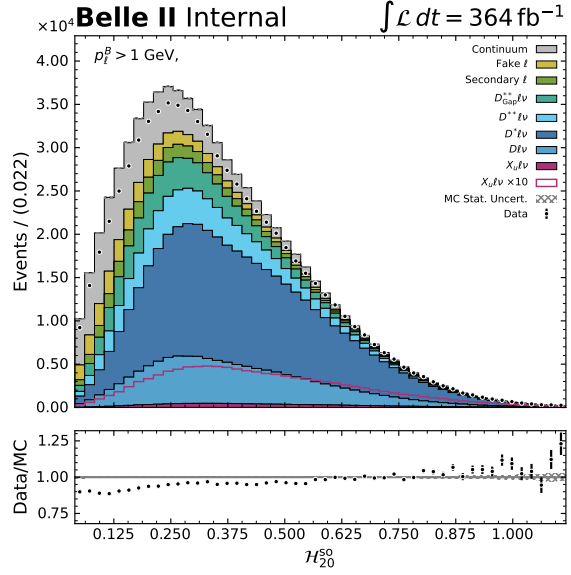
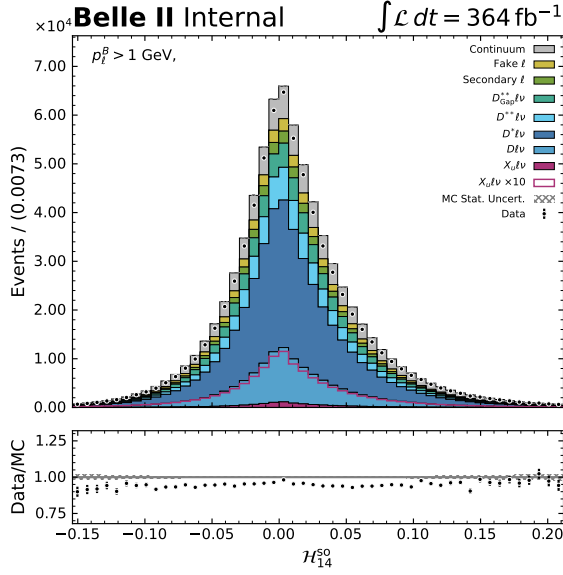


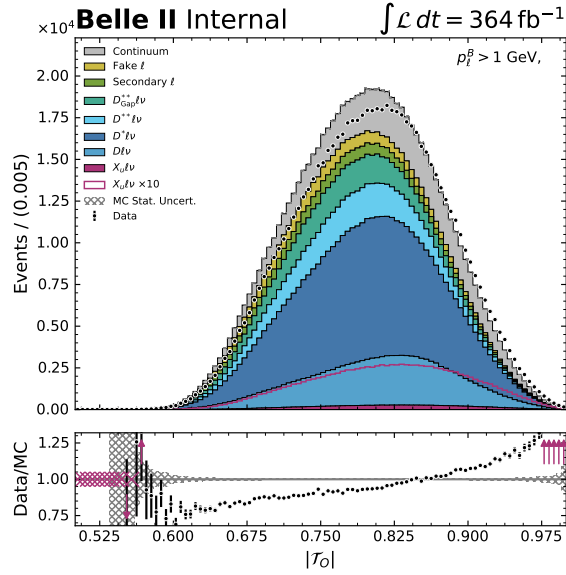
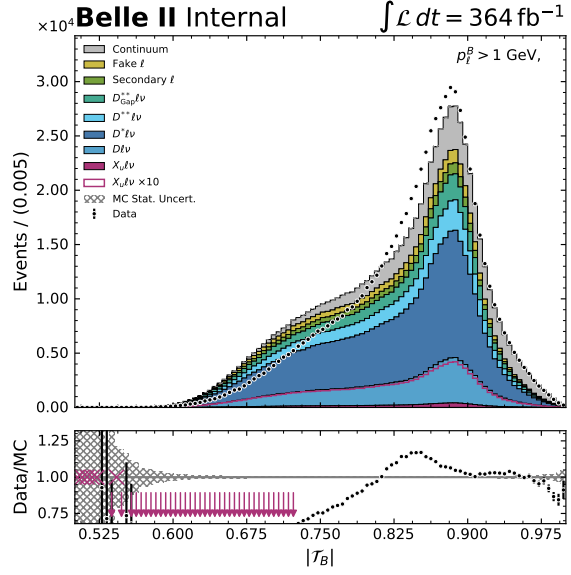
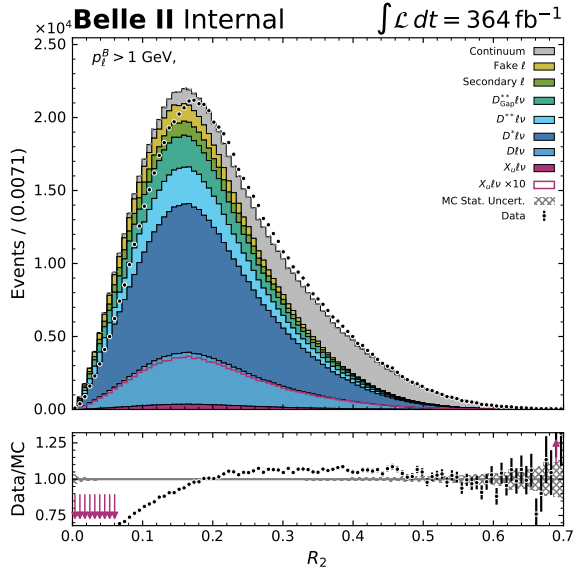


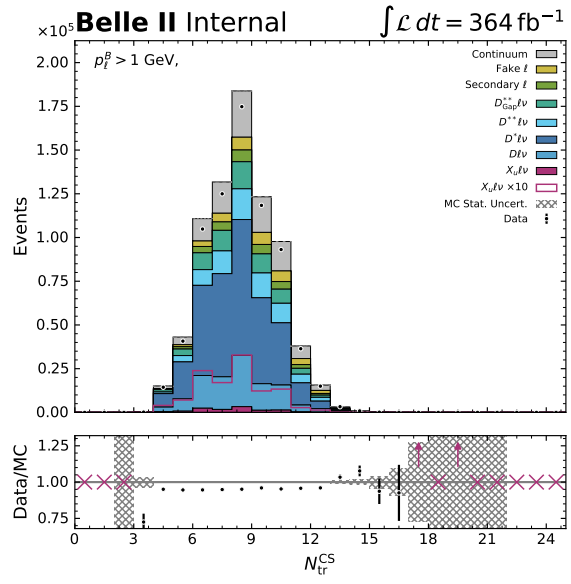
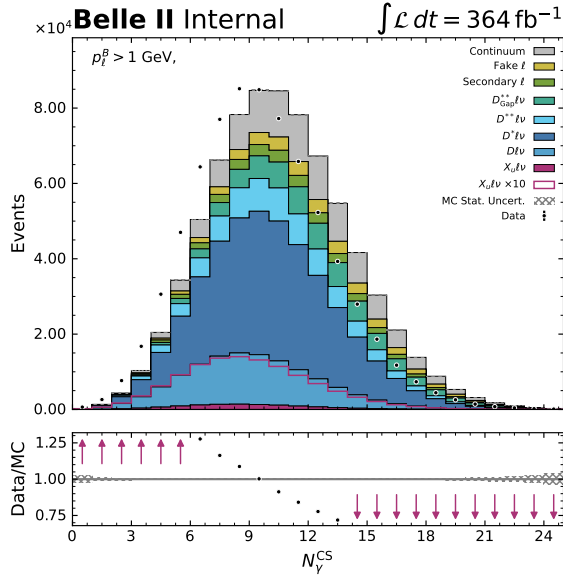






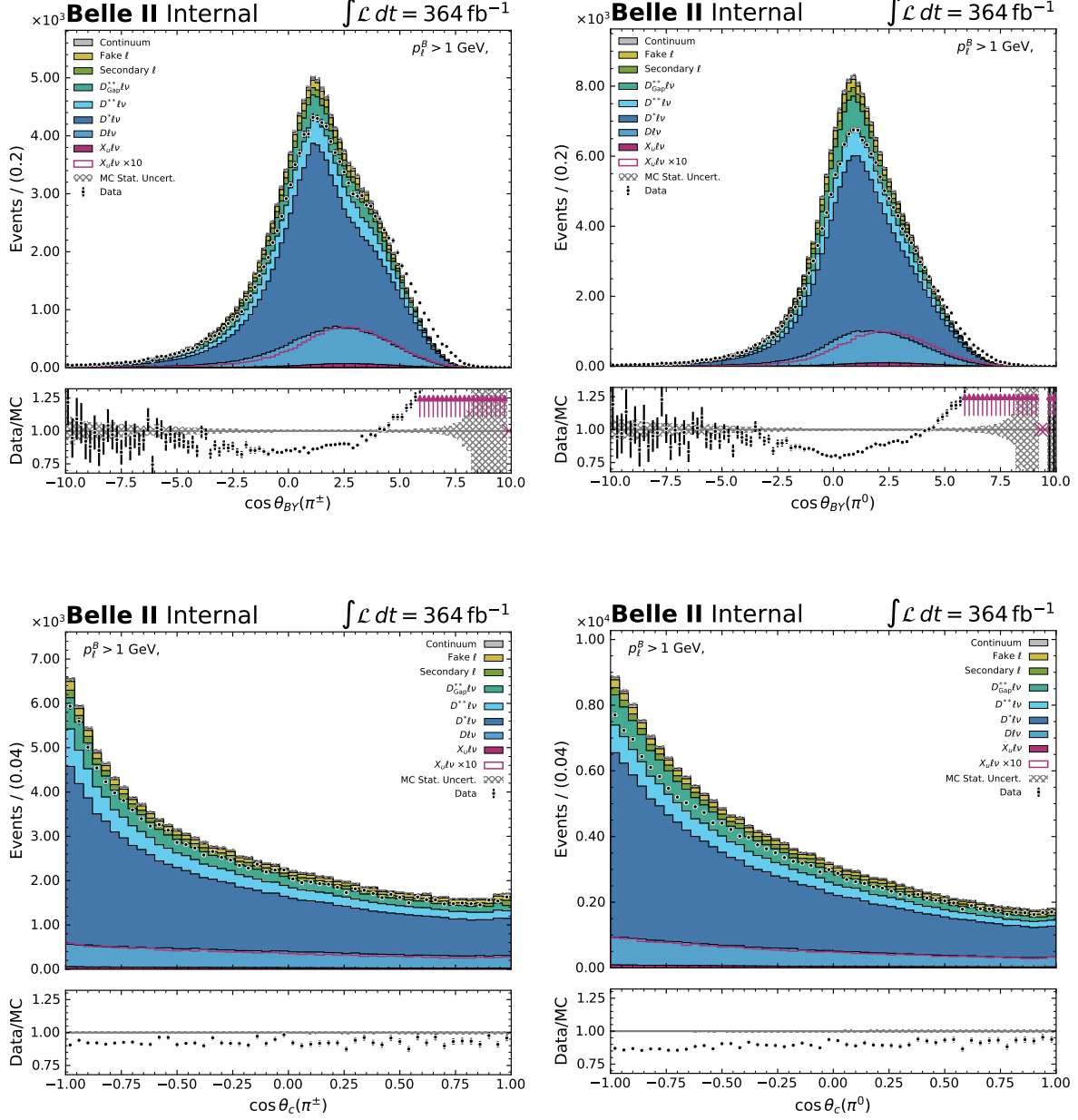


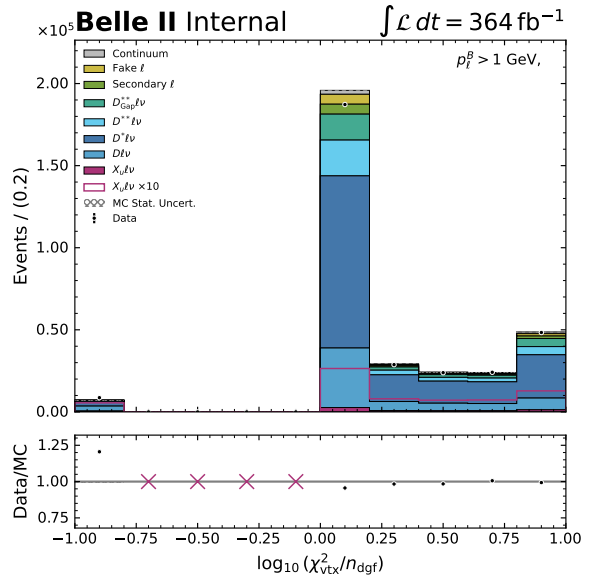
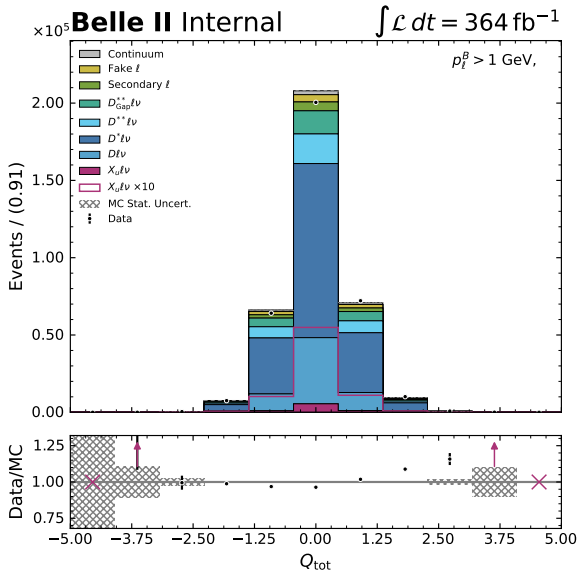
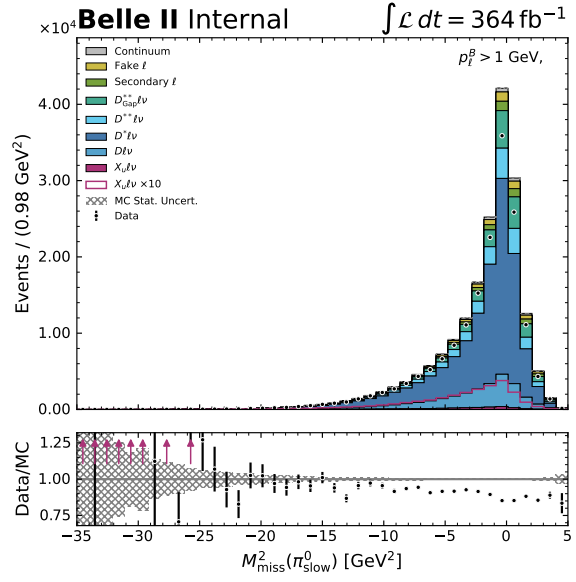
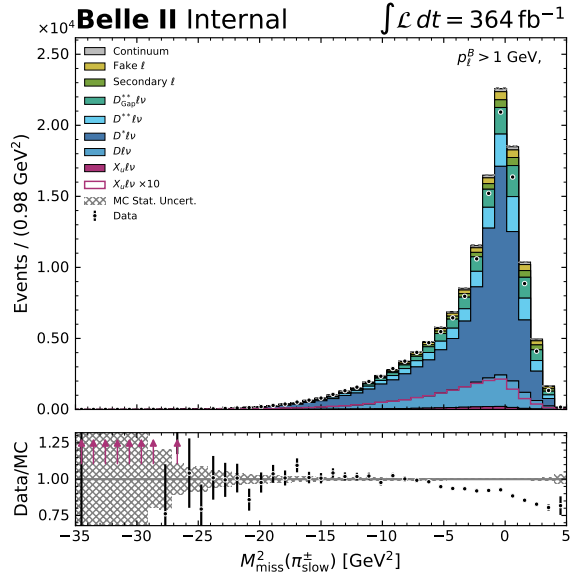


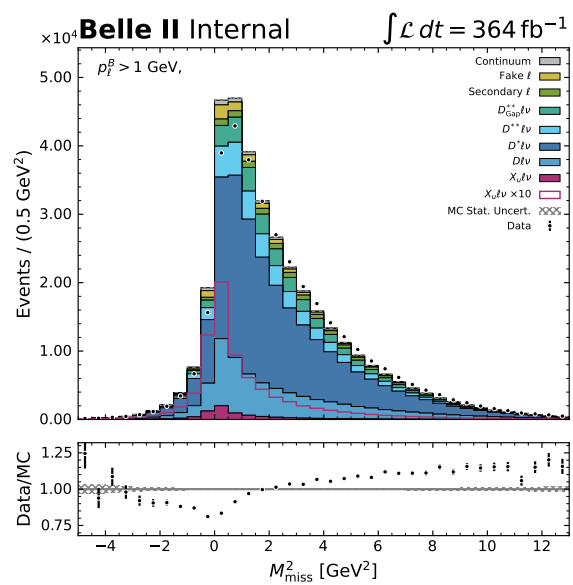


1976 C. $X_c\ell\nu$ SUPPRESSION MVA INPUT FEATURES

1977 The $X_c\ell\nu$ suppression MVA input features after reconstruction, preselection, continuum
 1978 suppression and K veto are shown here.







1979 **D. DATA - MC COMPARISON IN SB1 IN DIFFERENT CHANNELS**

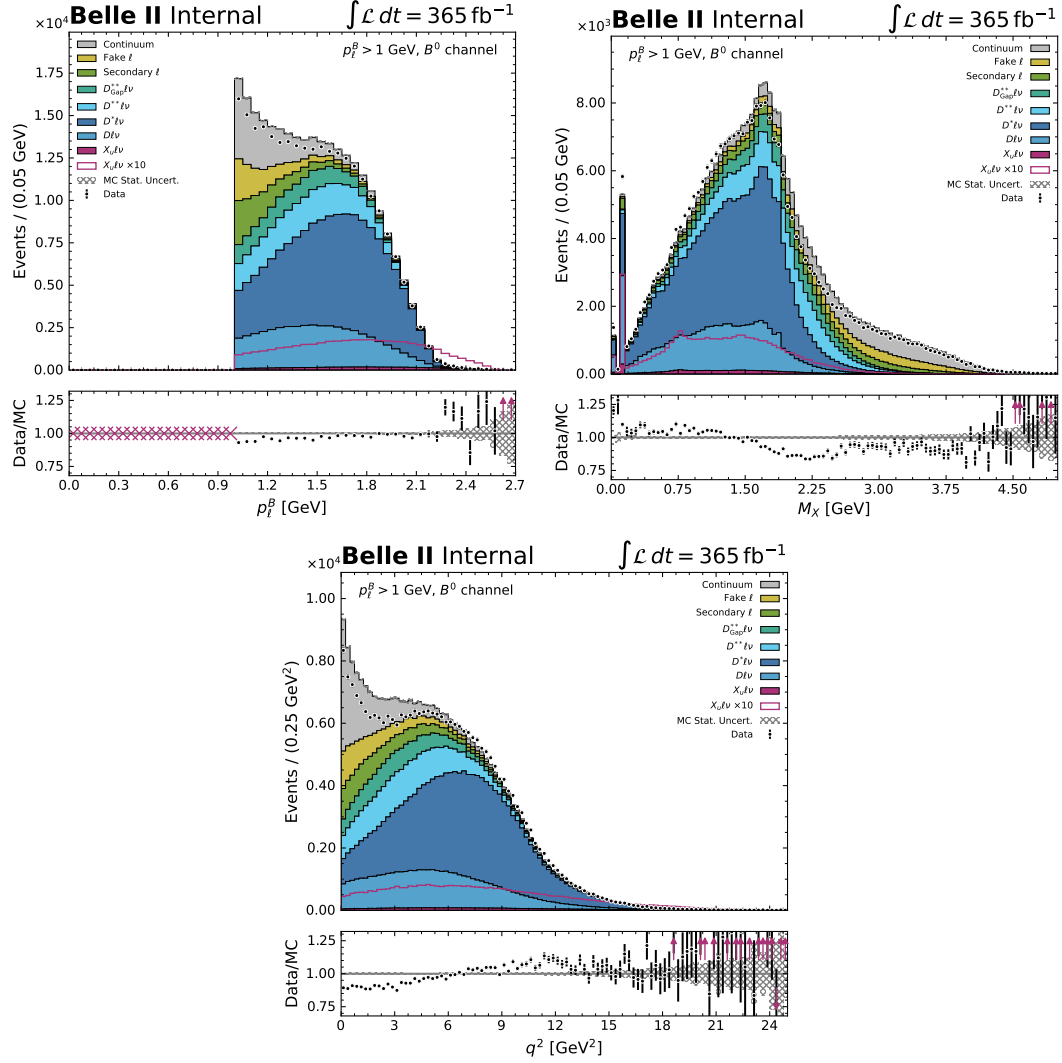


FIG. 84: Data-MC comparison of p_ℓ^B , M_X and q^2 in side-band 1 in the B^0 channel.

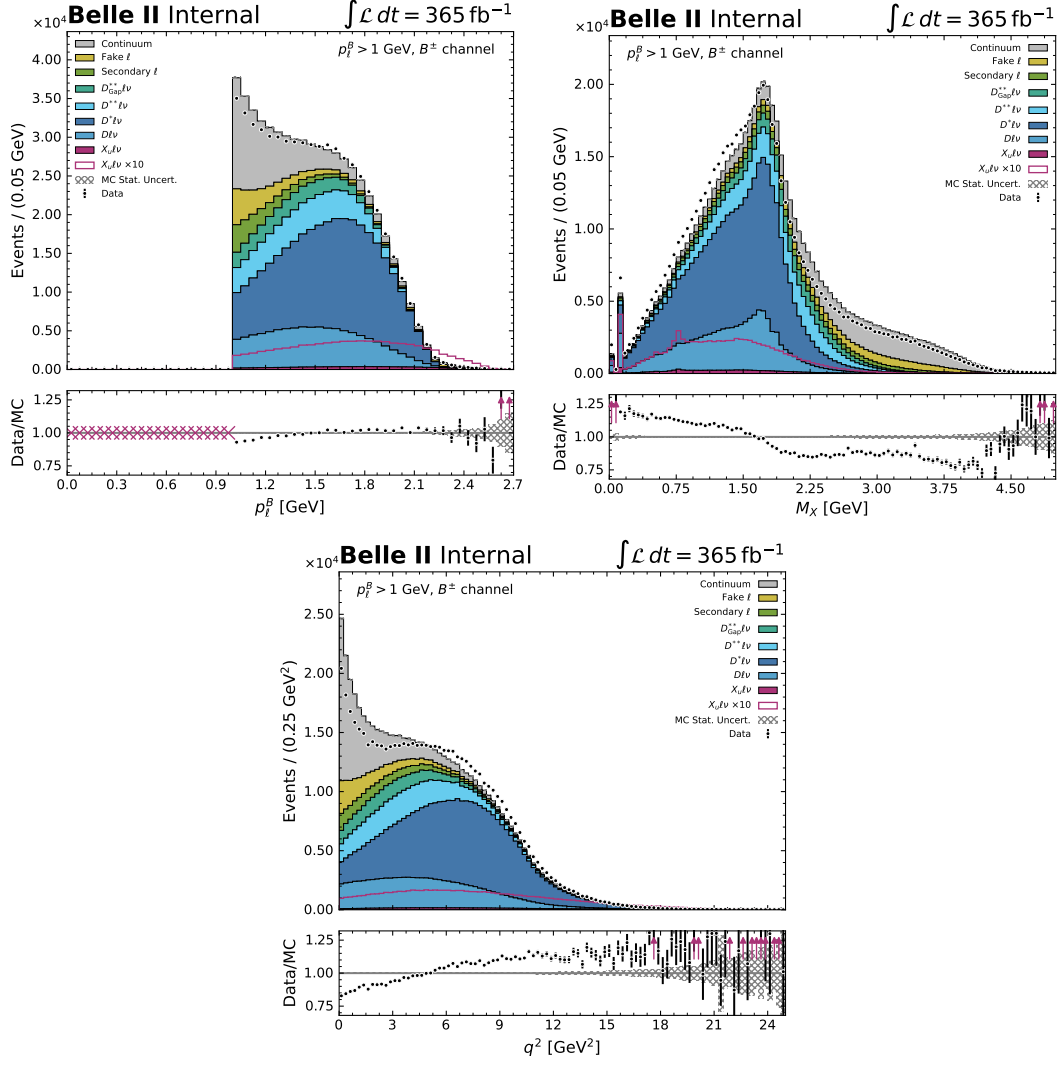


FIG. 85: Data-MC comparison of p_ℓ^B , M_X and q^2 in side-band 1 in the $B^\pm$ channel.

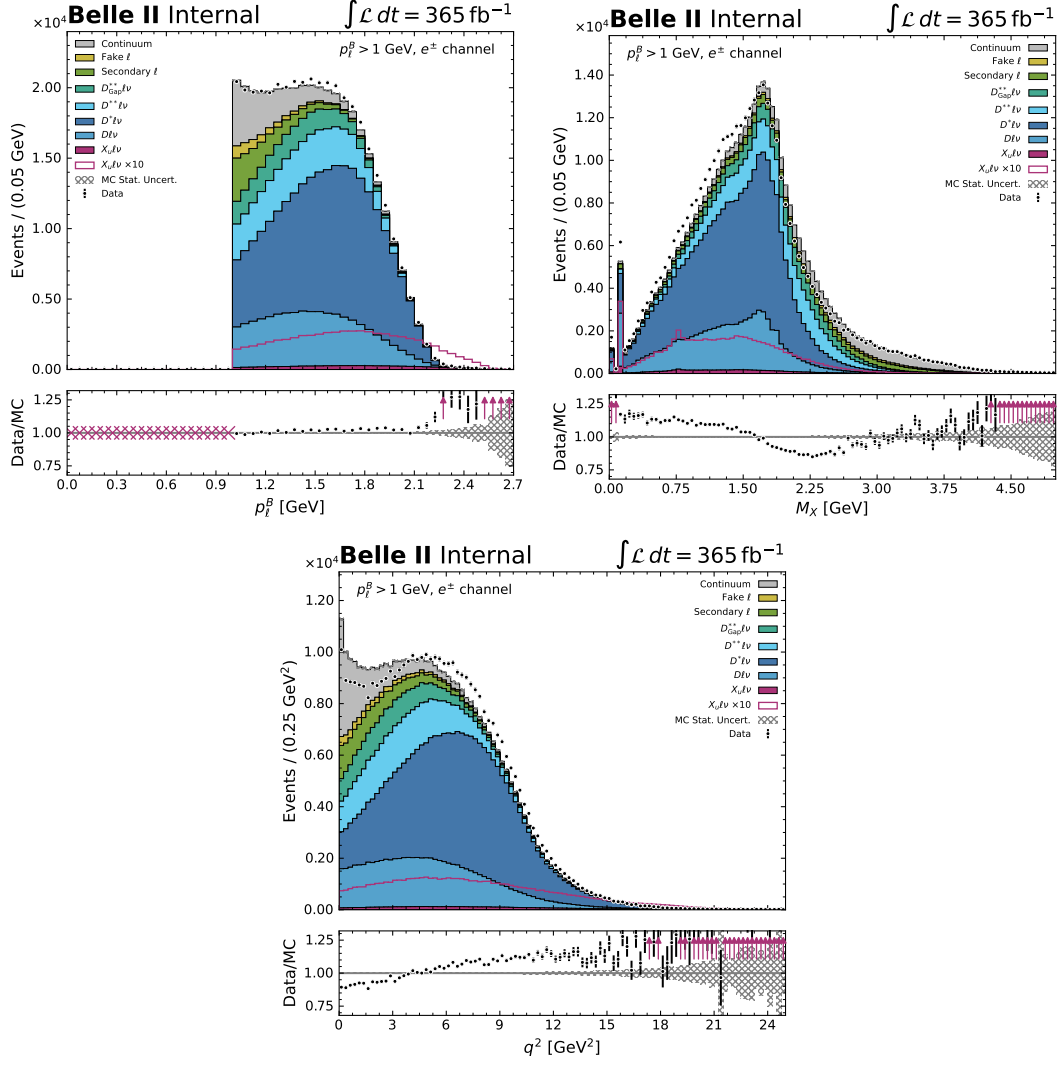


FIG. 86: Data-MC comparison of p_ℓ^B , M_X and q^2 in side-band 1 in the $e^\pm$ channel.

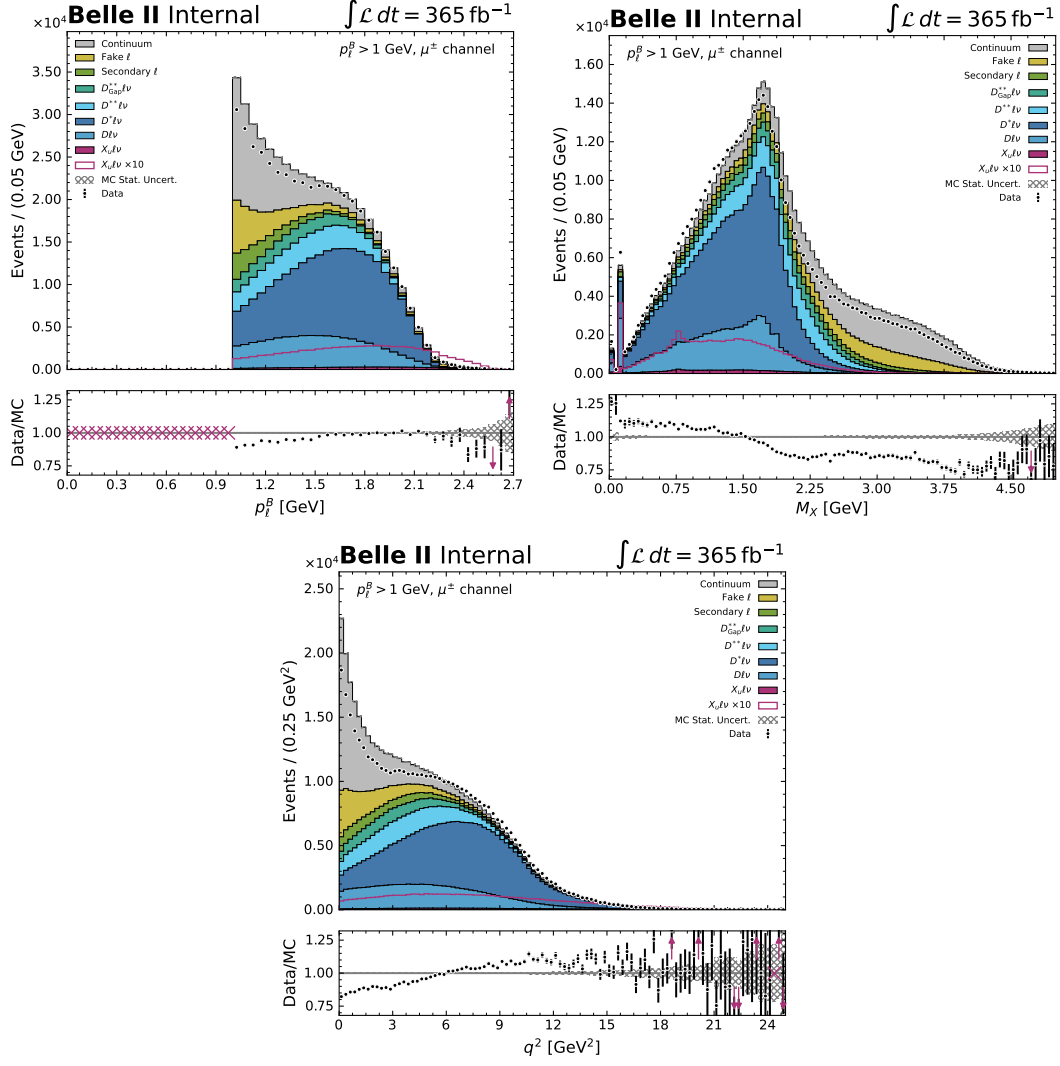


FIG. 87: Data-MC comparison of p_ℓ^B , M_X and q^2 in side-band 1 in the $\mu^\pm$ channel.

1980 **E. PYHF**

1981 **1. Notation**

- 1982 • b : bins
- 1983 • c : channels
 - 1984 – Signal region (SR)
 - 1985 – Control region (CR)
- 1986 • s : samples
 - 1987 – $X_u \ell \nu$ -in
 - 1988 – $X_u \ell \nu$ -out
 - 1989 – $X_c \ell \nu$
 - 1990 – Other backgrounds
- 1991 • ν : event rate (ν^0 : nominal rate)
- 1992 • μ : scale factor
- 1993 • γ : bin-wise scale factors
- 1994 • α : interpolation parameter
- 1995 • κ : multiplicative modifiers
- 1996 • Δ : additive modifiers
- 1997 • $\boldsymbol{\eta}$: unconstrained parameters
- 1998 • $\boldsymbol{\chi}$: constrained parameters
- 1999 • $\boldsymbol{n}$: event counts
- 2000 • $\boldsymbol{a}$: auxiliary measurements
- 2001 • Free-floating scale factors
 - 2002 – $X_u \ell \nu$ -in: 1 free-floating scale factor, not present in control region channel but
 - 2003 only in signal region channel
 - 2004 – $X_u \ell \nu$ -out: 1 free-floating scale factor, not present in control region channel but
 - 2005 only in signal region channel
 - 2006 – $X_c \ell \nu$: 1 free-floating scale factor and 1 free-floating scale factor per bin, present
 - 2007 in both channels
 - 2008 – Other backgrounds: 1 free-floating scale factor, present in both channels

2009 **2. Statistical model**

2010 A general statistical model can be written as follows:

$$f(\mathbf{n}, \mathbf{a} | \boldsymbol{\eta}, \boldsymbol{\chi}) = \prod_c \prod_b \text{Pois}(n_{cb} | \nu_{cb}(\boldsymbol{\eta}, \boldsymbol{\chi})) \prod_{\chi \in \boldsymbol{\chi}} c_\chi(a_\chi | \chi), \quad (18)$$

2011 where the first term corresponds to the simultaneous measurement of multiple channels and
2012 the second to constraint terms for auxiliary measurements. The event rates are defined as:

$$\nu_{cb} = \sum_{s \in \text{samples}} \nu_{scb}(\boldsymbol{\eta}, \boldsymbol{\chi}) = \sum_{s \in \text{samples}} \left(\prod_{\kappa \in \boldsymbol{\kappa}} \kappa_{scb}(\boldsymbol{\eta}, \boldsymbol{\chi}) \right) \cdot \left(\nu_{scb}^0(\boldsymbol{\eta}, \boldsymbol{\chi}) + \sum_{\Delta \in \boldsymbol{\Delta}} \Delta_{scb}(\boldsymbol{\eta}, \boldsymbol{\chi}) \right), \quad (19)$$

2013 where κ_{scb} and Δ_{scb} are multiplicative and additive modifiers respectively. They therefore
2014 correspond to normalisation and shape uncertainties respectively. Each systematic source
2015 in the fit is split in normalisation and shape variation. As explained in the main text, for
2016 the sample $s = X_c \ell \nu$, we incorporate binned free-floating normalisations. The event rates
2017 for this sample can therefore be written as:

$$\nu_{cb}^{X_c} = \left(\gamma_b^{X_c} \cdot \prod_{\kappa \in \boldsymbol{\kappa}} \kappa_{cb}^{X_c} \right) \cdot \left(\nu_{cb}^{0, X_c} + \sum_{\Delta \in \boldsymbol{\Delta}} \Delta_{cb}^{X_c} \right), \quad (20)$$

2018 The total $X_c \ell \nu$ event rate can be written:

$$\begin{aligned} \nu^{X_c} &= \sum_{b, c} \nu_{cb}^{X_c} \\ \Rightarrow \nu^{X_c} &= \sum_b \gamma_b^{X_c} \sum_c \left(\prod_{\kappa} \kappa_{cb}^{X_c} \right) \cdot \left(\nu_{cb}^{0, X_c} + \sum_{\Delta} \Delta_{cb}^{X_c} \right) \end{aligned} \quad (21)$$

2019 For the $X_u \ell \nu$ -in template, the event rate can be written similarly with a single unbinned
2020 free-floating normalisation. The signal template is fixed in the control region and its yields
2021 are negligible in this region compared to the background yields. Therefore, the signal event
2022 rate can be written as:

$$\nu^{X_u} = \mu^{X_u} \sum_b \left(\prod_{\kappa} \kappa_{\text{SR}, b}^{X_u} \right) \cdot \left(\nu_{\text{SR}, b}^{0, X_u} + \sum_{\Delta} \Delta_{\text{SR}, b}^{X_u} \right) \quad (22)$$

2023 The $X_u \ell \nu$ -out event rate can be written in the exact same way with its own associated
2024 free-floating normalisation and constrained modifiers. In some of the fits, this template is
2025 completely neglected. In other fits, the yields are small but not negligible which leads to
2026 large biases and uncertainties in the fit. In these cases the same free-floating normalisation
2027 μ^{X_u} is assigned to both templates. Finally, the other-background template is assigned a
2028 Gaussian-constrained multiplicative modifier (**normsys**).

2029 **3. Modifiers**

2030 For correlated shape and normalisation uncertainties, **pyhf** makes use of interpolating func-
2031 tions, f_p and g_p (for additive and multiplicative modifiers respectively), constructed from

a small number of evaluations of the expected rate. We summarise in the following table each type of modification and its associated constraint term that we implement in our `pyhf` setup.

Description	Modification	Constraint term
Correlated shape (histosys)	$\Delta_{scb}(\alpha) = f_p(\alpha)$	Gaussian with width given by the auxiliary data
Normalisation uncertainty (normsys)	$\kappa_{scb}(\alpha) = g_p(\alpha)$	Gaussian with width given by the auxiliary data
MC statistical uncertainty (statererror)	$\kappa_{scb} = \gamma_b$	$\prod_b$ Gaussians
Normalisation (normfactor)	$\kappa_{scb} = \mu$	Free-floating
Data-driven shape (shapefactor)	$\kappa_{scb} = \gamma_b$	Free-floating

TABLE XXXV: Modifiers and constraints implemented in the `pyhf` fit setup described in Section 8

Parameter	Modifier	Constraint	Regions	Type of variation	Number of Parameters
Slow π efficiency	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	$3 (\pi^\pm) + 3 (\pi^0)$
Tracking efficiency	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	1
$B \rightarrow \pi \ell \nu$ form factors	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	5
$B \rightarrow \rho/\omega \ell \nu$ form factors	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	11
$B \rightarrow \eta^{(\prime)} \ell \nu$ form factors	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	2
$B \rightarrow D^{(*)} \ell \nu$ form factors	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	9
Broad $B \rightarrow D^{**} \ell \nu$ form factors	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	3
Narrow $B \rightarrow D^{**} \ell \nu$ form factors	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	4
Lepton ID	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Toy	200 toys (truncated to $\mathcal{O}(10)$)
Kaon ID	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Toy	200 toys (truncated to $\mathcal{O}(10)$)
FEI	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	$11 (B^0) + 12 (B^+) \text{ (truncated to } \mathcal{O}(5))$
B branching fractions	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	$10 (B^0) + 12 (B^+)$
D branching fractions	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	234 (truncated to $\mathcal{O}(15)$)
DFN parameter variations	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	2
Inclusive model	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	One-sided	1
K_0^S efficiency	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	One-sided	1
$s\bar{s}$ fragmentation	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	1
MC Stats	κ	$\prod_b$ Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	1 for each region
$f^{\pm/0}$	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	1
Cont. normalisation calibration	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	1
Cont. shape calibration	$\Delta + \kappa$	Gauss	Signal + $\text{CR}_{0,low}$	Toy	200 (truncated to $\mathcal{O}(15)$)
Signal-in normalisation	μ^{X_u}	Unconstrained	Signal region	Two-sided	1
$B \rightarrow X_c \ell \nu$ shape	$\gamma_b^{X_c}$	Unconstrained	Signal + $\text{CR}_{0,low}$	Two-sided	1 per bin
$B \rightarrow X_c \ell \nu$ normalisation	κ	Gauss	Signal + $\text{CR}_{0,low}$ separately	Two-sided	1
Other backgrounds normalisation	κ	Gauss	Signal + $\text{CR}_{0,low}$	Two-sided	1

TABLE XXXVI: `pyhf` parameters

4. Postfit projections

In order to cross-check the results of the fit, we project them on variables not used for the fit. To do so, we use the cabinetry function `cabinetry.model_utils.match_fit_results` which takes as input a `pyhf` model (not fitted) produced for the variable on which to apply the fit and the fit results obtained for the chosen fit variable. The matching of the nuisance parameters is done by matching NP with the same name. However, the transfer of some of the nuisance parameters is not trivial so they are modified as:

- 2042 • up/down variations: these NP depend on a single interpolation parameter so they can
2043 trivially be matched and they are therefore not modified
- 2044 • toy variations: see Section 8.3.1 for details on how these NP are implemented in the
2045 fit. In the transfer fit, the variations in the eigendirections that are generated in the
2046 original fit are taken and added as $\pm\sigma$ weights to the dataset, then treated as up/down
2047 variations for transfer fits
- 2048 • normalisation factors: represented by a single free-floating number so their transfer is
2049 trivial
- 2050 • shape factors: they are given as a single free-floating number to the fit which internally
2051 splits them as one normalisation per bin. To translate them properly, we split the
2052 $B \rightarrow X_c \ell \nu$ sample into subsamples, one per bin. Each is given a norm factor with the
2053 same name as the nuisance parameter associated with that bin in the original fit. The
2054 fit then matches each shape factor from the original fit to a normalisation factor in
2055 the new pyhf model

2056 One important caveat is that the MC statistical uncertainty cannot be transferred since
2057 the binning and the number of events per bin cannot be matched between two different
2058 variables. We show in Figure 88 an example of such a transfer. We use as nominal fit, the
2059 $\text{CR}_{K,low}/\text{CR}_{K,high}$ fit 1 with $E_\ell^B:q^2$ which we transfer to M_X . The $B \rightarrow X_c \ell \nu$ template is
2060 split in 12 separate templates, one for each of the 12 $E_\ell^B:q^2$ bins in the original fit.

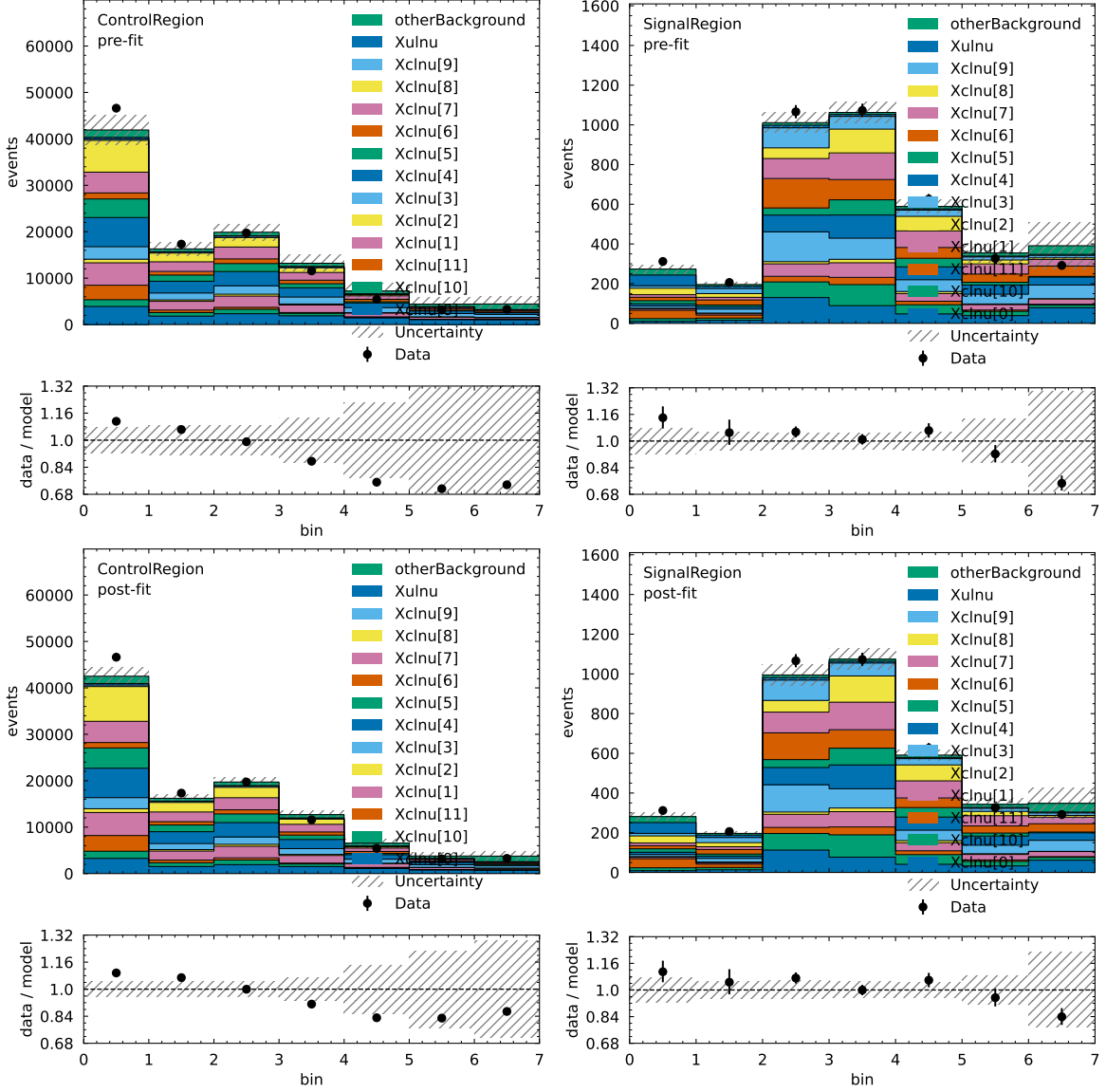


FIG. 88: Projection of the $CR_{K,low} / CR_{K,high}$ (CR/SR) $E_{\ell}^B:q^2$ fit projected on M_X .

2062 F. PLOTS FROM FITS TO ALL VARIABLES IN ALL REGIONS

2063 In this Section, we add plots from all fits described in Section 8. The pre- and post-fit
 2064 distributions are shown together with the ranking of the 30 nuisance parameters with largest
 2065 impact, the toy pull distribution, the linearity test and the correlation matrix (only entries
 2066 with a value larger than or equal to 0.20 are shown).

2067 1. q^2 fit

2068 1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

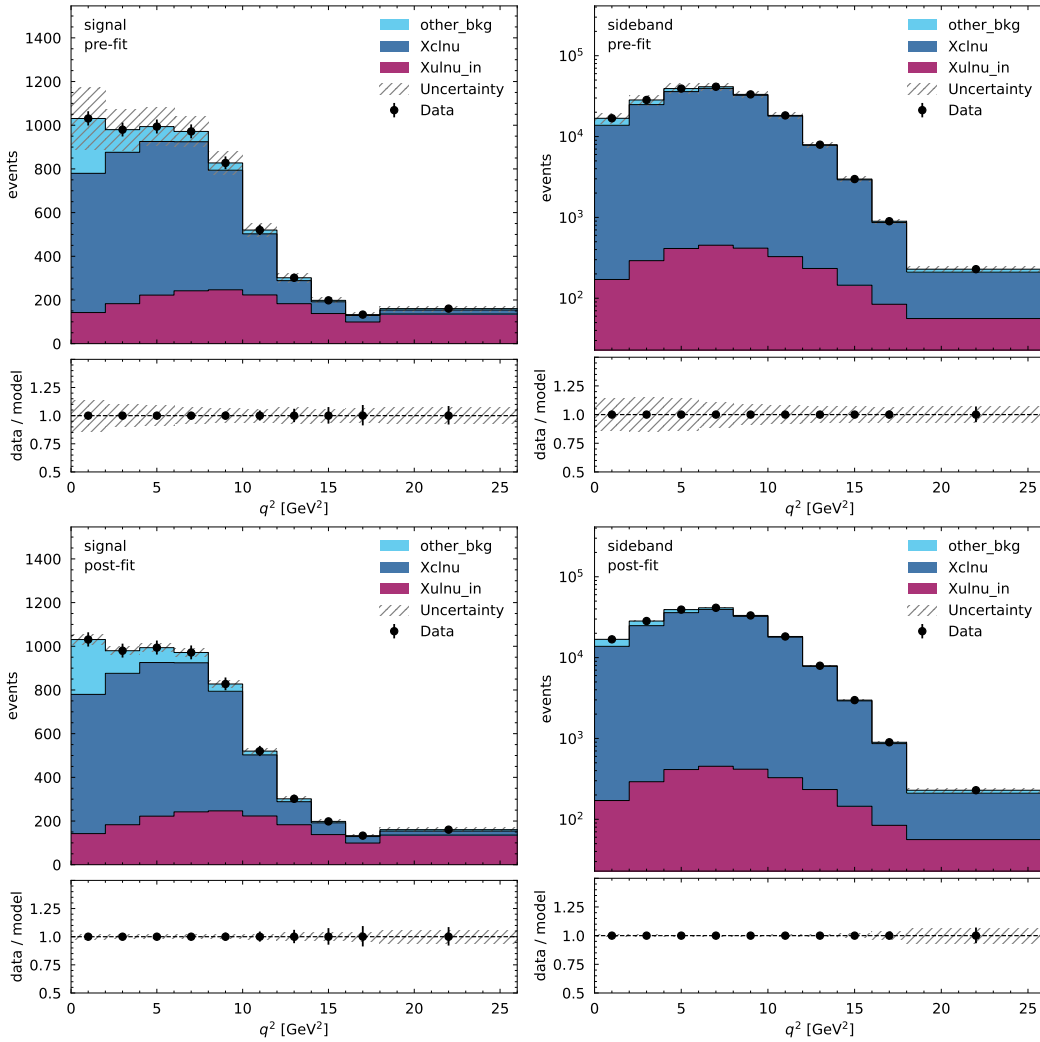


FIG. 89: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

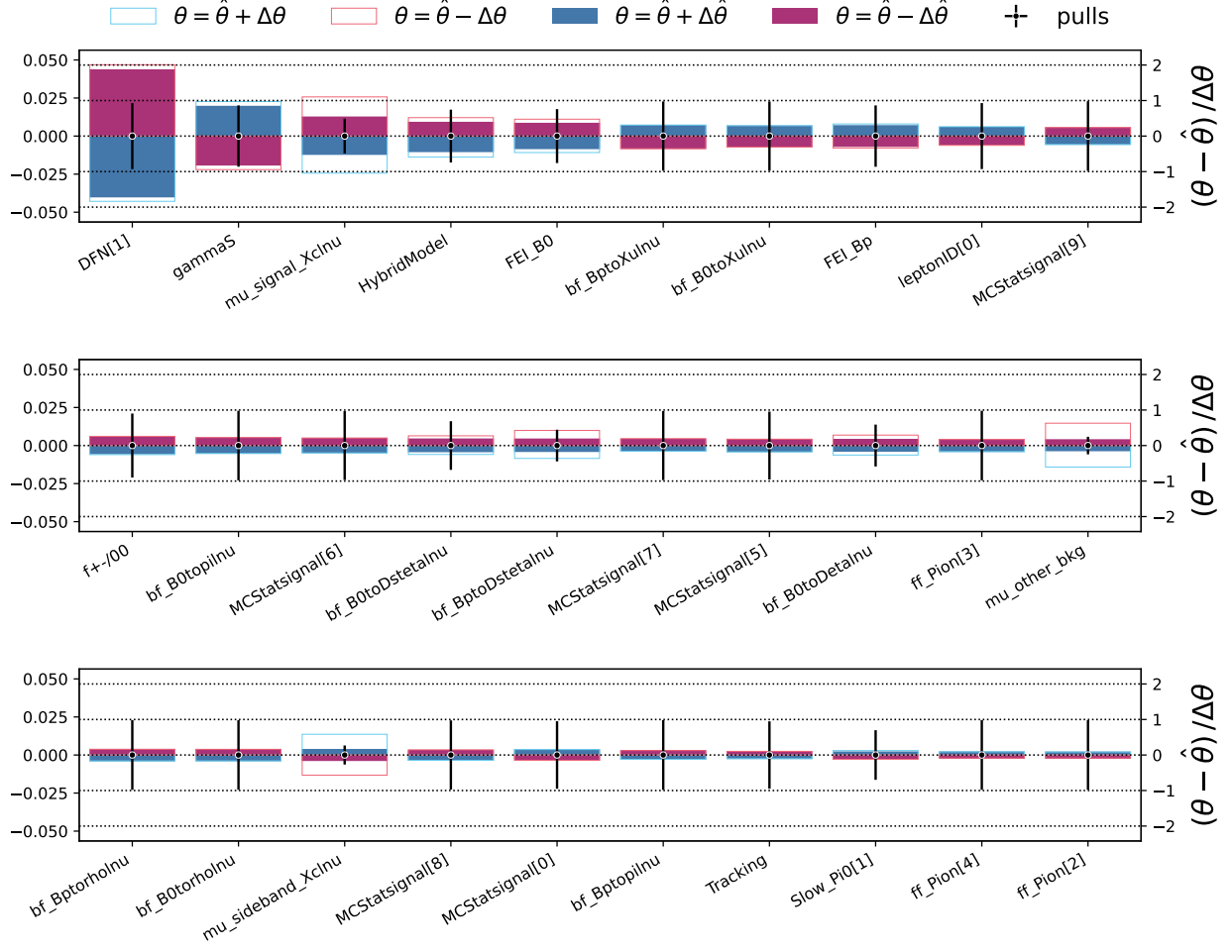


FIG. 90: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

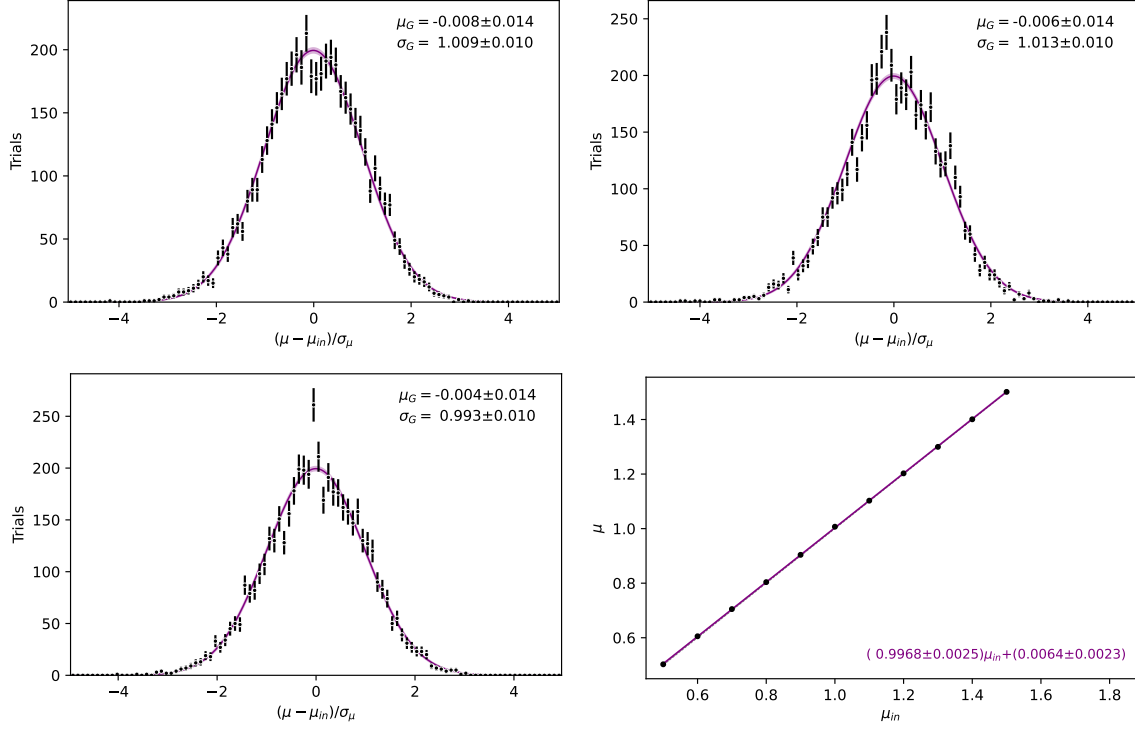
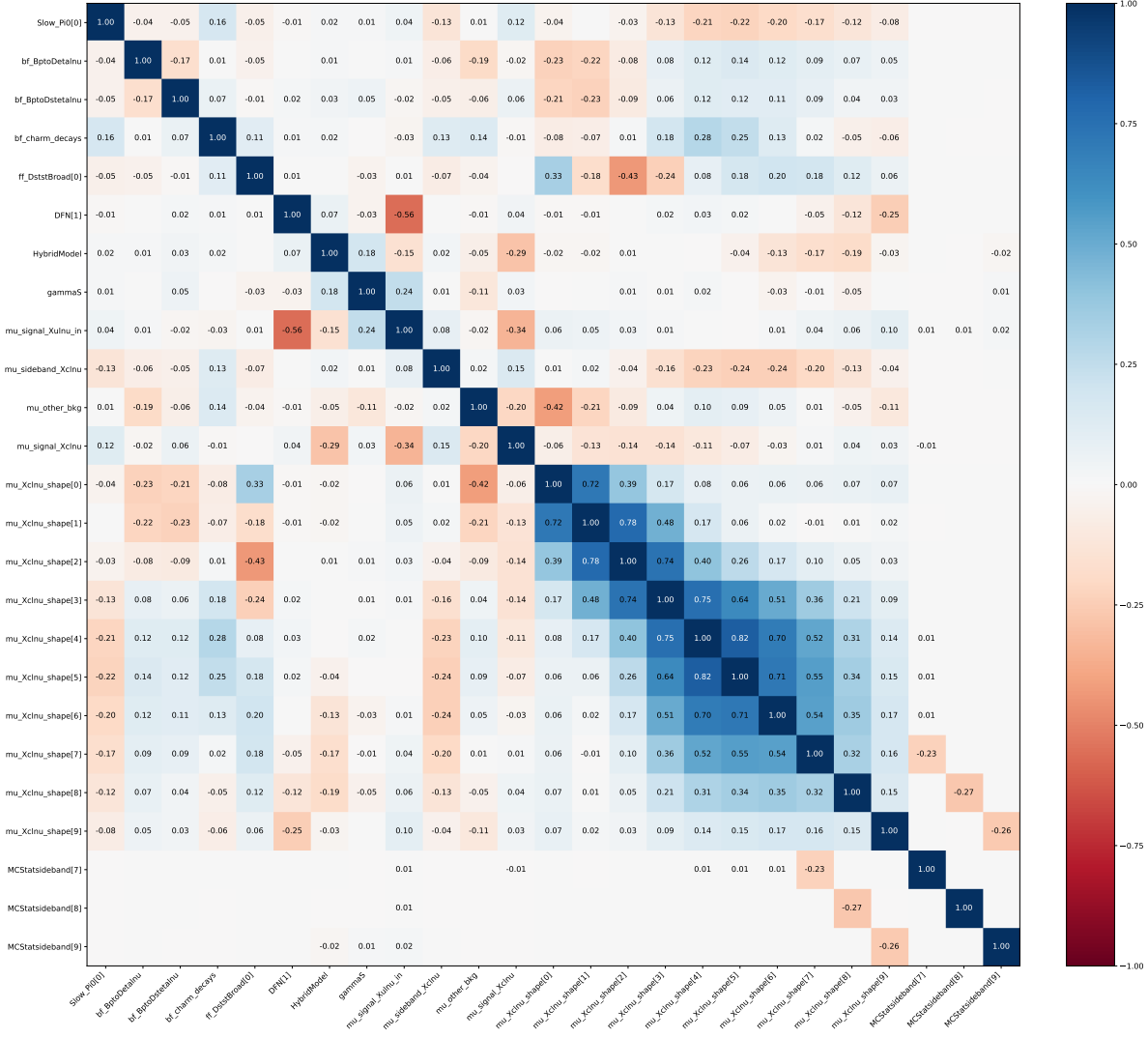


FIG. 91: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal pulls; top right: $X_c \ell \nu$ factor pulls (bin 1); bottom left: $X_c \ell \nu$ factor pulls (bin 9); bottom right: linearity test



2069 2. Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$, $q^2 > 8 \text{ GeV}^2$, Experimental cut: $p_\ell^B > 1.0$
2070 GeV

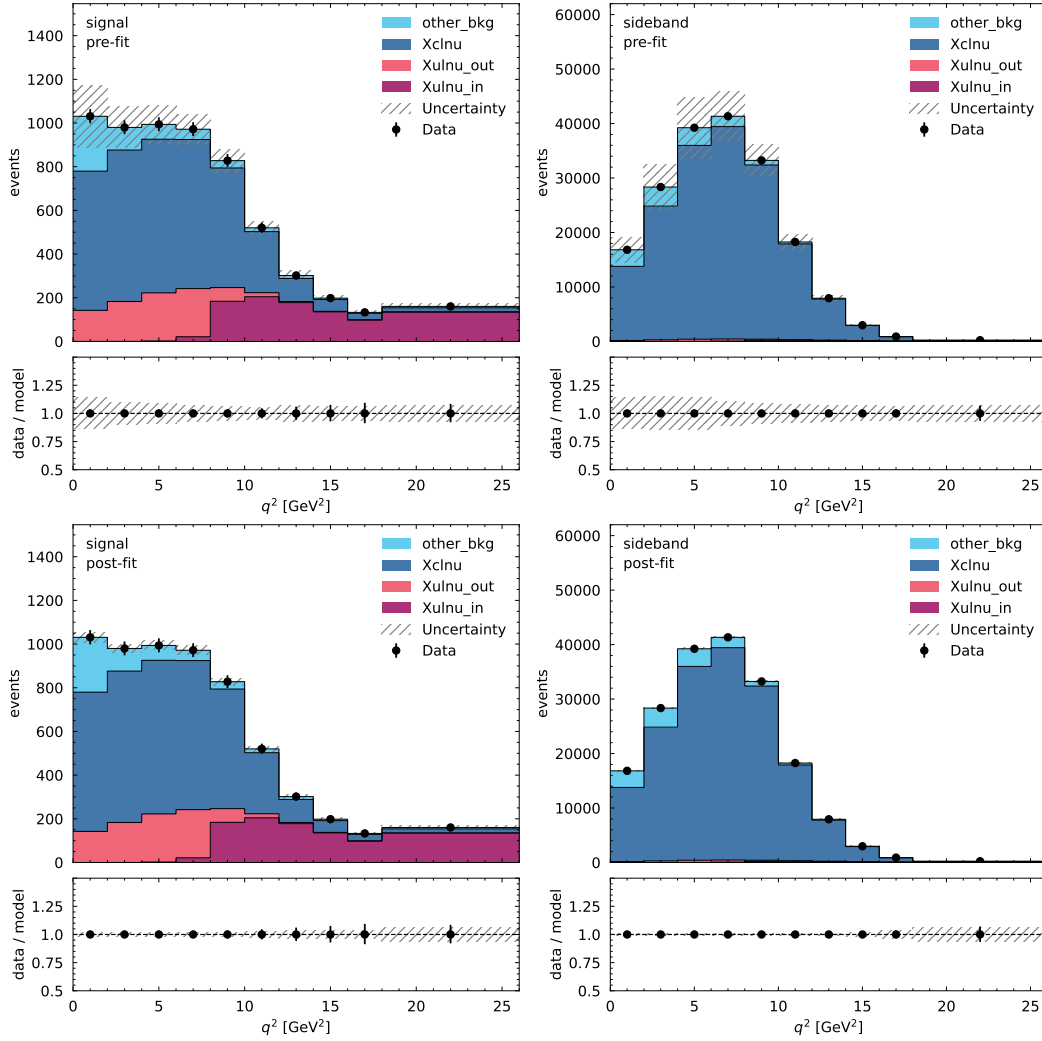


FIG. 93: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$, $q^2 > 8 \text{ GeV}^2$, Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$

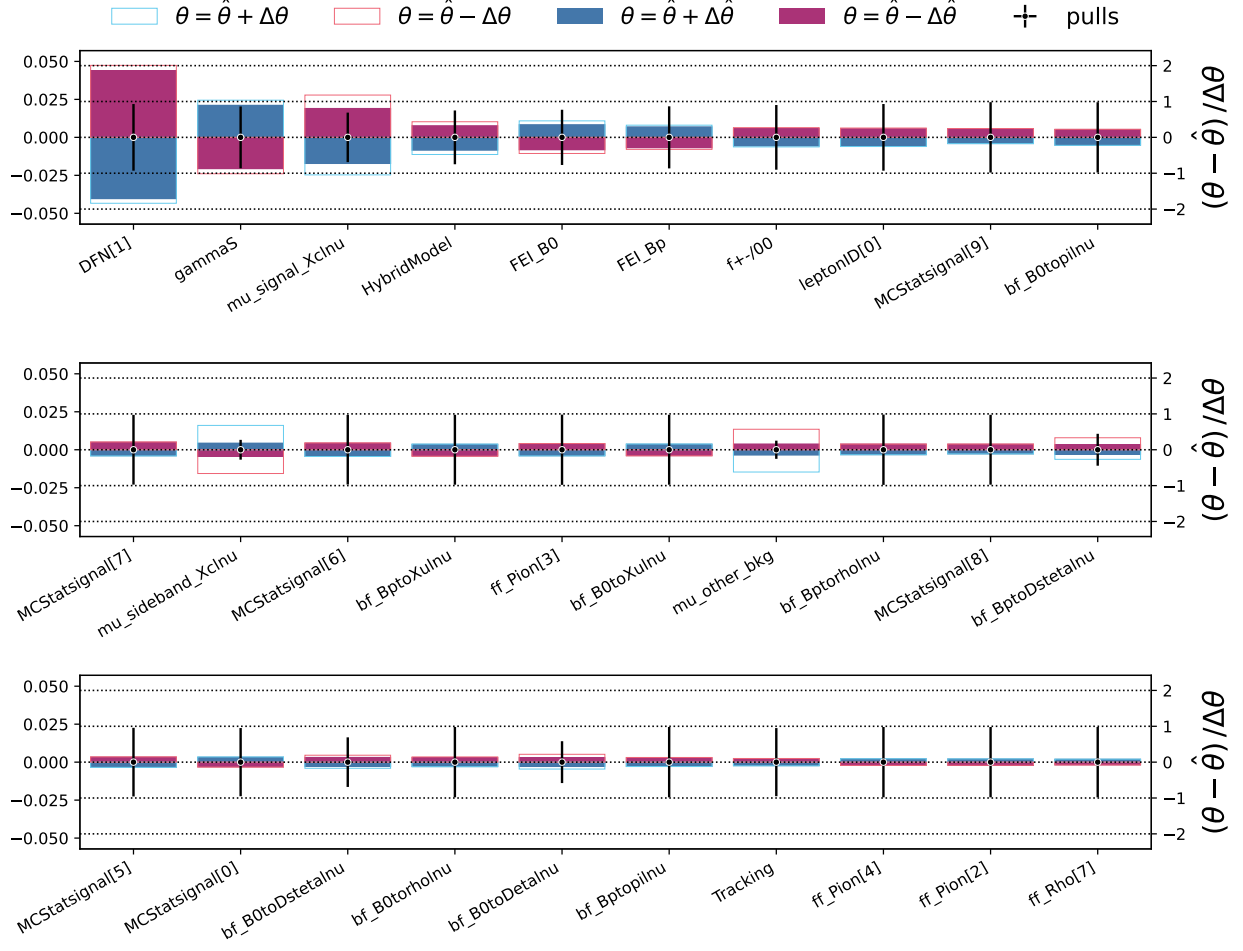


FIG. 94: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV²,
Experimental cut: $p_\ell^B > 1.0$ GeV

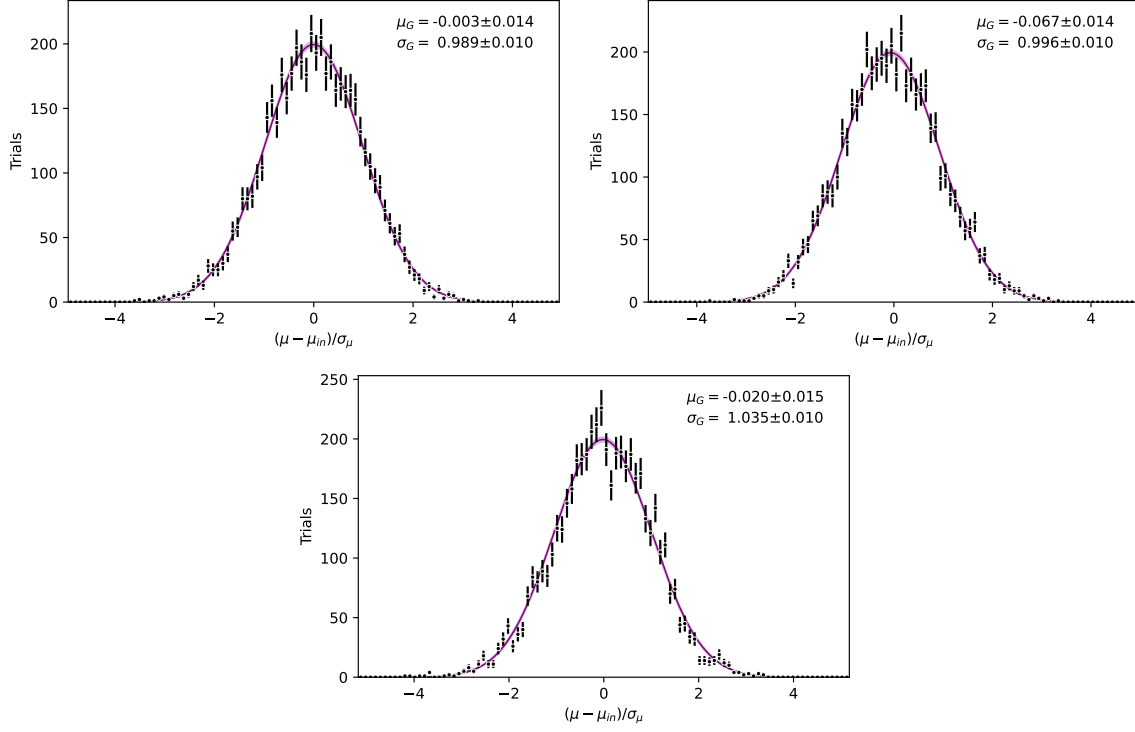
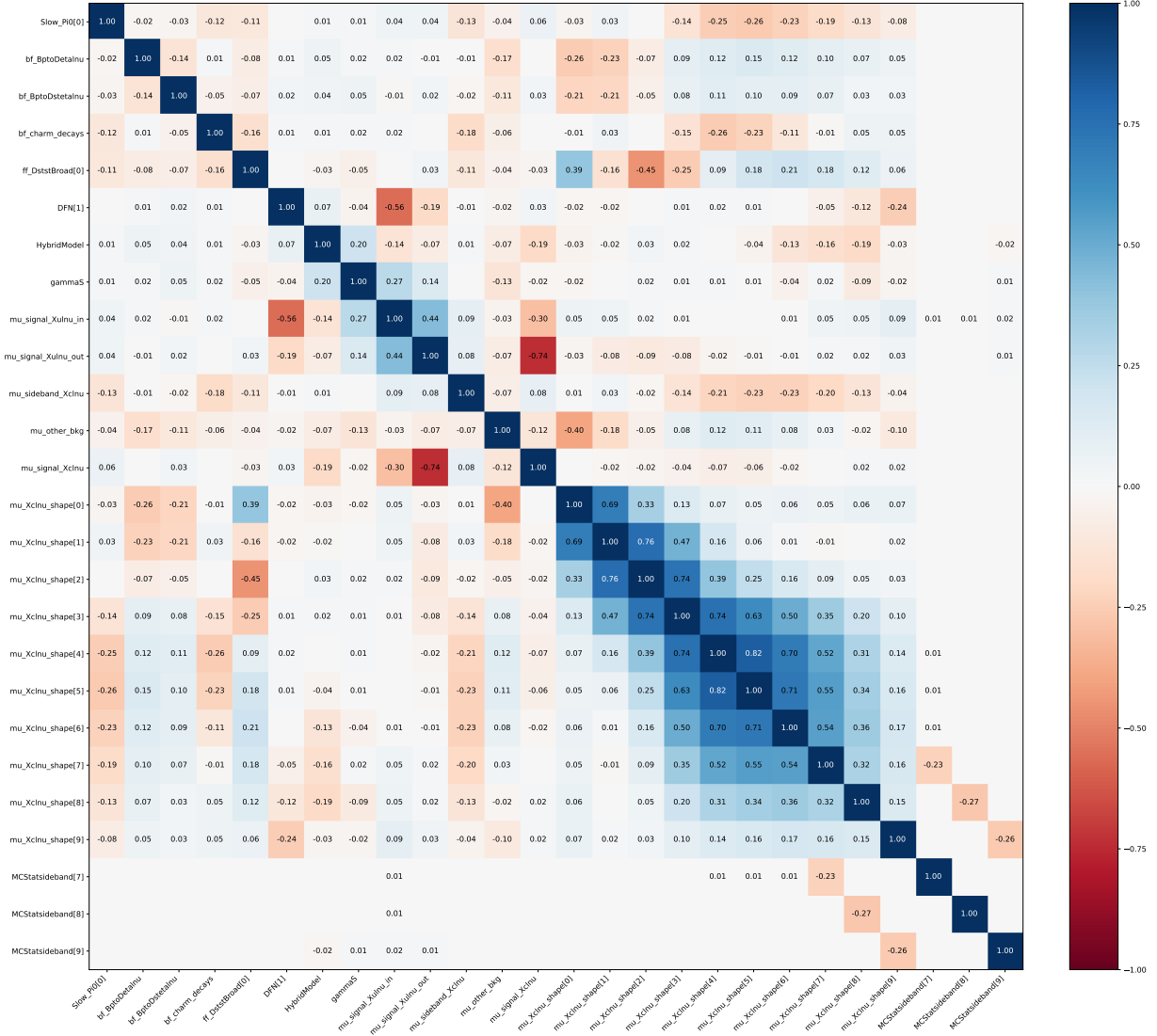


FIG. 95: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV², Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal-in pulls; top right: signal-out pulls; bottom left: $X_{c\ell\nu}$ factor pulls (bin 1); bottom right: $X_{c\ell\nu}$ factor pulls (bin 9)



2071 3. Phase-space region: $p_\ell^B > 2.1 \text{ GeV}$, Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$

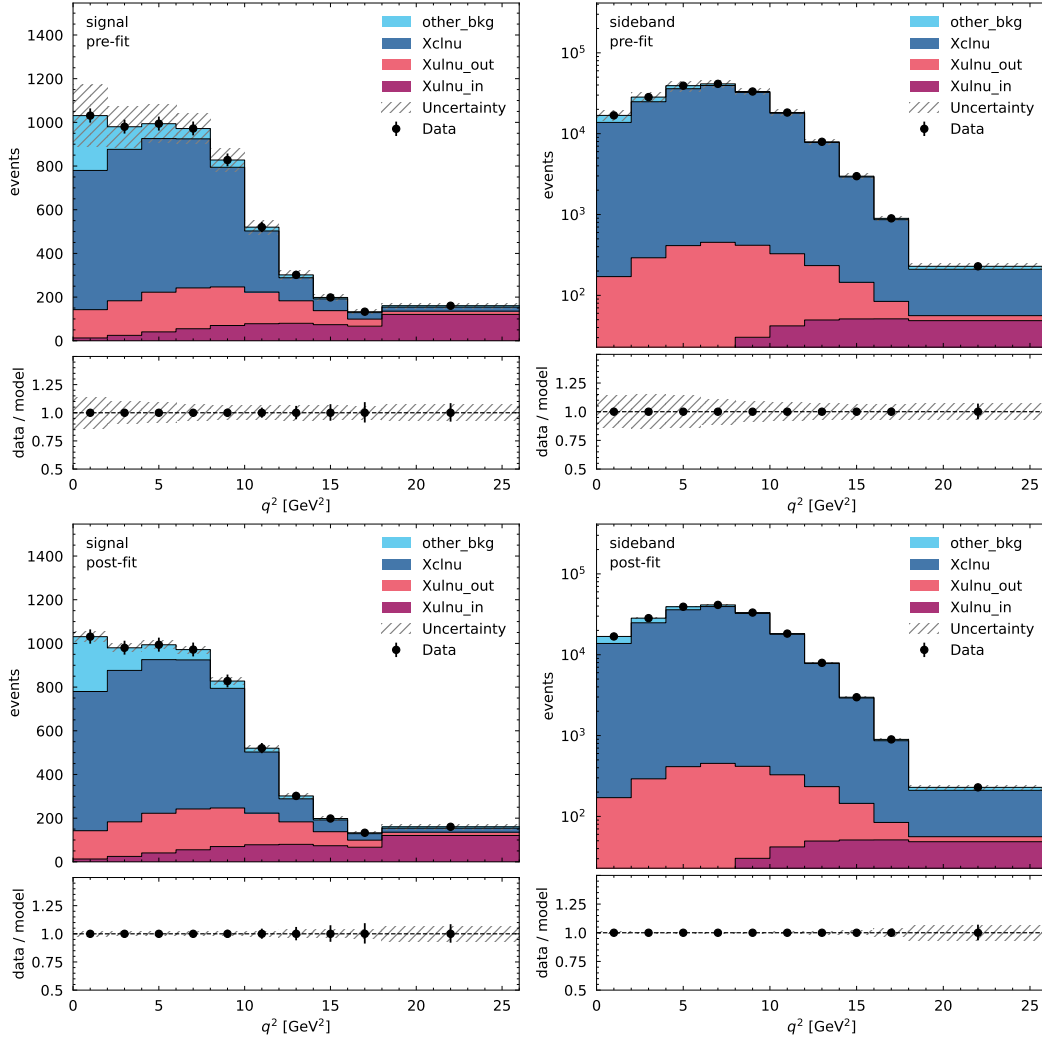


FIG. 97: Variable: q^2 , Phase-space region: $p_\ell^B > 2.1 \text{ GeV}$, Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$

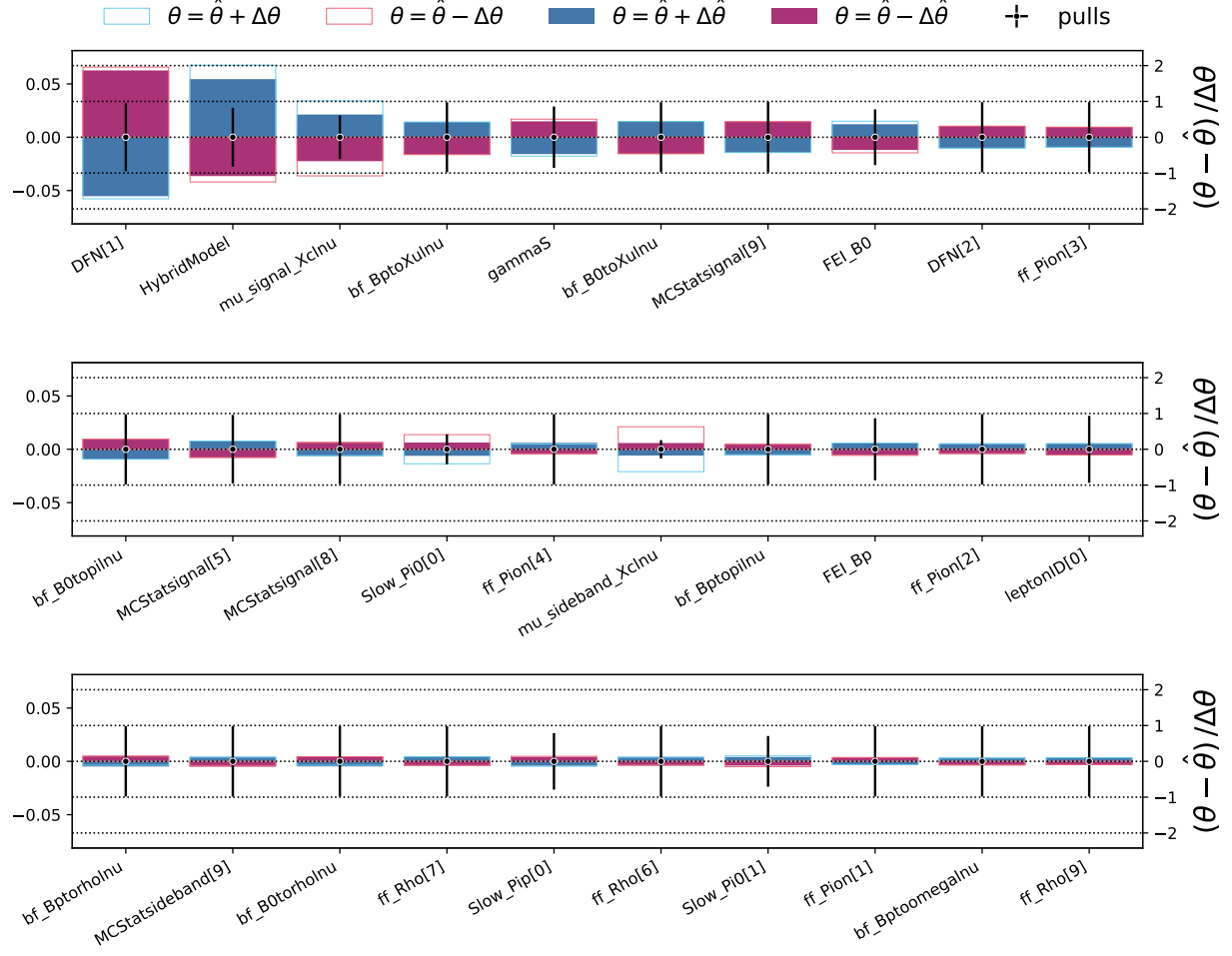


FIG. 98: Variable: q^2 , Phase-space region: $p_\ell^B > 2.1$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

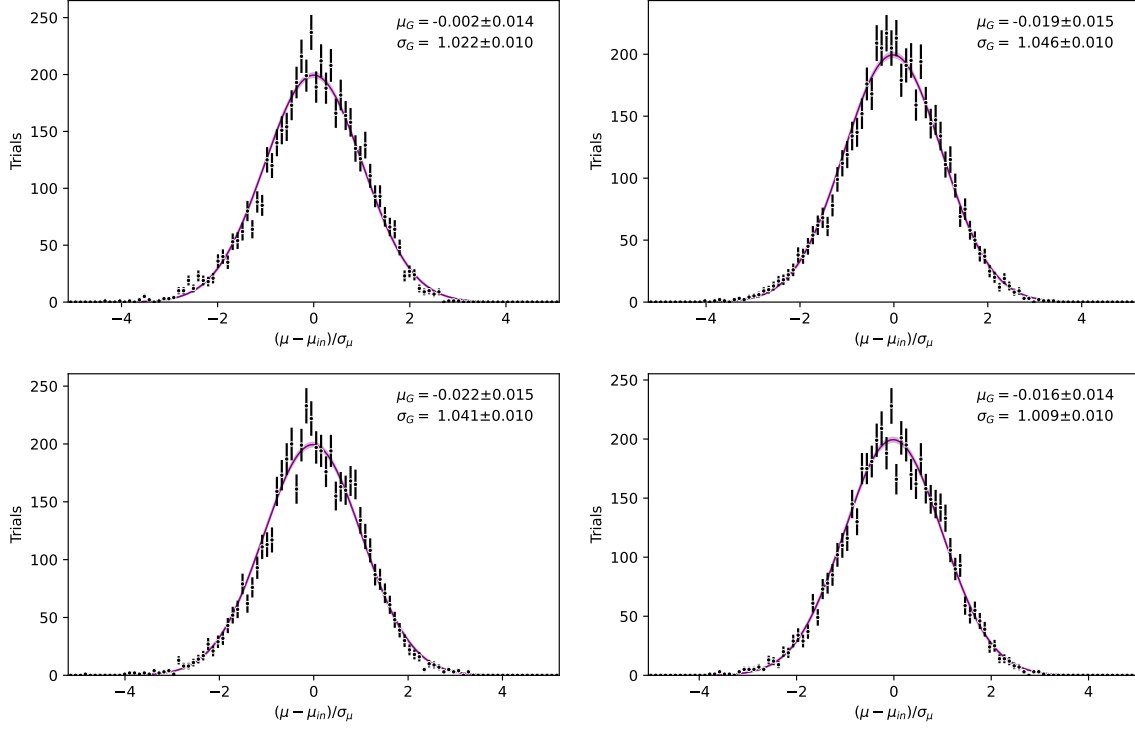
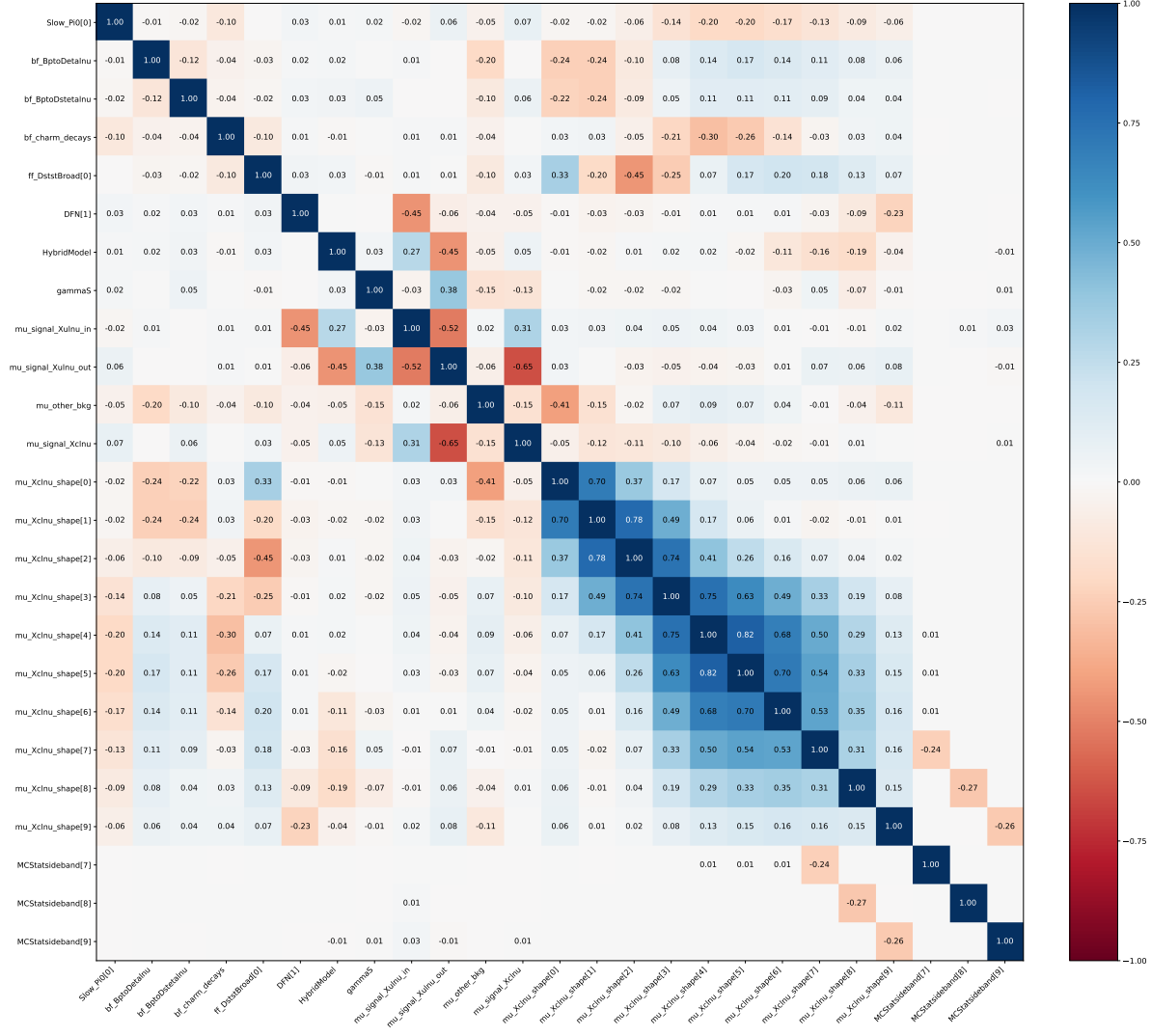


FIG. 99: Variable: q^2 , Phase-space region: $p_\ell^B > 2.1$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal-in pulls; top right: signal-out pulls; bottom left: $X_{c\ell\nu}$ factor pulls (bin 1); bottom right: $X_{c\ell\nu}$ factor pulls (bin 9)



2072 **2. p_ℓ^B fit**

2073 1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

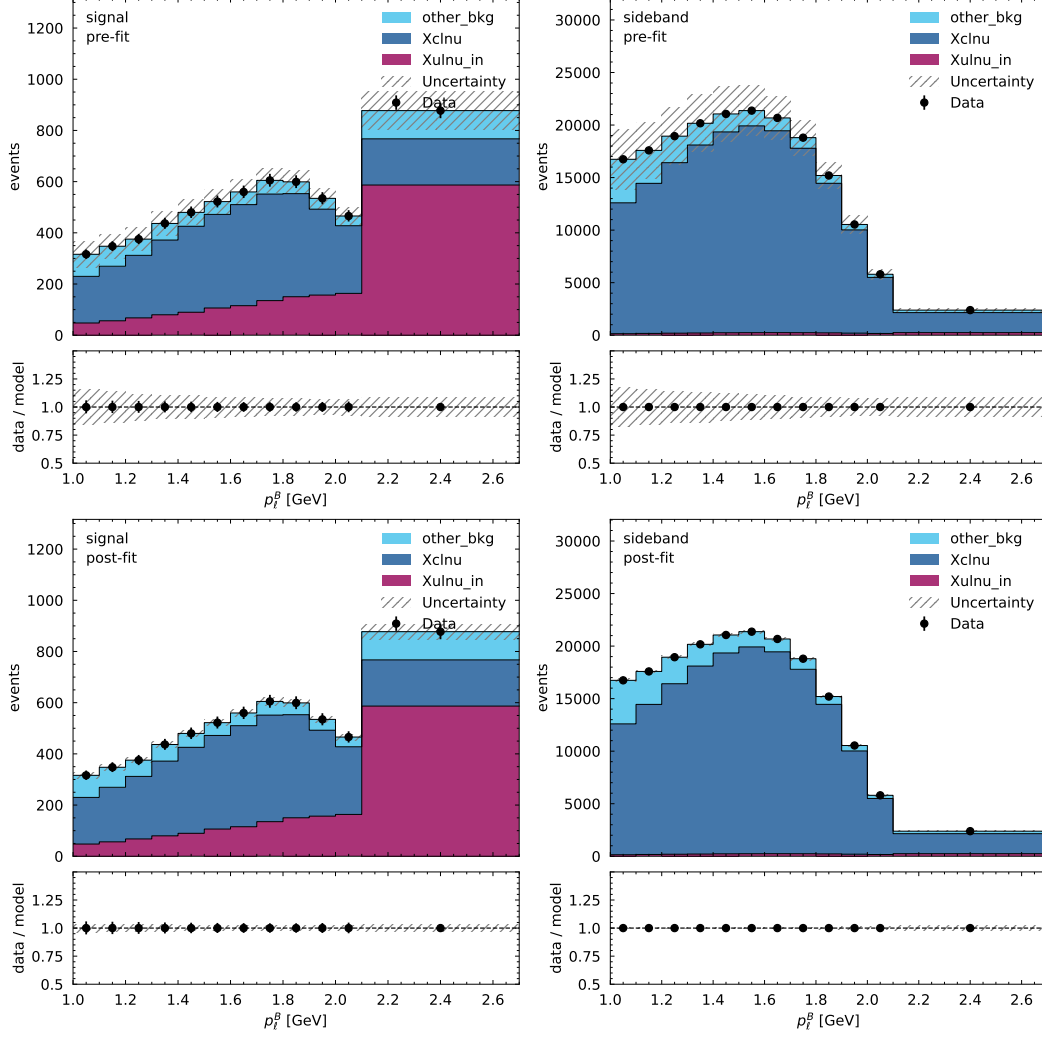


FIG. 101: Variable: p_ℓ^B , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

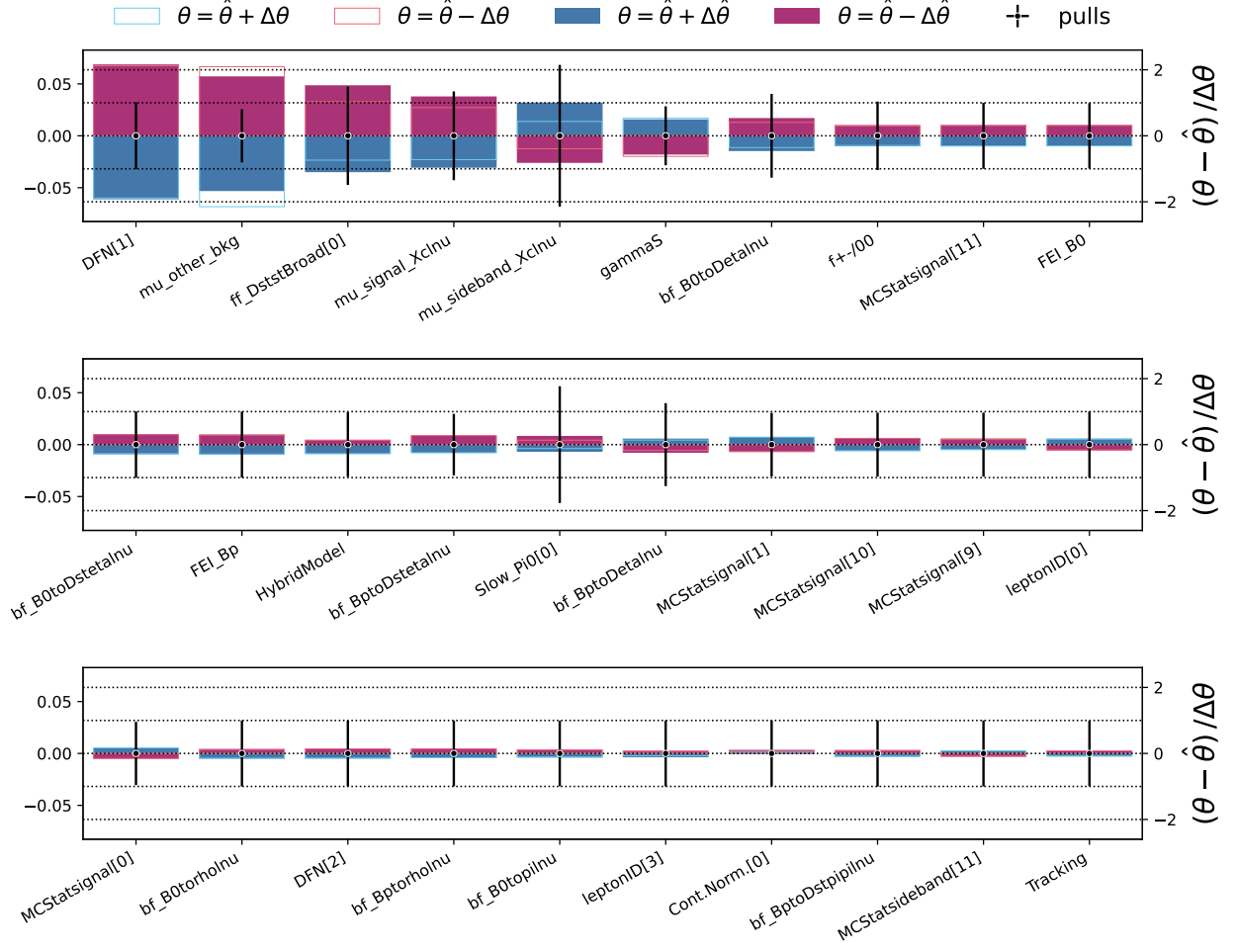


FIG. 102: Variable: p_ℓ^B , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

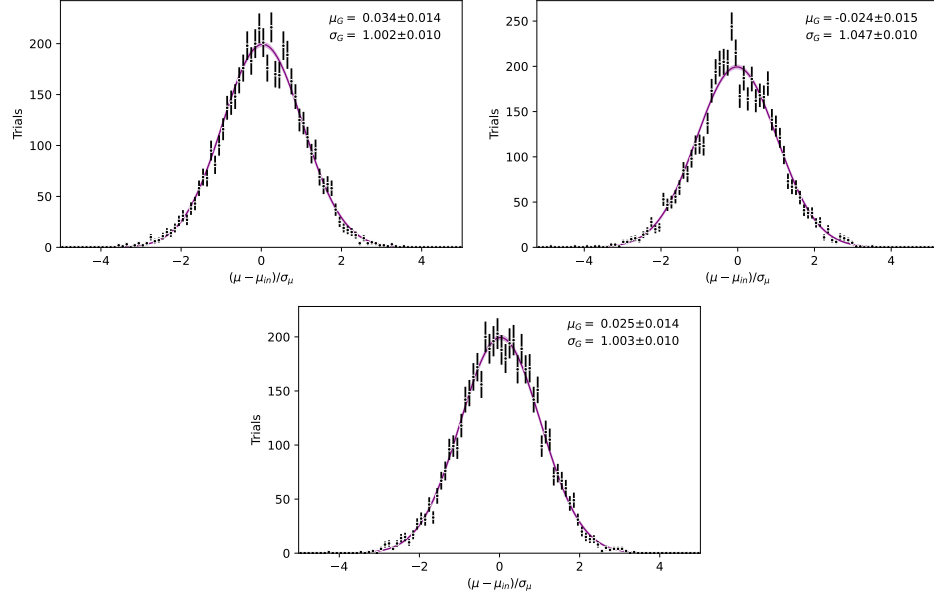
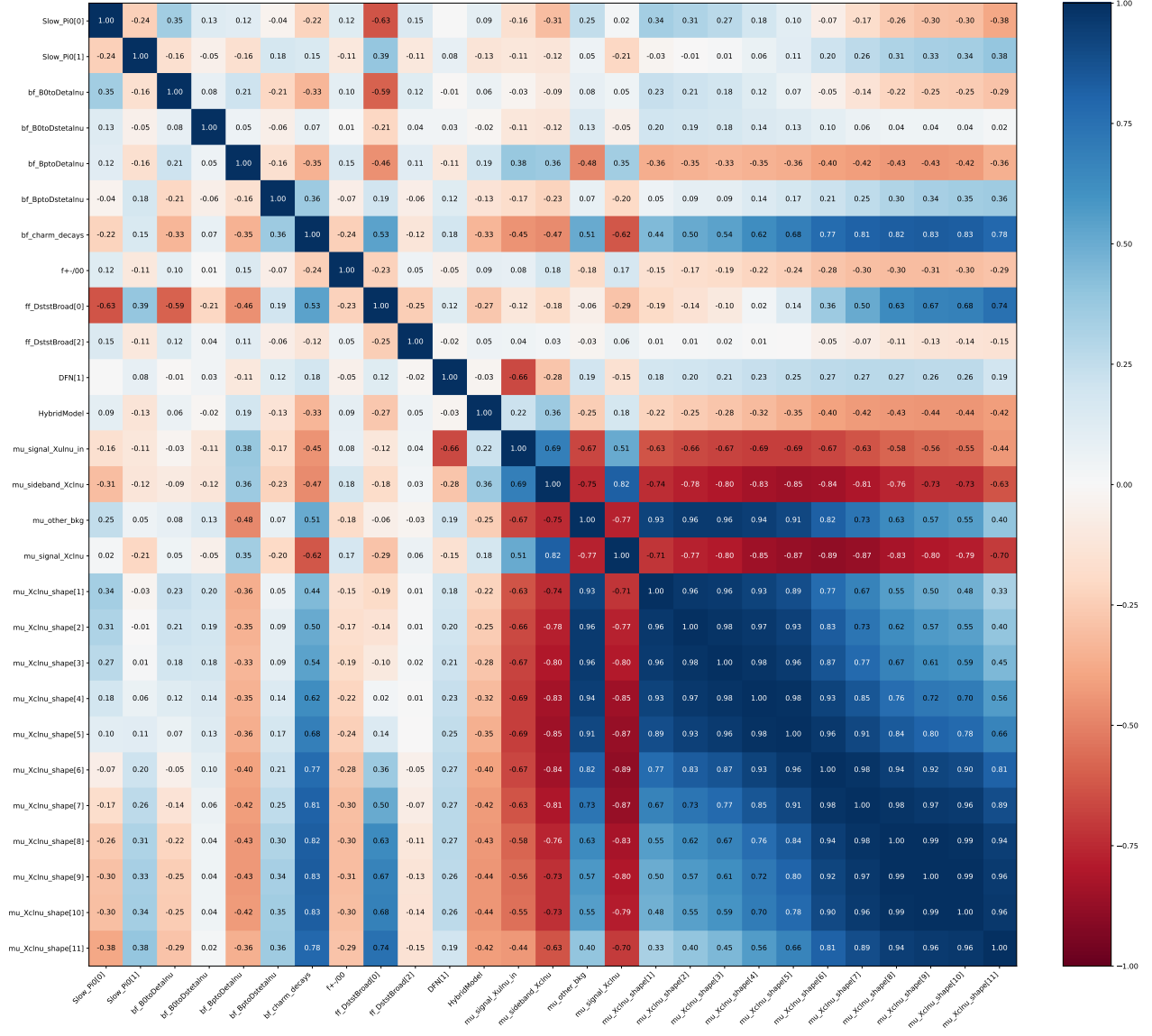


FIG. 103: Variable: p_ℓ^B , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal pulls; top right: $X_c \ell \nu$ factor pulls (bin 1); bottom: $X_c \ell \nu$ factor pulls (bin 11)



2074 2. Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$, $q^2 > 8 \text{ GeV}^2$, Experimental cut: $p_\ell^B > 1.0$
2075 GeV , $M_X < 1.7 \text{ GeV}$, $q^2 > 8 \text{ GeV}^2$

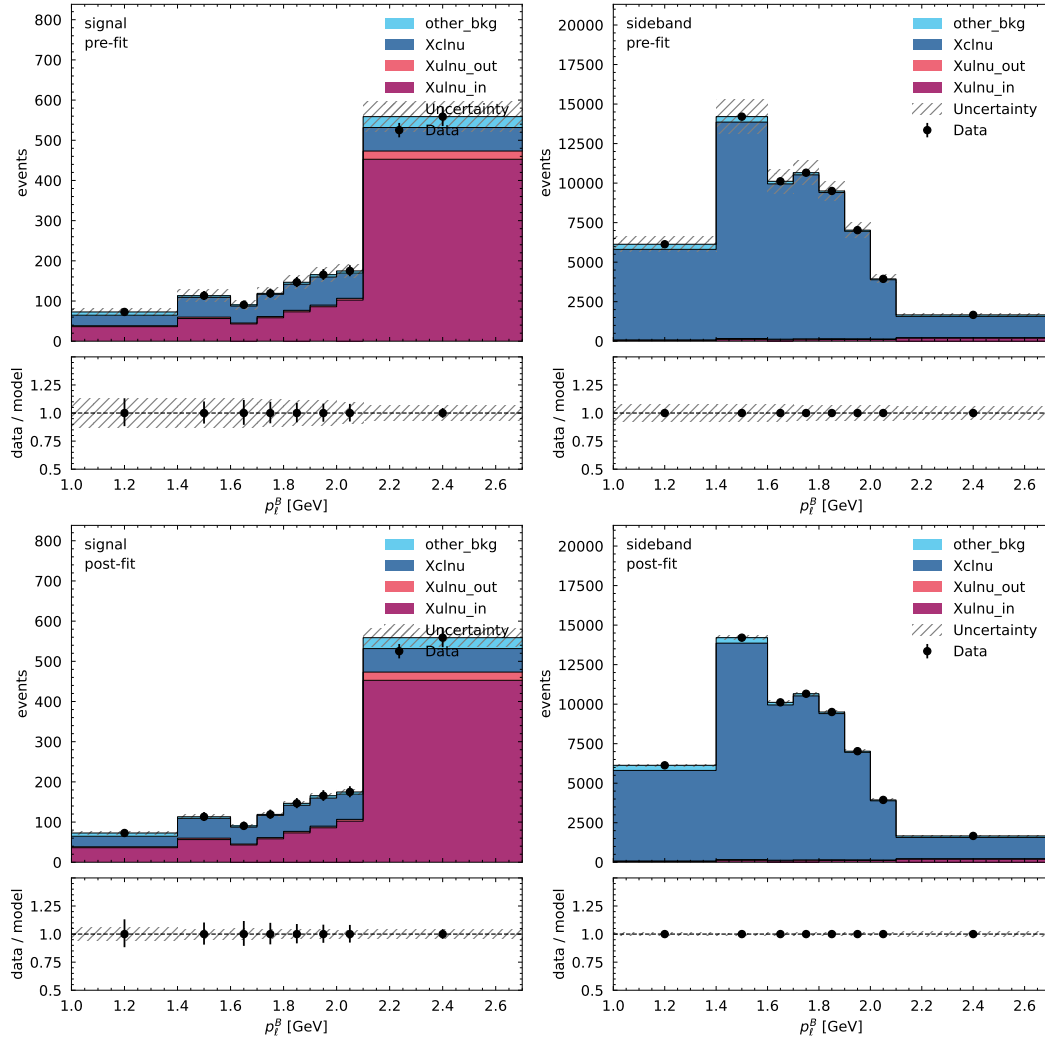


FIG. 105: Variable: p_ℓ^B , Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$, $q^2 > 8 \text{ GeV}^2$,
Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$, $q^2 > 8 \text{ GeV}^2$

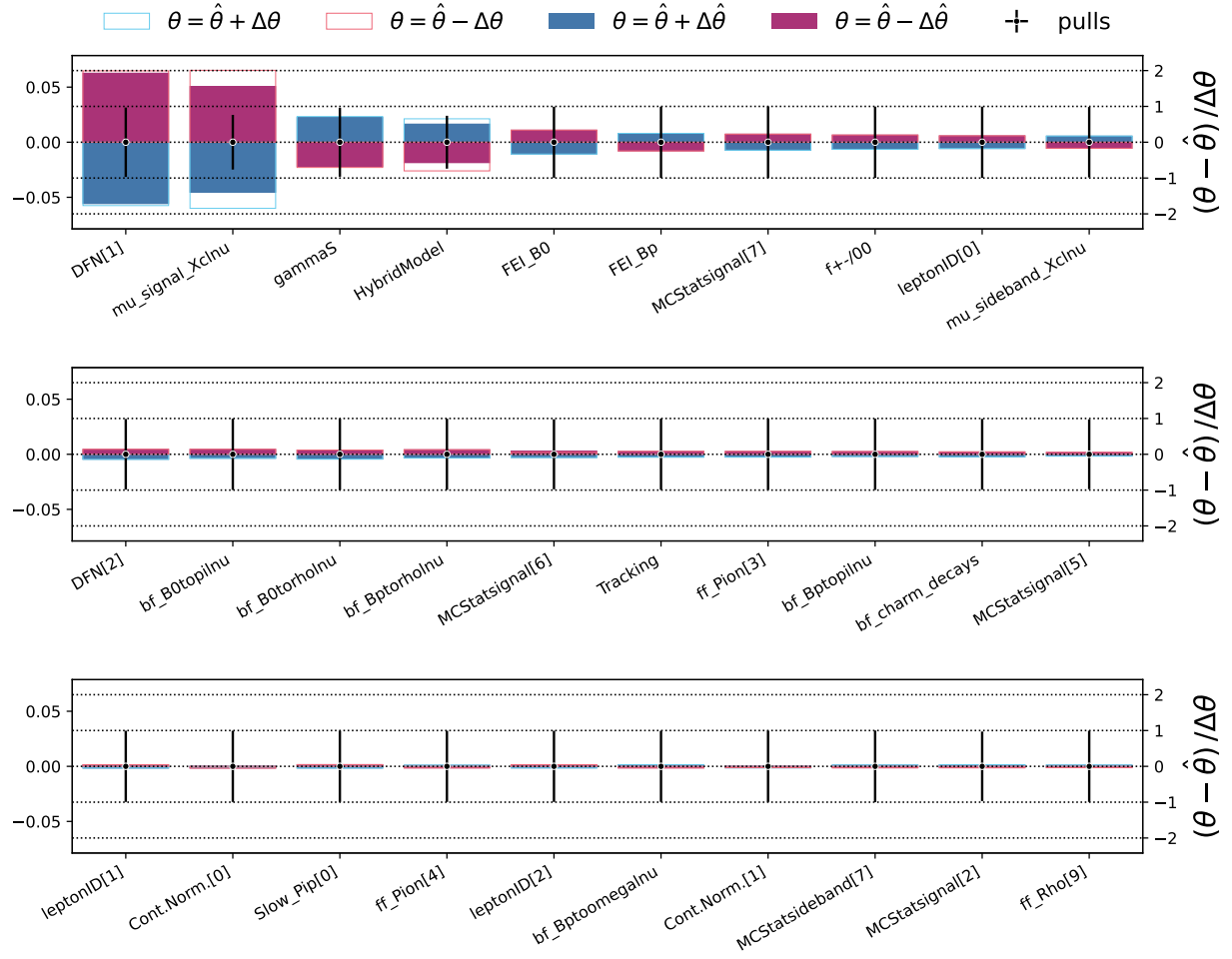


FIG. 106: Variable: p_ℓ^B , Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV²,
Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV²

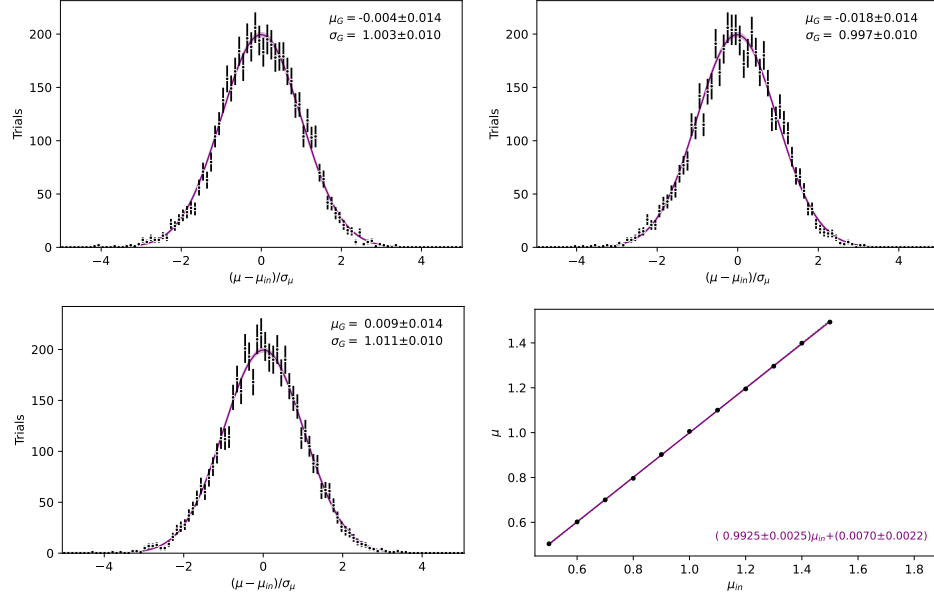


FIG. 107: Variable: p_ℓ^B , Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV², Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV². Top left: signal pulls; top right: $X_c\ell\nu$ factor pulls (bin 1); bottom left: $X_c\ell\nu$ factor pulls (bin 9); bottom right: linearity check

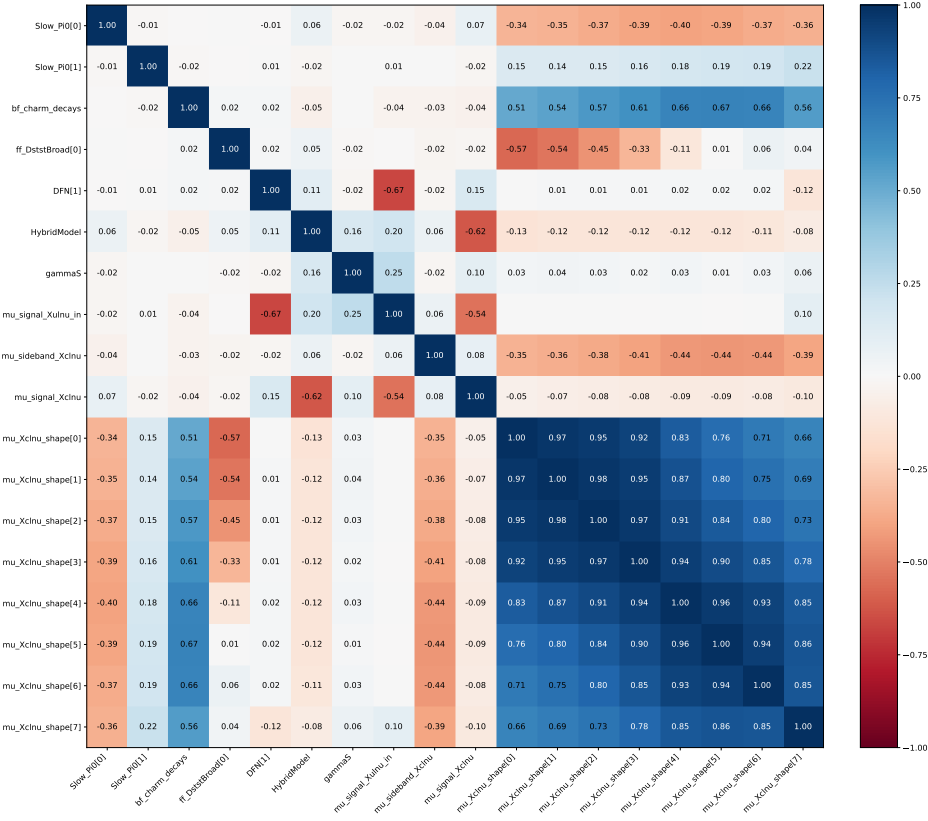


FIG. 108: Variable: p_ℓ^B , Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV², Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, $q^2 > 8$ GeV²

2076 **3. M_X fit**

2077 1. Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$

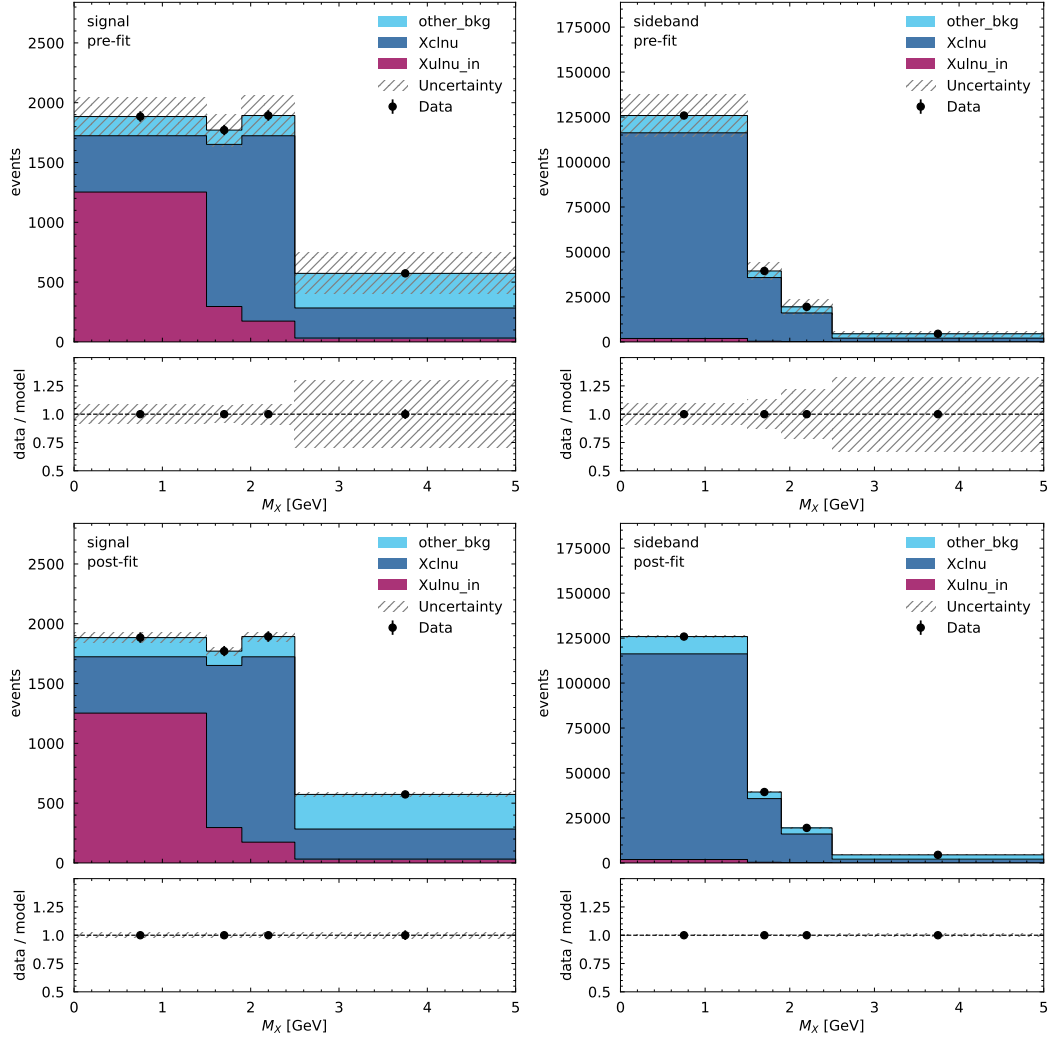


FIG. 109: Variable: M_X , Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$

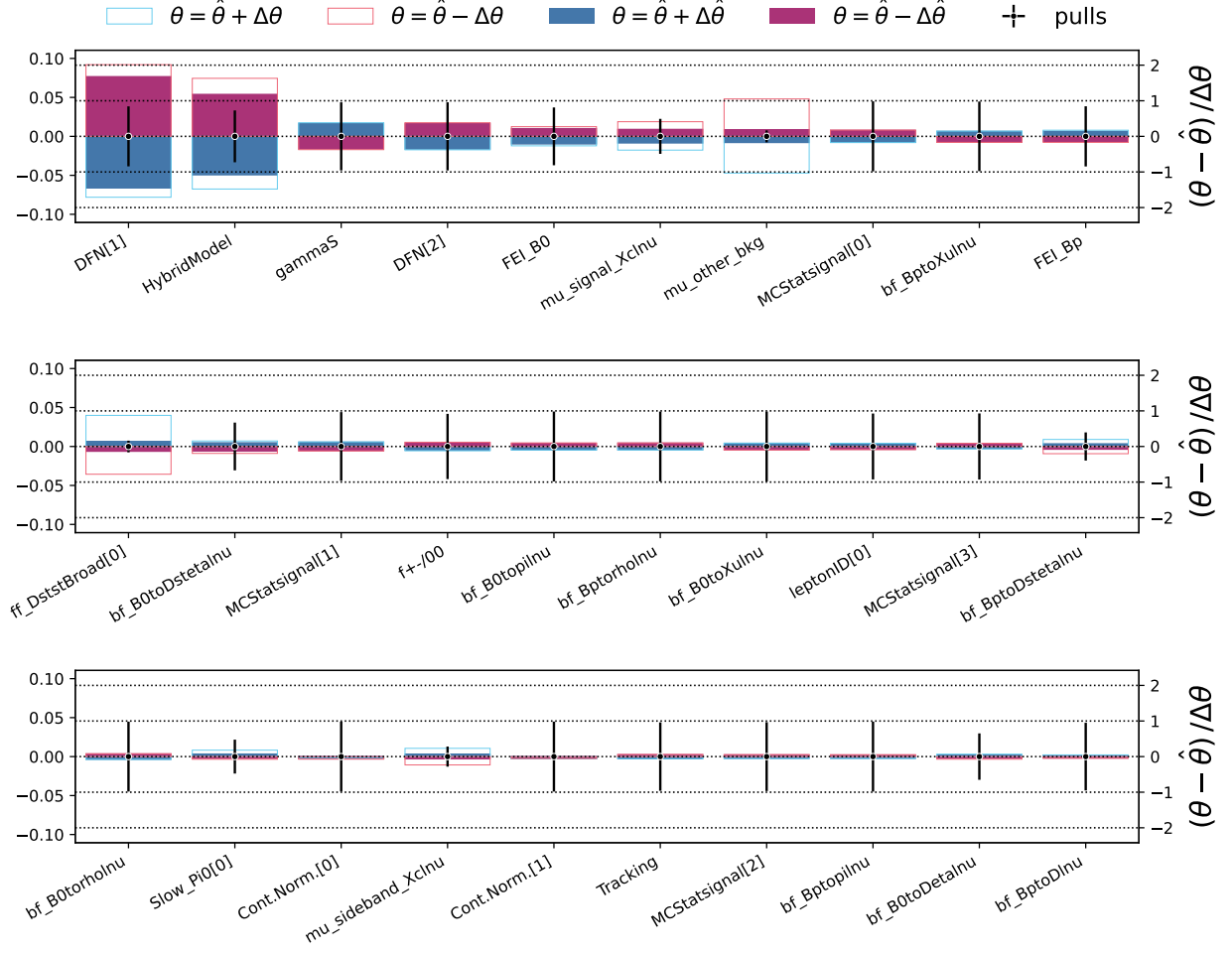


FIG. 110: Variable: M_X , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

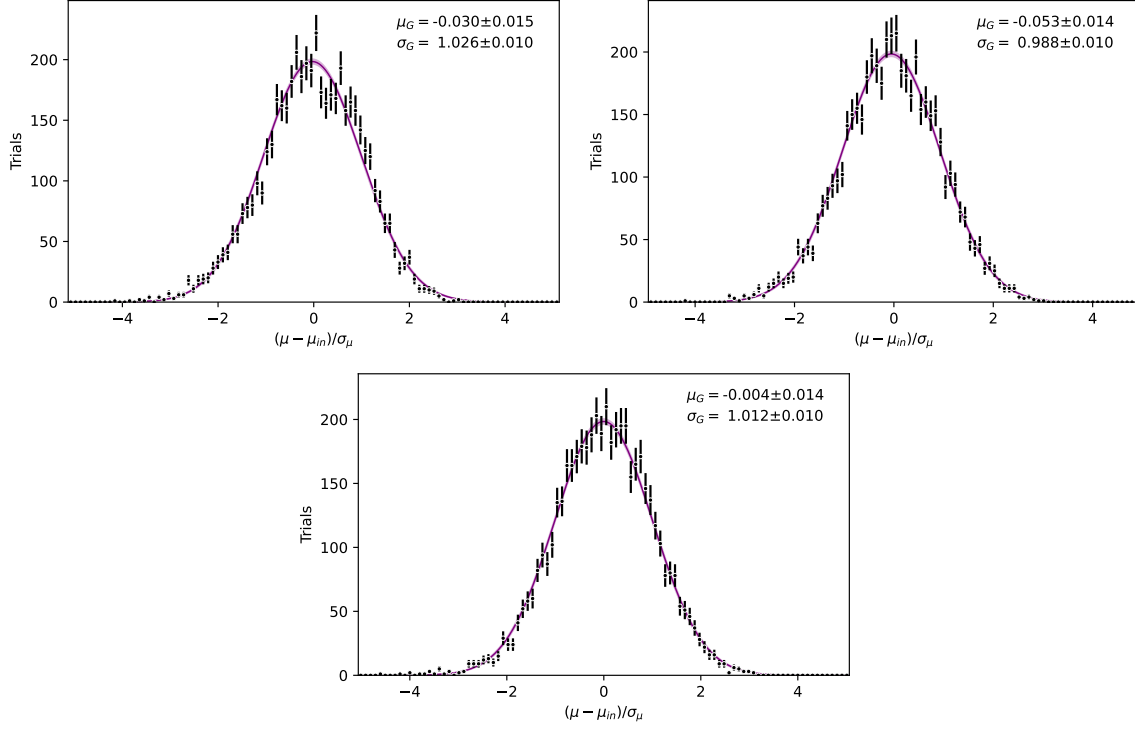


FIG. 111: Variable: M_X , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal pulls; top right: $X_{\ell}\nu$ factor pulls (bin 1); bottom: $X_{\ell}\nu$ factor pulls (bin 3)

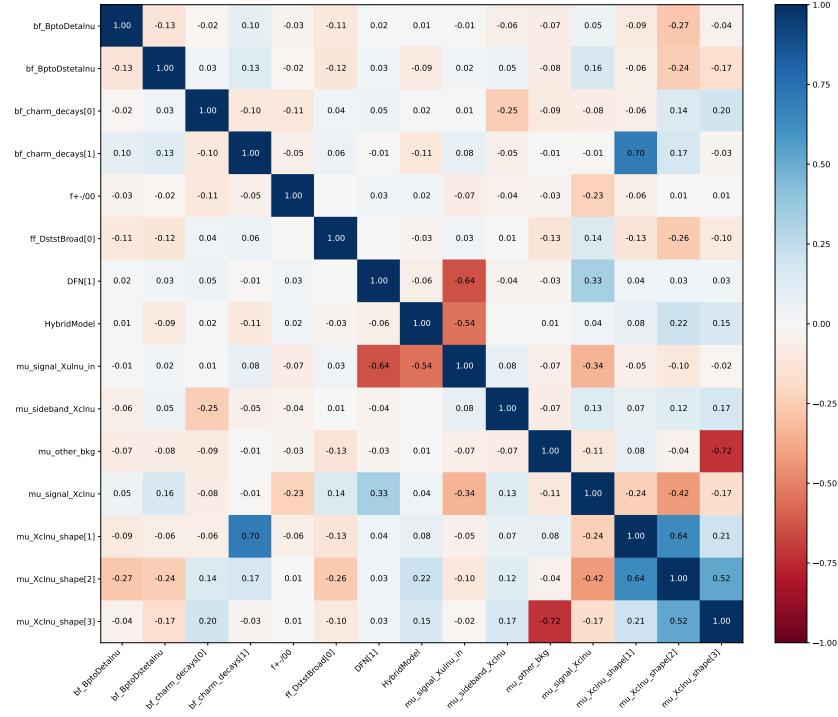


FIG. 112: Variable: M_X , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

2078 **4. $M_X:q^2$ fit**

2079 The two-dimensional bins of $M_X:q^2$ are flattened resulting in a total of 17 bins simply labelled
 2080 1 to 17.

2081 1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

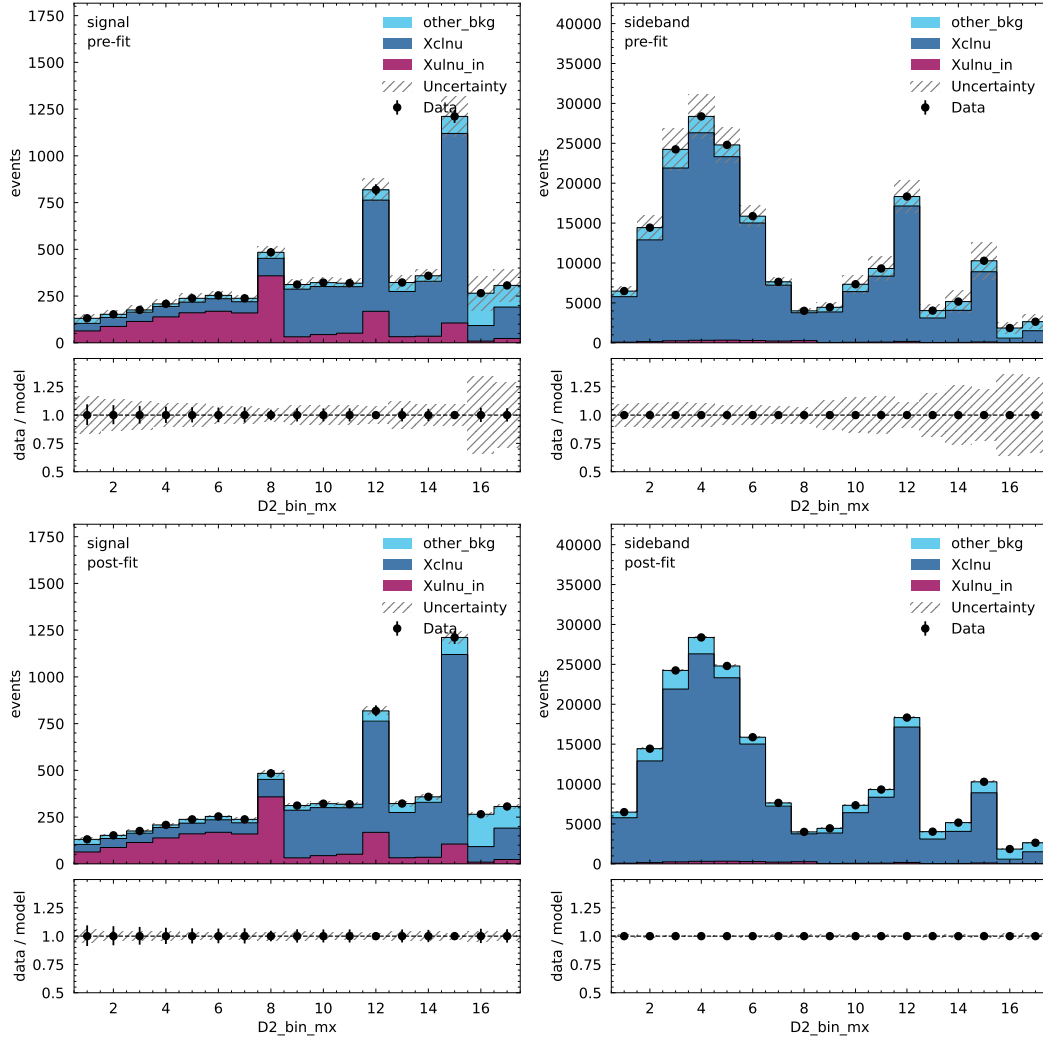


FIG. 113: Variable: $M_X:q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

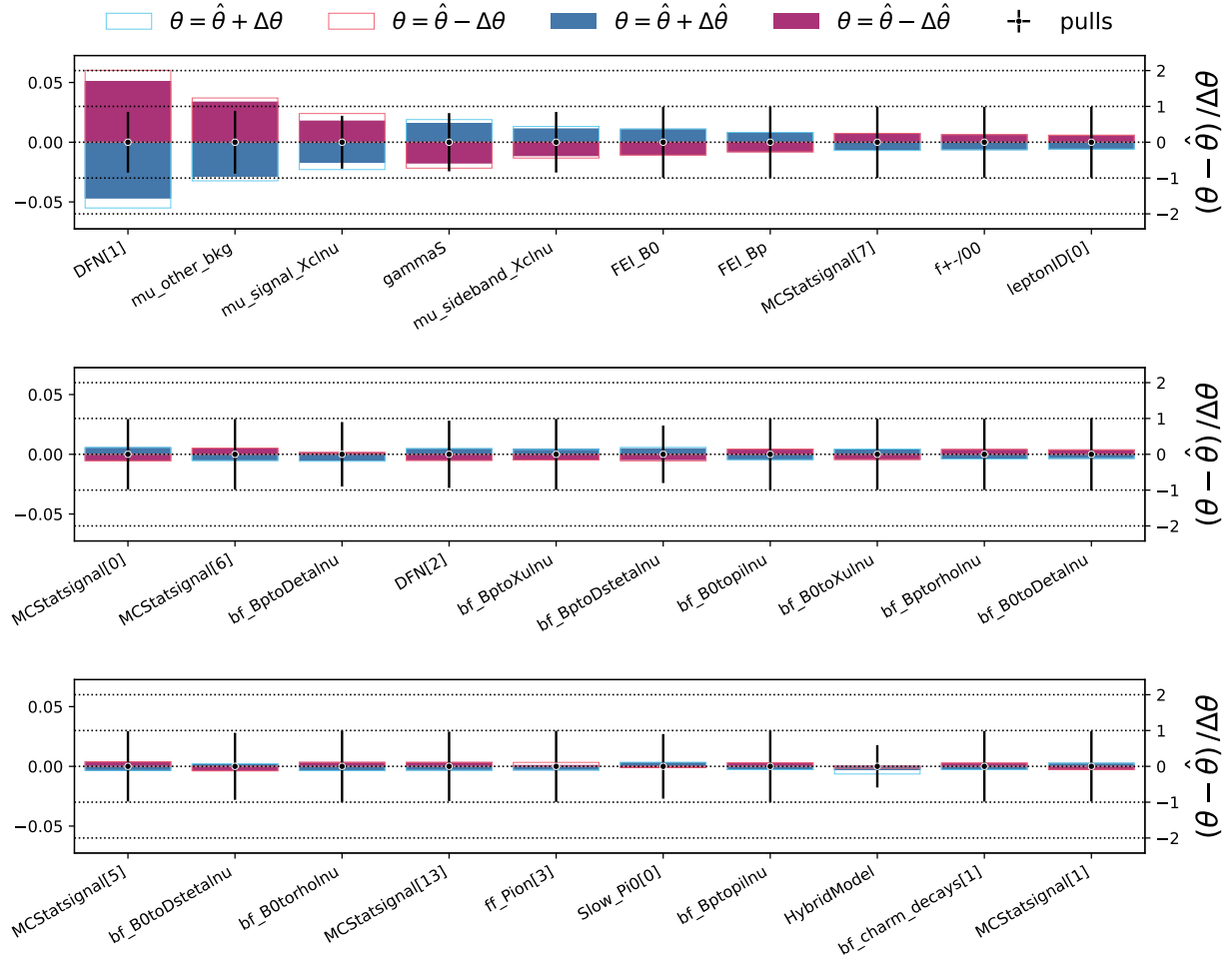


FIG. 114: Variable: $M_X:q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

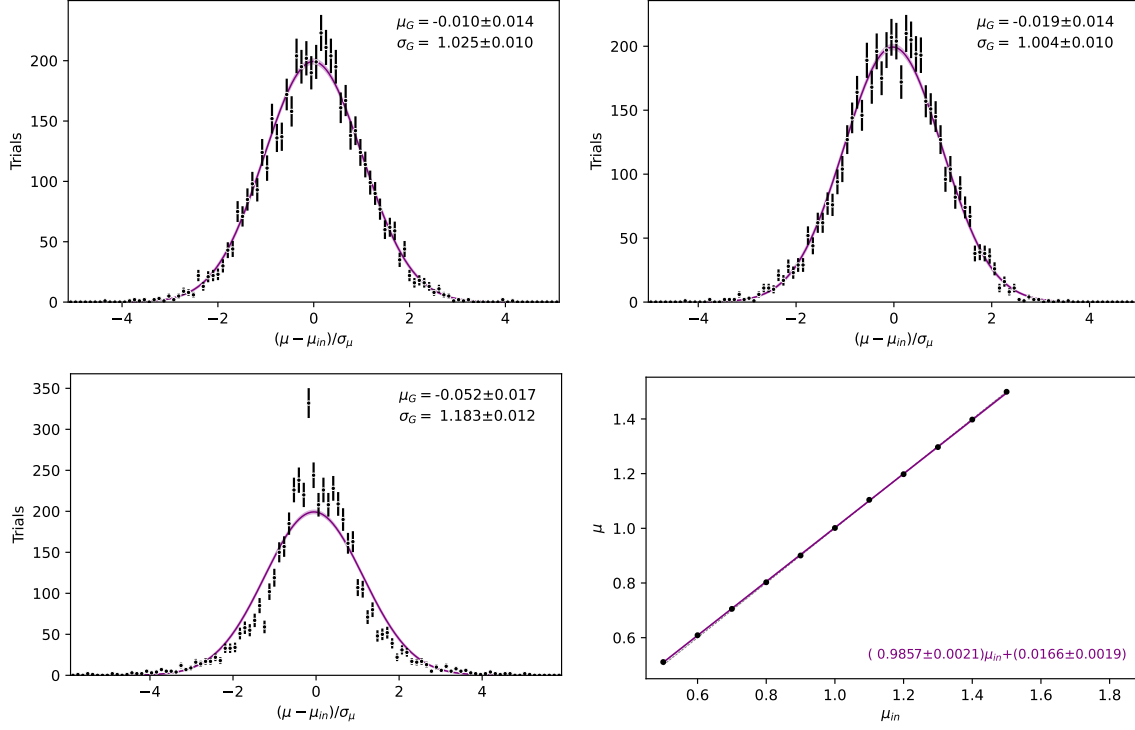
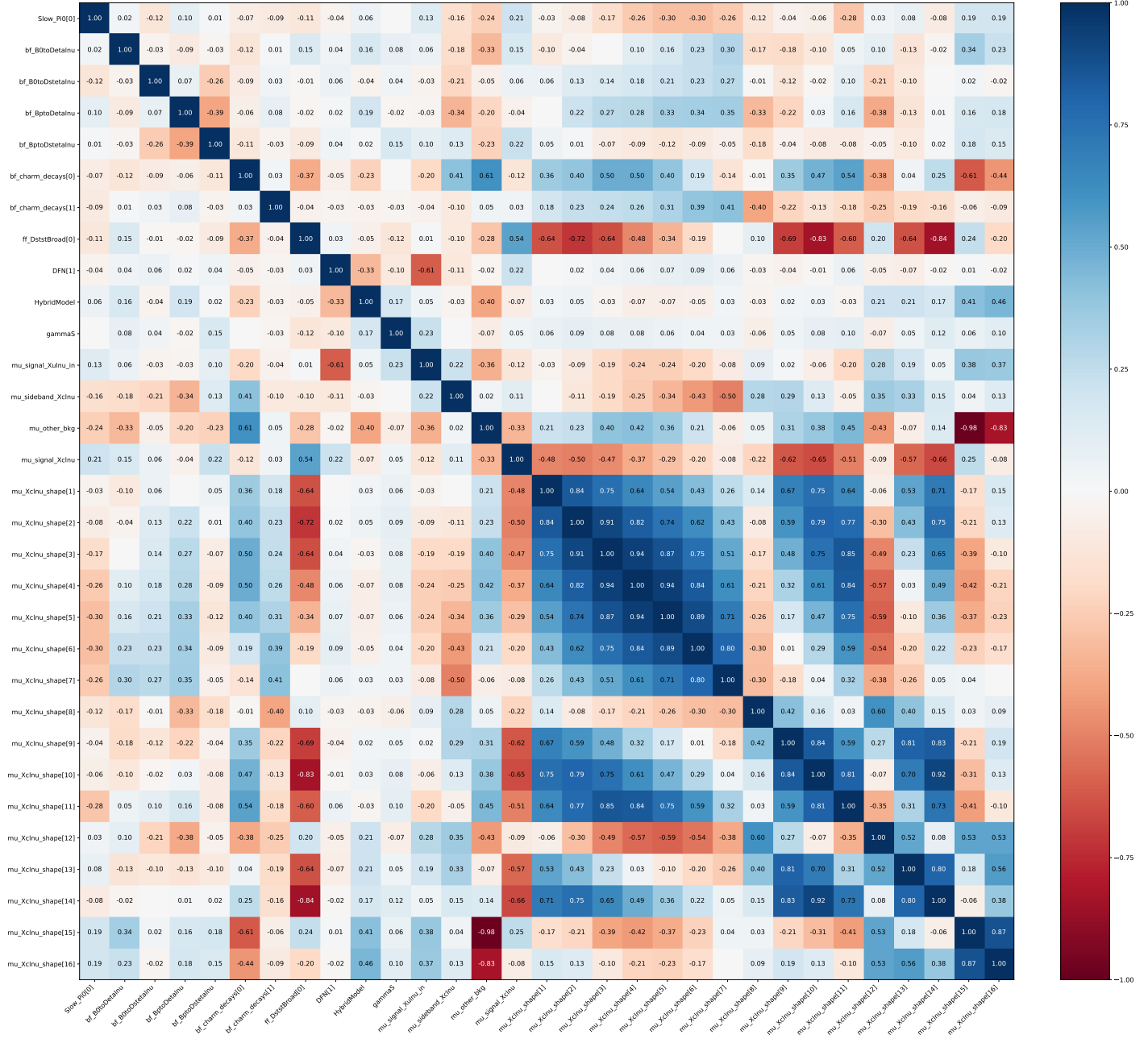


FIG. 115: Variable: $M_X:q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal pulls; top right: $X_{c\ell\nu}$ factor pulls (bin 1); bottom left: $X_{c\ell\nu}$ factor pulls (bin 16); bottom right: linearity test



2082 **5. $p_\ell^B : q^2$ fit**

2083 The two-dimensional bins of $p_\ell^B : q^2$ are flattened resulting in a total of 25 bins simply labelled
 2084 1 to 25.

2085 1. Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

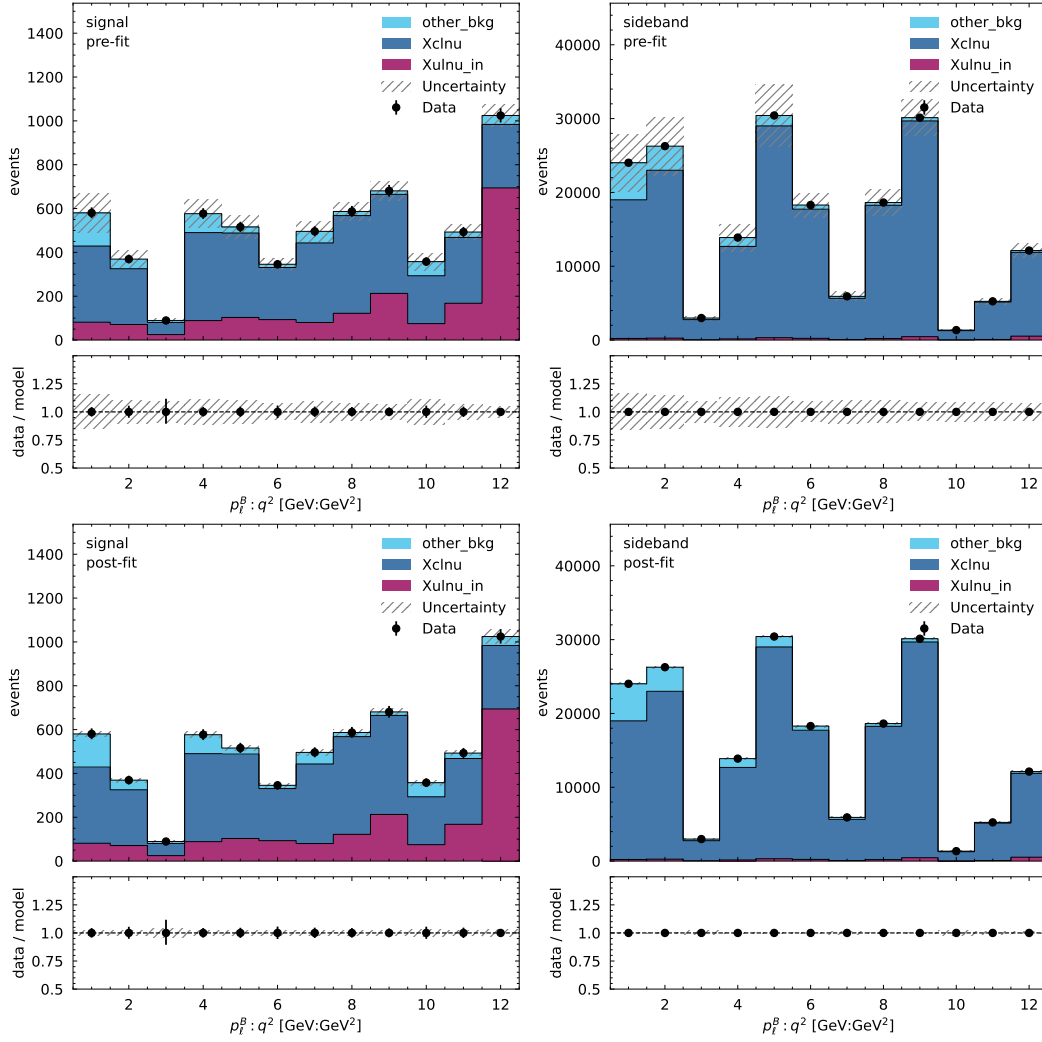


FIG. 117: Variable: $p_\ell^B : q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

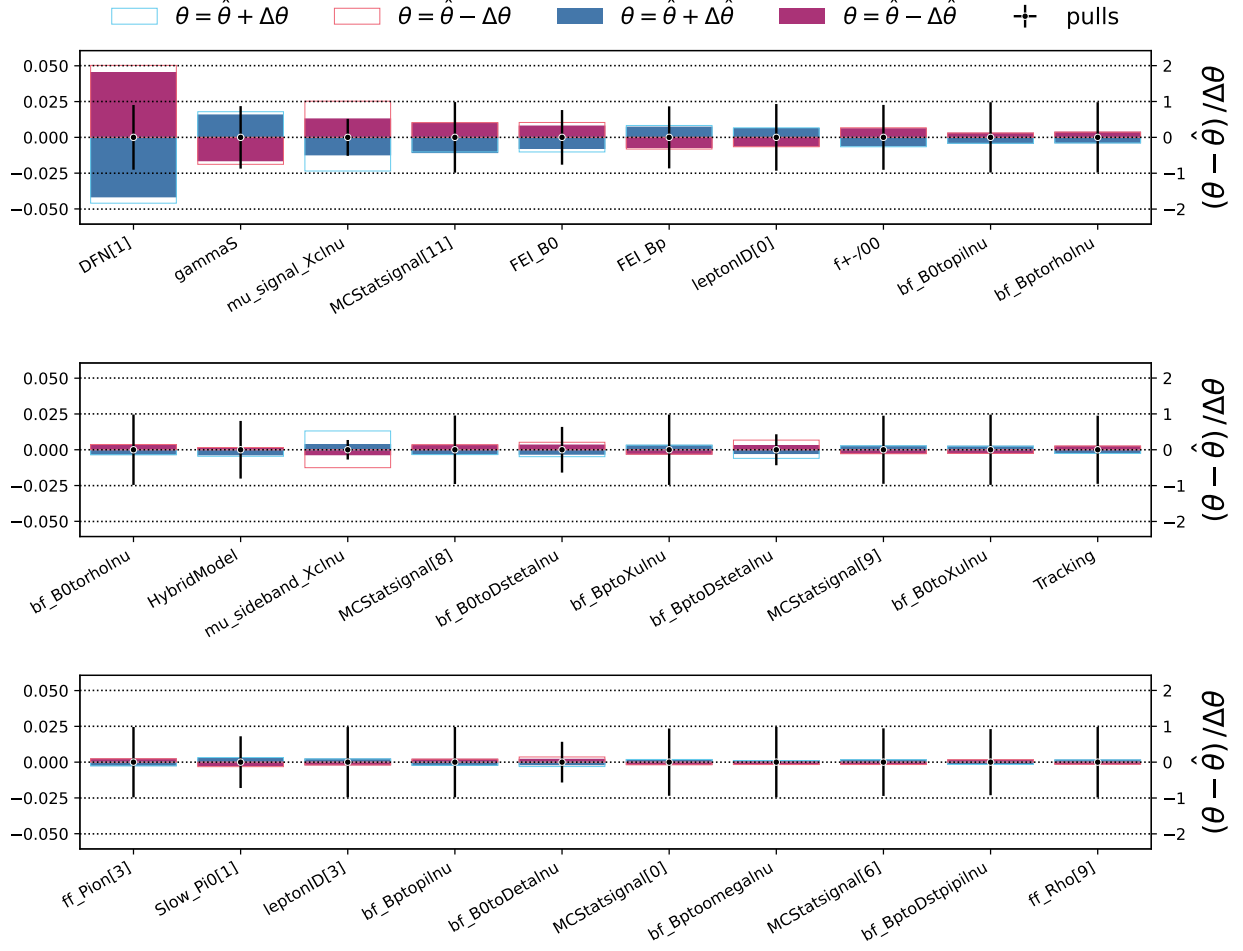


FIG. 118: Variable: $p_\ell^B:q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

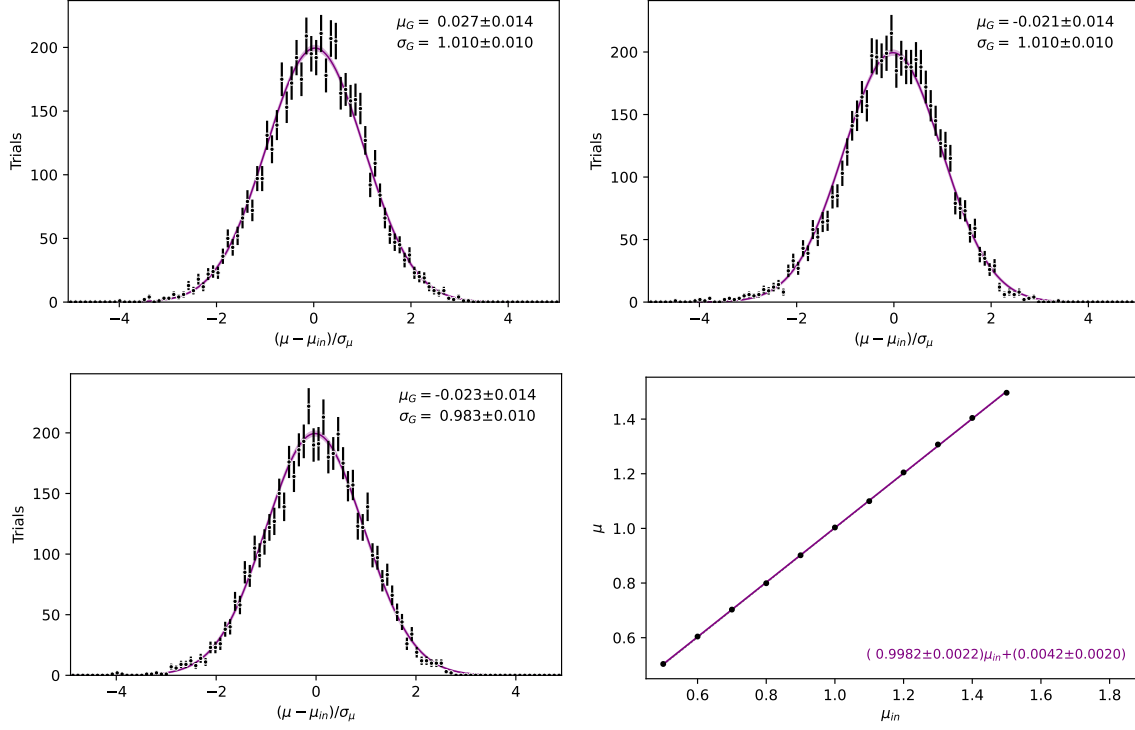


FIG. 119: Variable: $p_\ell^B:q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal pulls; top right: $X_{e\ell\nu}$ factors pulls (bin 1); bottom left: $X_{e\ell\nu}$ factor pulls (bin 24); bottom right: linearity test



2086 2. Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$, Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$,
 2087 $M_X < 1.7 \text{ GeV}$

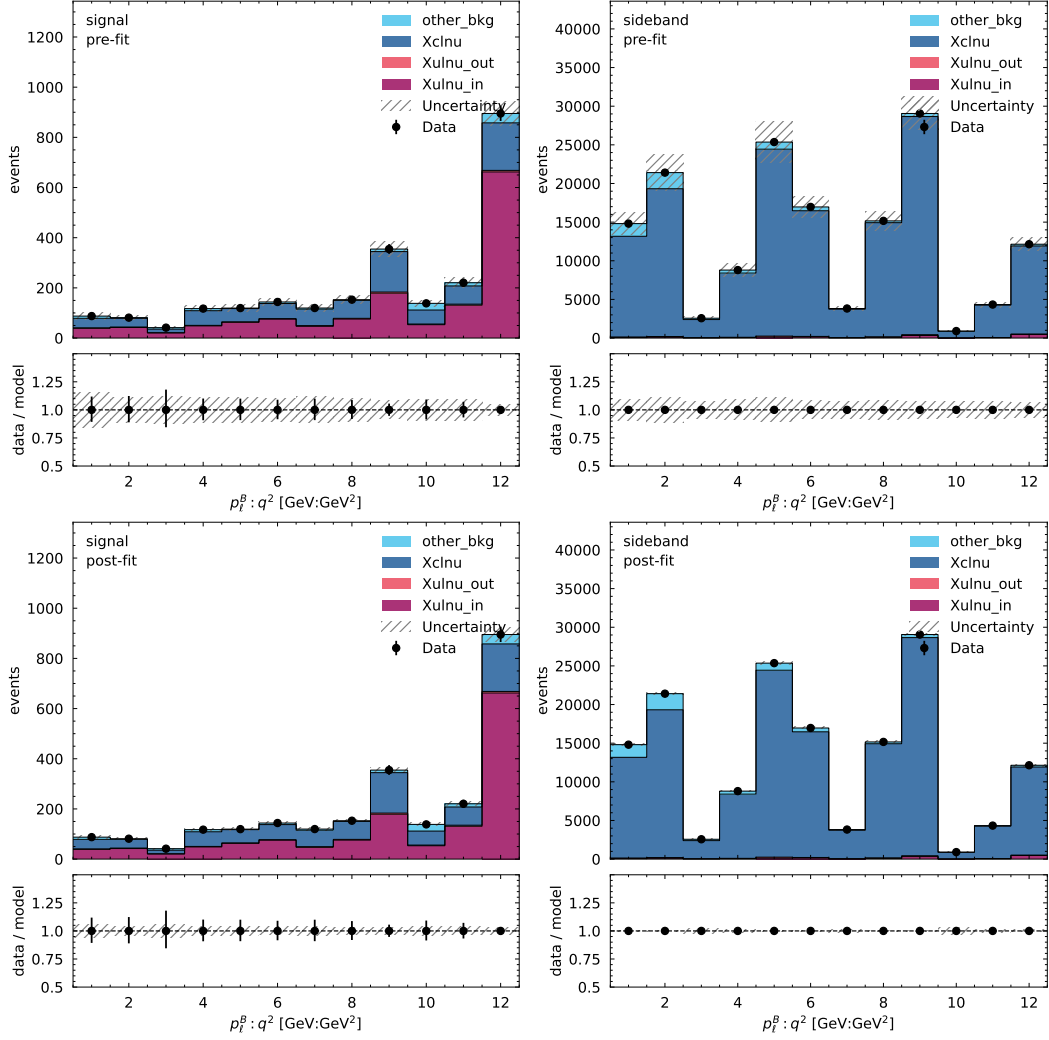


FIG. 121: Variable: $p_\ell^B : q^2$, Phase-space region: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$,
 Experimental cut: $p_\ell^B > 1.0 \text{ GeV}$, $M_X < 1.7 \text{ GeV}$

2089 plb

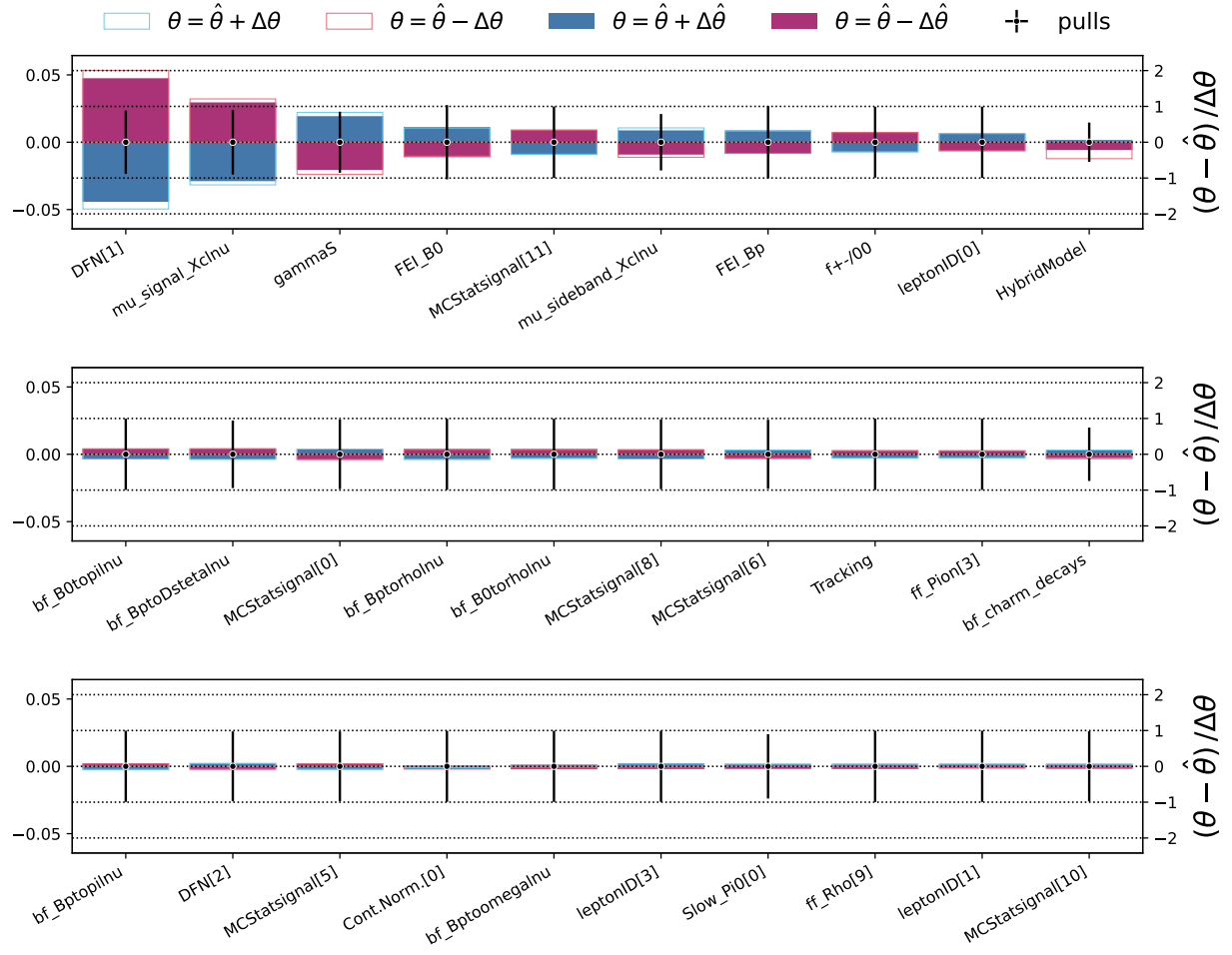


FIG. 122: Variable: $p_\ell^B:q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV,
Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV

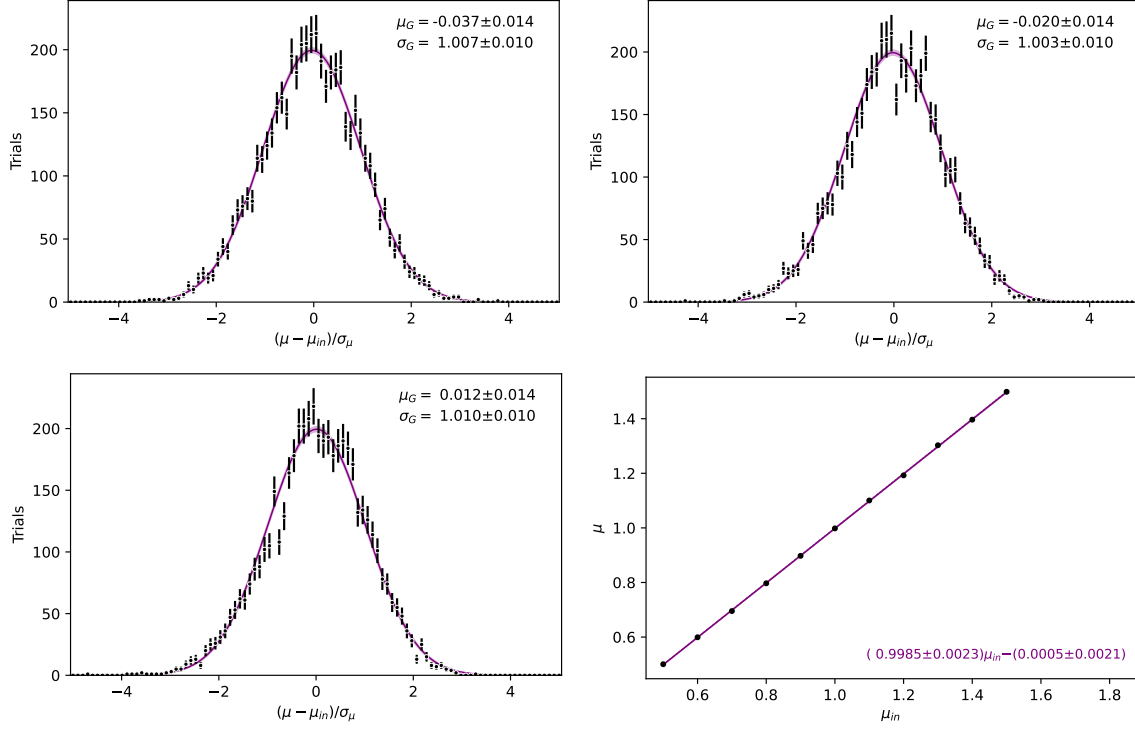
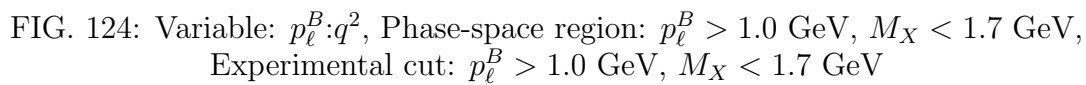


FIG. 123: Variable: $p_\ell^B:q^2$, Phase-space region: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV. Top left: signal pulls; top right: $X_{\ell\nu}$ factor pulls (bin 3); bottom left: $X_{\ell\nu}$ factor pulls (10); bottom right: linearity test



2091 K_L are poorly detected by the KLM detector and most of them are missed. As they are
 2092 produced in charm decays, they can therefore lead to mismodelling of the $X_c \ell \nu$ component
 2093 in our study. As shown in Section 7 and in particular in Fig. 39, in our case they are not
 2094 expected to represent the main source of data-MC disagreement. However, the K_L rates in
 2095 the signal region and orthogonal $\text{CR}_{0,low}$, could be different. As we use the later region to
 2096 correct the $X_c \ell \nu$ component in the former region, this difference could (at least partially)
 2097 invalidate the correction procedure. This difference is therefore discussed in this Section.

2098

2099 By comparing Figures 125 to 128, the following observations can be made. First of all,
 2100 events with K_L^0 seem to be well rejected by the $X_c \ell \nu$ MVA classifier. However, since we can
 2101 only reject K^+ and K_S^0 , the K veto used to define the signal region doesn't reject events
 2102 where K_L^0 are found. Being able to reject K_L^0 could help reduce the $X_c \ell \nu$ background even
 2103 further. The different relative rates of K_L events in the signal and $\text{CR}_{0,low}$ region used as
 2104 control region in the signal extraction procedure (*c.f* Section 8), could lead to a different
 2105 data-MC disagreement between the two regions. In the signal region, the K_L event yields
 2106 are about 20% of the $B \rightarrow X_u \ell \nu$ yields and the relative rate of K_L events is about 3 times
 2107 higher in $\text{CR}_{0,low}$ than in SR so the K_L modelling could impact the $X_c \ell \nu$ correction. In the
 2108 uncertainties, we factor in the different $X_c \ell \nu$ composition between the CR and the SR as
 2109 discussed in Section 8.8 which should partially cover the K_L rate difference between these
 2110 two regions. In order to complete the study made in Section 7.3.1, we repeat the pseudo-data
 2111 fit described in Section 8.8 by up-scaling K_L events by 30%. The branching fraction in that
 2112 case is measured to be $\Delta \mathcal{B}(B \rightarrow X_u \ell \nu) = (1.58 \pm 0.14) \cdot 10^{-3}$ which differs by about 1%
 2113 from the Asimov result $\Delta \mathcal{B}(B \rightarrow X_u \ell \nu) = (1.60 \pm 0.14) \cdot 10^{-3}$. The difference is negligible
 2114 compared to the total uncertainty.

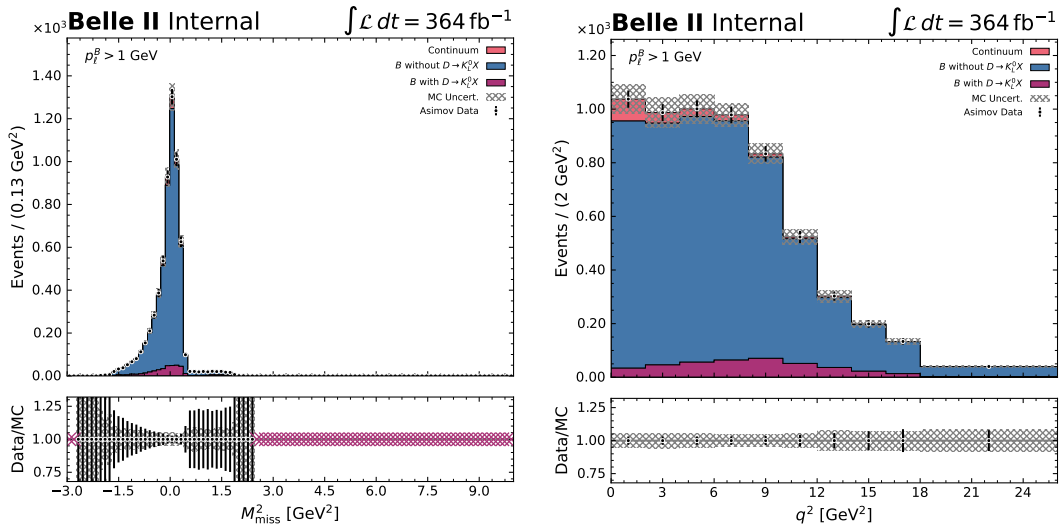


FIG. 125: M_{miss}^2 (left) and q^2 (right) distributions in the signal region split into the following three components: continuum, B events where a $D \rightarrow K_L^0$ decay is found and B events where no $D \rightarrow K_L^0$ decay is found.

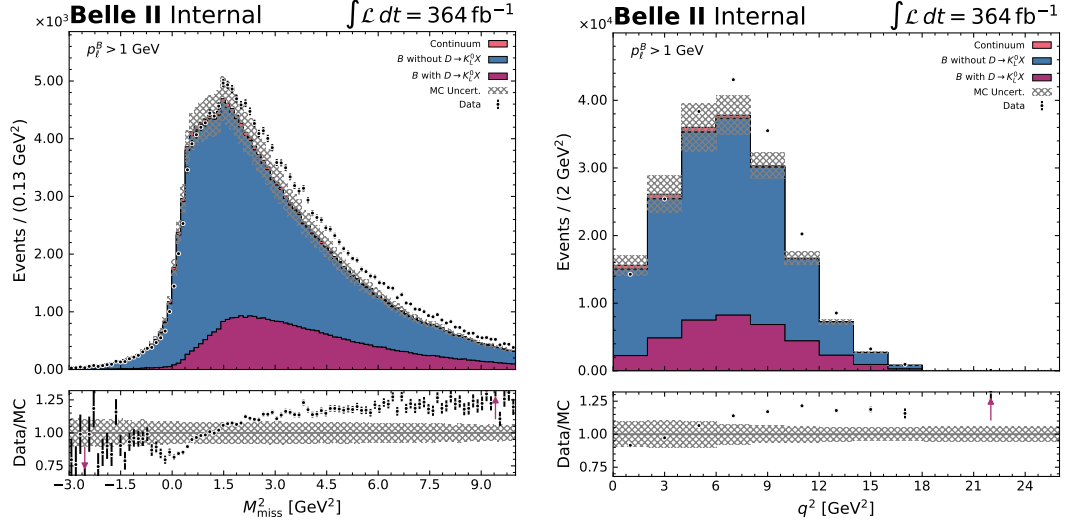


FIG. 126: M_{miss}^2 (left) and q^2 (right) distributions in $\text{CR}_{0,\text{low}}$ split into the following three components: continuum, B events where a $D \rightarrow K_L^0 X$ decay is found and B events where no $D \rightarrow K_L^0$ decay is found.

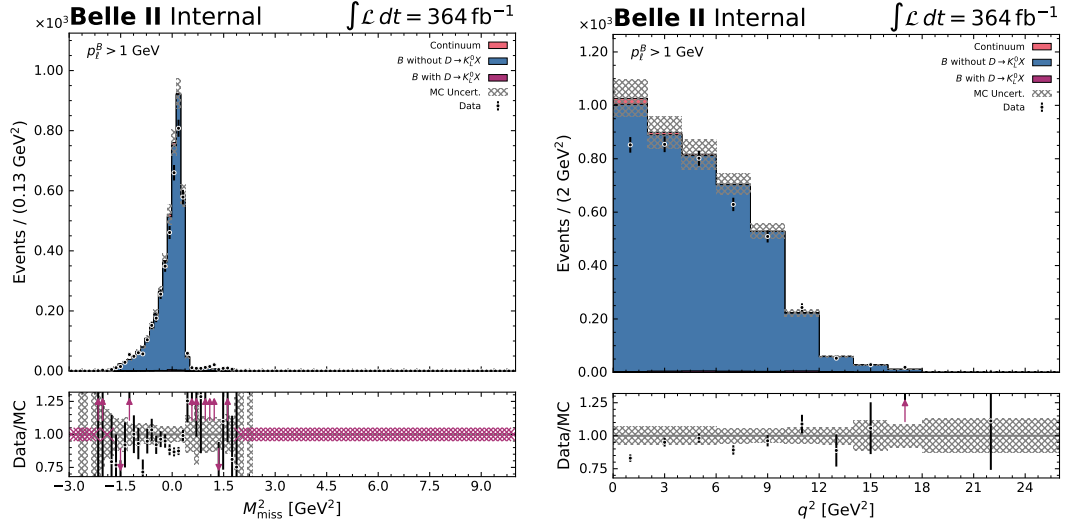


FIG. 127: M_{miss}^2 (left) and q^2 (right) distributions in $\text{CR}_{K,\text{high}}$ split into the following three components: continuum, B events where a $D \rightarrow K_L^0 X$ decay is found and B events where no $D \rightarrow K_L^0$ decay is found.

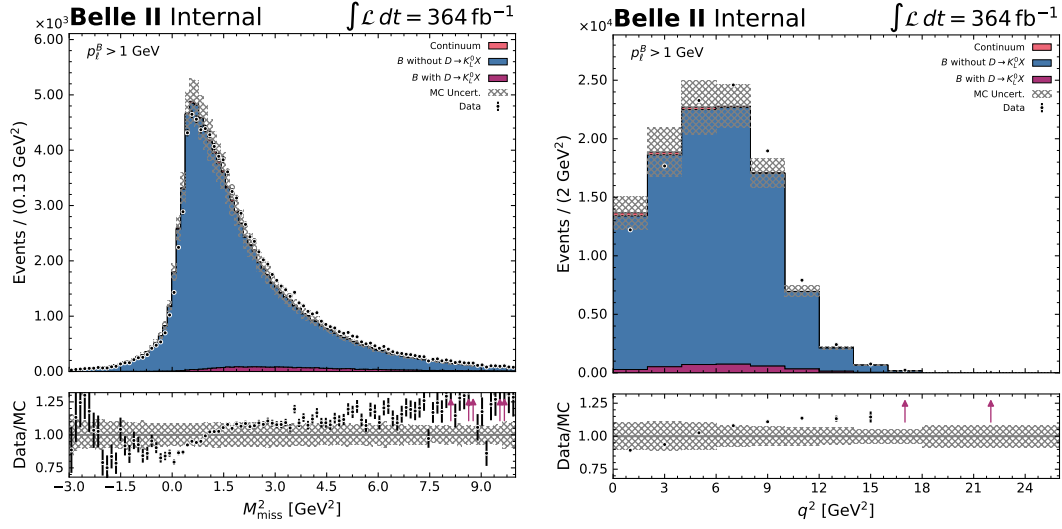


FIG. 128: M_{miss}^2 (left) and q^2 (right) distributions in $\text{CR}_{K,\text{low}}$ split into the following three components: continuum, B events where a $D \rightarrow K_L^0$ decay is found and B events where no $D \rightarrow K_L^0$ decay is found.

2115 H. HADRON ID

2116 We choose the kaon/pion ID likelihood-based score rather than the neural network one as
 2117 the relative amount of fakes is much lower. We show below the efficiency and fake rate for
 2118 kaons and pions for both types of scores. We use the nominal sample after preselection.
 2119 Only tracks covered by the ROE mask are considered.

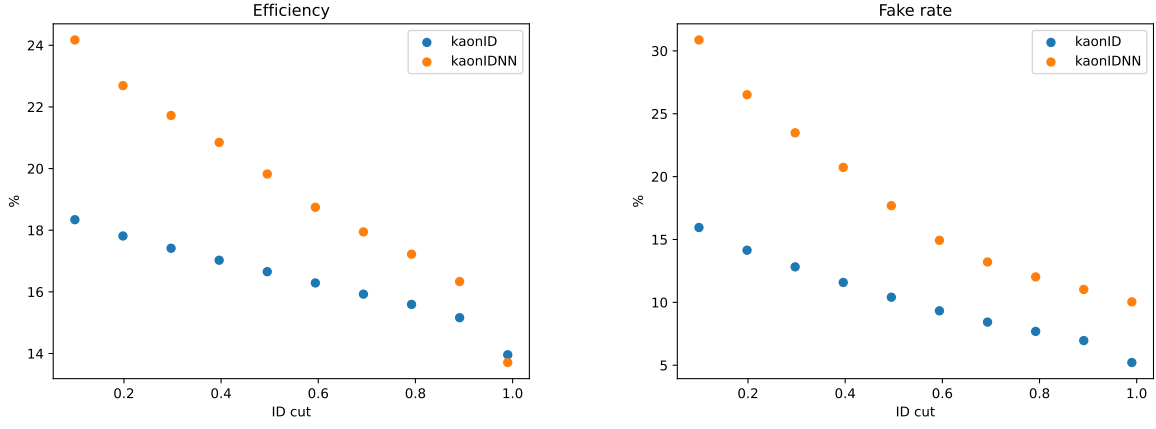


FIG. 129: Efficiency and fake rate for different likelihood-based (kaonID) and neural-network-based (kaonIDNN) $K^\pm$ identification score cuts

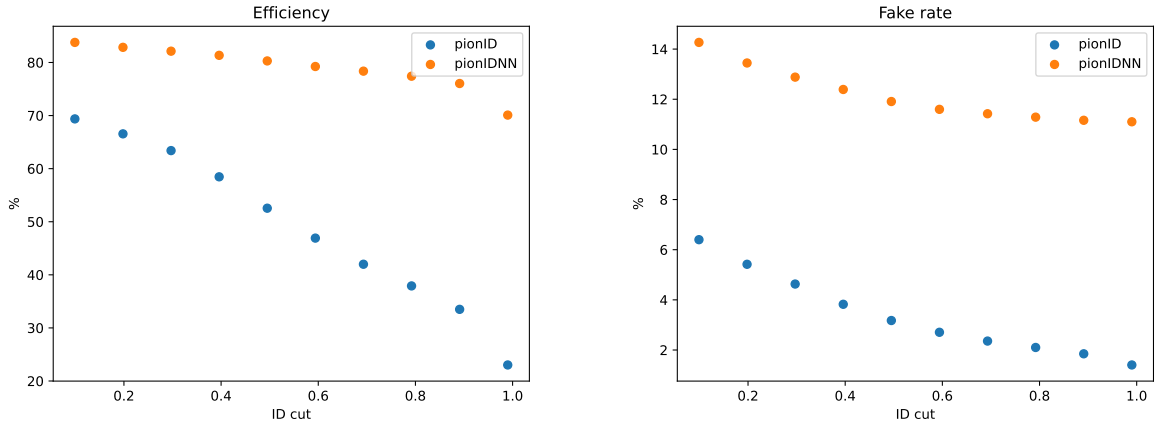


FIG. 130: Efficiency and fake rate for different likelihood-based (pionID) and neural-network-based (pionIDNN) $\pi^\pm$ identification score cuts

2120 I. FITS WITH CONTROL REGIONS

2121 1. Fit 3

2122 We repeat for fit 3 the test that was detailed in Section 8 10 for fit 1. The $\text{CR}'_{0,\text{low}} X_c \ell \nu$
 2123 normalisation is equal to 1.16 ± 0.04 and the $\text{VR}'_2 X_c \ell \nu$ normalisation is equal to 1.18 ± 0.04 .
 2124 The obtained p -value is 0.62.

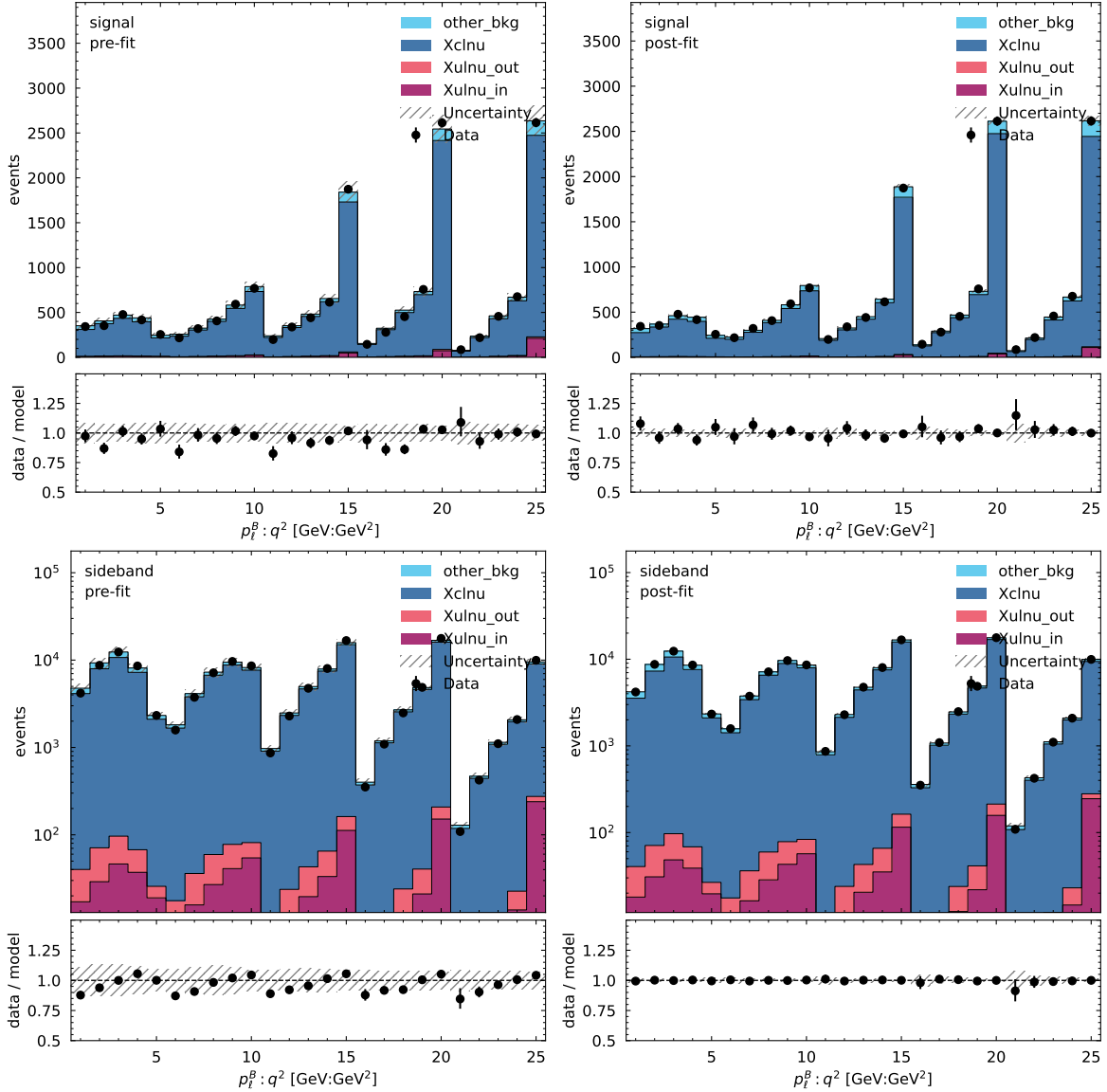


FIG. 131: Pre- (left) and post-fit (right) q^2 distributions in VR'_2 (top) and $\text{CR}'_{0,\text{low}}$ (bottom) for fit 3.

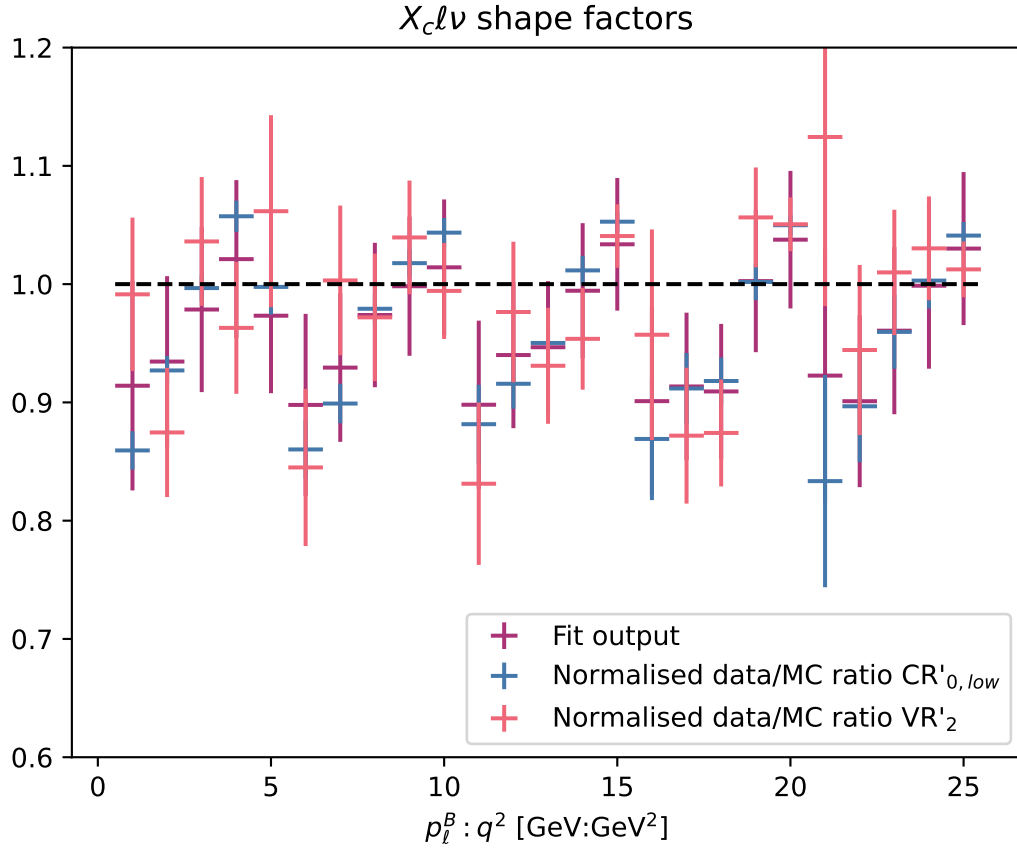


FIG. 132: Comparison between the normalised $X_{c\ell\nu}$ data/MC ratio for regions $\text{CR}'_{0,low}$ and VR'_2 and the fit output $X_{c\ell\nu}$ shape factors.

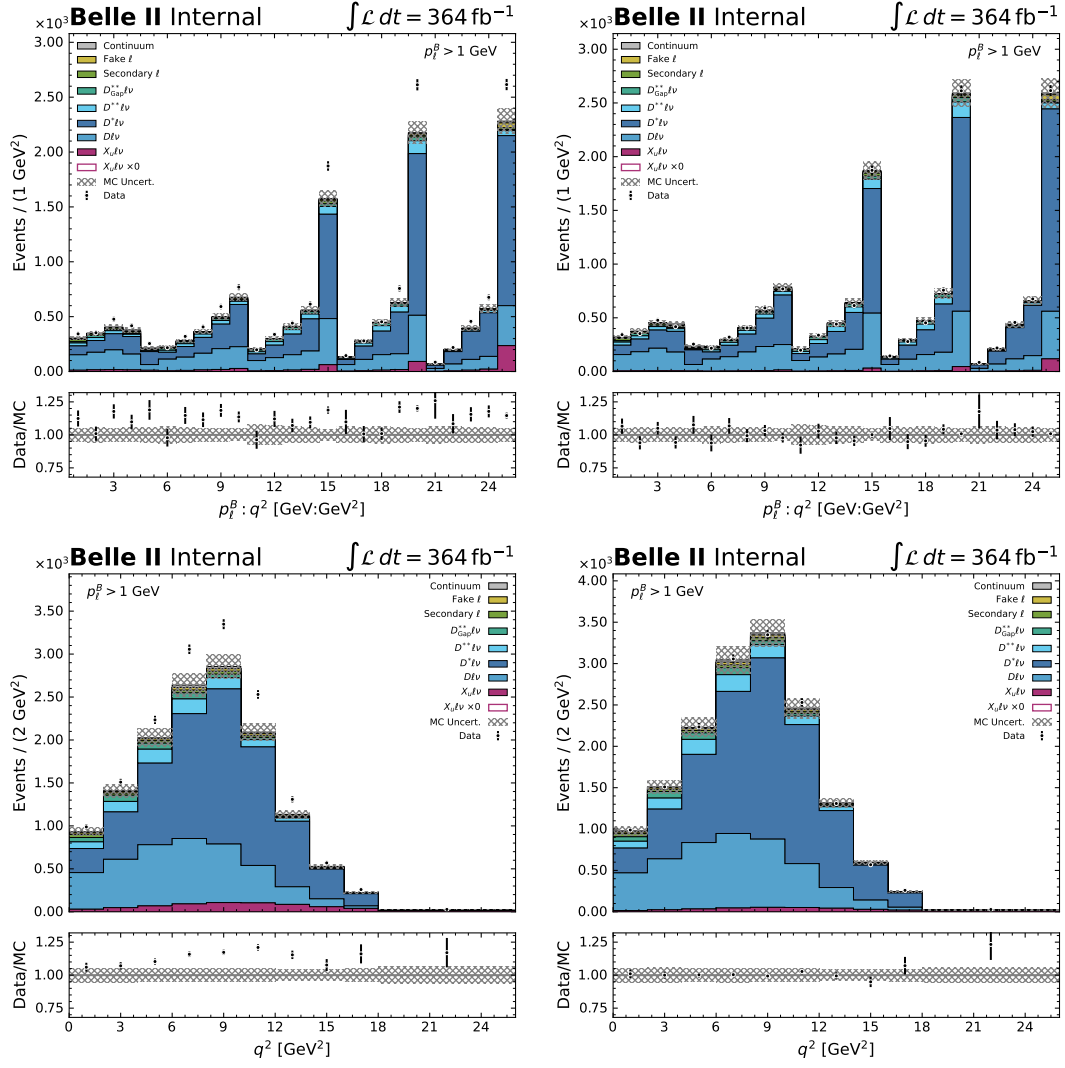


FIG. 133: 2D $p_\ell^B : q^2$ (top) and q^2 (bottom) distributions in VR'₂ before (left) and after (right) applying the $X_c l \nu$ normalisation and shape corrections.

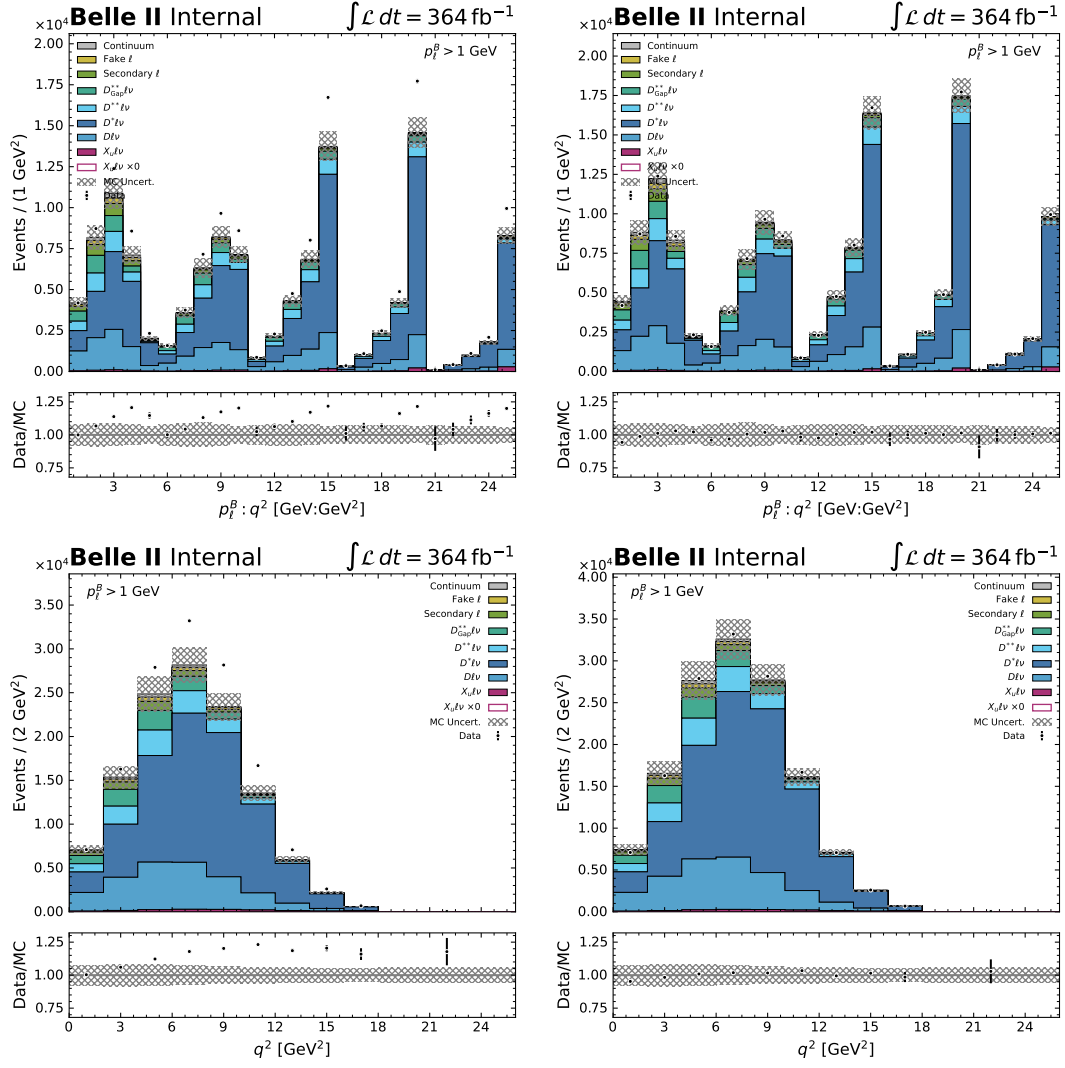


FIG. 134: 2D $p_\ell^B : q^2$ (top) and q^2 (bottom) distributions in $\text{CR}'_{0,low}$ before (left) and after (right) applying the $X_c \ell \nu$ normalisation and shape corrections.

2125 **2. Fit 5**

2126 We repeat for fit 5 the test that was detailed in Section 8 10 for fit 1. The $\text{CR}'_{0,low} X_c \ell \nu$
 2127 normalisation is equal to 1.22 ± 0.02 and the $\text{VR}'_2 X_c \ell \nu$ normalisation is equal to 1.24 ± 0.06 .
 2128 The obtained p -value is 0.64.

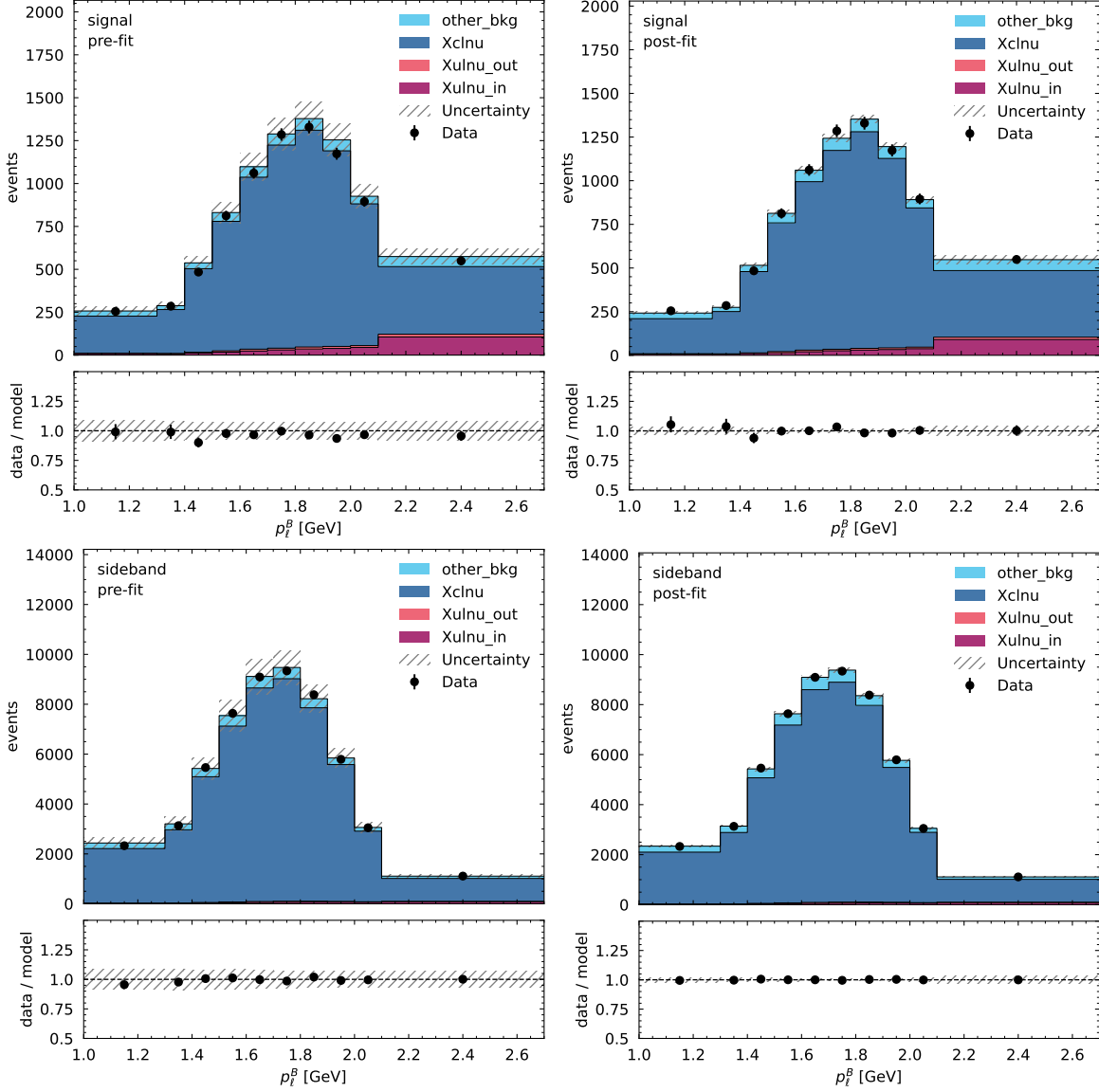


FIG. 135: Pre- (left) and post-fit (right) q^2 distributions in VR'_2 (top) and $\text{CR}'_{0,low}$ (bottom) for fit 5.

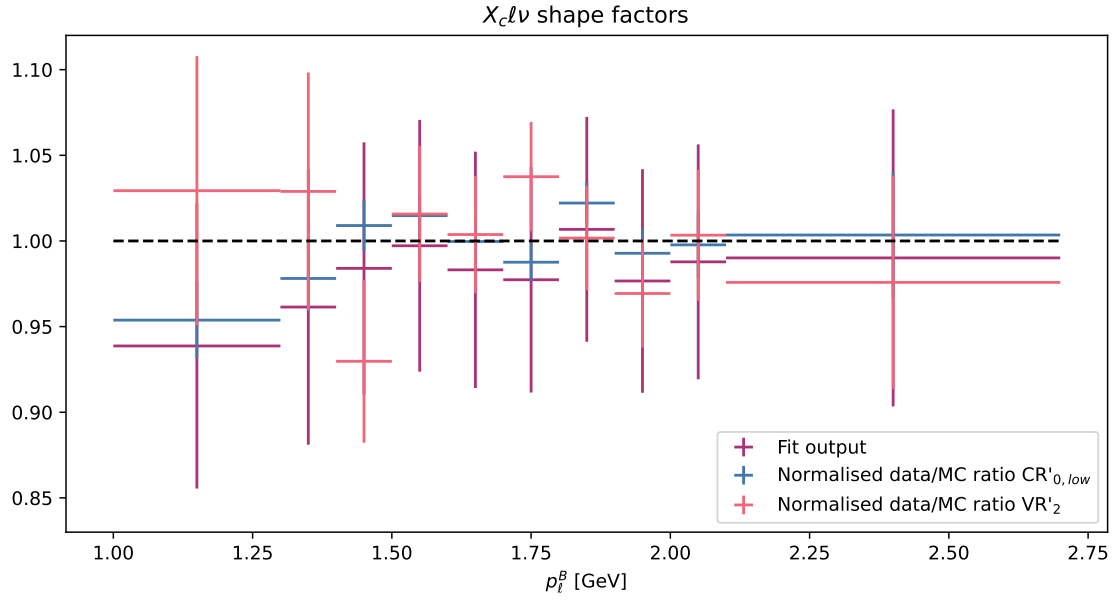


FIG. 136: Comparison between the normalised $X_{e\ell\nu}$ data/MC ratio for regions $CR'_{0,low}$ and VR'_2 and the fit output $X_{e\ell\nu}$ shape factors.

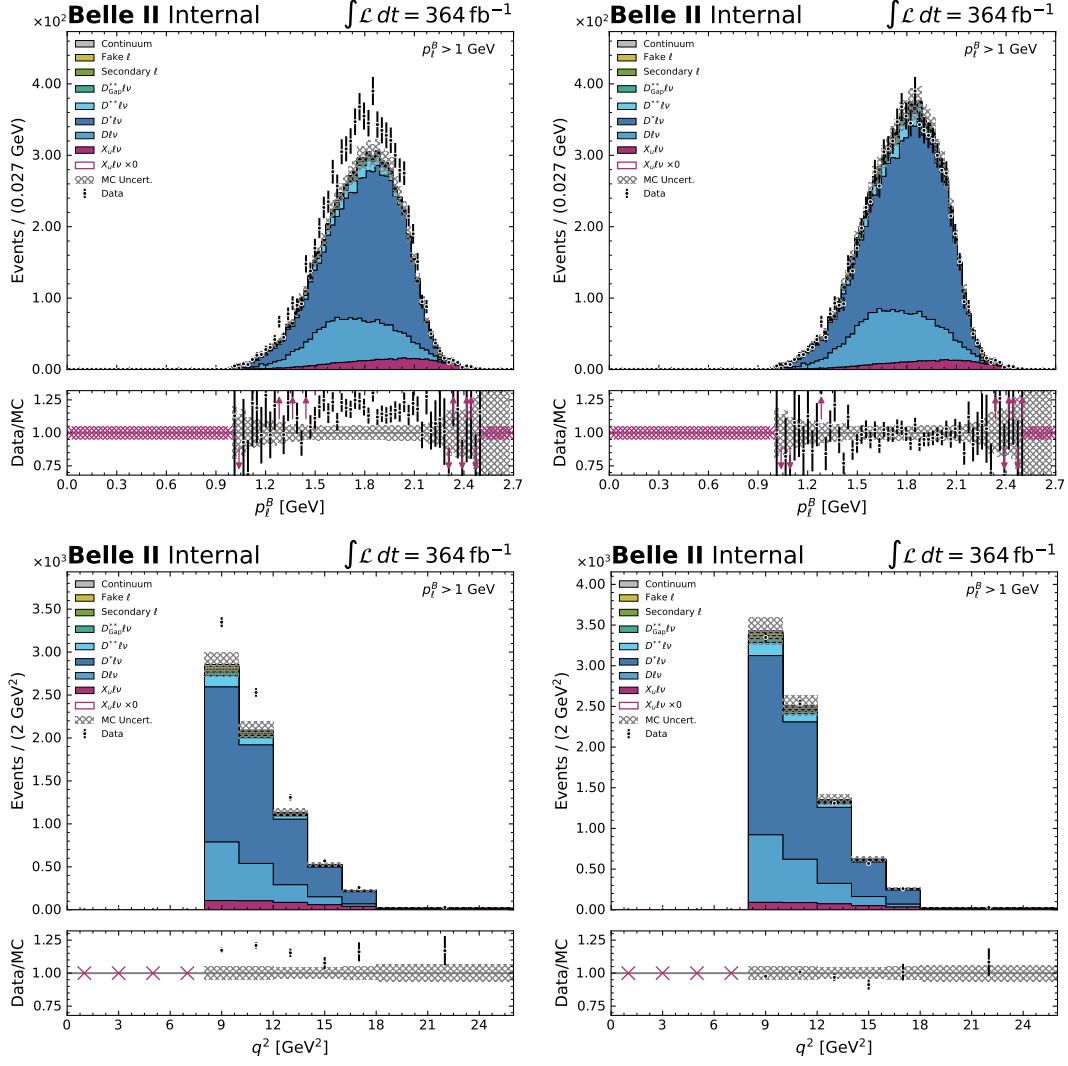


FIG. 137: p_ℓ^B (top) and q^2 (bottom) distributions in VR'_2 before (left) and after (right) applying the $X_c \ell \nu$ normalisation and shape corrections.

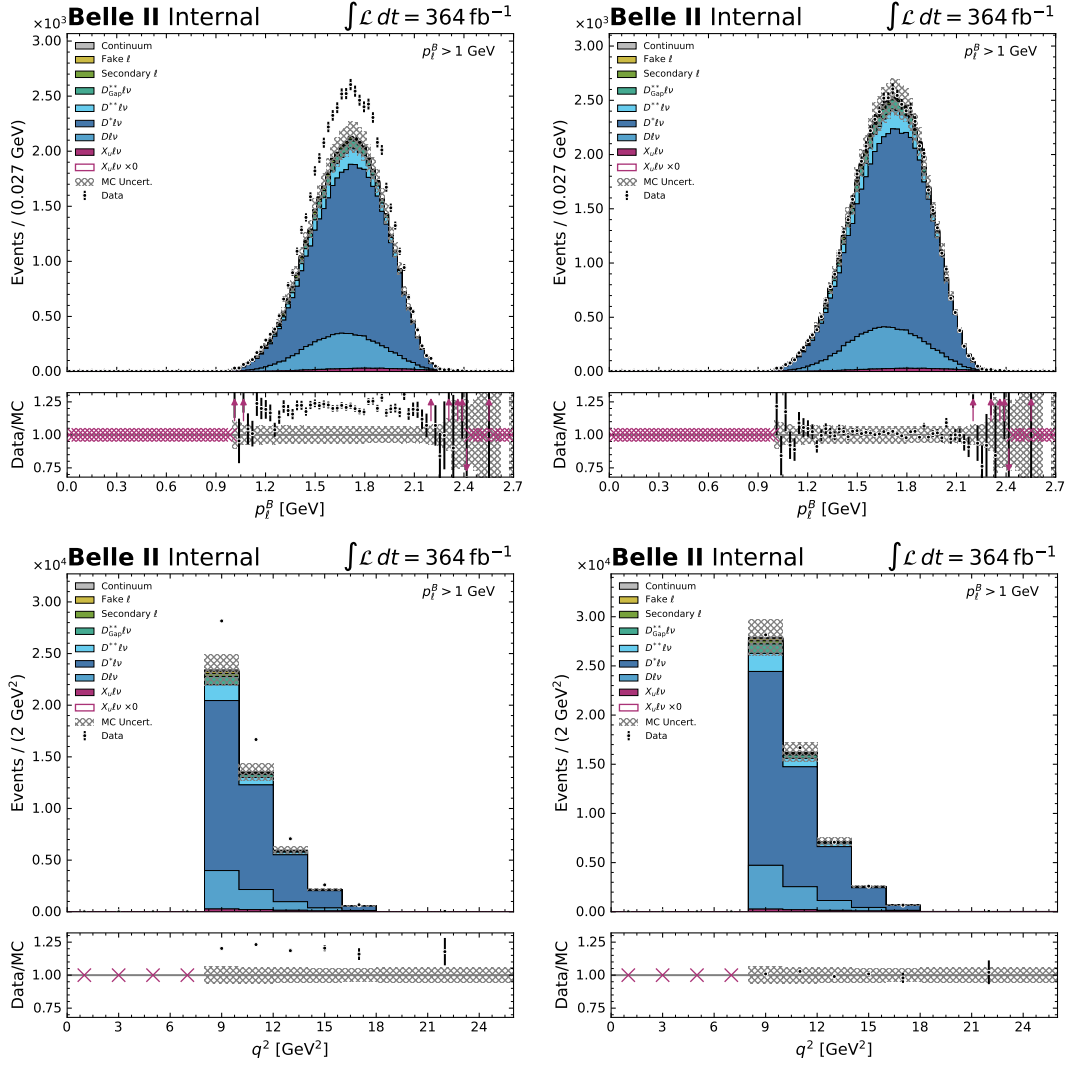
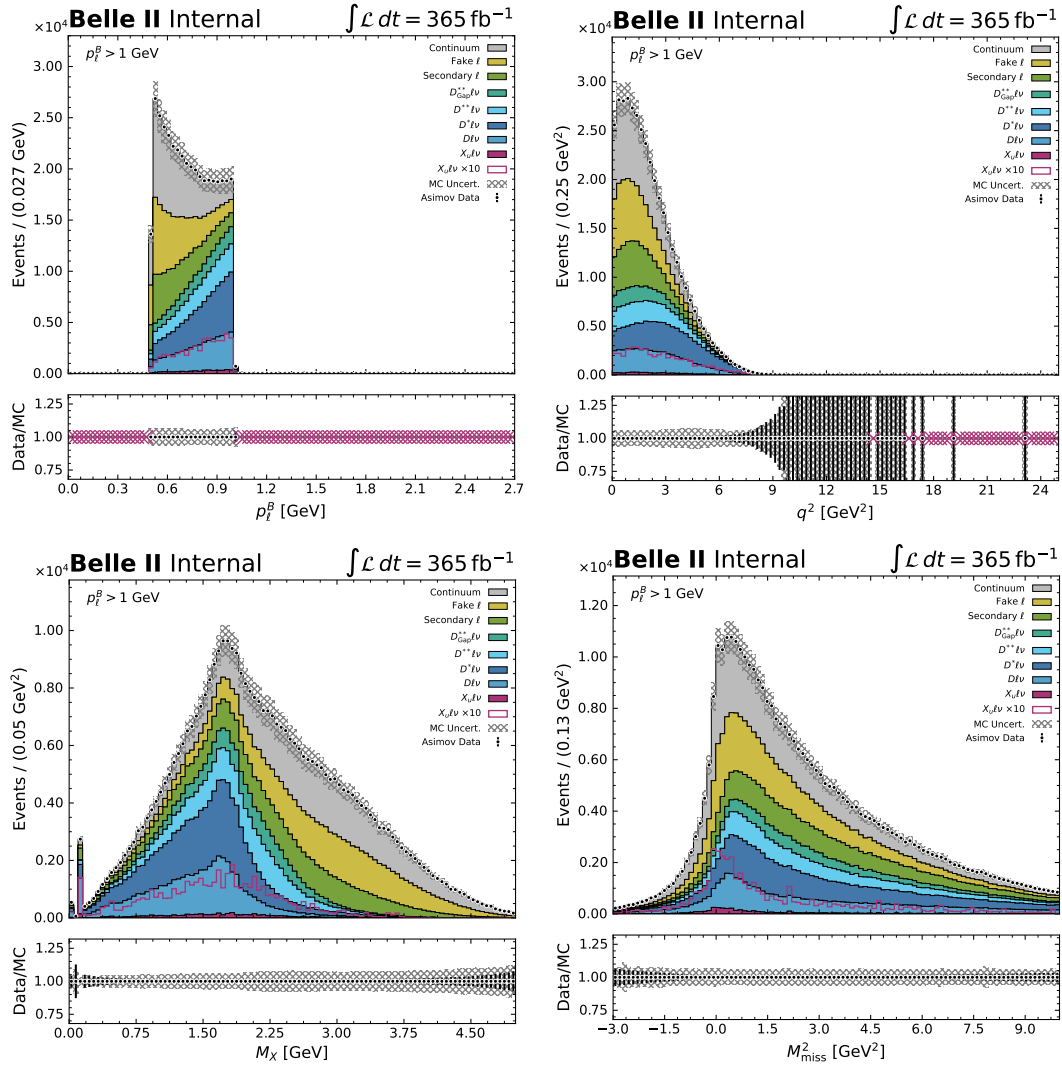


FIG. 138: p_l^B (top) and q^2 (bottom) distributions in $\text{CR}'_{0,low}$ before (left) and after (right) applying the $X_c l \nu$ normalisation and shape corrections.

2129 J. LOW-LEPTON-MOMENTUM REGION

2130 We show here kinematical distributions for a region with $0.5 < p_\ell^B < 1.0$ GeV after applying
 2131 all preselections.



2132 K. FIT WITH SPLIT LEPTON FLAVOUR

2133 We present here the results of fit 1 with split lepton flavour.

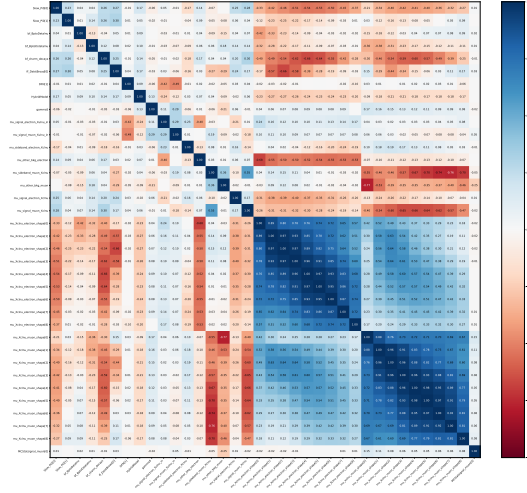


FIG. 139: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

2134 1. Electron

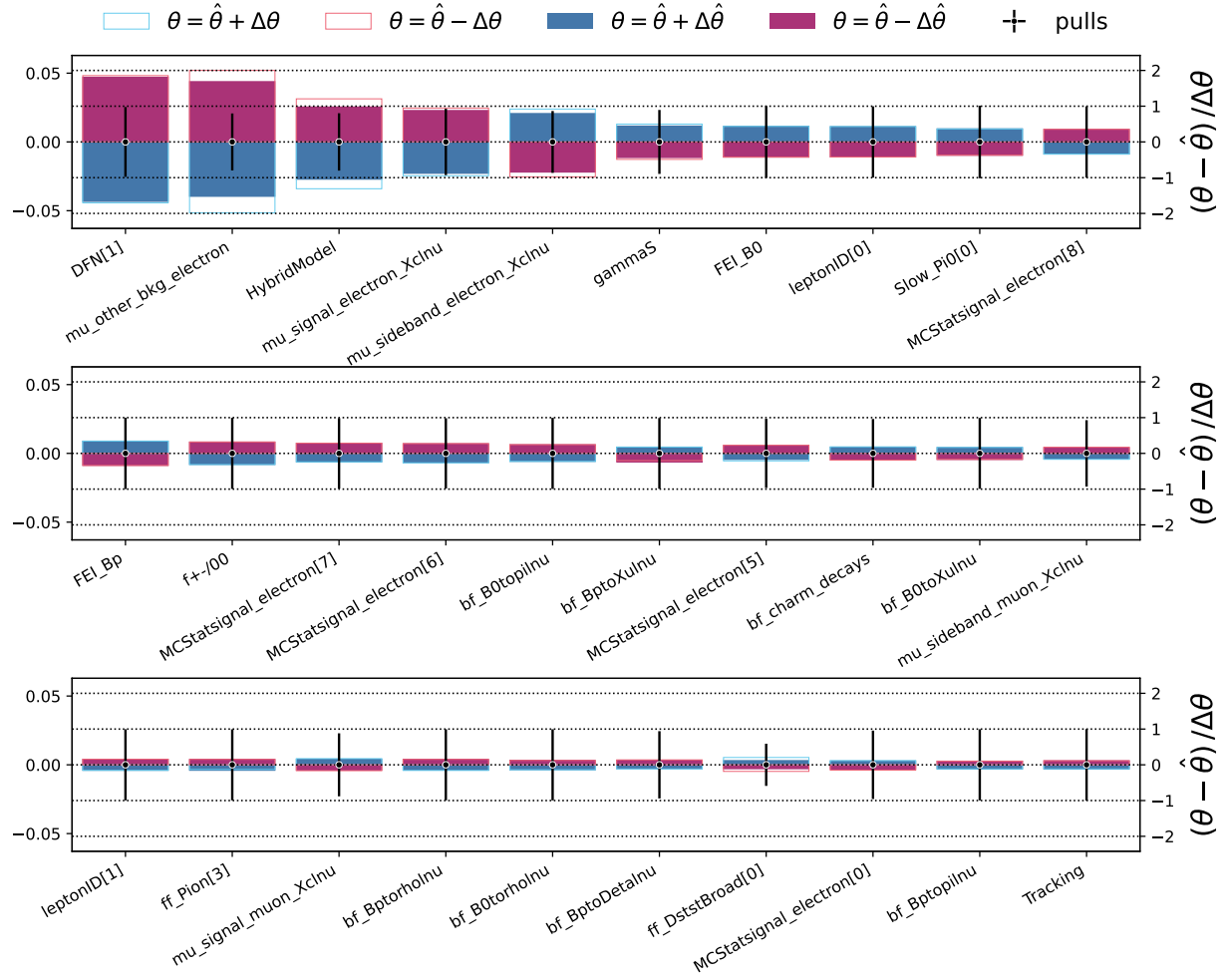


FIG. 140: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

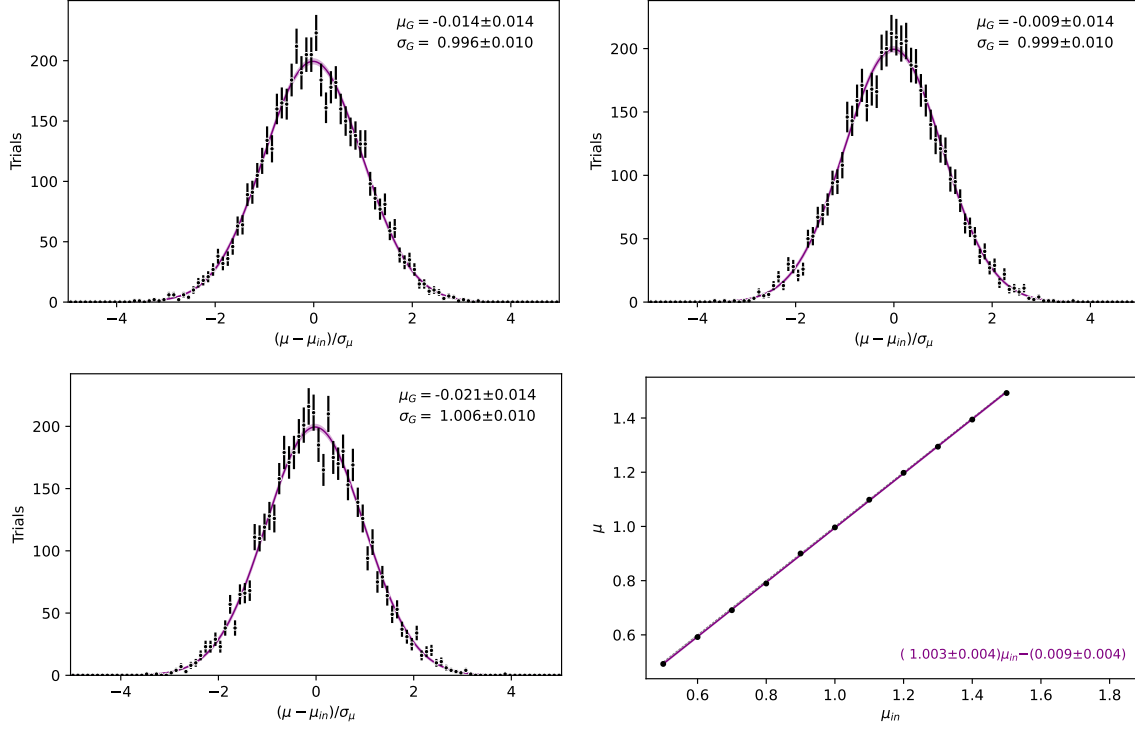


FIG. 141: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal pulls; top right: $X_{c\ell\nu}$ factor pulls (bin 1); bottom left: $X_{c\ell\nu}$ factor pulls (bin 6); bottom right: linearity test

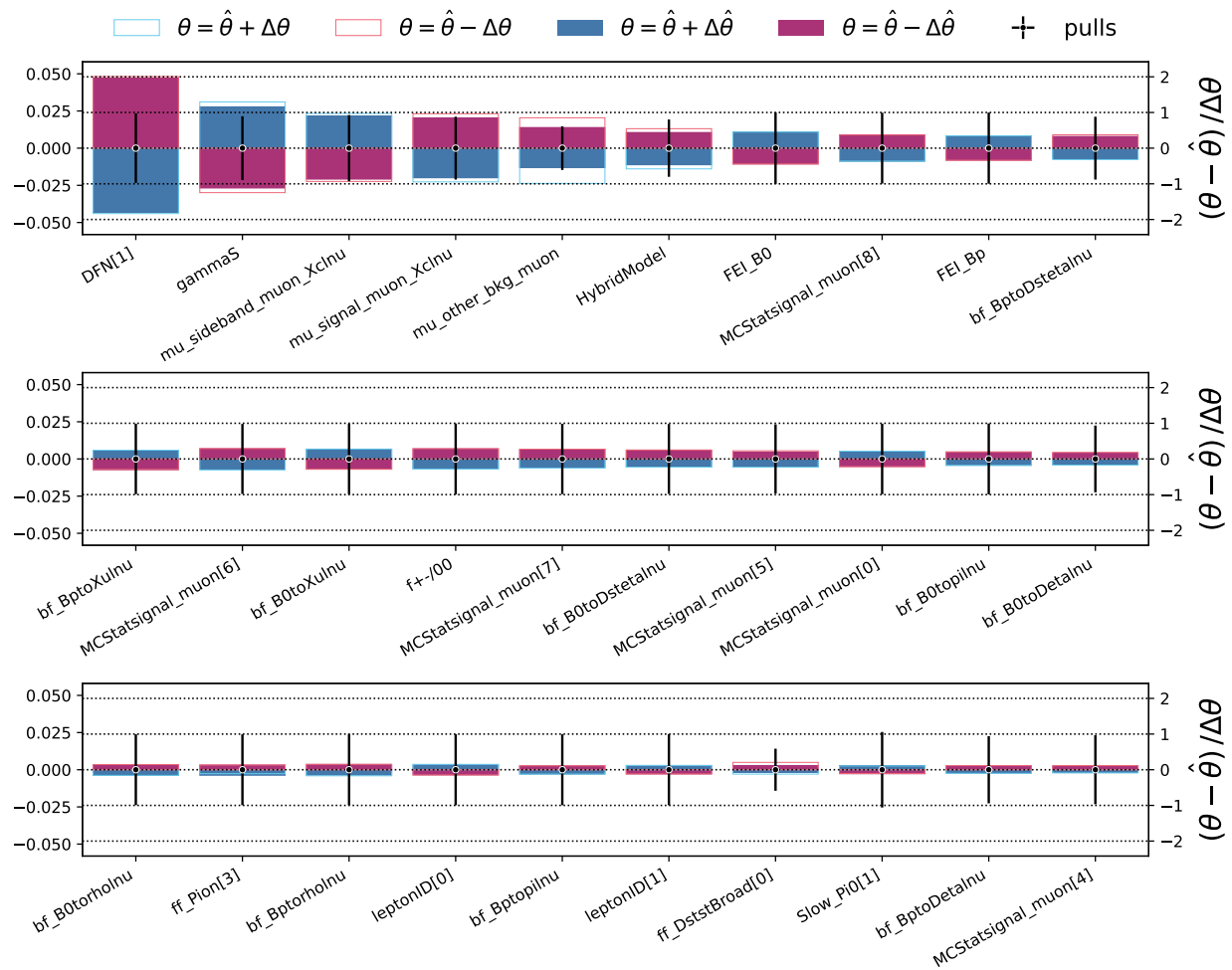


FIG. 142: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV

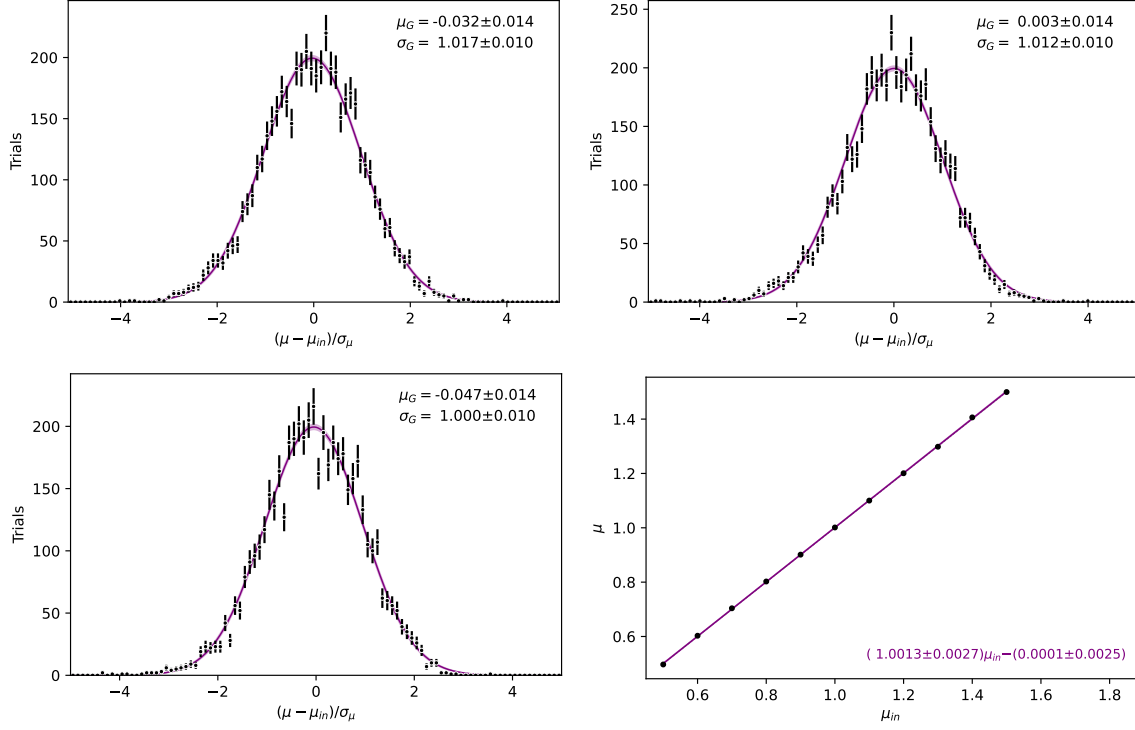
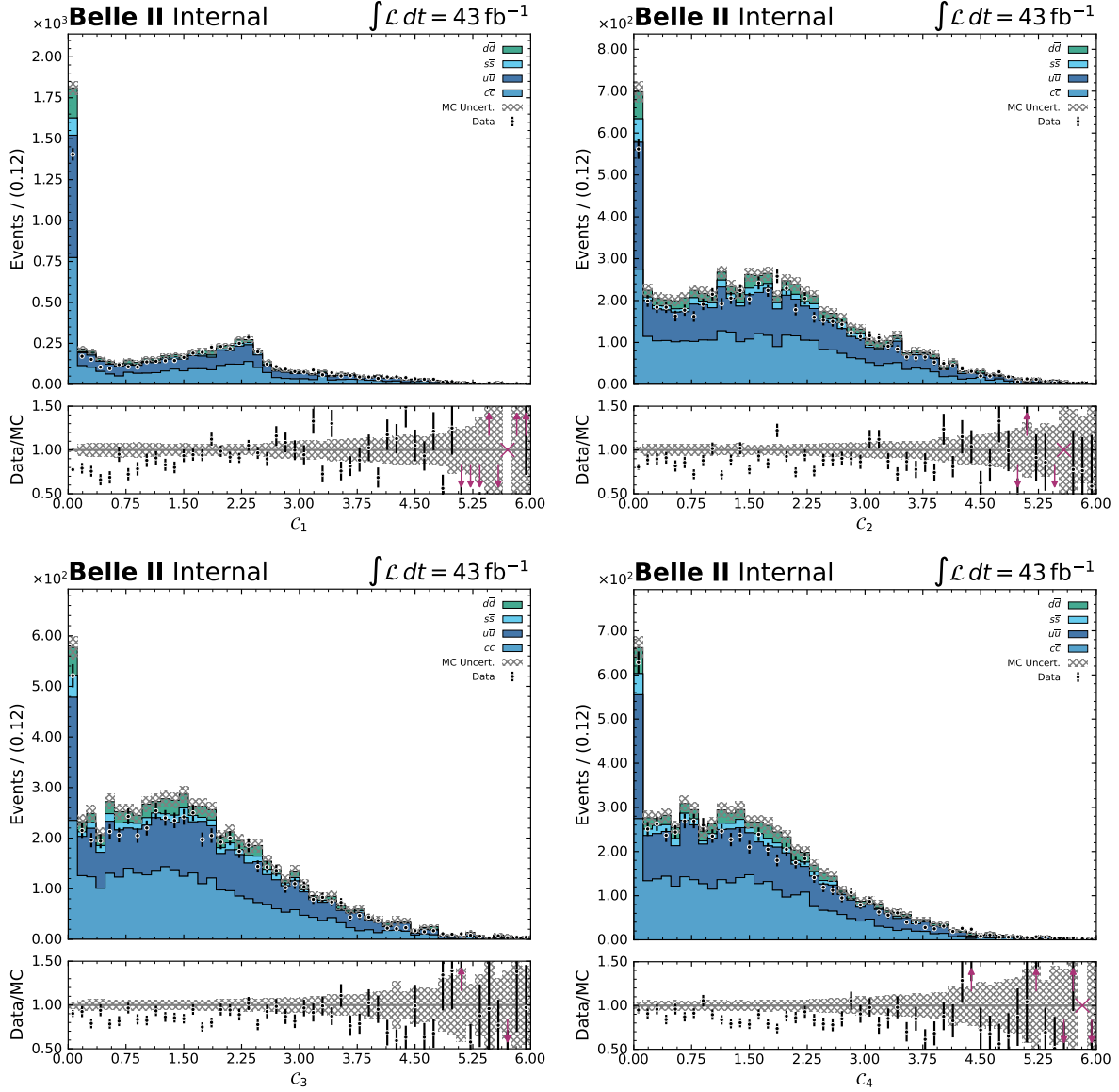


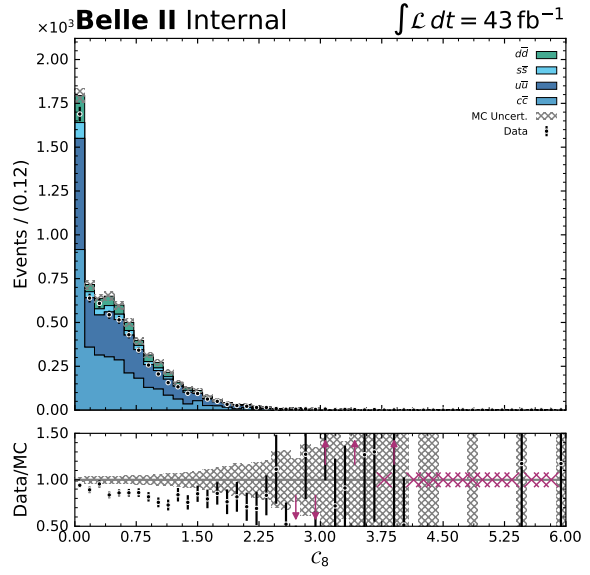
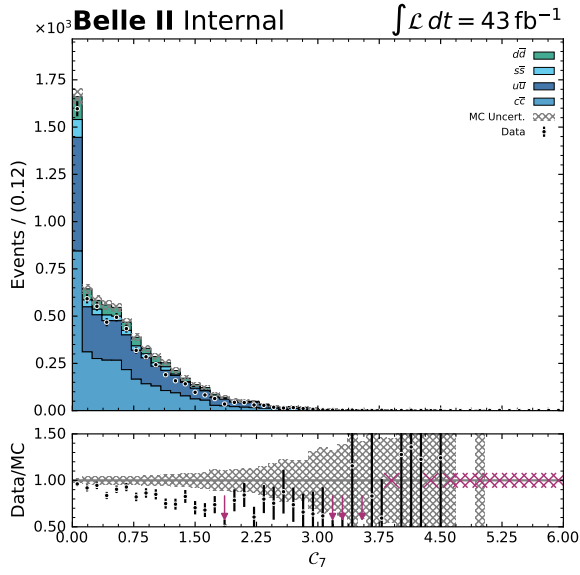
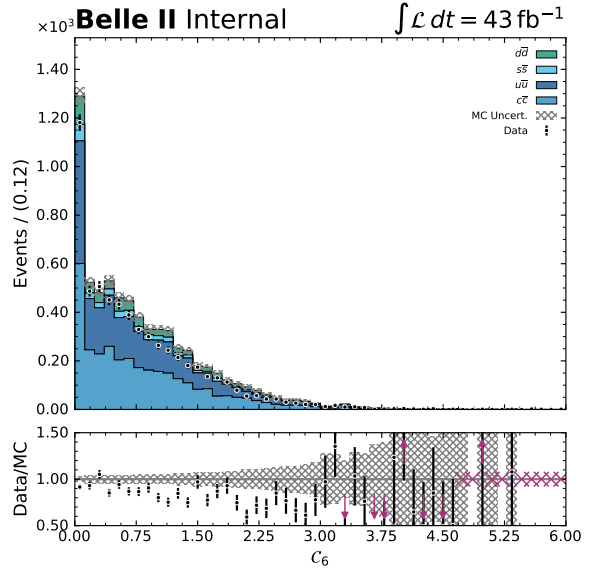
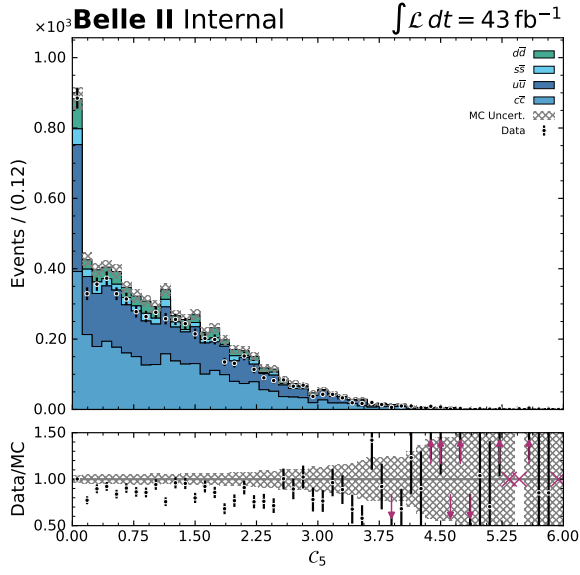
FIG. 143: Variable: q^2 , Phase-space region: $p_\ell^B > 1.0$ GeV, Experimental cut: $p_\ell^B > 1.0$ GeV. Top left: signal pulls; top right: $X_{c\ell\nu}$ factor pulls (bin 1); bottom left: $X_{c\ell\nu}$ factor pulls (bin 6); bottom right: linearity test

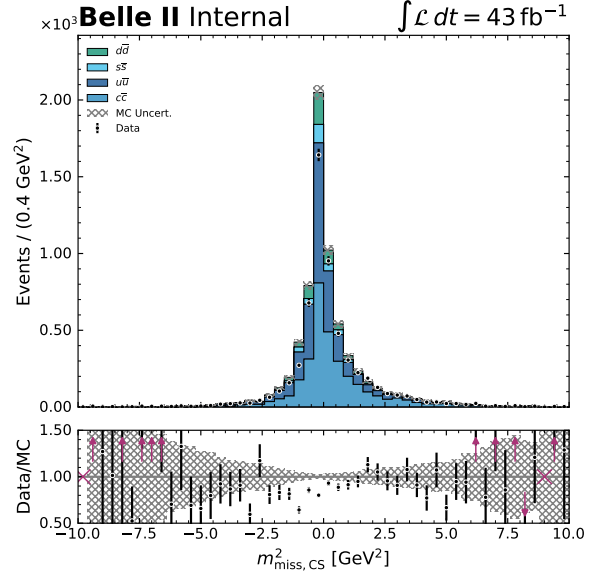
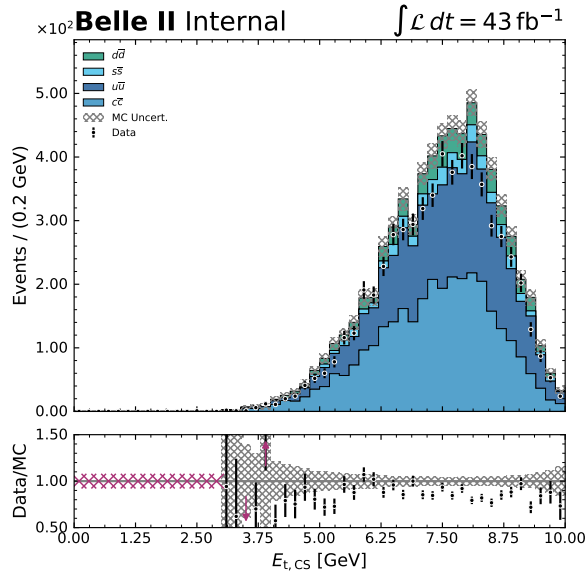
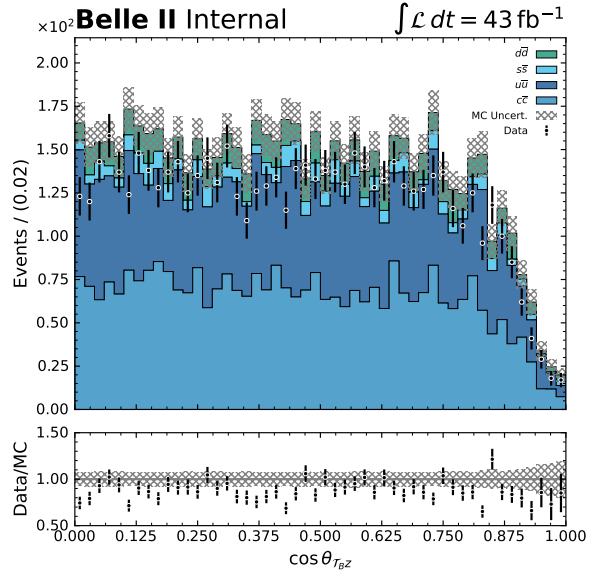
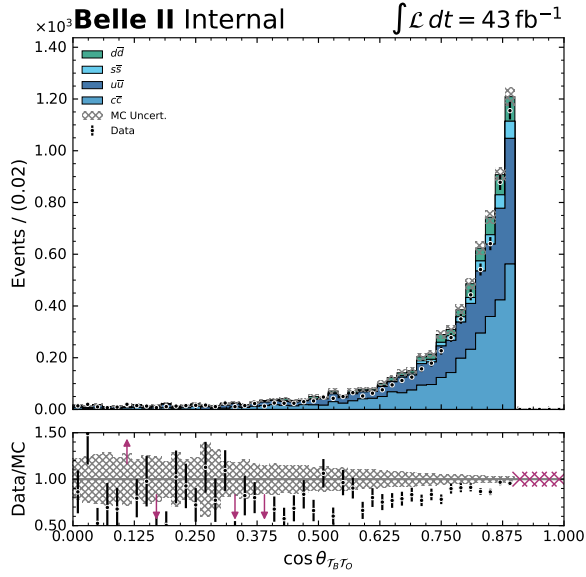
2136 L. CONTINUUM REWEIGHTING VARIABLES

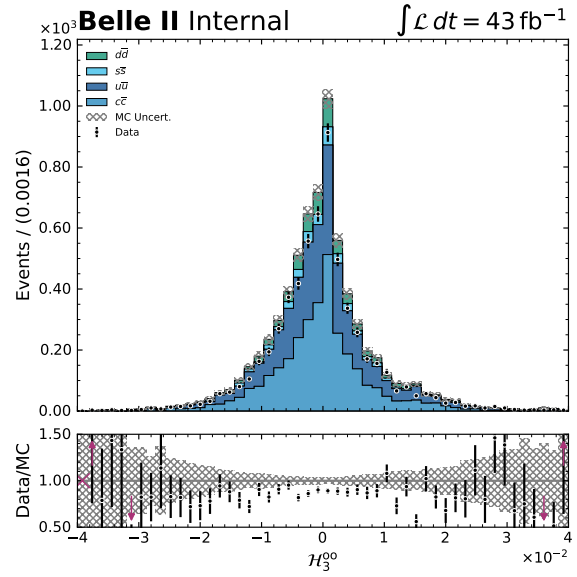
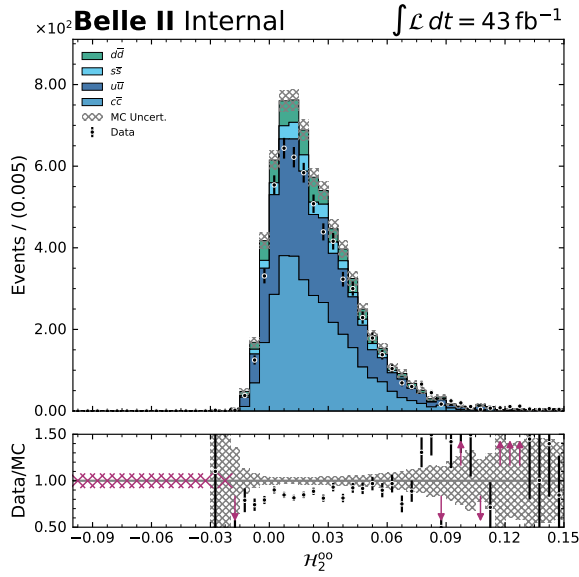
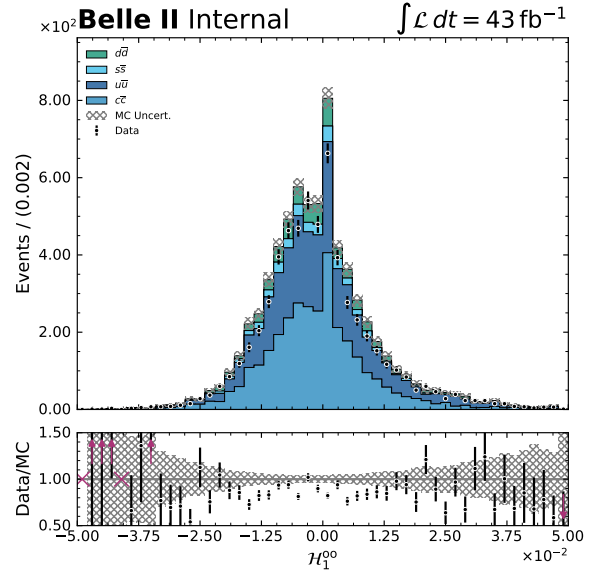
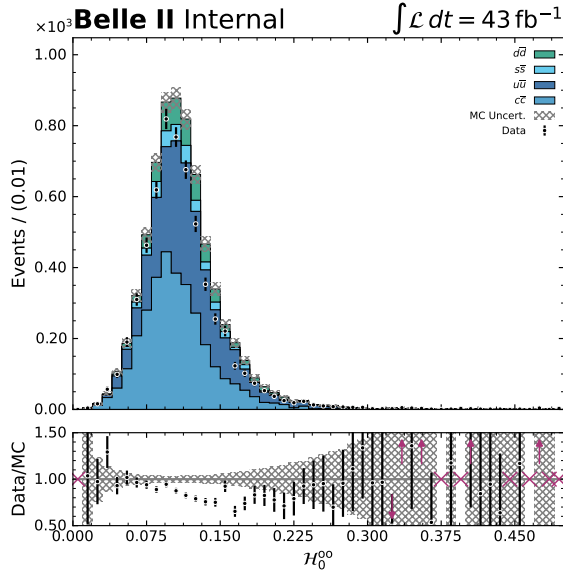
2137 The variables used for continuum reweighting before and after reweighting are shown here.
 2138 A few variables not used for the training are also shown to illustrate the correction.

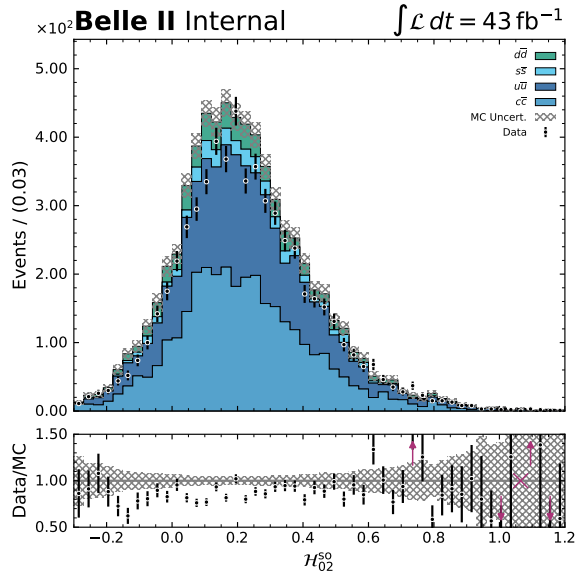
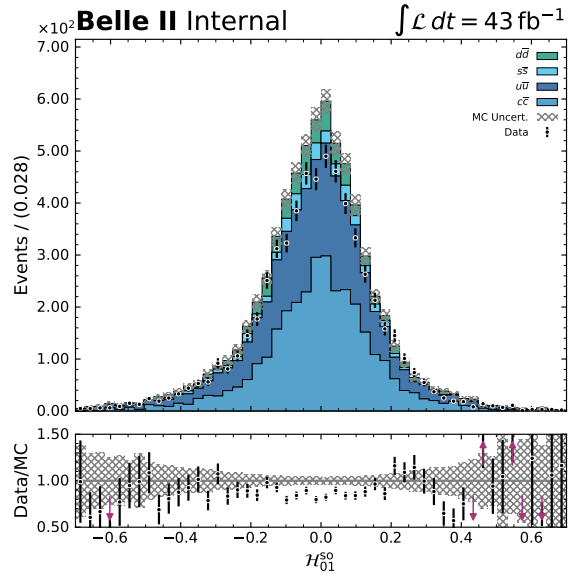
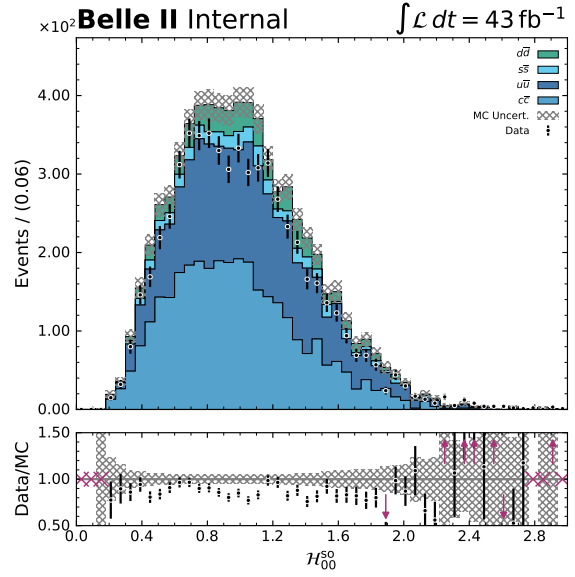
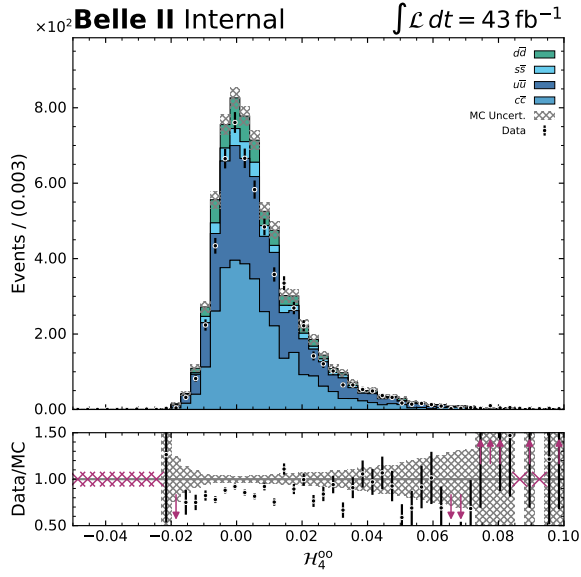
2139 1. Before applying corrections

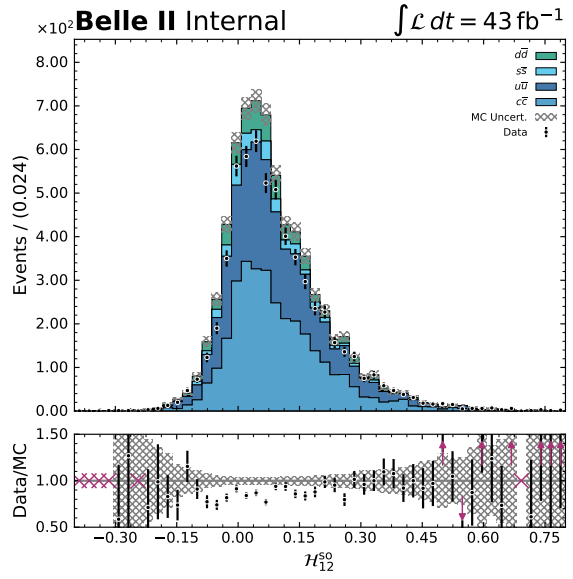
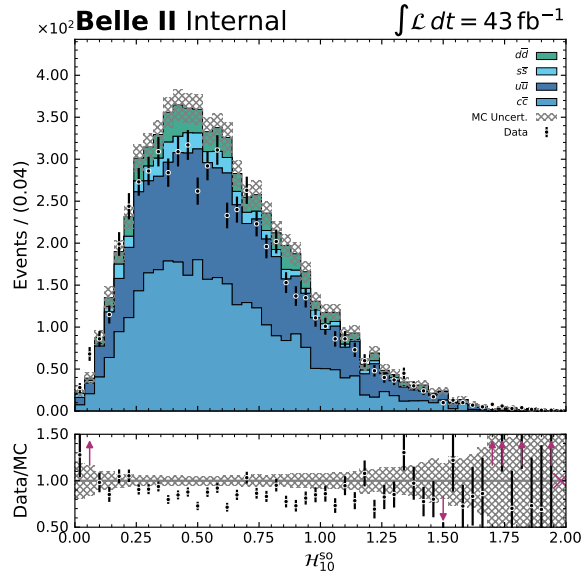
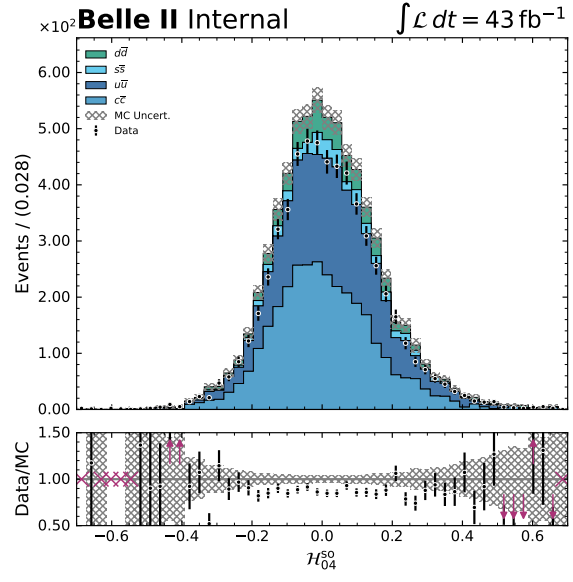
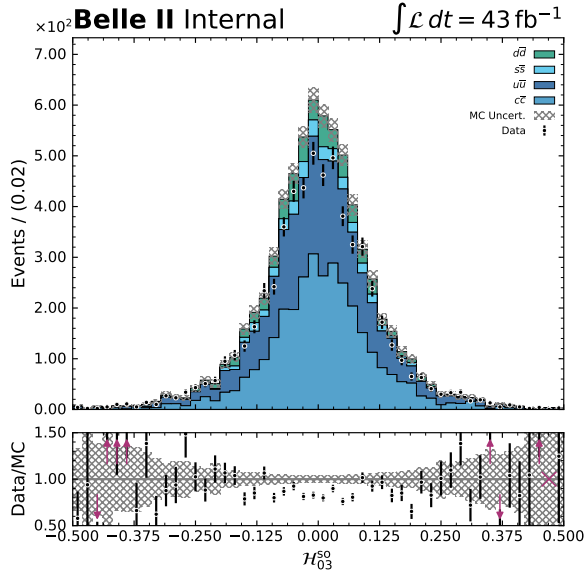


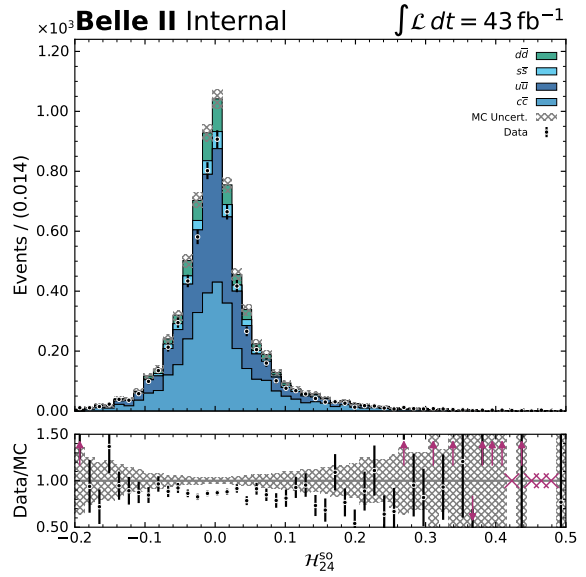
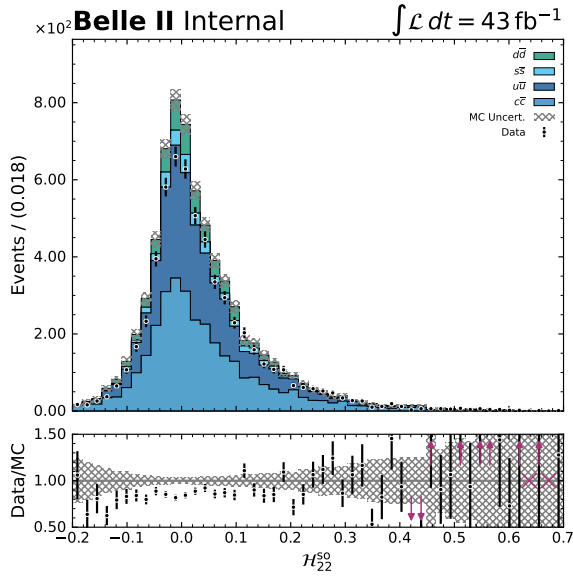
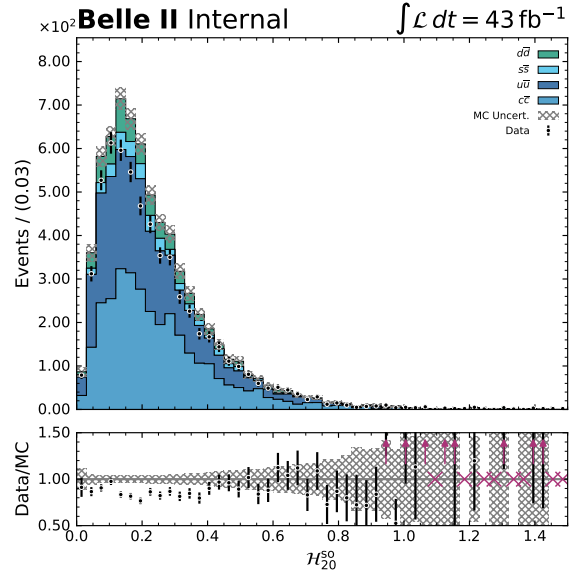
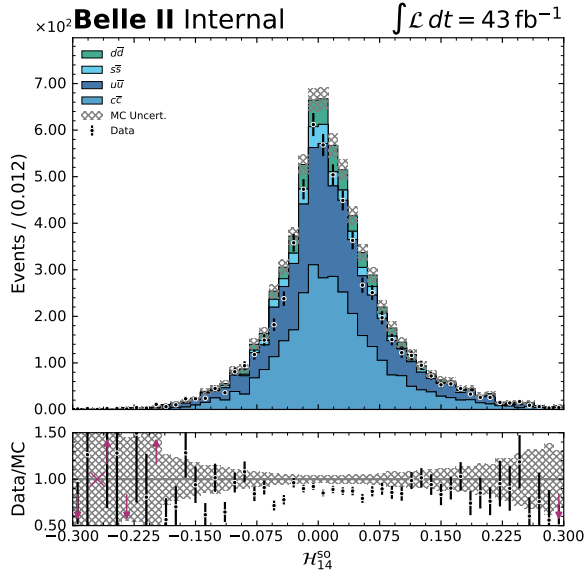


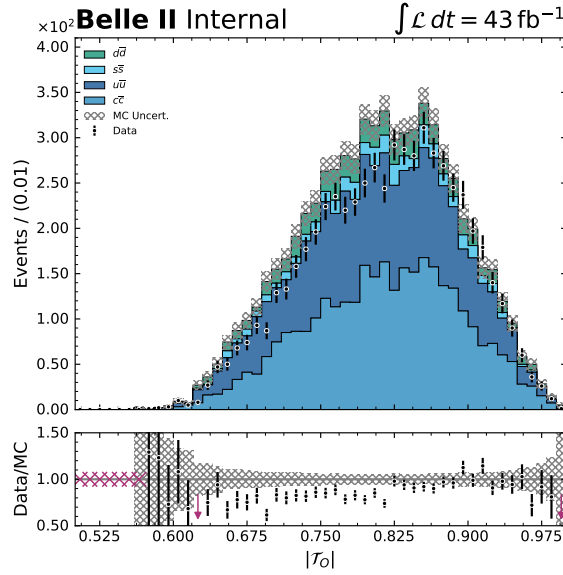
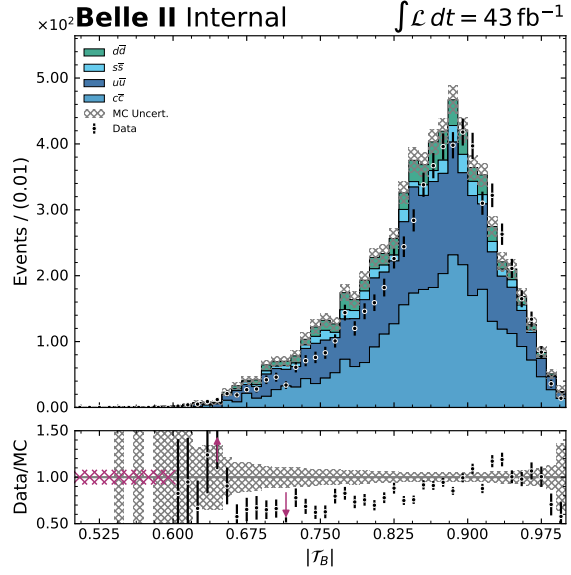
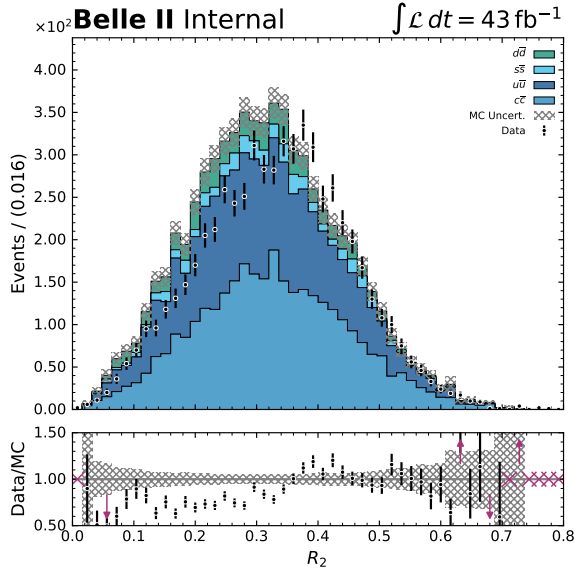


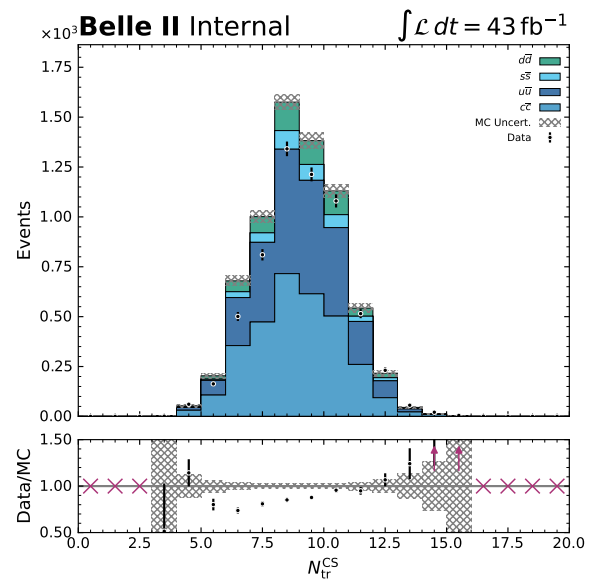
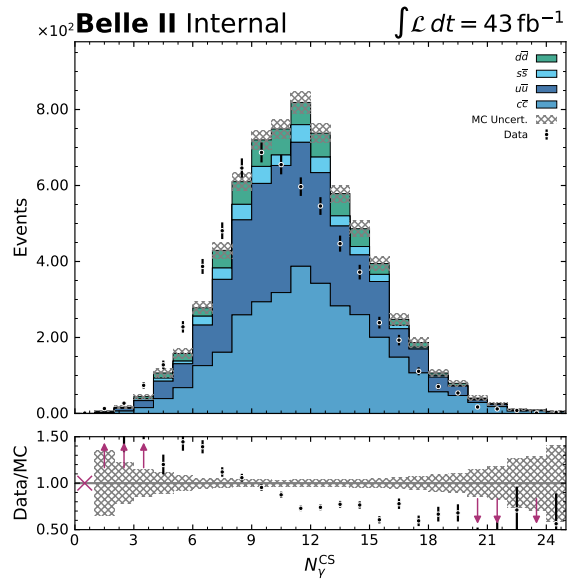


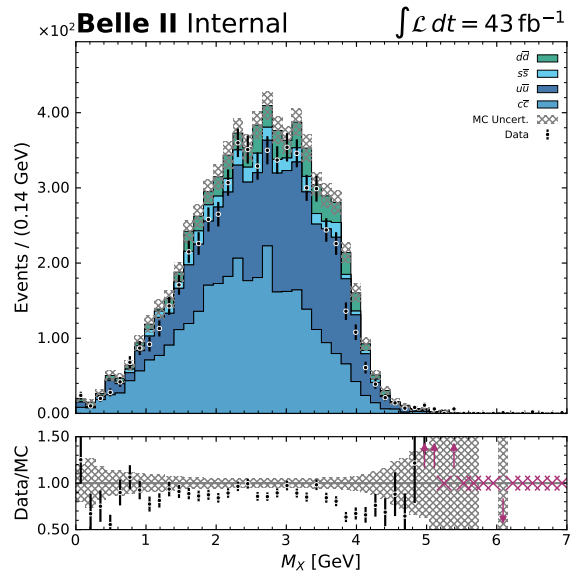
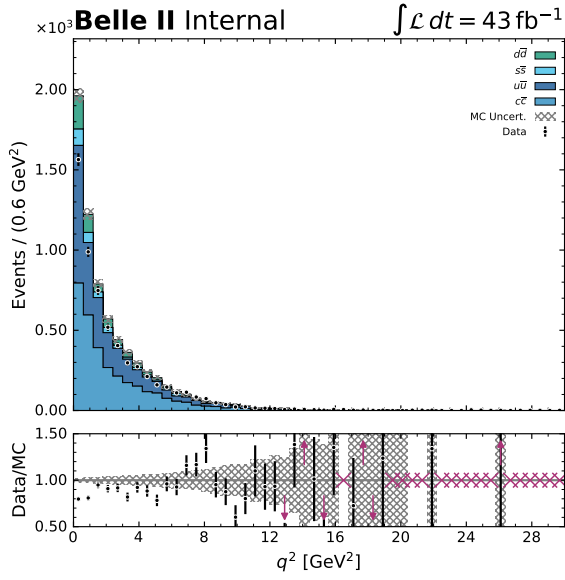
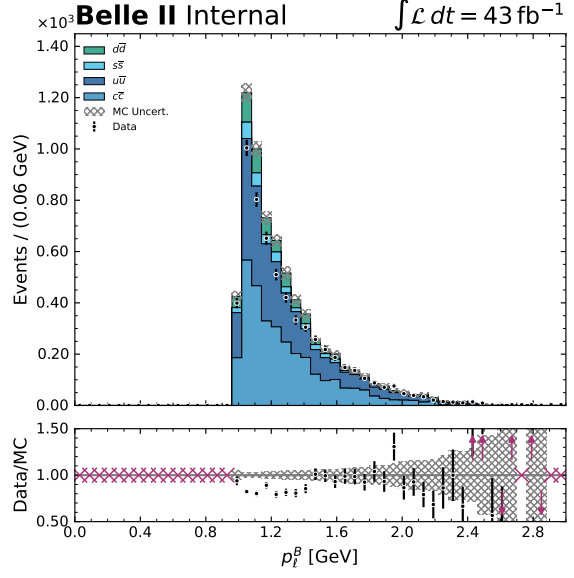
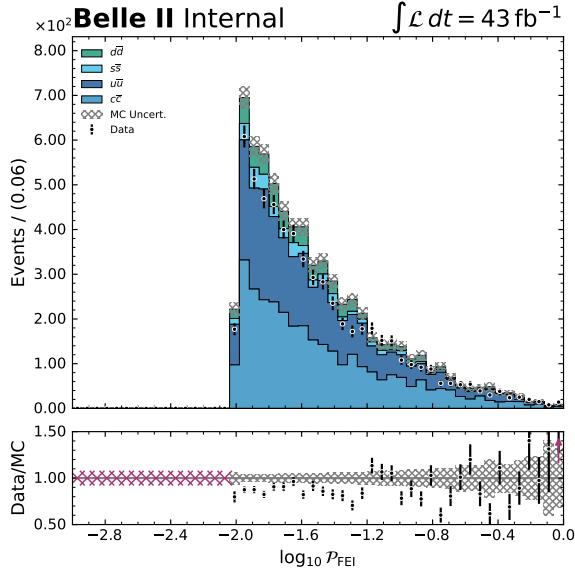




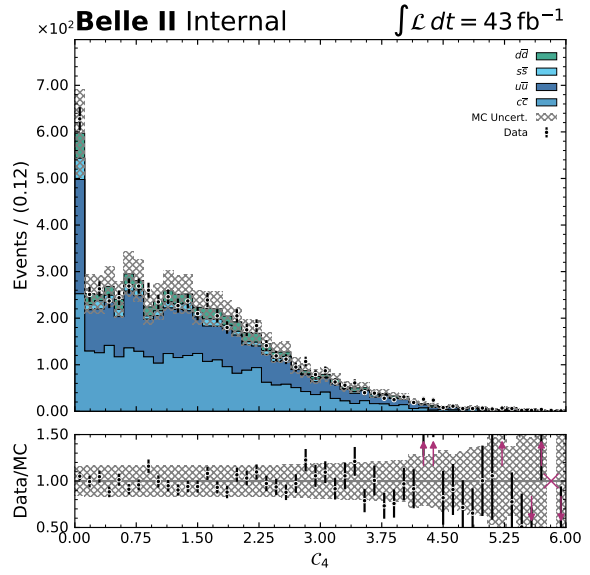
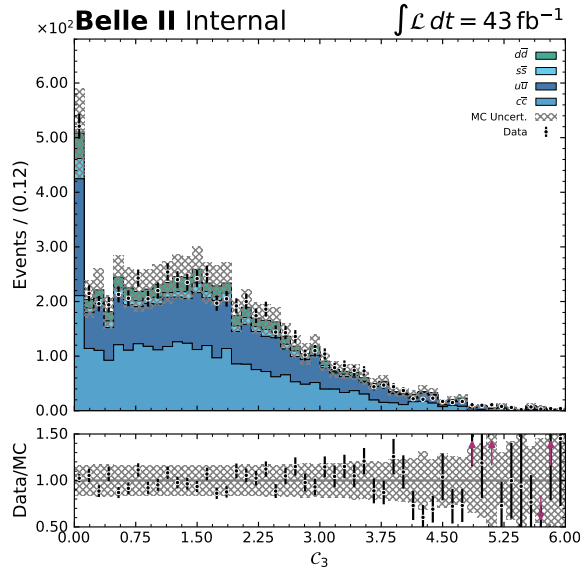
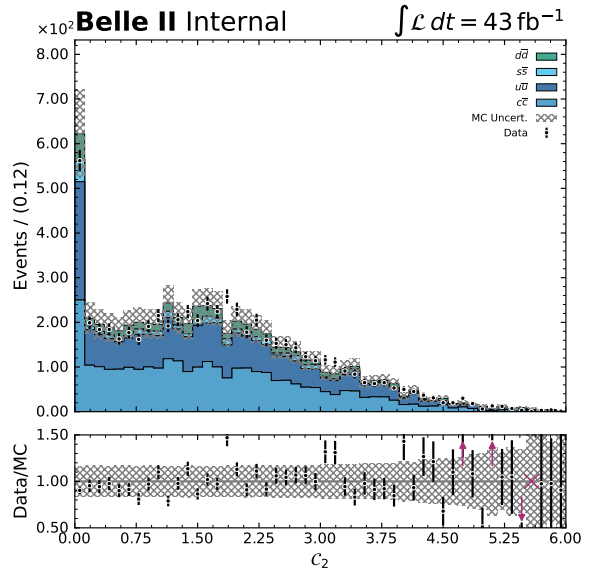
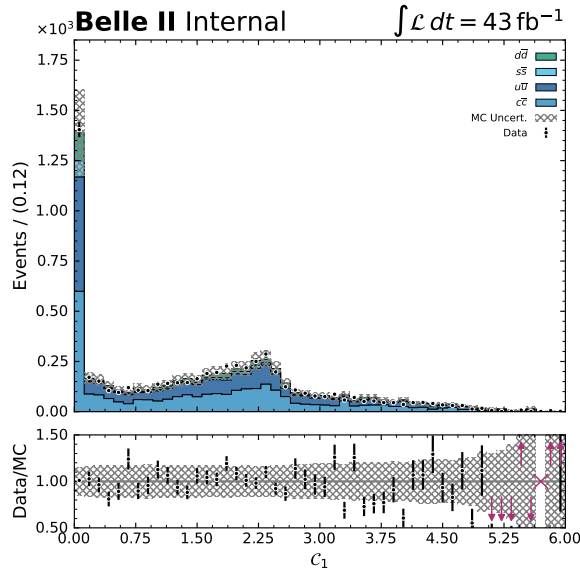


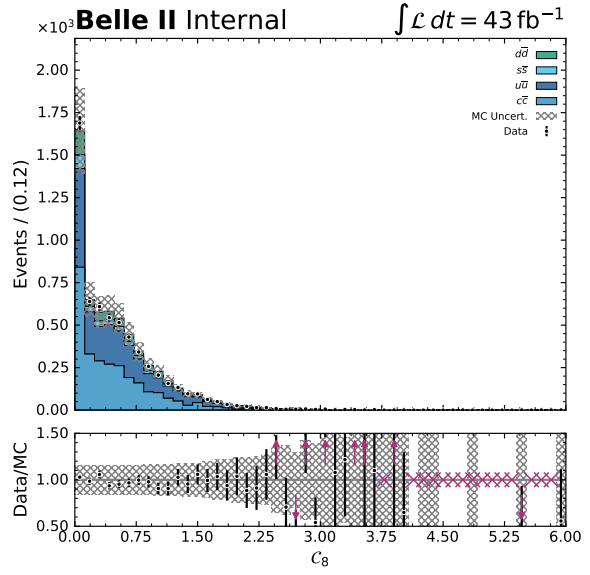
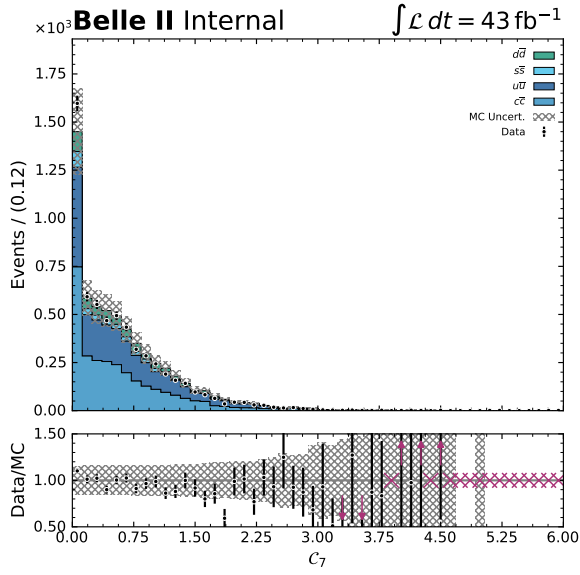
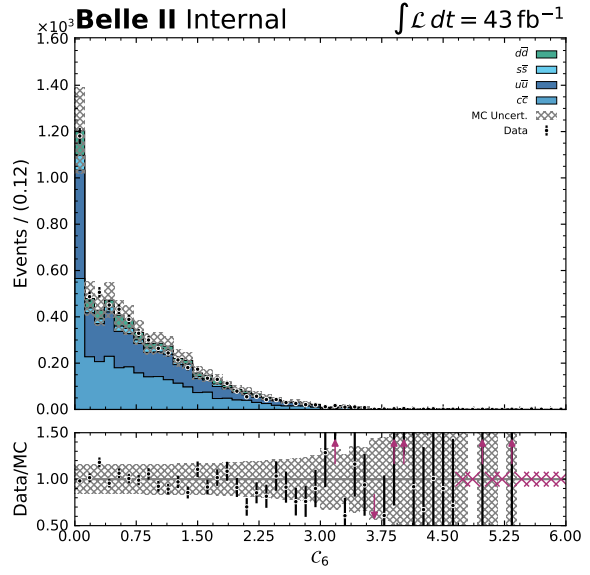
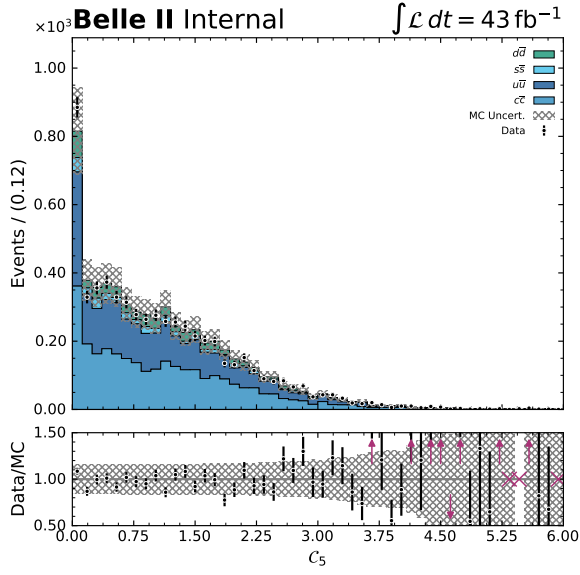


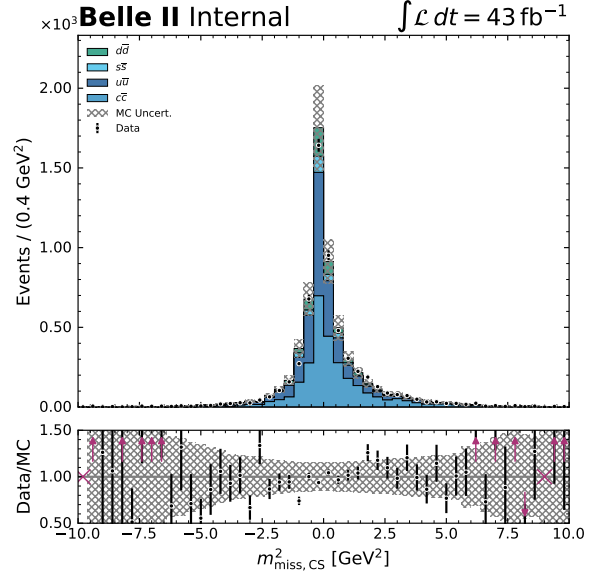
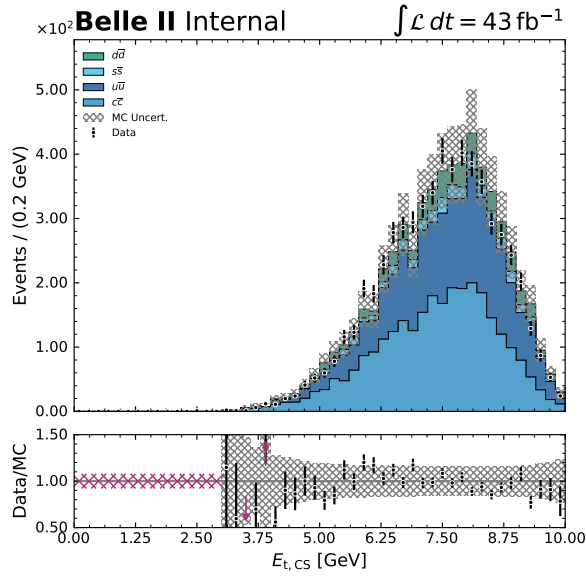
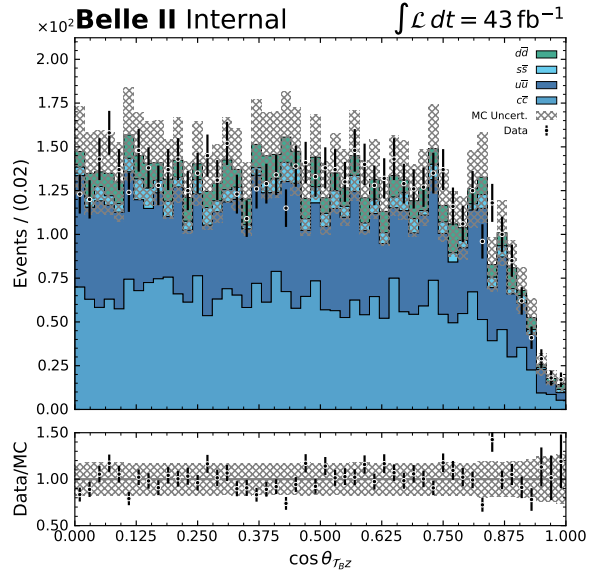
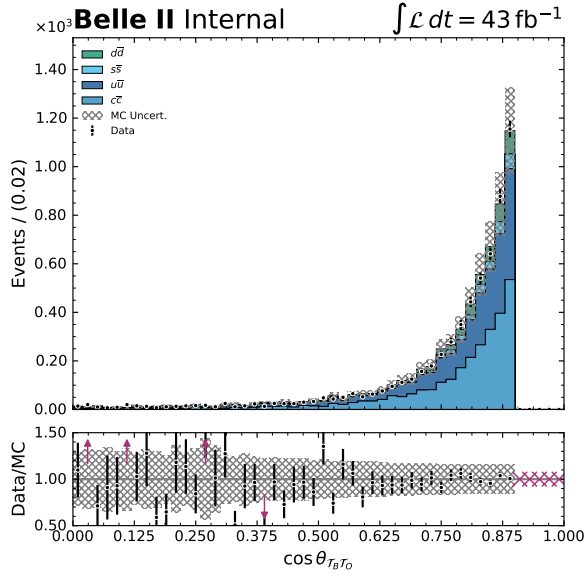


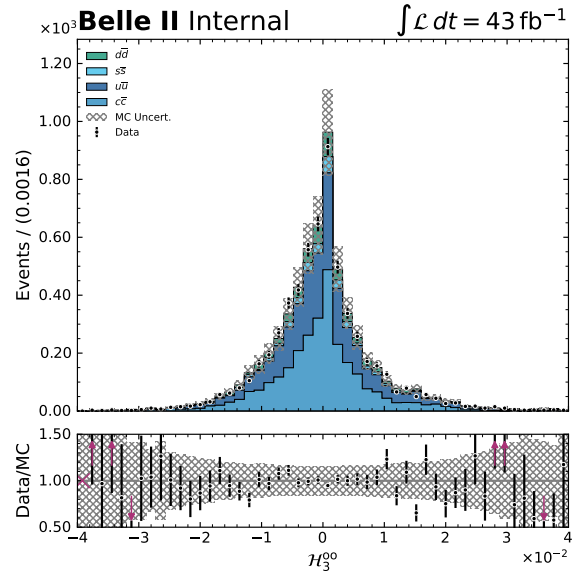
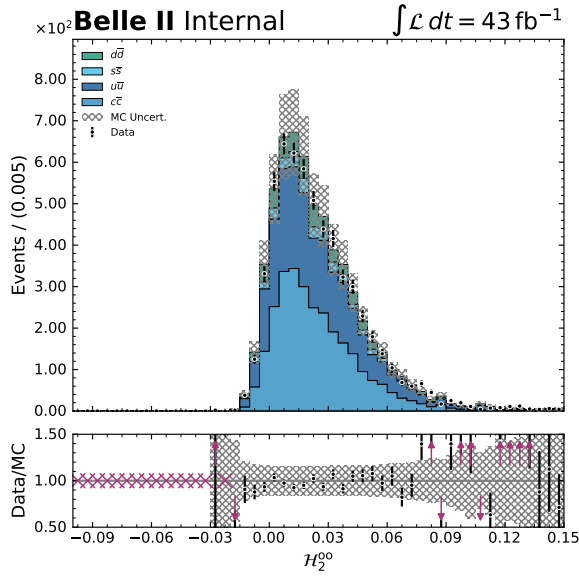
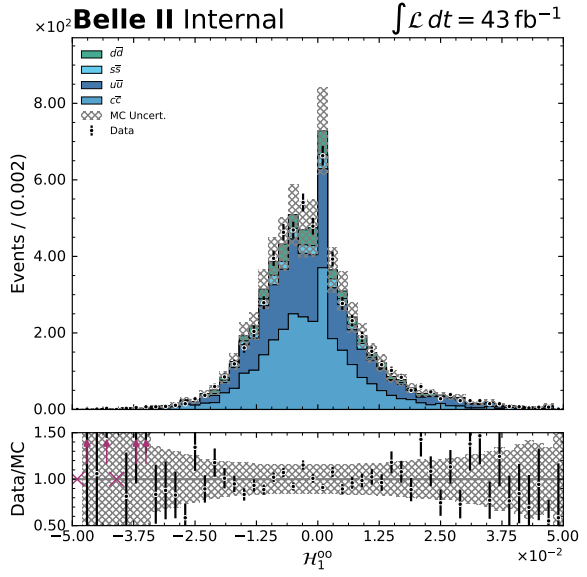
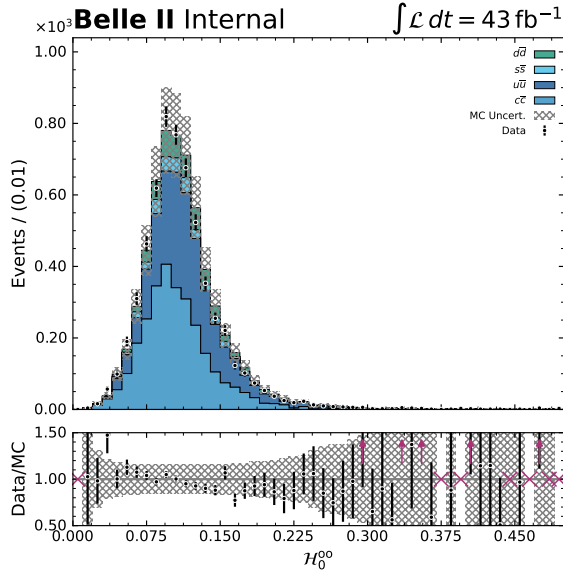


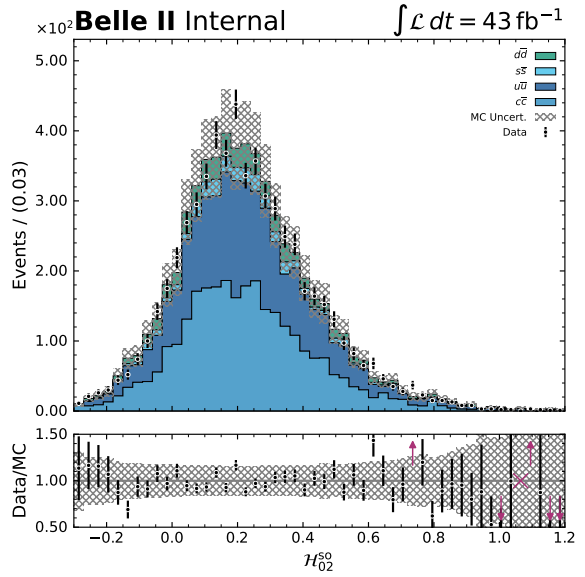
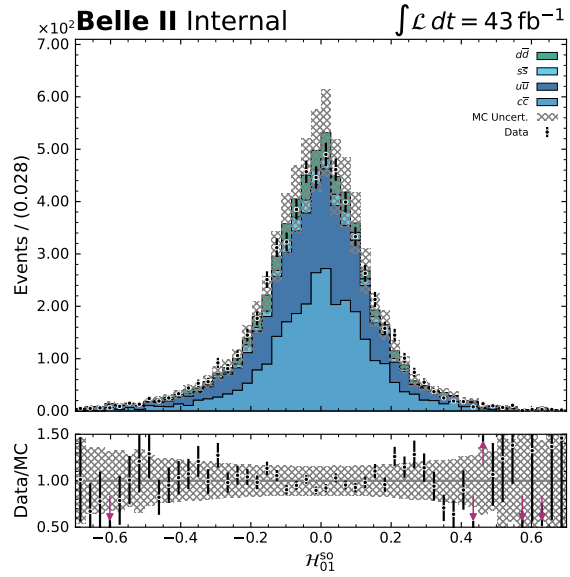
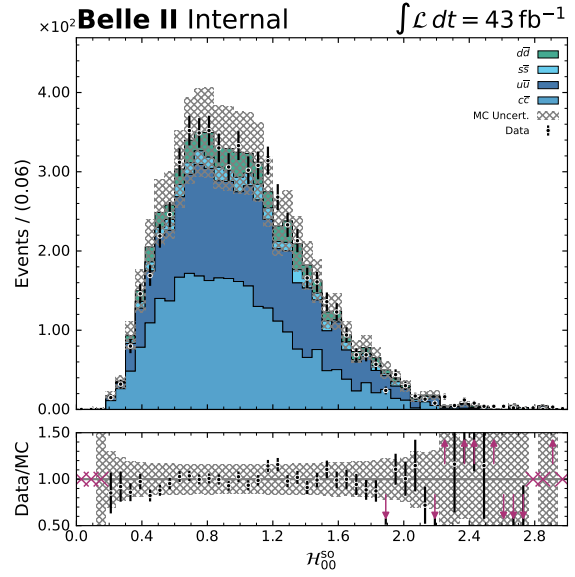
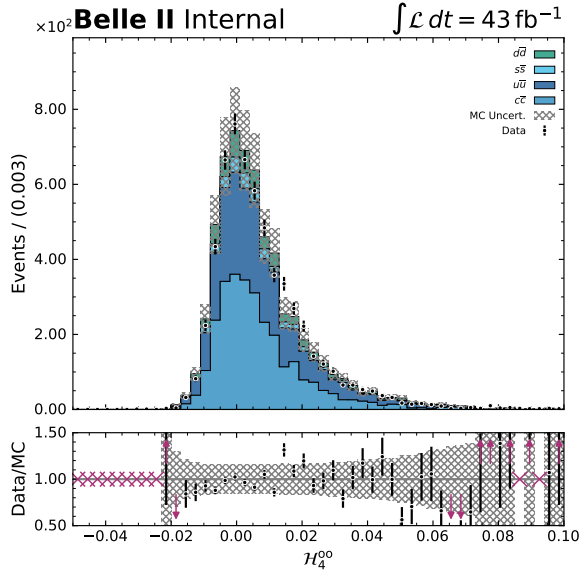
2140 **2. After applying corrections**

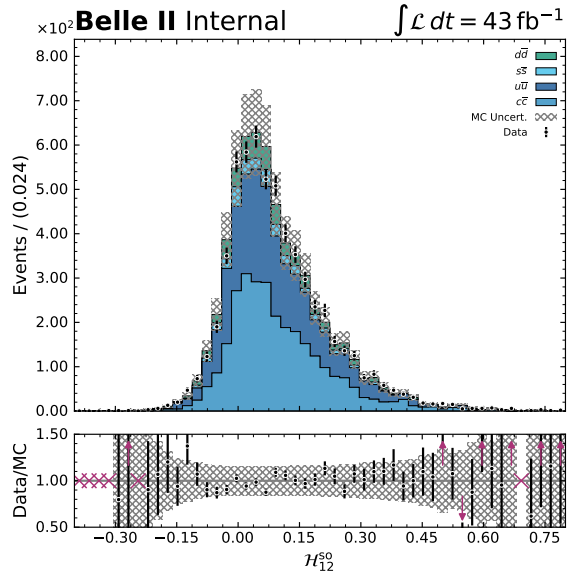
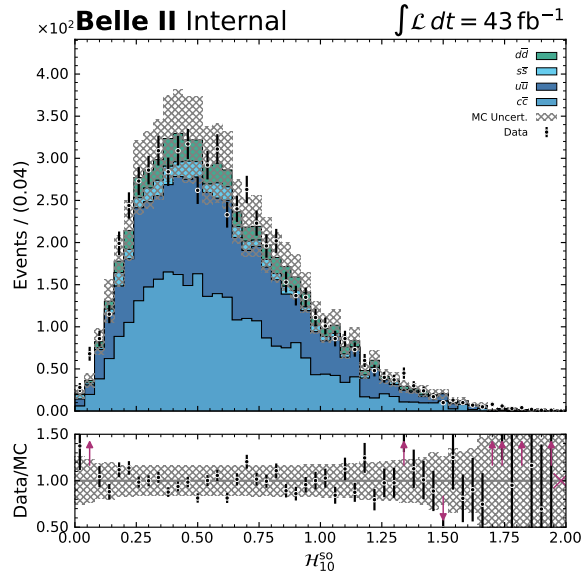
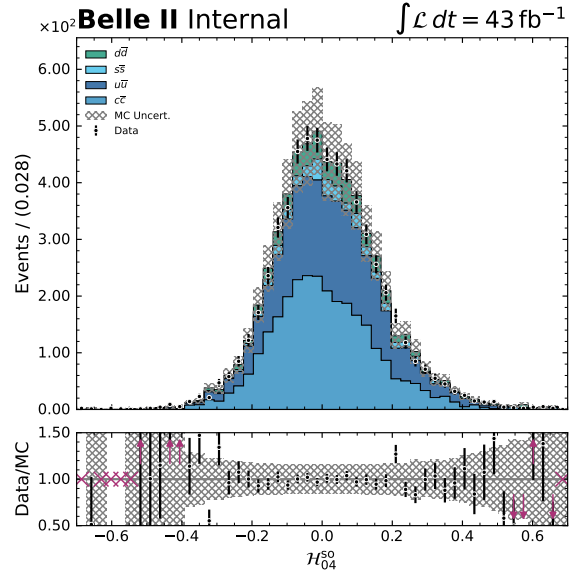
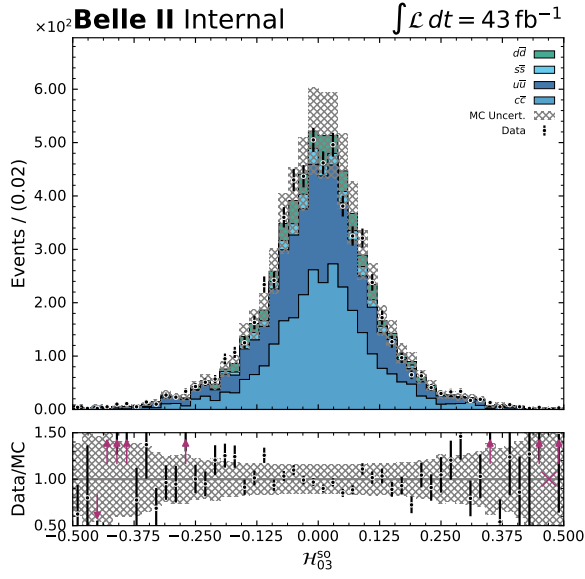


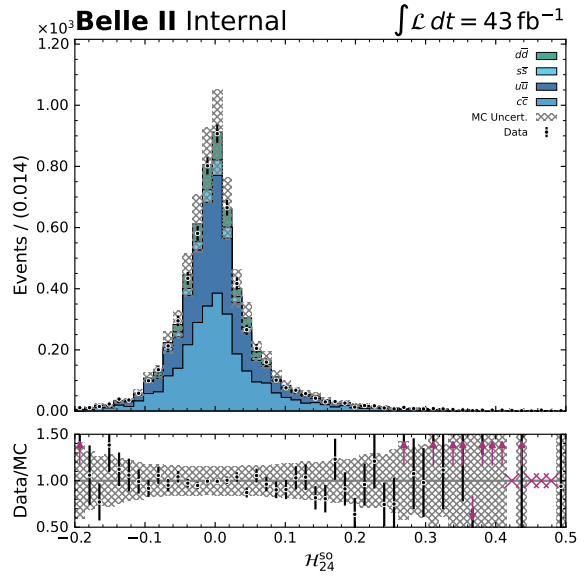
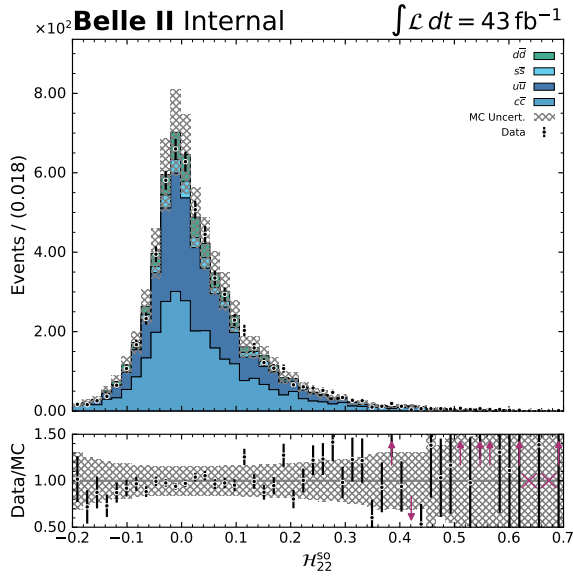
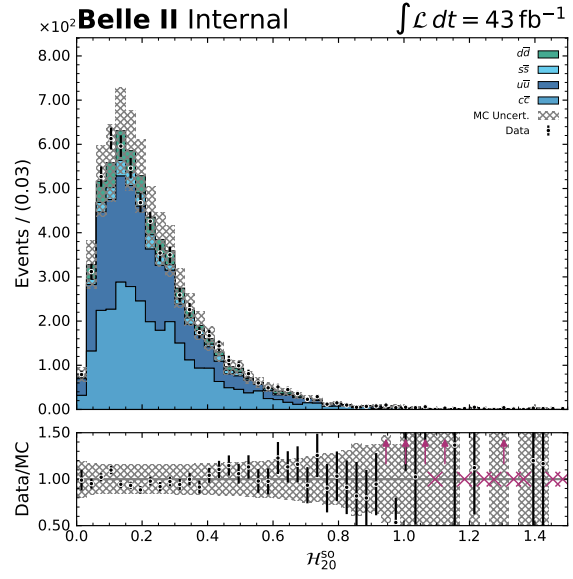
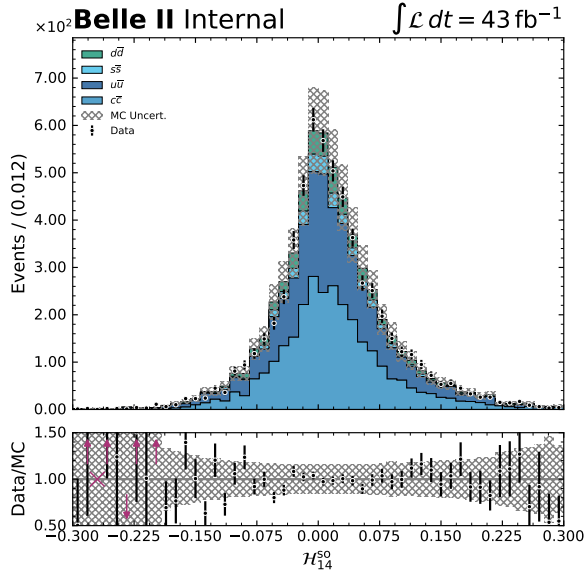


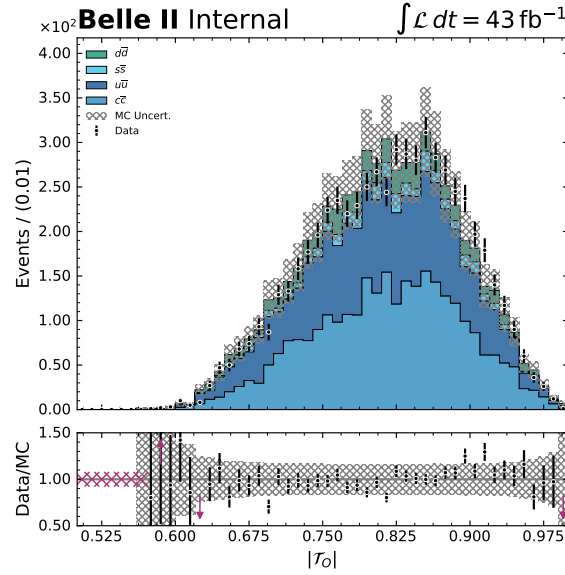
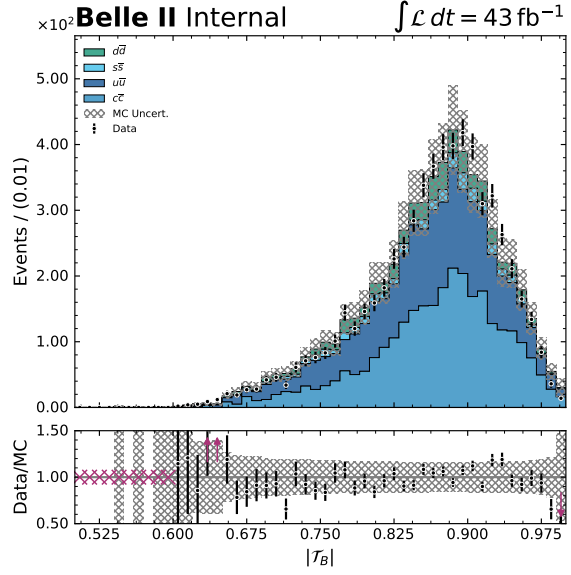
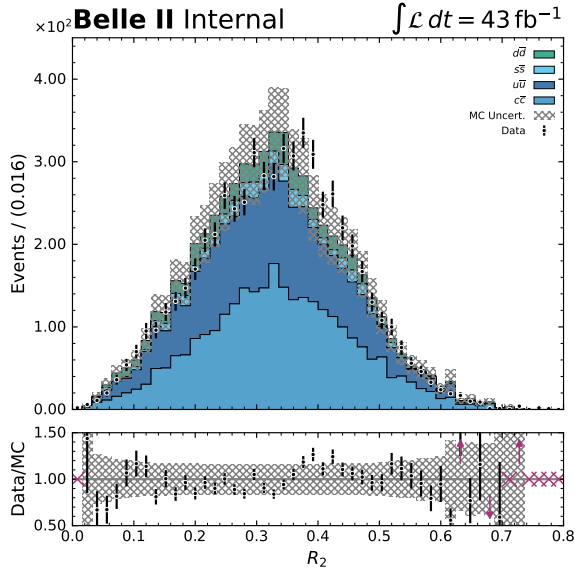


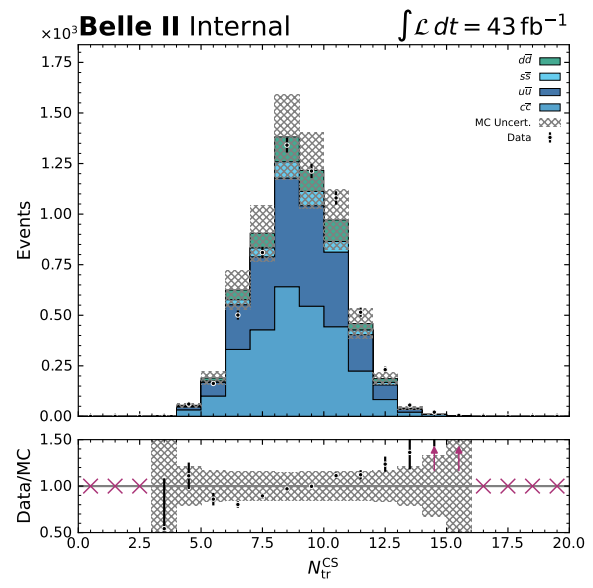
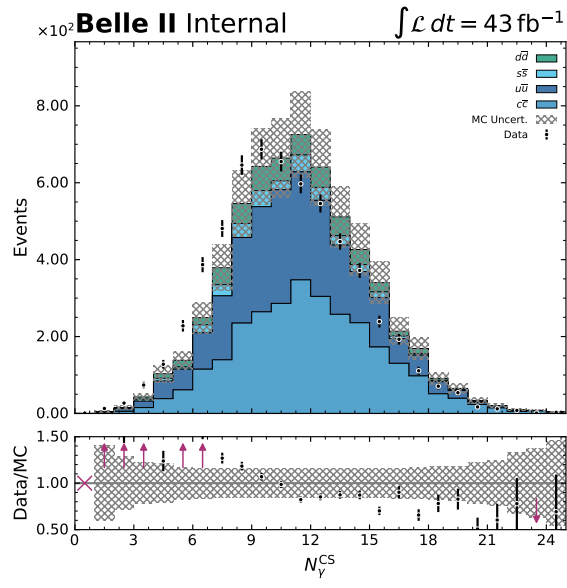


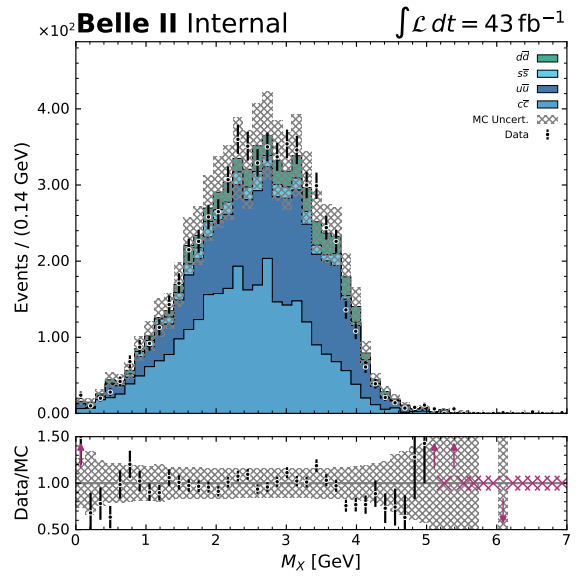
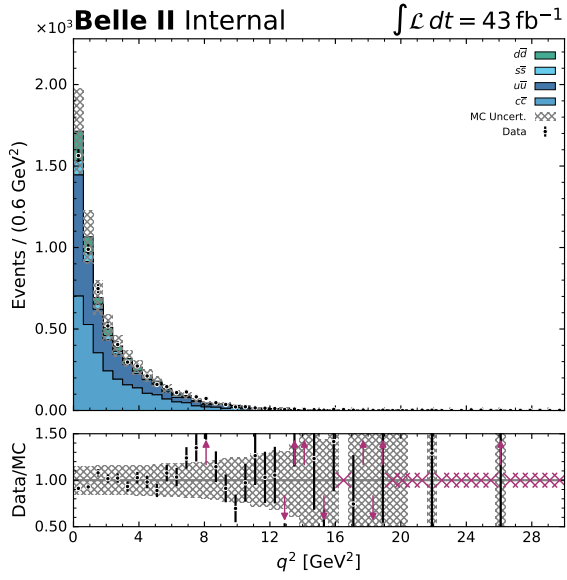
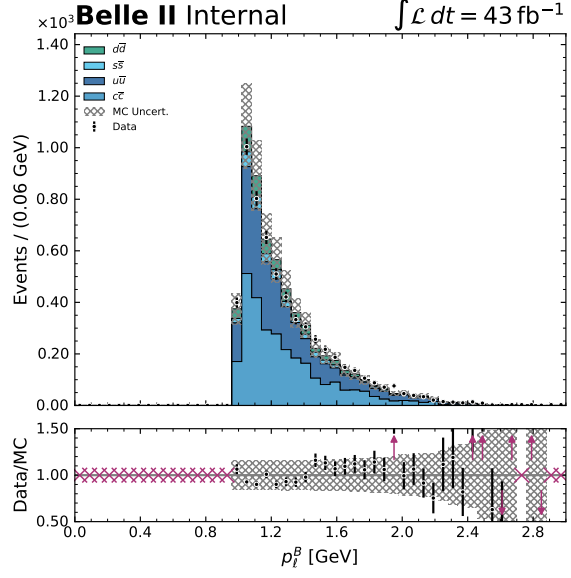
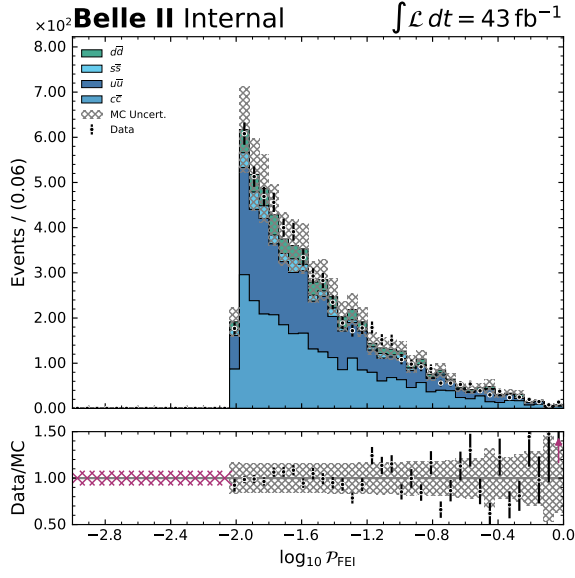












2141 M. TRACK AND CLUSTER MULTIPLICITIES

2142 We show in this Appendix the track and cluster multiplicities for X_u and X_c events in the
 2143 $B \rightarrow X\ell\nu$ region and the signal region.

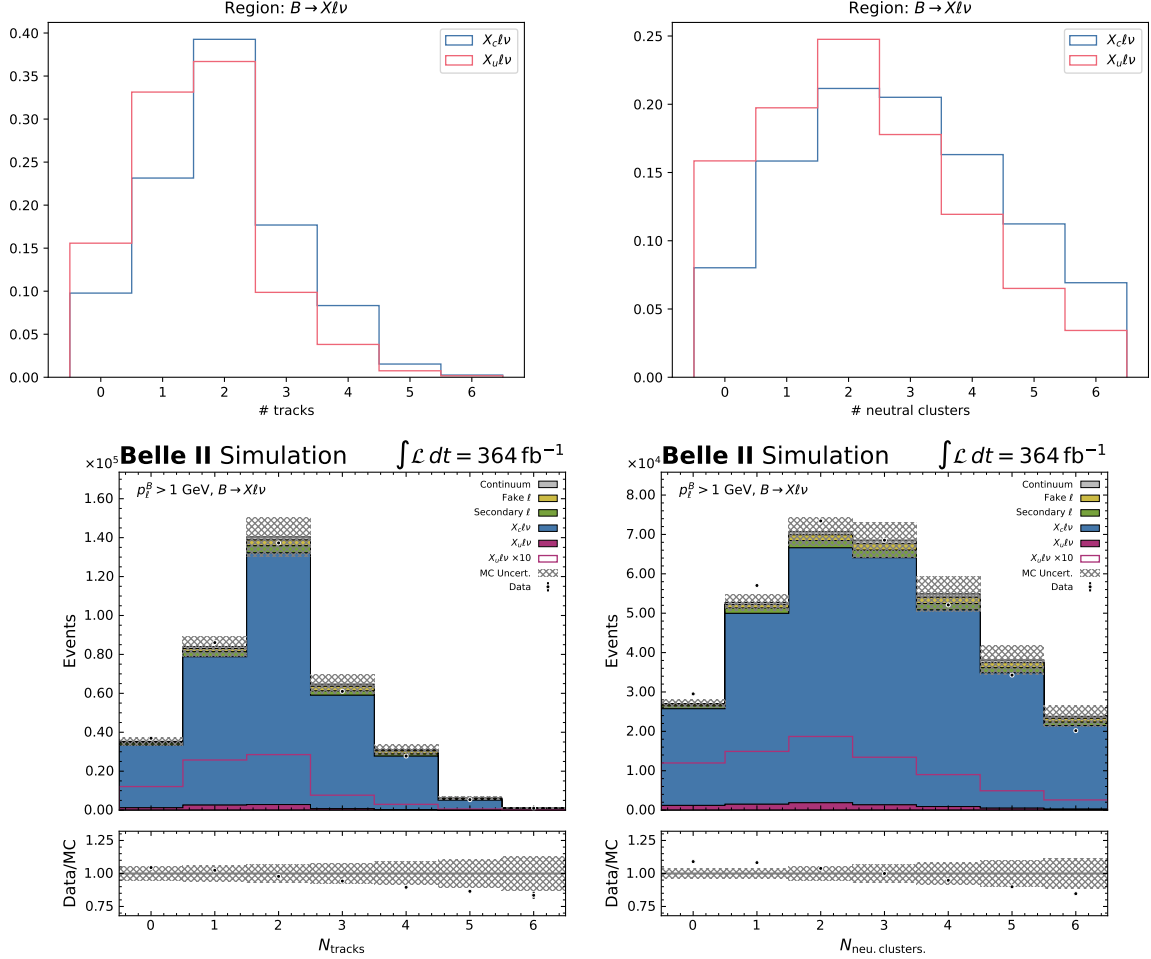


FIG. 144: The track (left) and neutral cluster (right) multiplicities in the $B \rightarrow X\ell\nu$ region. The top plot only compares the normalised MC $B \rightarrow X_c\ell\nu$ and $B \rightarrow X_u\ell\nu$ components.

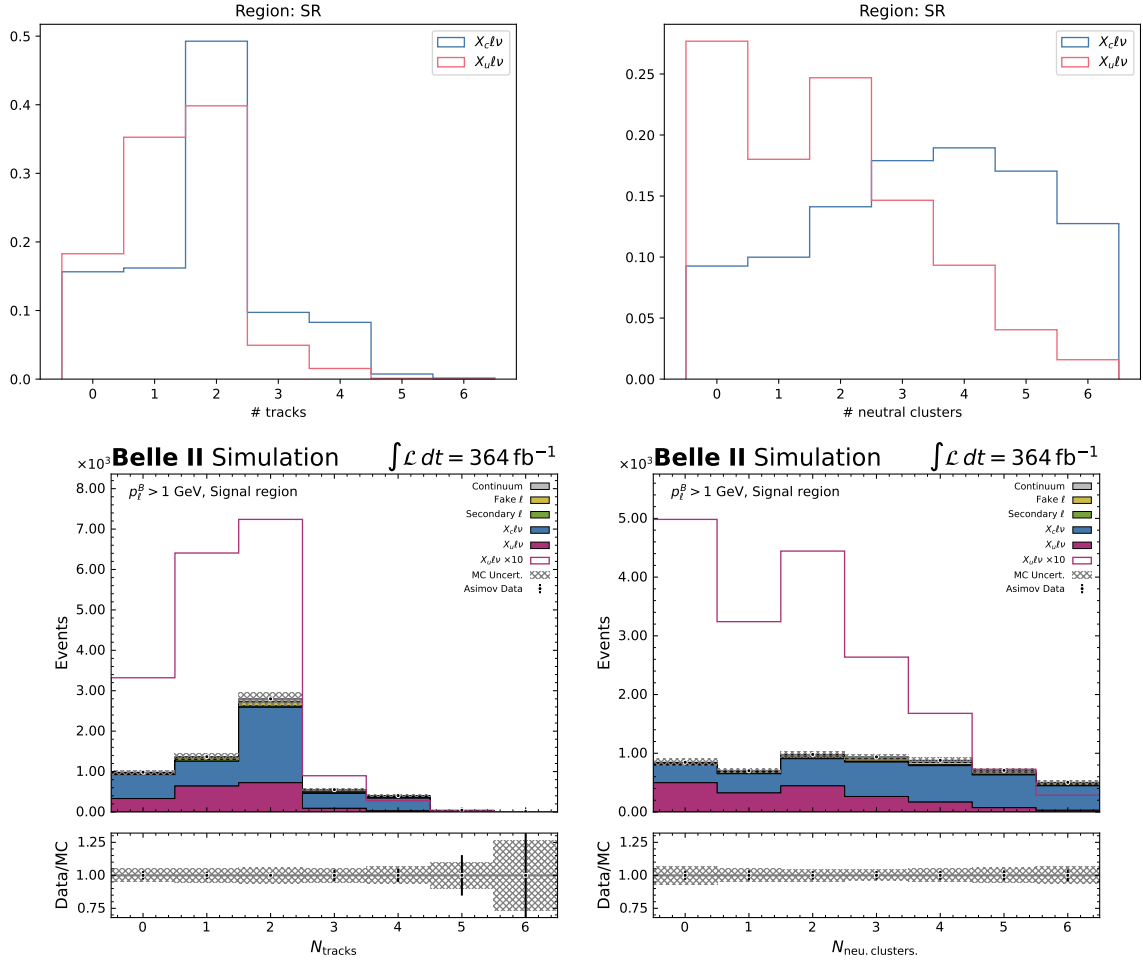


FIG. 145: The track (left) and neutral cluster (right) multiplicities in the signal region. The top plot only compares the normalised MC $B \rightarrow X_c \ell \nu$ and $B \rightarrow X_u \ell \nu$ components.

2144 N. LOW FEI CUT STUDIES

2145 We document in this Section a few tests performed on a sample with an FEI cut $\mathcal{P}_{\text{FEI}} > 0.001$
 2146 instead of $\mathcal{P}_{\text{FEI}} > 0.01$ as it is set in our nominal sample. We show in Figures 146 and 147
 2147 the data-MC agreement for the three main kinematical variables for the $B \rightarrow X\ell\nu$ and
 2148 $\text{CR}_{0,\text{low}}$ regions respectively. We show in Figure 148 the residuals for q^2 for different FEI
 2149 cuts. Finally, we performed a fit using the low FEI cut sample by repeating exactly the same
 2150 procedure as described in Section 8 for the nominal fit. The pre- and post-fit q^2 distributions
 2151 are shown in Figure 149. From the Asimov fit, the obtained uncertainty is comparable to
 2152 the one obtained in our nominal fit.

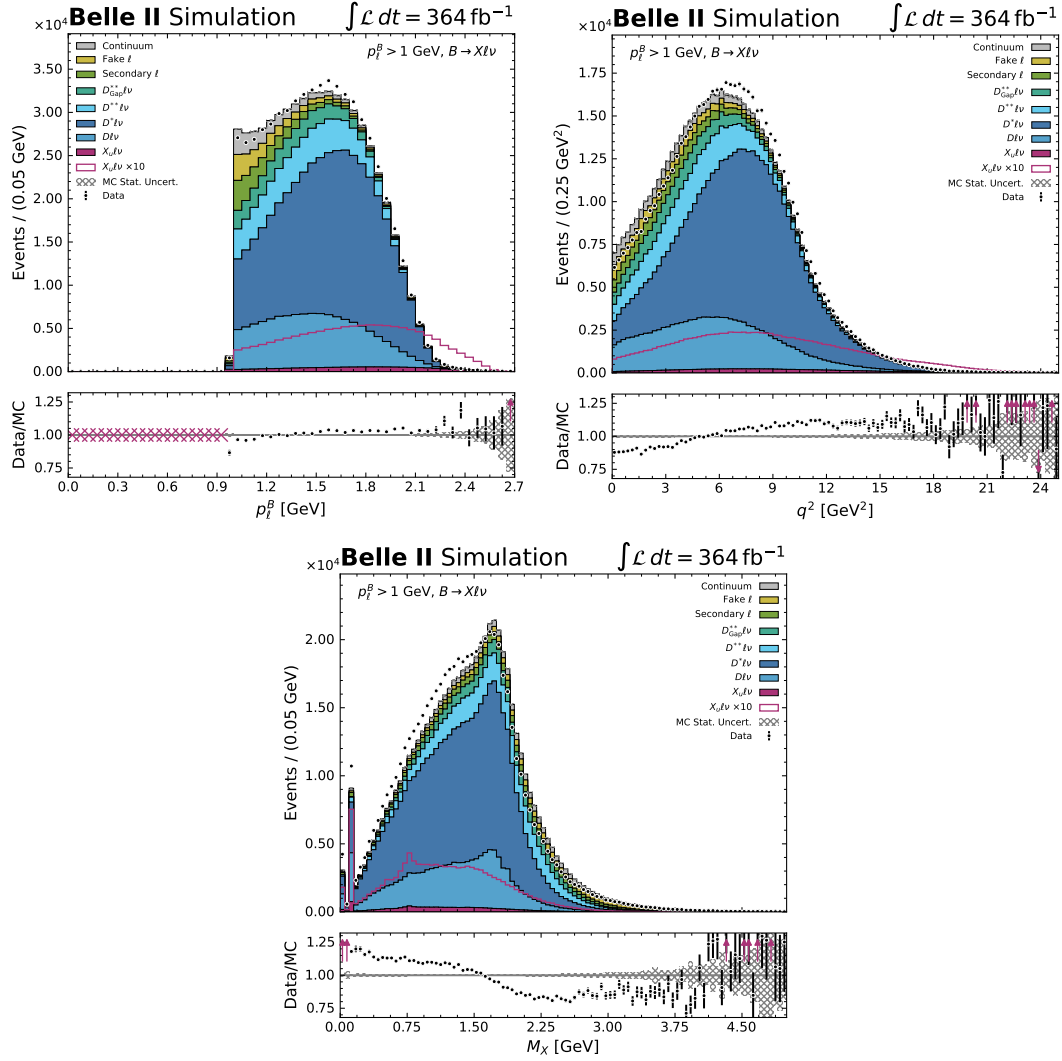


FIG. 146: Kinematic variables in $B \rightarrow X\ell\nu$ region

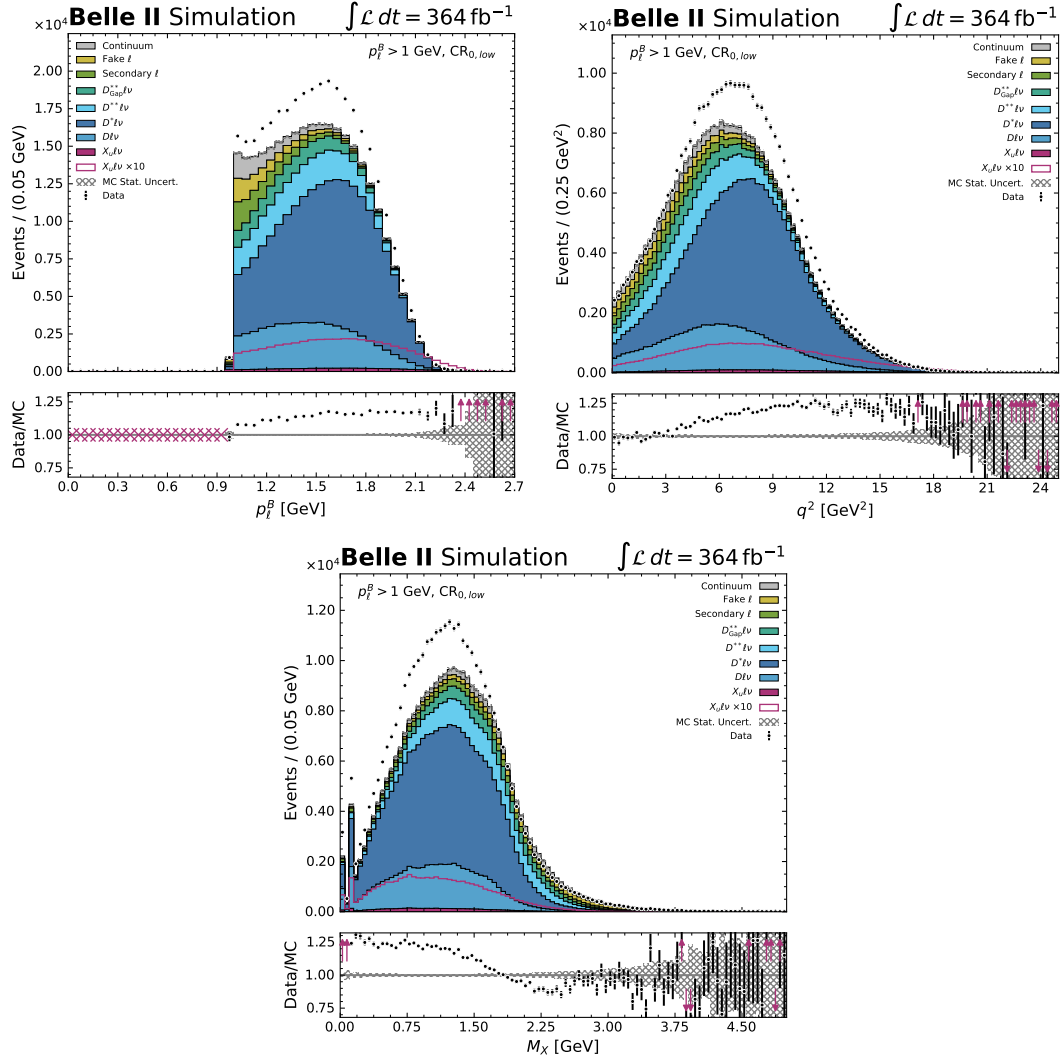


FIG. 147: Kinematic variables in $B \rightarrow X \ell \nu$ region

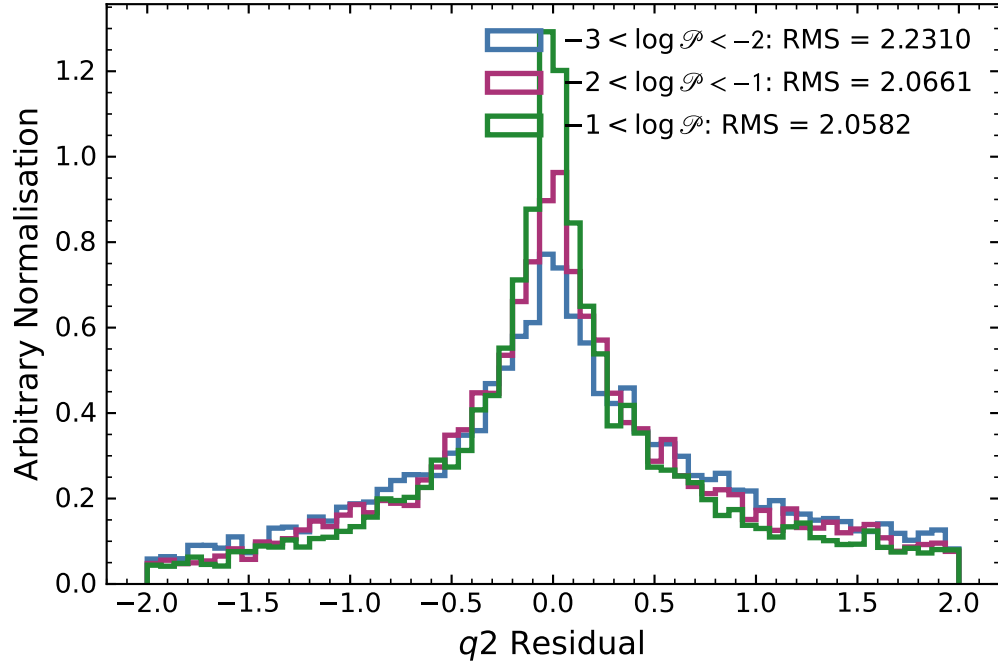


FIG. 148: q^2 residuals for different FEI score windows

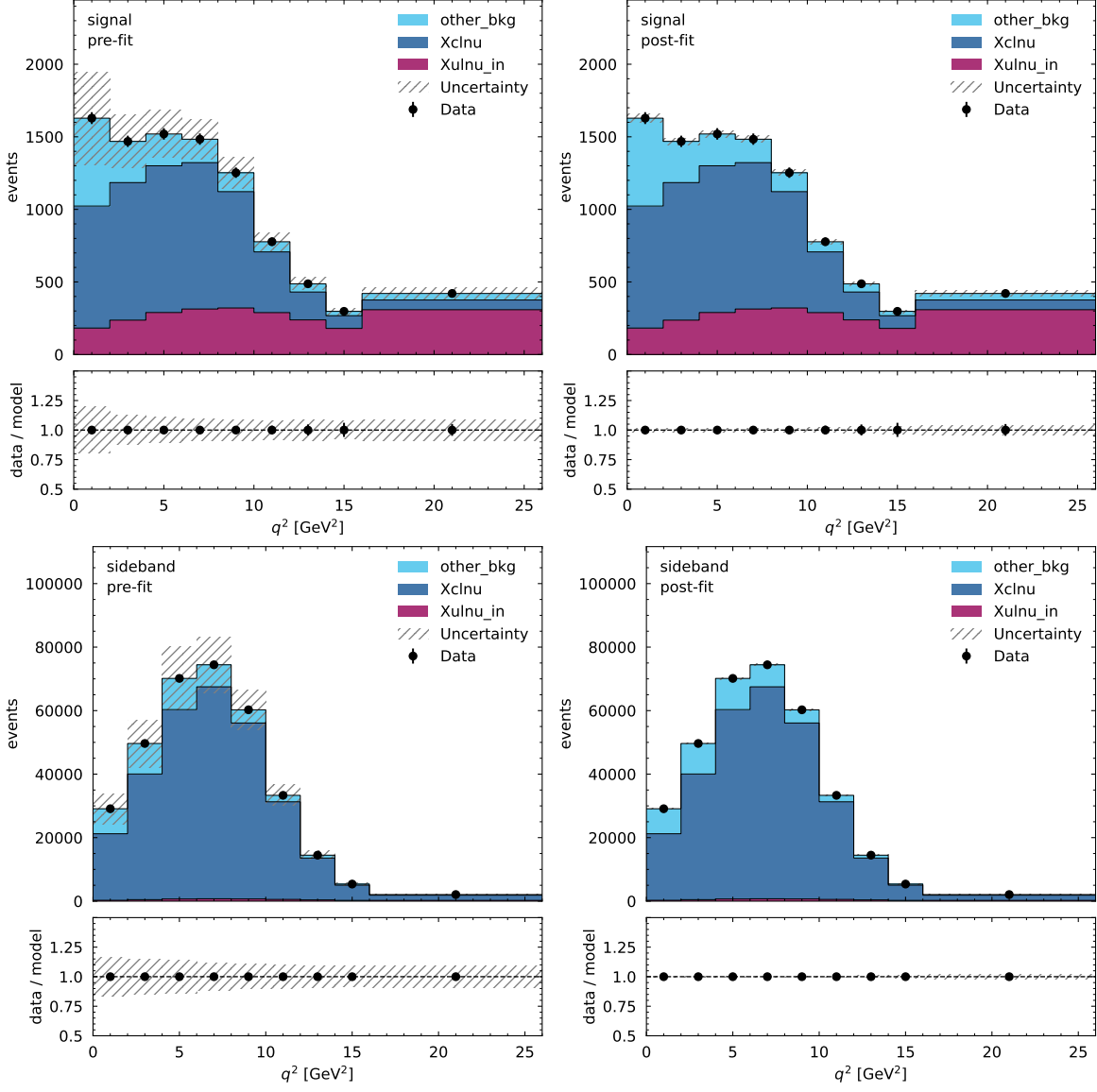


FIG. 149: Signal region (top) and control region (bottom) pre- (left) and post-fit (right) q^2 distributions

2153 O. STRANGE QUARK FRAGMENTATION

2154 1. γ_s

2155 We use the two most recent measurements of γ_s from JADE [11] (0.27 ± 0.06) and TASSO [12]
 2156 (0.35 ± 0.05). To cover the full range of values covered by these two measurements we
 2157 choose a 1σ range of 0.30 ± 0.09 (the value of 0.30 is also used at LHCb though at much
 2158 higher energies). After discussing directly with the Pythia authors we summarise here
 2159 the conclusions. The default Pythia value of 0.217 comes from the Monash tune which is
 2160 dominated by LEP data (the γ_s parameter is called `StringFlav:probStoUD` in Pythia 8.3).
 2161 In Figure 150 (left) it can be seen that this value yields a good agreement for the average
 2162 kaon multiplicity for high energies. However, at low energies the data undershoots the
 2163 Pythia prediction. The same distribution with a value of $\gamma_s = 0.35$ is shown in Figure 150
 2164 (right). The plot suggests that a more correct value should rather be somewhere in the
 2165 range 0.25 - 0.30 for low energies. However, the key issue is that there are no recent reliable
 2166 measurements for γ_s for low energies. In our case, the hadronisation processes occur below
 2167 5 GeV, mostly in the range 1-3 GeV. Therefore, based on all the available information the
 2168 most reasonable choice is to use a rather broad range between 0.21 - 0.39 which covers all
 2169 the different values quoted here.

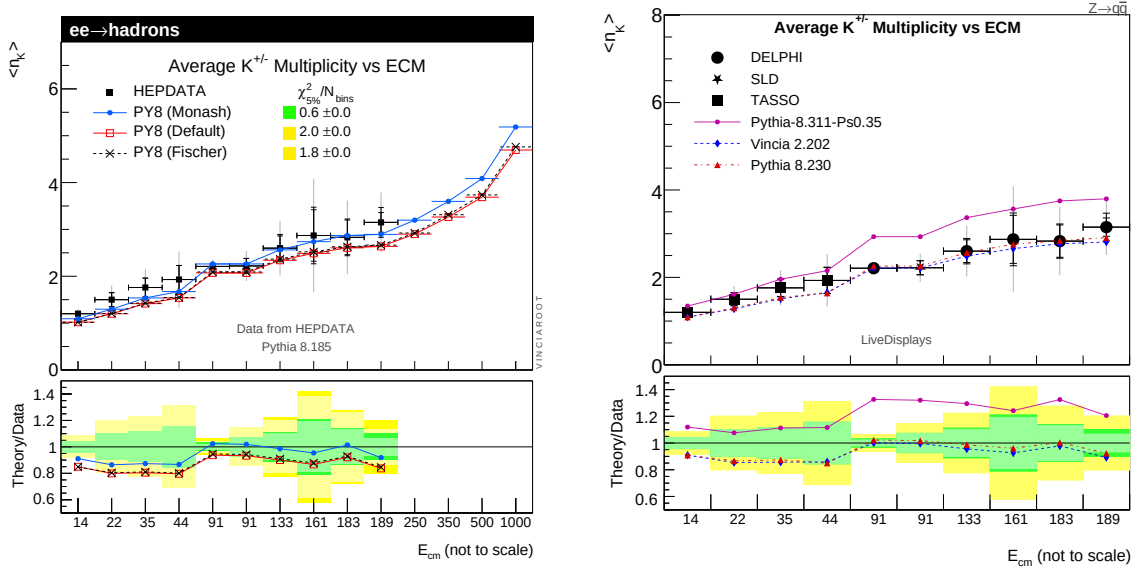


FIG. 150: Kaon multiplicity at different energies for different measurements and Pythia tunes. On the left plot the Pythia 8.3 parameter `StringFlav:probStoUD` (γ_s) is set to 0.217 and on the right it is set to 0.35.

2170 2. Fraction of kaons before/after hybridisation

2171 Kaons in $B \rightarrow X_u \ell \nu$ decays are only produced in non-resonant events which are scaled down
 2172 by the hybrid procedure and replaced by resonant events. Therefore, the number of kaon
 2173 events is reduced by the hybrid procedure which can impact the reconstruction efficiency of
 2174 $B \rightarrow X_u \ell \nu$ events since kaons are rejected in the signal regions used in our fits. The impact

2175 of such an effect will be limited by the hadronic mass necessary to produce a pair of kaons
2176 (~ 1 GeV whereas the threshold to produce the lowest non-resonant hadronic system is
2177 twice the pion mass so ~ 0.3 GeV). Kaon events represent about 5%-15% of all $B \rightarrow X_u \ell \nu$
2178 events in all fit regions.

2179

2180 After applying the hybrid procedure, the number of kaon events among all inclusive
2181 $B \rightarrow X_u \ell \nu$ events is reduced by 10%, 20% and 30% in fit regions 1, 3 and 5, respec-
2182 tively. As described in the previous section, we vary the value of γ_s , and therefore the
2183 number of kaon events, by 30% which covers the shifts caused by the hybrid method.

2184

2185 Besides, the kaon event shift could also be different for charged and neutral B decays
2186 because of the kaon charges. Before applying the hybrid method to the inclusive sample
2187 of $B \rightarrow X_u \ell \nu$ events, there's about 1-2% more kaon events in B^+ decays than in B^0 de-
2188 cays depending on the region. Afterwards, this difference shifts to 2-8% and so the hybrid
2189 weights introduce a difference of about 2-5%. This effect is larger in the broader regions of
2190 phase-space. The unbalance introduced by this effect is of the same order as the uncertainty
2191 assigned on $f^{\pm/00}$. It is also worth noting that some of the $B \rightarrow X_u \ell \nu$ modelling uncertain-
2192 ties are quite conservative, the pion multiplicity in regions 3 and 5 in particular. Therefore,
2193 any effect caused by the shift in the number of kaons due to the hybrid procedure is already
2194 well covered by the assigned uncertainties.

2195 P. VARIABLE DEPENDENT EFFICIENCY

2196 We show here the MC signal efficiency as a function of p_ℓ^B , M_X and q^2 at the generator level.
 2197 The efficiency is defined as the fraction of MC generated signal events that are selected by
 2198 the signal region fit 1 cuts (see the main text for a detail description of all these selections).
 2199 The selection $p_\ell^B > 1.0$ GeV is already applied. The efficiency is shown in percentage.

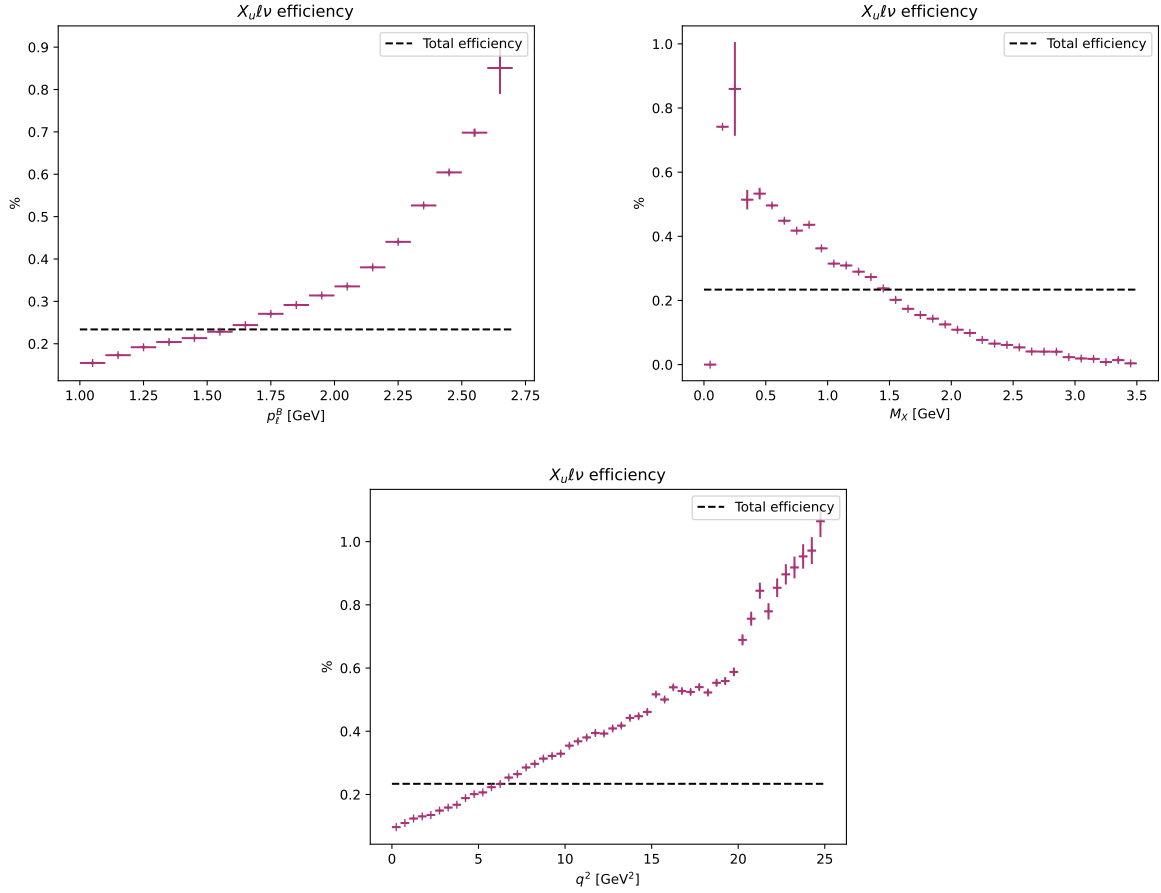


FIG. 151: Total generator level signal efficiency

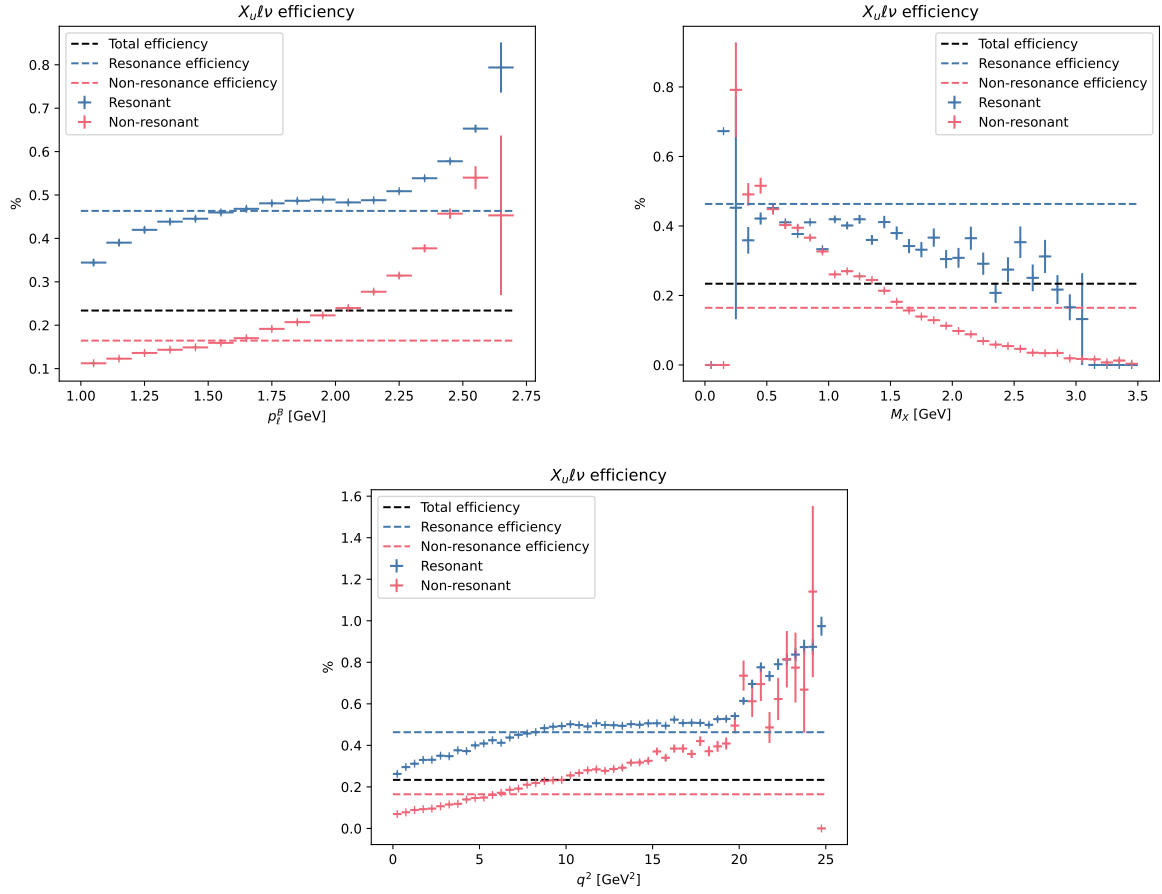


FIG. 152: Total generator level resonant and non-resonant signal efficiency

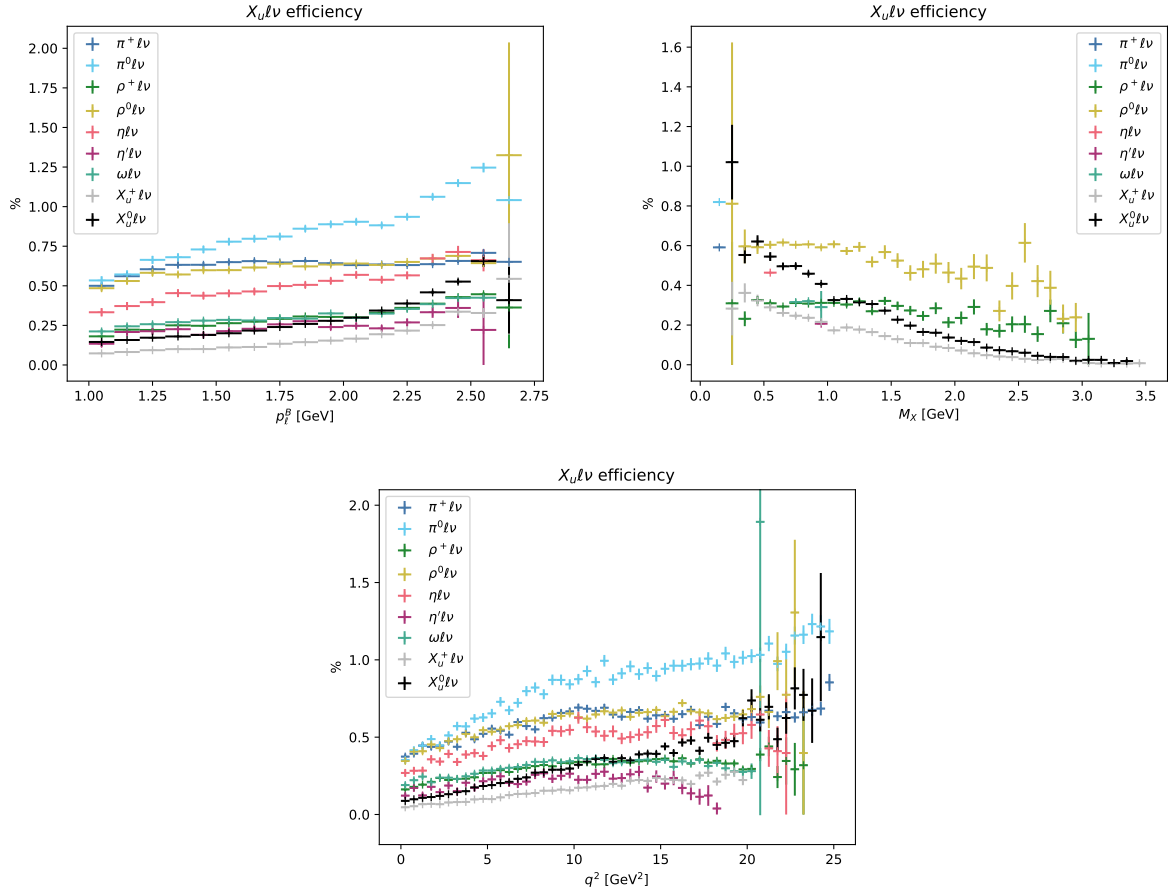
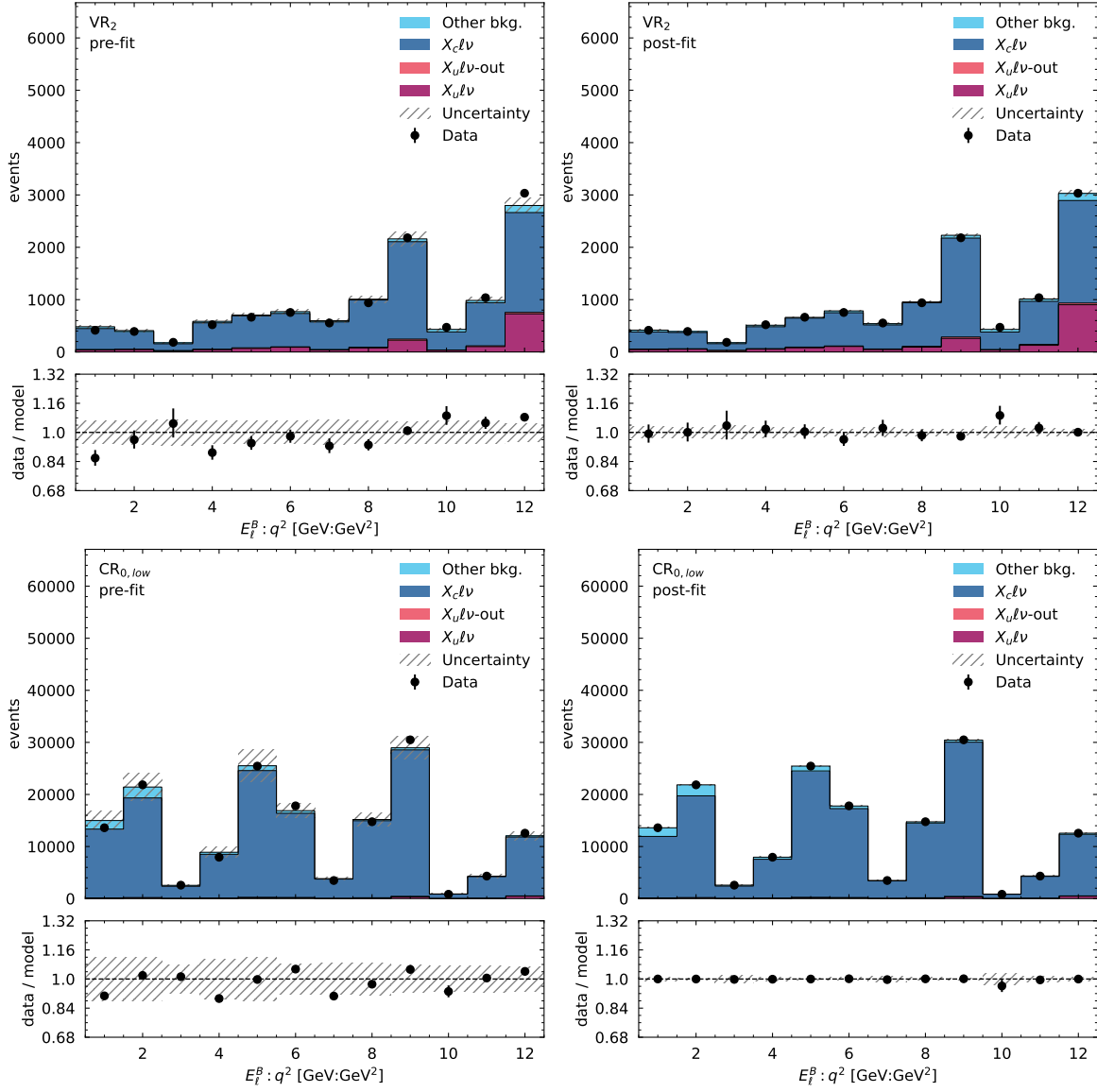


FIG. 153: Total generator level signal efficiency broken down in different resonant and non-resonant modes

FIG. 154: Pre and postfit $E_\ell^B : q^2$ distributions of the $\text{CR}_{0,\text{low}}$ - VR₂ fit 3.

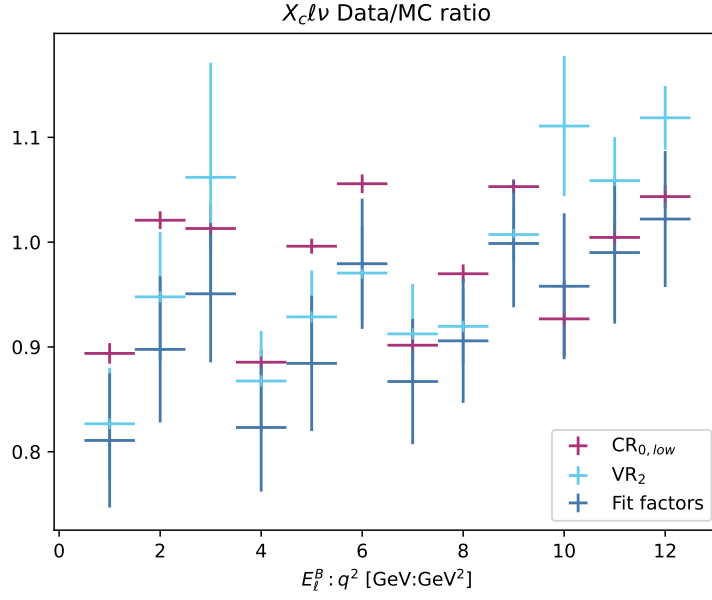


FIG. 156: Comparison of the normalised prefit (fit 3) $X_c \ell \nu$ data/MC ratios in $CR_{0,low}$ and VR_2 with statistical uncertainties and $X_c \ell \nu$ shape factors with full fit uncertainties.

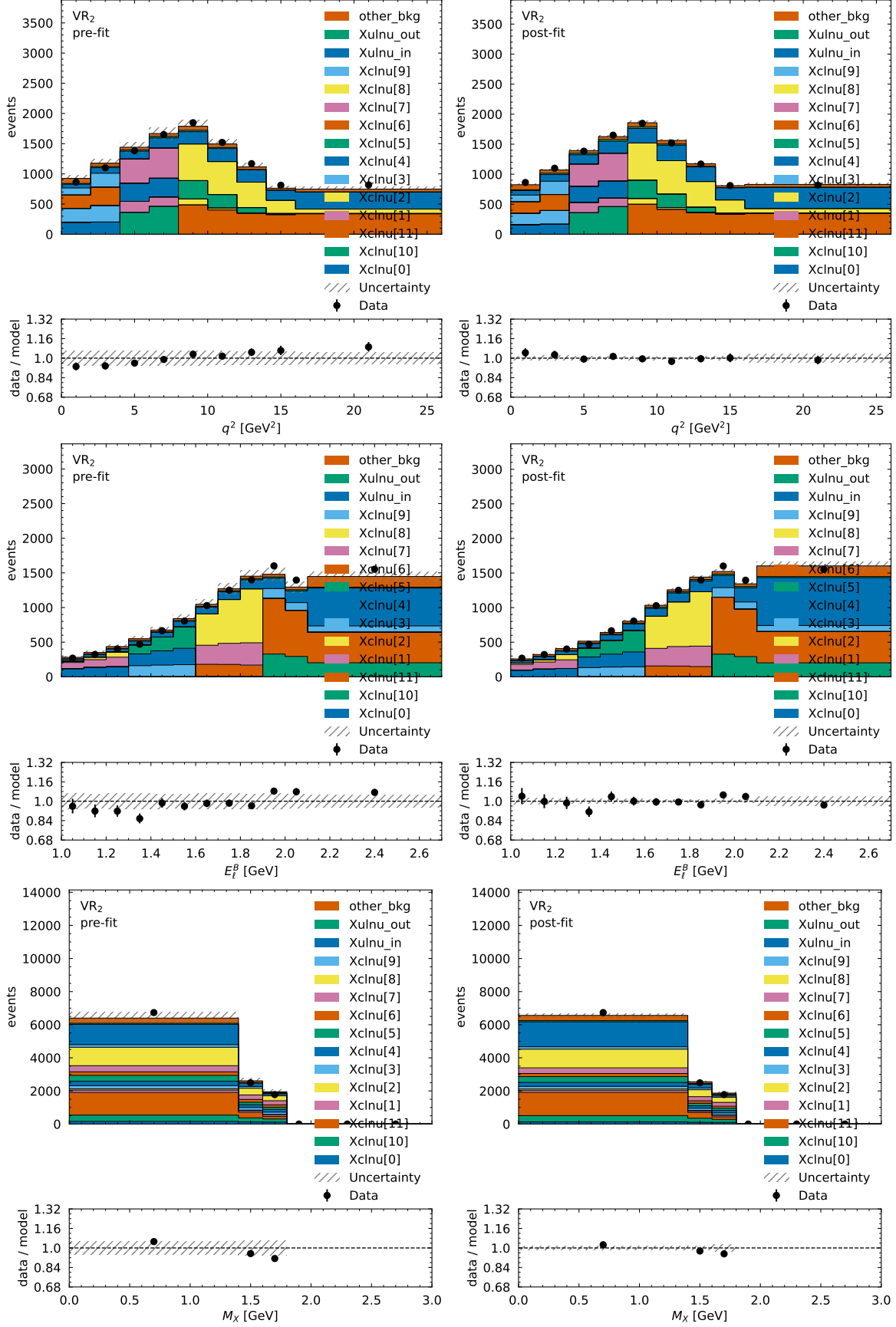
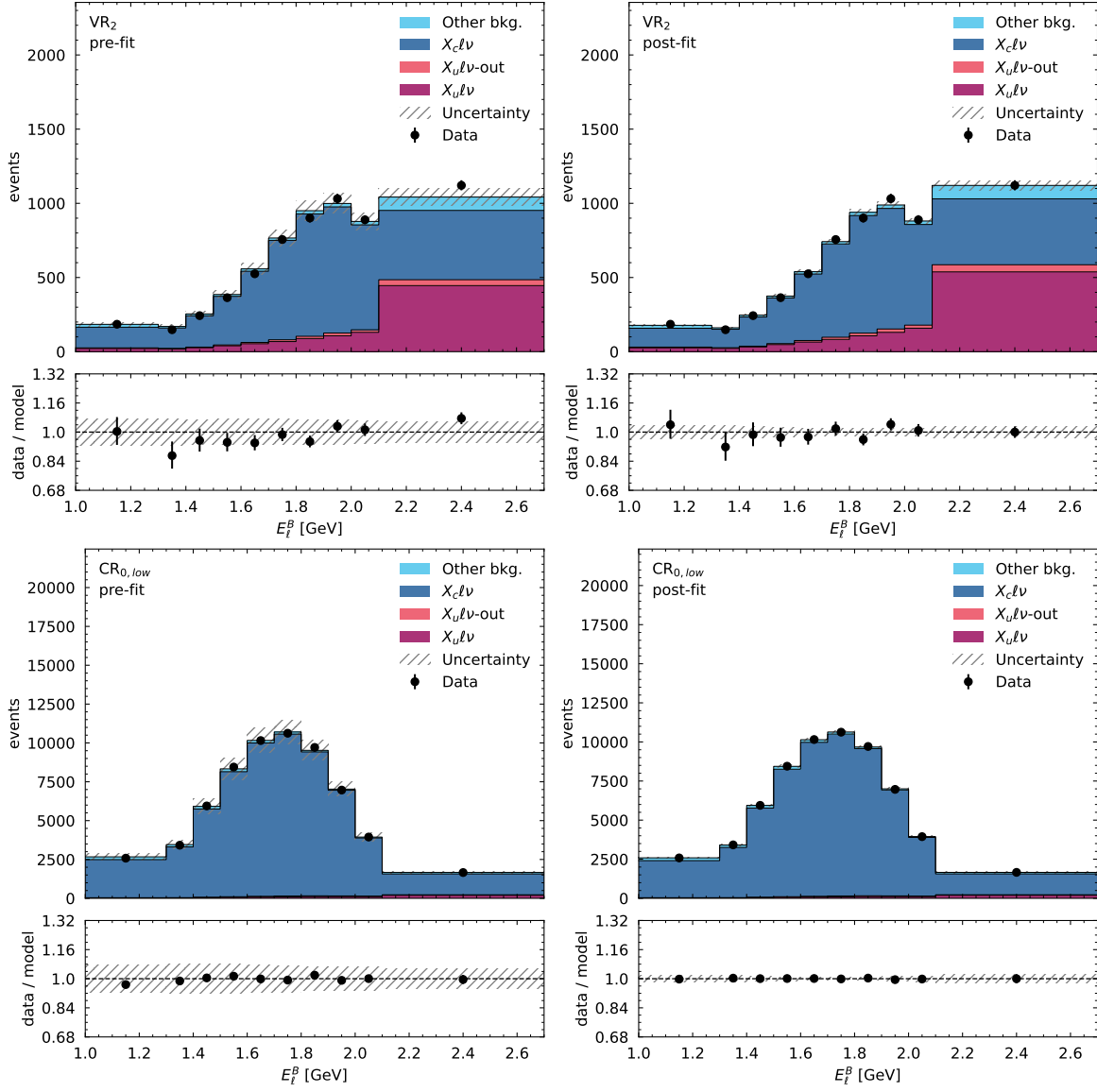


FIG. 157: Projections of the $E_\ell^B:q^2$ fit 3 on E_ℓ^B , q^2 and M_X .

FIG. 158: Pre and postfit E_{ℓ}^B distributions of the $CR_{0,low}$ - VR_2 fit 5.

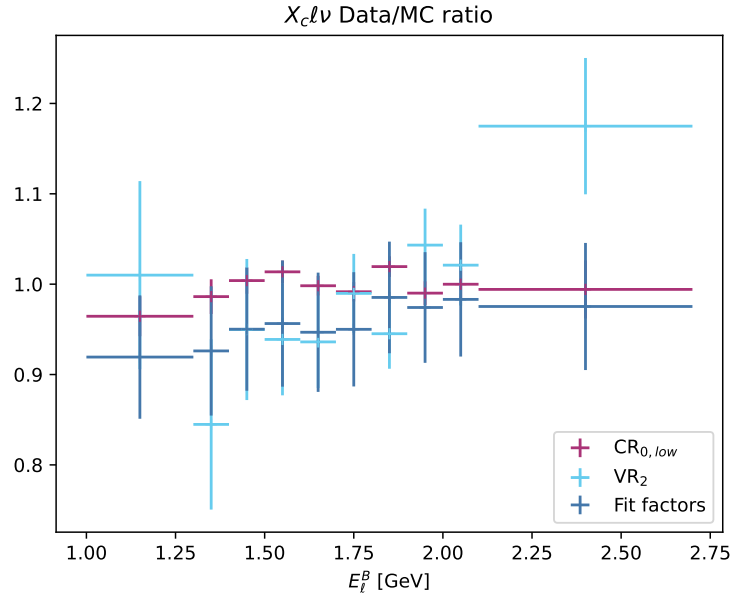


FIG. 160: Comparison of the normalised prefit (fit 5) $X_{c\ell\nu}$ data/MC ratios in $CR_{0,low}$ and VR_2 with statistical uncertainties and $X_{c\ell\nu}$ shape factors with full fit uncertainties.

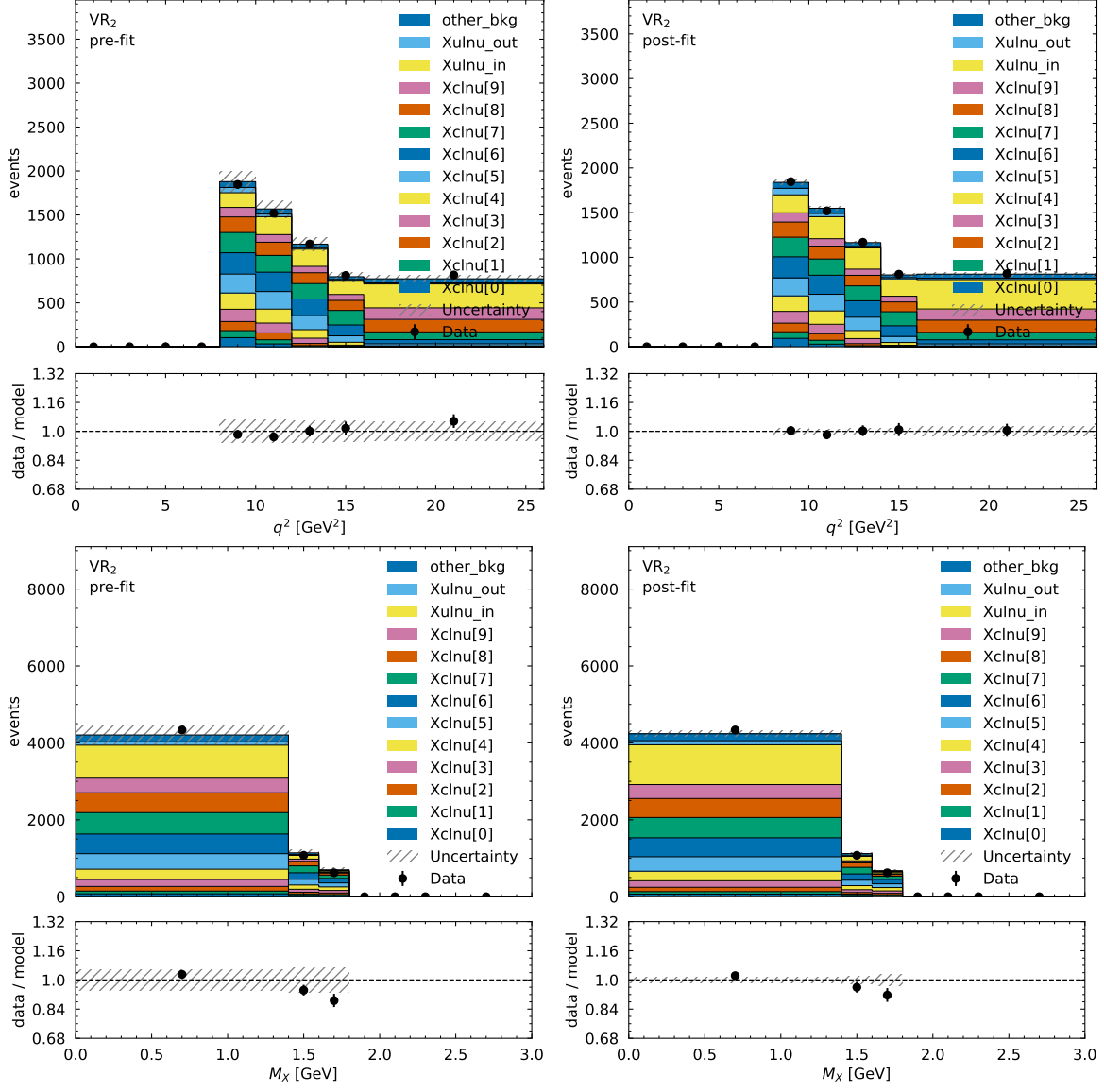


FIG. 161: Projections of the E_ℓ^B fit 5 on q^2 and M_X .

2203 R. SIGNAL REGION DISTRIBUTIONS

2204 1. Splits in lepton flavour

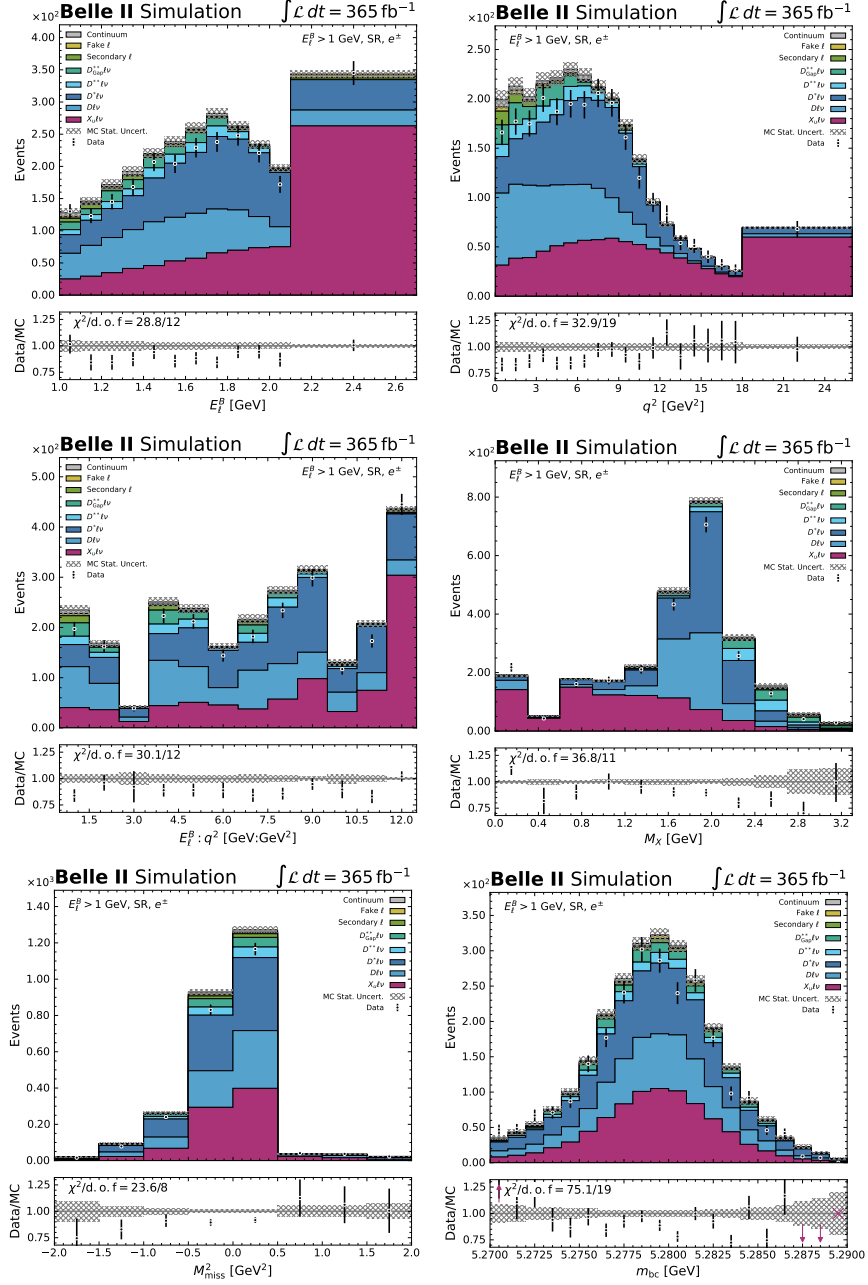


FIG. 162: Signal region distributions for $X_u e \nu$.

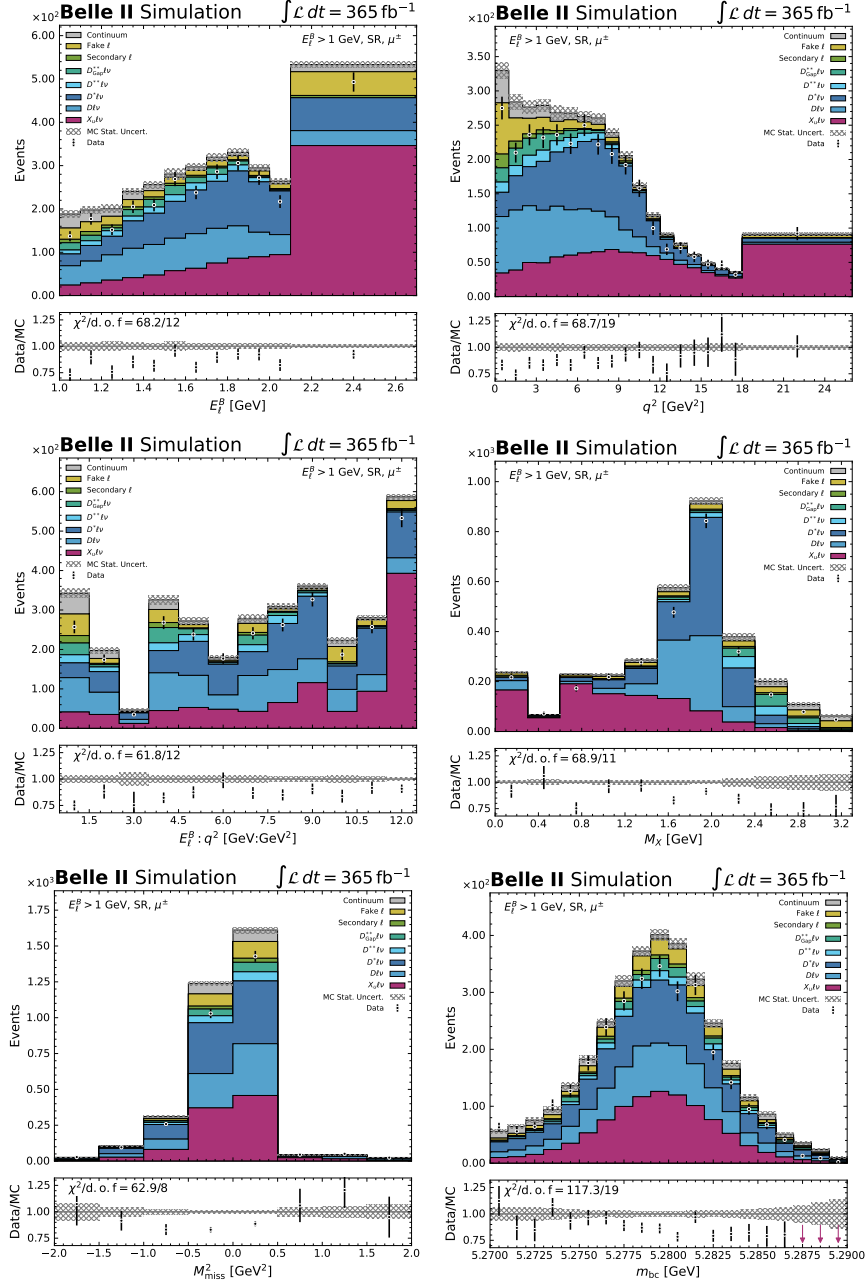


FIG. 163: Signal region distributions $X_u \mu \nu$.

2205 **2. Fit 3 region: $E_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV**

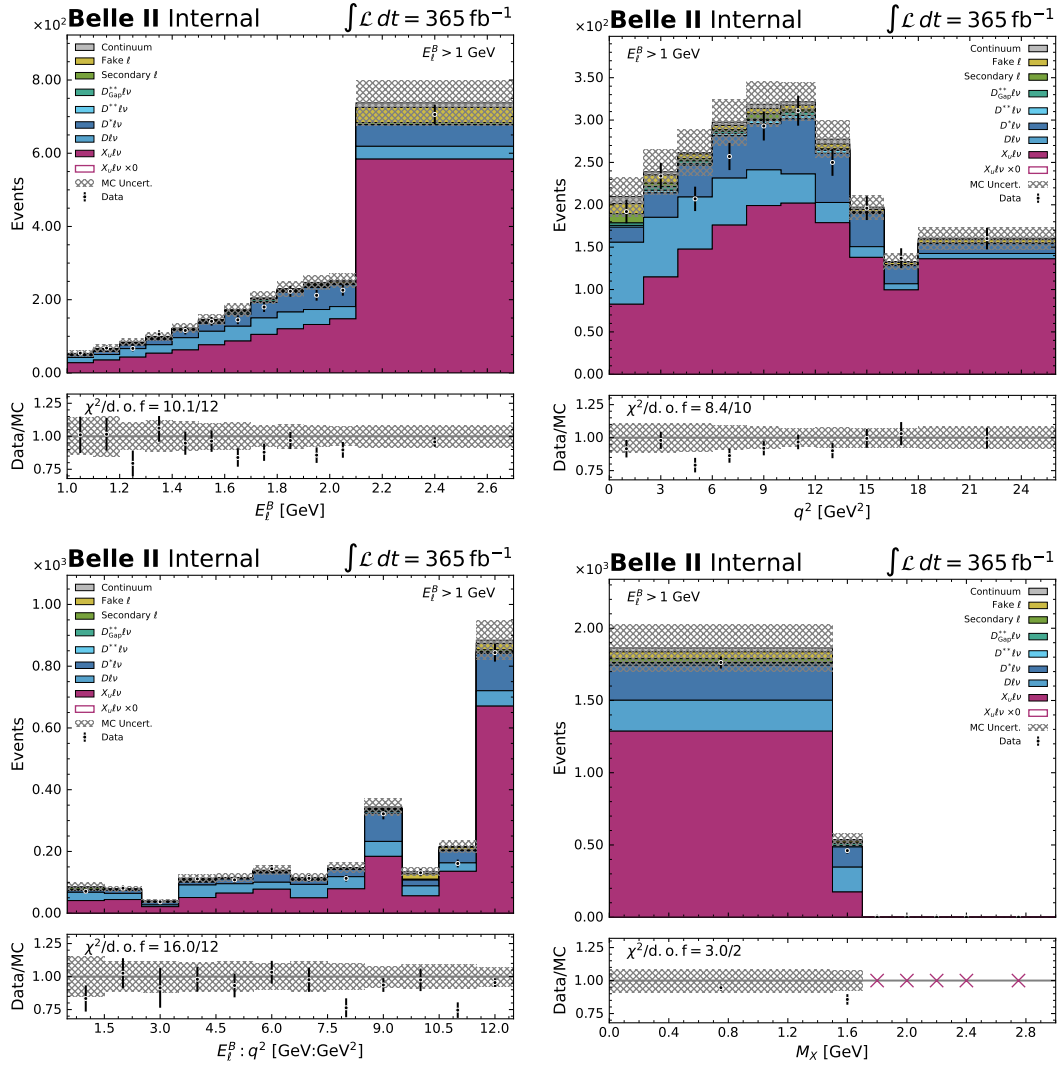


FIG. 164: Signal region distributions for fit 3.

2206 **3. Fit 5 region: $E_\ell^B > 1.0$ GeV, $M_X < 1.7$ GeV**

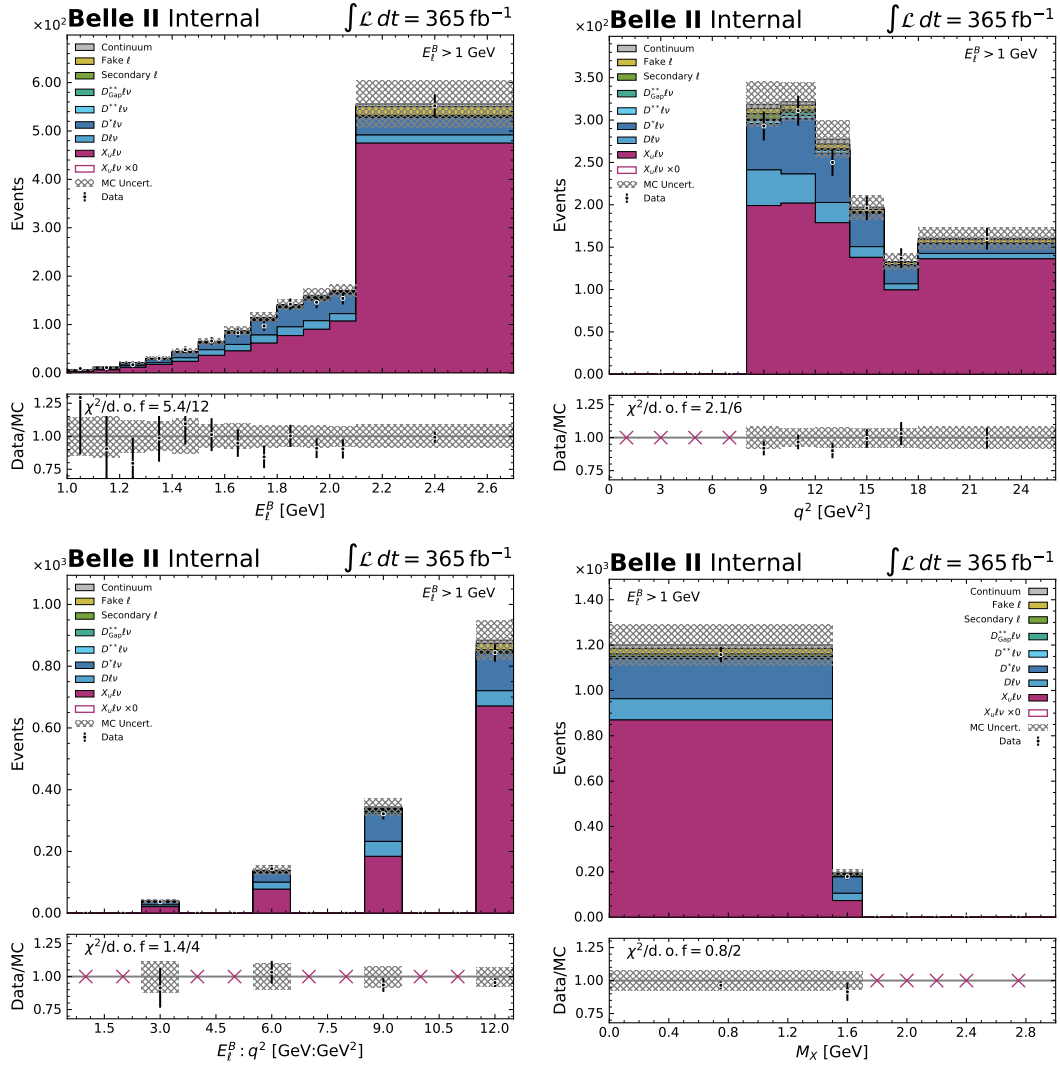


FIG. 165: Signal region distributions for fit 5.

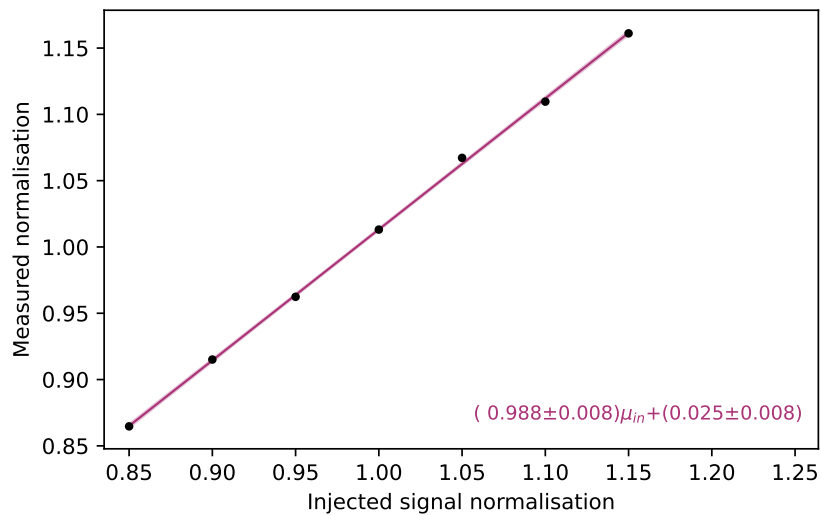


FIG. 166: Distribution of output vs input normalisations of the signal injected $\text{CR}_{K,low} - \text{CR}_{K,high}$ fit setup described in the text.

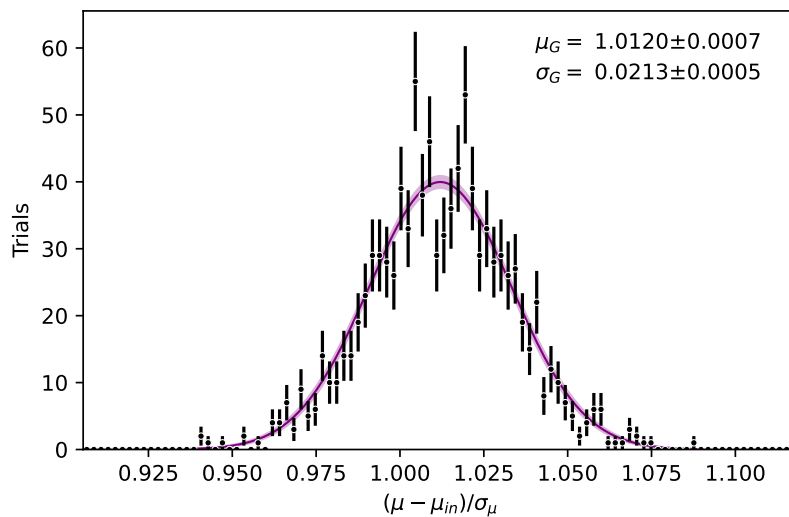


FIG. 167: Distribution of output normalisations with 1,000 toys created by varying the data yields in $\text{CR}_{K,high}$ within their statistical uncertainties. The normalisation of the signal injected in data is chosen to be 1.

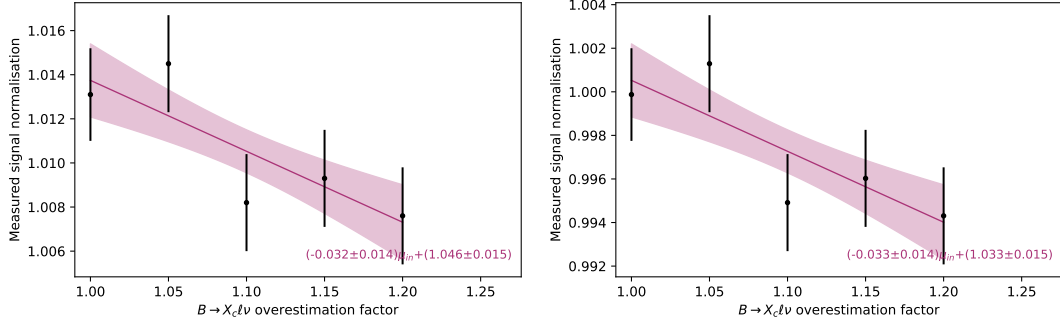


FIG. 168: Distribution of output signal normalisations vs $X_c \ell \nu$ overestimation factor in the signal injected $\text{CR}_{K,low} - \text{CR}_{K,high}$ fit setup described in the text. In the right-hand plot, the first source of bias described above has been corrected for. The normalisation of the injected signal is equal to 1.

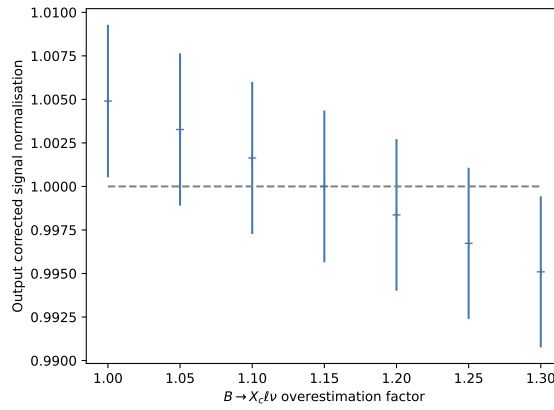


FIG. 169: Output signal normalisation corrected by the two corrections described in the text vs different $X_c \ell \nu$ overestimation factors. The uncertainty is the one added for the second source of bias.

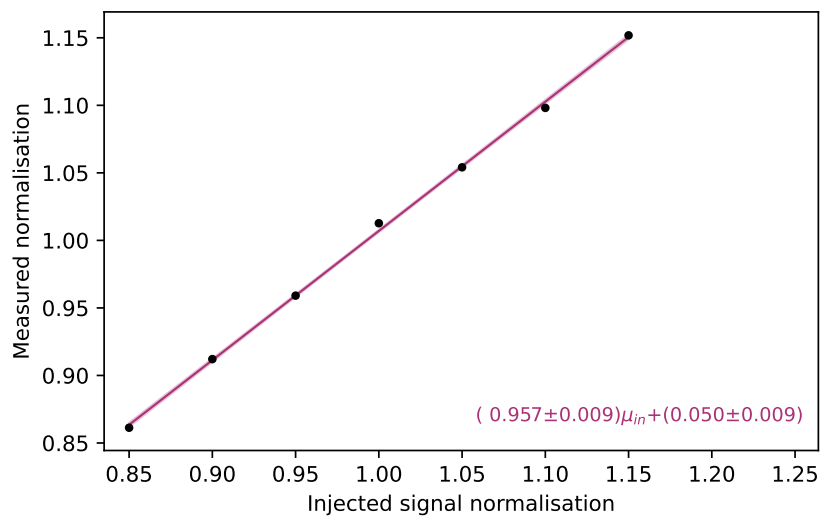


FIG. 170: Distribution of output vs input normalisations of the signal injected $\text{CR}_{K,low} - \text{CR}_{K,high}$ fit setup described in the text.

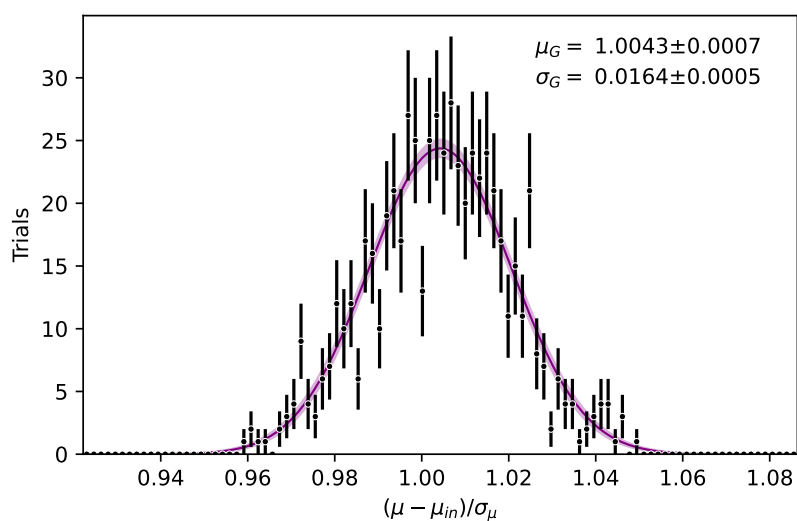


FIG. 171: Distribution of output normalisations with 1,000 toys created by varying the data yields in $\text{CR}_{K,high}$ within their statistical uncertainties. The normalisation of the signal injected in data is chosen to be 1.

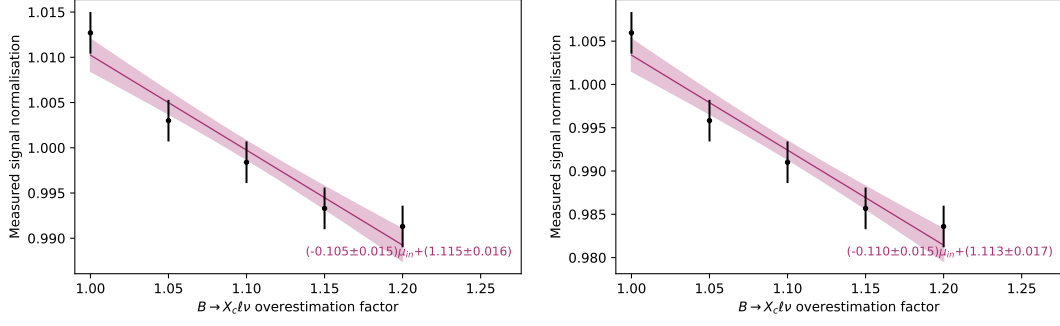


FIG. 172: Distribution of output signal normalisations vs $X_c \ell \nu$ overestimation factor in the signal injected $\text{CR}_{K,low} - \text{CR}_{K,high}$ fit setup described in the text. In the right-hand plot, the first source of bias described above has been corrected for. The normalisation of the injected signal is equal to 1.

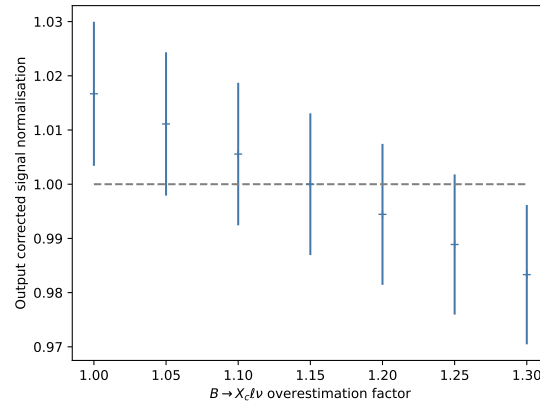


FIG. 173: Output signal normalisation corrected by the two corrections described in the text vs different $X_c \ell \nu$ overestimation factors. The uncertainty is the one added for the second source of bias.

-
- [1] S. Choudhury et al., Belle Collaboration, *Measurement of the B^+/B^0 production ratio in e^+e^- collisions at the $\Upsilon(4S)$ resonance using $B \rightarrow J/\psi(\ell\ell)K$ decays at Belle*, Phys. Rev. D **107** (Feb, 2023) L031102. <https://link.aps.org/doi/10.1103/PhysRevD.107.L031102>.
- [2] M. Hohmann et al., *Inclusive search for the Weak Annihilation contribution to Charmless Semileptonic B Decays at Belle II*, Belle II Intern. Note (2023) .
<https://docs.belle2.org/record/3927/files/BELLE2-NOTE-PH-2023-060.pdf>.
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